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Artificial Intelligence Risk Management Framework (AI RMF 1.0)

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Update Schedule and Versions

The Artificial Intelligence Risk Management Framework (AI RMF) is intended to be a living document.

NIST will review the content and usefulness of the Framework regularly to determine if an update is appropriate; a review with formal input from the AI community is expected to take place no later than 2028. The Framework will employ a two-number versioning system to track and identify major and minor changes. The first number will represent the generation of the AI RMF and its companion documents (e.g., 1.0) and will change only with major revisions. Minor revisions will be tracked using “n” after the generation number (e.g., 1.1). All changes will be tracked using a Version Control Table which identifies the history, including version number, date of change, and description of change. NIST plans to update the AI RMF Playbook frequently. Comments on the AI RMF Playbook may be sent via email to AIframework@nist.gov at any time and will be reviewed and integrated on a semi-annual basis.

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Executive Summary

Artificial intelligence (AI) technologies have significant potential to transform society and people's lives – from commerce and health to transportation and cybersecurity to the environment and our planet. AI technologies can drive inclusive economic growth and support scientific advancements that improve the conditions of our world. AI technologies, however, also pose risks that can negatively impact individuals, groups, organizations, communities, society, the environment, and the planet. Like risks for other types of technology, AI risks can emerge in a variety of ways and can be characterized as long- or short-term, high- or low-probability, systemic or localized, and high- or low-impact.

The AI RMF refers to an *AI system* as an engineered or machine-based system that can, for a given set of objectives, generate outputs such as predictions, recommendations, or decisions influencing real or virtual environments. AI systems are designed to operate with varying levels of autonomy (Adapted from: OECD Recommendation on AI:2019; ISO/IEC 22989:2022).

While there are myriad standards and best practices to help organizations mitigate the risks of traditional software or information-based systems, the risks posed by AI systems are in many ways unique (See Appendix B). AI systems, for example, may be trained on data that can change over time, sometimes significantly and unexpectedly, affecting system functionality and trustworthiness in ways that are hard to understand. AI systems and the contexts in which they are deployed are frequently complex, making it difficult to detect and respond to failures when they occur. AI systems are inherently socio-technical in nature, meaning they are influenced by societal dynamics and human behavior. AI risks – and benefits – can emerge from the interplay of technical aspects combined with societal factors related to how a system is used, its interactions with other AI systems, who operates it, and the social context in which it is deployed.

These risks make AI a uniquely challenging technology to deploy and utilize both for organizations and within society. Without proper controls, AI systems can amplify, perpetuate, or exacerbate inequitable or undesirable outcomes for individuals and communities. With proper controls, AI systems can mitigate and manage inequitable outcomes.

AI risk management is a key component of responsible development and use of AI systems. Responsible AI practices can help align the decisions about AI system design, development, and uses with intended aim and values. Core concepts in responsible AI emphasize human centrality, social responsibility, and sustainability. AI risk management can drive responsible uses and practices by prompting organizations and their internal teams who design, develop, and deploy AI to think more critically about context and potential or unexpected negative and positive impacts. Understanding and managing the risks of AI systems will help to enhance trustworthiness, and in turn, cultivate public trust.

Social responsibility can refer to the organization’s responsibility “for the impacts of its decisions and activities on society and the environment through transparent and ethical behavior” (ISO 26000:2010). *Sustainability* refers to the “state of the global system, including environmental, social, and economic aspects, in which the needs of the present are met without compromising the ability of future generations to meet their own needs” (ISO/IEC TR 24368:2022). Responsible AI is meant to result in technology that is also equitable and accountable. The expectation is that organizational practices are carried out in accord with “*professional responsibility*,” defined by ISO as an approach that “aims to ensure that professionals who design, develop, or deploy AI systems and applications or AI-based products or systems, recognize their unique position to exert influence on people, society, and the future of AI” (ISO/IEC TR 24368:2022).

As directed by the National Artificial Intelligence Initiative Act of 2020 (P.L. 116-283), the goal of the AI RMF is to offer a resource to the organizations designing, developing, deploying, or using AI systems to help manage the many risks of AI and promote trustworthy and responsible development and use of AI systems. The Framework is intended to be **voluntary**, rights-preserving, non-sector-specific, and use-case agnostic, providing flexibility to organizations of all sizes and in all sectors and throughout society to implement the approaches in the Framework.

The Framework is designed to equip organizations and individuals – referred to here as *AI actors* – with approaches that increase the trustworthiness of AI systems, and to help foster the responsible design, development, deployment, and use of AI systems over time. AI actors are defined by the Organisation for Economic Co-operation and Development (OECD) as “those who play an active role in the AI system lifecycle, including organizations and individuals that deploy or operate AI” [OECD (2019) Artificial Intelligence in Society—OECD iLibrary] (See Appendix A).

The AI RMF is intended to be practical, to adapt to the AI landscape as AI technologies continue to develop, and to be operationalized by organizations in varying degrees and capacities so society can benefit from AI while also being protected from its potential harms.

The Framework and supporting resources will be updated, expanded, and improved based on evolving technology, the standards landscape around the world, and AI community experience and feedback. NIST will continue to align the AI RMF and related guidance with applicable international standards, guidelines, and practices. As the AI RMF is put into use, additional lessons will be learned to inform future updates and additional resources.

The Framework is divided into two parts. Part 1 discusses how organizations can frame the risks related to AI and describes the intended audience. Next, AI risks and trustworthiness are analyzed, outlining the characteristics of trustworthy AI systems, which include

valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy enhanced, and fair with their harmful biases managed.

Part 2 comprises the “Core” of the Framework. It describes four specific functions to help organizations address the risks of AI systems in practice. These functions – **GOVERN**, **MAP**, **MEASURE**, and **MANAGE** – are broken down further into categories and subcategories. While **GOVERN** applies to all stages of organizations’ AI risk management processes and procedures, the **MAP**, **MEASURE**, and **MANAGE** functions can be applied in AI system-specific contexts and at specific stages of the AI lifecycle.

Additional resources related to the Framework are included in the AI RMF Playbook, which is available via the NIST AI RMF website:

<https://www.nist.gov/itl/ai-risk-management-framework>.

Development of the AI RMF by NIST in collaboration with the private and public sectors is directed and consistent with its broader AI efforts called for by [the National AI Initiative Act of 2020](#), [the National Security Commission on Artificial Intelligence recommendations](#), and [the Plan for Federal Engagement in Developing Technical Standards and Related Tools](#). Engagement with the AI community during this Framework’s development – via responses to a formal Request for Information, three widely attended workshops, public comments on a concept paper and two drafts of the Framework, discussions at multiple public forums, and many small group meetings – has informed development of the AI RMF 1.0 as well as AI research and development and evaluation conducted by NIST and others. Priority research and additional guidance that will enhance this Framework will be captured in an associated AI Risk Management Framework Roadmap to which NIST and the broader community can contribute.

Part 1: Foundational Information

1. Framing Risk

AI risk management offers a path to minimize potential negative impacts of AI systems, such as threats to civil liberties and rights, while also providing opportunities to maximize positive impacts. Addressing, documenting, and managing AI risks and potential negative impacts effectively can lead to more trustworthy AI systems.

1.1 Understanding and Addressing Risks, Impacts, and Harms

In the context of the AI RMF, *risk* refers to the composite measure of an event’s probability of occurring and the magnitude or degree of the consequences of the corresponding event. The impacts, or consequences, of AI systems can be positive, negative, or both and can result in opportunities or threats (Adapted from: ISO 31000:2018). When considering the negative impact of a potential event, risk is a function of 1) the negative impact, or magnitude of harm, that would arise if the circumstance or event occurs and 2) the likelihood of occurrence (Adapted from: OMB Circular A-130:2016). Negative impact or harm can be experienced by individuals, groups, communities, organizations, society, the environment, and the planet.

“Risk management refers to coordinated activities to direct and control an organization with regard to risk” (Source: ISO 31000:2018).

While risk management processes generally address negative impacts, this Framework offers approaches to minimize anticipated negative impacts of AI systems *and* identify opportunities to maximize positive impacts. Effectively managing the risk of potential harms could lead to more trustworthy AI systems and unleash potential benefits to people (individuals, communities, and society), organizations, and systems/ecosystems. Risk management can enable AI developers and users to understand impacts and account for the inherent limitations and uncertainties in their models and systems, which in turn can improve overall system performance and trustworthiness and the likelihood that AI technologies will be used in ways that are beneficial.

The AI RMF is designed to address new risks as they emerge. This flexibility is particularly important where impacts are not easily foreseeable and applications are evolving. While some AI risks and benefits are well-known, it can be challenging to assess negative impacts and the degree of harms. Figure 1 provides examples of potential harms that can be related to AI systems.

AI risk management efforts should consider that humans may assume that AI systems work – and work well – in *all* settings. For example, whether correct or not, AI systems are often perceived as being more objective than humans or as offering greater capabilities than general software.



Fig. 1. Examples of potential harms related to AI systems. Trustworthy AI systems and their responsible use can mitigate negative risks and contribute to benefits for people, organizations, and ecosystems.

1.2 Challenges for AI Risk Management

Several challenges are described below. They should be taken into account when managing risks in pursuit of AI trustworthiness.

1.2.1 Risk Measurement

AI risks or failures that are not well-defined or adequately understood are difficult to measure quantitatively or qualitatively. The inability to appropriately measure AI risks does not imply that an AI system necessarily poses either a high or low risk. Some risk measurement challenges include:

Risks related to third-party software, hardware, and data: Third-party data or systems can accelerate research and development and facilitate technology transition. They also may complicate risk measurement. Risk can emerge both from third-party data, software or hardware itself and how it is used. Risk metrics or methodologies used by the organization developing the AI system may not align with the risk metrics or methodologies used by the organization *deploying or operating* the system. Also, the organization developing the AI system may not be transparent about the risk metrics or methodologies it used. Risk measurement and management can be complicated by how customers use or integrate third-party data or systems into AI products or services, particularly without sufficient internal governance structures and technical safeguards. Regardless, all parties and AI actors should manage risk in the AI systems they develop, deploy, or use as standalone or integrated components.

Tracking emergent risks: Organizations' risk management efforts will be enhanced by identifying and tracking emergent risks and considering techniques for measuring them.

AI system impact assessment approaches can help AI actors understand potential impacts or harms within specific contexts.

Availability of reliable metrics: The current lack of consensus on robust and verifiable measurement methods for risk and trustworthiness, and applicability to different AI use cases, is an AI risk measurement challenge. Potential pitfalls when seeking to measure negative risk or harms include the reality that development of metrics is often an institutional endeavor and may inadvertently reflect factors unrelated to the underlying impact. In addition, measurement approaches can be oversimplified, gamed, lack critical nuance, become relied upon in unexpected ways, or fail to account for differences in affected groups and contexts.

Approaches for measuring impacts on a population work best if they recognize that contexts matter, that harms may affect varied groups or sub-groups differently, and that communities or other sub-groups who may be harmed are not always direct users of a system.

Risk at different stages of the AI lifecycle: Measuring risk at an earlier stage in the AI lifecycle may yield different results than measuring risk at a later stage; some risks may be latent at a given point in time and may increase as AI systems adapt and evolve. Furthermore, different AI actors across the AI lifecycle can have different risk perspectives. For example, an AI developer who makes AI software available, such as pre-trained models, can have a different risk perspective than an AI actor who is responsible for deploying that pre-trained model in a specific use case. Such deployers may not recognize that their particular uses could entail risks which differ from those perceived by the initial developer. All involved AI actors share responsibilities for designing, developing, and deploying a trustworthy AI system that is fit for purpose.

Risk in real-world settings: While measuring AI risks in a laboratory or a controlled environment may yield important insights pre-deployment, these measurements may differ from risks that emerge in operational, real-world settings.

Inscrutability: Inscrutable AI systems can complicate risk measurement. Inscrutability can be a result of the opaque nature of AI systems (limited explainability or interpretability), lack of transparency or documentation in AI system development or deployment, or inherent uncertainties in AI systems.

Human baseline: Risk management of AI systems that are intended to augment or replace human activity, for example decision making, requires some form of baseline metrics for comparison. This is difficult to systematize since AI systems carry out different tasks – and perform tasks differently – than humans.

1.2.2 Risk Tolerance

While the AI RMF can be used to prioritize risk, it does not prescribe risk tolerance. *Risk tolerance* refers to the organization's or AI actor's (see Appendix A) readiness to bear the risk in order to achieve its objectives. Risk tolerance can be influenced by legal or regulatory requirements (Adapted from: ISO GUIDE 73). Risk tolerance and the level of risk that is acceptable to organizations or society are highly contextual and application and use-case specific. Risk tolerances can be influenced by policies and norms established by AI system owners, organizations, industries, communities, or policy makers. Risk tolerances are likely to change over time as AI systems, policies, and norms evolve. Different organizations may have varied risk tolerances due to their particular organizational priorities and resource considerations.

Emerging knowledge and methods to better inform harm/cost-benefit tradeoffs will continue to be developed and debated by businesses, governments, academia, and civil society. To the extent that challenges for specifying AI risk tolerances remain unresolved, there may be contexts where a risk management framework is not yet readily applicable for mitigating negative AI risks.

The Framework is intended to be flexible and to augment existing risk practices which should align with applicable laws, regulations, and norms. Organizations should follow existing regulations and guidelines for risk criteria, tolerance, and response established by organizational, domain, discipline, sector, or professional requirements. Some sectors or industries may have established definitions of harm or established documentation, reporting, and disclosure requirements. Within sectors, risk management may depend on existing guidelines for specific applications and use case settings. Where established guidelines do not exist, organizations should define reasonable risk tolerance. Once tolerance is defined, this AI RMF can be used to manage risks and to document risk management processes.

1.2.3 Risk Prioritization

Attempting to eliminate negative risk entirely can be counterproductive in practice because not all incidents and failures can be eliminated. Unrealistic expectations about risk may lead organizations to allocate resources in a manner that makes risk triage inefficient or impractical or wastes scarce resources. A risk management culture can help organizations recognize that not all AI risks are the same, and resources can be allocated purposefully. Actionable risk management efforts lay out clear guidelines for assessing trustworthiness of each AI system an organization develops or deploys. Policies and resources should be prioritized based on the assessed risk level and potential impact of an AI system. The extent to which an AI system may be customized or tailored to the specific context of use by the AI deployer can be a contributing factor.

When applying the AI RMF, risks which the organization determines to be highest for the AI systems within a given context of use call for the most urgent prioritization and most thorough risk management process. In cases where an AI system presents unacceptable negative risk levels – such as where significant negative impacts are imminent, severe harms are actually occurring, or catastrophic risks are present – development and deployment should cease in a safe manner until risks can be sufficiently managed. If an AI system’s development, deployment, and use cases are found to be low-risk in a specific context, that may suggest potentially lower prioritization.

Risk prioritization may differ between AI systems that are designed or deployed to directly interact with humans as compared to AI systems that are not. Higher initial prioritization may be called for in settings where the AI system is trained on large datasets comprised of sensitive or protected data such as personally identifiable information, or where the outputs of the AI systems have direct or indirect impact on humans. AI systems designed to interact only with computational systems and trained on non-sensitive datasets (for example, data collected from the physical environment) may call for lower initial prioritization. Nonetheless, regularly assessing and prioritizing risk based on context remains important because non-human-facing AI systems can have downstream safety or social implications.

Residual risk – defined as risk remaining after risk treatment (Source: ISO GUIDE 73) – directly impacts end users or affected individuals and communities. Documenting residual risks will call for the system provider to fully consider the risks of deploying the AI product and will inform end users about potential negative impacts of interacting with the system.

1.2.4 Organizational Integration and Management of Risk

AI risks should not be considered in isolation. Different AI actors have different responsibilities and awareness depending on their roles in the lifecycle. For example, organizations developing an AI system often will not have information about how the system may be used. AI risk management should be integrated and incorporated into broader enterprise risk management strategies and processes. Treating AI risks along with other critical risks, such as cybersecurity and privacy, will yield a more integrated outcome and organizational efficiencies.

The AI RMF may be utilized along with related guidance and frameworks for managing AI system risks or broader enterprise risks. Some risks related to AI systems are common across other types of software development and deployment. Examples of overlapping risks include: privacy concerns related to the use of underlying data to train AI systems; the energy and environmental implications associated with resource-heavy computing demands; security concerns related to the confidentiality, integrity, and availability of the system and its training and output data; and general security of the underlying software and hardware for AI systems.

Organizations need to establish and maintain the appropriate accountability mechanisms, roles and responsibilities, culture, and incentive structures for risk management to be effective. Use of the AI RMF alone will not lead to these changes or provide the appropriate incentives. Effective risk management is realized through organizational commitment at senior levels and may require cultural change within an organization or industry. In addition, small to medium-sized organizations managing AI risks or implementing the AI RMF may face different challenges than large organizations, depending on their capabilities and resources.

2. Audience

Identifying and managing AI risks and potential impacts – both positive and negative – requires a broad set of perspectives and actors across the AI lifecycle. Ideally, AI actors will represent a diversity of experience, expertise, and backgrounds and comprise demographically and disciplinarily diverse teams. The AI RMF is intended to be used by AI actors across the AI lifecycle and dimensions.

The OECD has developed a framework for classifying AI lifecycle activities according to five key socio-technical dimensions, each with properties relevant for AI policy and governance, including risk management [OECD (2022) OECD Framework for the Classification of AI systems — OECD Digital Economy Papers]. Figure 2 shows these dimensions, slightly modified by NIST for purposes of this framework. The NIST modification highlights the importance of test, evaluation, verification, and validation (TEVV) processes throughout an AI lifecycle and generalizes the operational context of an AI system.

AI dimensions displayed in Figure 2 are the Application Context, Data and Input, AI Model, and Task and Output. AI actors involved in these dimensions who perform or manage the design, development, deployment, evaluation, and use of AI systems and drive AI risk management efforts are the *primary* AI RMF audience.

Representative AI actors across the lifecycle dimensions are listed in Figure 3 and described in detail in Appendix A. Within the AI RMF, all AI actors work together to manage risks and achieve the goals of trustworthy and responsible AI. AI actors with TEVV-specific expertise are integrated throughout the AI lifecycle and are especially likely to benefit from the Framework. Performed regularly, TEVV tasks can provide insights relative to technical, societal, legal, and ethical standards or norms, and can assist with anticipating impacts and assessing and tracking emergent risks. As a regular process within an AI lifecycle, TEVV allows for both mid-course remediation and post-hoc risk management.

The People & Planet dimension at the center of Figure 2 represents human rights and the broader well-being of society and the planet. The AI actors in this dimension comprise a separate AI RMF audience who *informs* the primary audience. These AI actors may include trade associations, standards developing organizations, researchers, advocacy groups,



Fig. 2. Lifecycle and Key Dimensions of an AI System. Modified from OECD (2022) [OECD Framework for the Classification of AI systems — OECD Digital Economy Papers](#). The two inner circles show AI systems' key dimensions and the outer circle shows AI lifecycle stages. Ideally, risk management efforts start with the Plan and Design function in the application context and are performed throughout the AI system lifecycle. See Figure 3 for representative AI actors.

environmental groups, civil society organizations, end users, and potentially impacted individuals and communities. These actors can:

- assist in providing context and understanding potential and actual impacts;
- be a source of formal or quasi-formal norms and guidance for AI risk management;
- designate boundaries for AI operation (technical, societal, legal, and ethical); and
- promote discussion of the tradeoffs needed to balance societal values and priorities related to civil liberties and rights, equity, the environment and the planet, and the economy.

Successful risk management depends upon a sense of collective responsibility among AI actors shown in Figure 3. The AI RMF functions, described in Section 5, require diverse perspectives, disciplines, professions, and experiences. Diverse teams contribute to more open sharing of ideas and assumptions about the purposes and functions of technology – making these implicit aspects more explicit. This broader collective perspective creates opportunities for surfacing problems and identifying existing and emergent risks.



Fig. 3. AI actors across AI lifecycle stages. See Appendix A for detailed descriptions of AI actor tasks, including details about testing, evaluation, verification, and validation tasks. Note that AI actors in the AI Model dimension (Figure 2) are separated as a best practice, with those building and using the models separated from those verifying and validating the models.

3. AI Risks and Trustworthiness

For AI systems to be trustworthy, they often need to be responsive to a multiplicity of criteria that are of value to interested parties. Approaches which enhance AI trustworthiness can reduce negative AI risks. This Framework articulates the following **characteristics** of trustworthy AI and offers guidance for addressing them. Characteristics of trustworthy AI systems include: **valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, and fair with harmful bias managed**. Creating trustworthy AI requires balancing each of these characteristics based on the AI system's context of use. While all characteristics are socio-technical system attributes, accountability and transparency also relate to the processes and activities internal to an AI system and its external setting. Neglecting these characteristics can increase the probability and magnitude of negative consequences.



Fig. 4. Characteristics of trustworthy AI systems. Valid & Reliable is a necessary condition of trustworthiness and is shown as the base for other trustworthiness characteristics. Accountable & Transparent is shown as a vertical box because it relates to all other characteristics.

Trustworthiness characteristics (shown in Figure 4) are inextricably tied to social and organizational behavior, the datasets used by AI systems, selection of AI models and algorithms and the decisions made by those who build them, and the interactions with the humans who provide insight from and oversight of such systems. Human judgment should be employed when deciding on the specific metrics related to AI trustworthiness characteristics and the precise threshold values for those metrics.

Addressing AI trustworthiness characteristics individually will not ensure AI system trustworthiness; tradeoffs are usually involved, rarely do all characteristics apply in every setting, and some will be more or less important in any given situation. Ultimately, trustworthiness is a social concept that ranges across a spectrum and is only as strong as its weakest characteristics.

When managing AI risks, organizations can face difficult decisions in balancing these characteristics. For example, in certain scenarios tradeoffs may emerge between optimizing for interpretability and achieving privacy. In other cases, organizations might face a tradeoff between predictive accuracy and interpretability. Or, under certain conditions such as data sparsity, privacy-enhancing techniques can result in a loss in accuracy, affecting decisions

about fairness and other values in certain domains. Dealing with tradeoffs requires taking into account the decision-making context. These analyses can highlight the existence and extent of tradeoffs between different measures, but they do not answer questions about how to navigate the tradeoff. Those depend on the values at play in the relevant *context* and should be resolved in a manner that is both transparent and appropriately justifiable.

There are multiple approaches for enhancing contextual awareness in the AI lifecycle. For example, subject matter experts can assist in the evaluation of TEVV findings and work with product and deployment teams to align TEVV parameters to requirements and deployment conditions. When properly resourced, increasing the breadth and diversity of input from interested parties and relevant AI actors throughout the AI lifecycle can enhance opportunities for informing contextually sensitive evaluations, and for identifying AI system benefits and positive impacts. These practices can increase the likelihood that risks arising in social contexts are managed appropriately.

Understanding and treatment of trustworthiness characteristics depends on an AI actor's particular role within the AI lifecycle. For any given AI system, an AI designer or developer may have a different perception of the characteristics than the deployer.

Trustworthiness characteristics explained in this document influence each other. Highly secure but unfair systems, accurate but opaque and uninterpretable systems, and inaccurate but secure, privacy-enhanced, and transparent systems are all undesirable. A comprehensive approach to risk management calls for balancing tradeoffs among the trustworthiness characteristics. It is the joint responsibility of all AI actors to determine whether AI technology is an appropriate or necessary tool for a given context or purpose, and how to use it responsibly. The decision to commission or deploy an AI system should be based on a contextual assessment of trustworthiness characteristics and the relative risks, impacts, costs, and benefits, and informed by a broad set of interested parties.

3.1 Valid and Reliable

Validation is the “confirmation, through the provision of objective evidence, that the requirements for a specific intended use or application have been fulfilled” (Source: ISO 9000:2015). Deployment of AI systems which are inaccurate, unreliable, or poorly generalized to data and settings beyond their training creates and increases negative AI risks and reduces trustworthiness.

Reliability is defined in the same standard as the “ability of an item to perform as required, without failure, for a given time interval, under given conditions” (Source: ISO/IEC TS 5723:2022). Reliability is a goal for overall correctness of AI system operation under the conditions of expected use and over a given period of time, including the entire lifetime of the system.

Accuracy and robustness contribute to the validity and trustworthiness of AI systems, and can be in tension with one another in AI systems.

Accuracy is defined by ISO/IEC TS 5723:2022 as “closeness of results of observations, computations, or estimates to the true values or the values accepted as being true.” Measures of accuracy should consider computational-centric measures (e.g., false positive and false negative rates), human-AI teaming, and demonstrate external validity (generalizable beyond the training conditions). Accuracy measurements should always be paired with clearly defined and realistic test sets – that are representative of conditions of expected use – and details about test methodology; these should be included in associated documentation. Accuracy measurements may include disaggregation of results for different data segments.

Robustness or *generalizability* is defined as the “ability of a system to maintain its level of performance under a variety of circumstances” (Source: ISO/IEC TS 5723:2022). Robustness is a goal for appropriate system functionality in a broad set of conditions and circumstances, including uses of AI systems not initially anticipated. Robustness requires not only that the system perform exactly as it does under expected uses, but also that it should perform in ways that minimize potential harms to people if it is operating in an unexpected setting.

Validity and reliability for deployed AI systems are often assessed by ongoing testing or monitoring that confirms a system is performing as intended. Measurement of validity, accuracy, robustness, and reliability contribute to trustworthiness and should take into consideration that certain types of failures can cause greater harm. AI risk management efforts should prioritize the minimization of potential negative impacts, and may need to include human intervention in cases where the AI system cannot detect or correct errors.

3.2 Safe

AI systems should “not under defined conditions, lead to a state in which human life, health, property, or the environment is endangered” (Source: ISO/IEC TS 5723:2022). Safe operation of AI systems is improved through:

- responsible design, development, and deployment practices;
- clear information to deployers on responsible use of the system;
- responsible decision-making by deployers and end users; and
- explanations and documentation of risks based on empirical evidence of incidents.

Different types of safety risks may require tailored AI risk management approaches based on context and the severity of potential risks presented. Safety risks that pose a potential risk of serious injury or death call for the most urgent prioritization and most thorough risk management process.

Employing safety considerations during the lifecycle and starting as early as possible with planning and design can prevent failures or conditions that can render a system dangerous. Other practical approaches for AI safety often relate to rigorous simulation and in-domain testing, real-time monitoring, and the ability to shut down, modify, or have human intervention into systems that deviate from intended or expected functionality.

AI safety risk management approaches should take cues from efforts and guidelines for safety in fields such as transportation and healthcare, and align with existing sector- or application-specific guidelines or standards.

3.3 Secure and Resilient

AI systems, as well as the ecosystems in which they are deployed, may be said to be *resilient* if they can withstand unexpected adverse events or unexpected changes in their environment or use – or if they can maintain their functions and structure in the face of internal and external change and degrade safely and gracefully when this is necessary (Adapted from: ISO/IEC TS 5723:2022). Common security concerns relate to adversarial examples, data poisoning, and the exfiltration of models, training data, or other intellectual property through AI system endpoints. AI systems that can maintain confidentiality, integrity, and availability through protection mechanisms that prevent unauthorized access and use may be said to be *secure*. Guidelines in the [NIST Cybersecurity Framework](#) and [Risk Management Framework](#) are among those which are applicable here.

Security and resilience are related but distinct characteristics. While resilience is the ability to return to normal function after an unexpected adverse event, security includes resilience but also encompasses protocols to avoid, protect against, respond to, or recover from attacks. Resilience relates to robustness and goes beyond the provenance of the data to encompass unexpected or adversarial use (or abuse or misuse) of the model or data.

3.4 Accountable and Transparent

Trustworthy AI depends upon accountability. Accountability presupposes transparency. *Transparency* reflects the extent to which information about an AI system and its outputs is available to individuals interacting with such a system – regardless of whether they are even aware that they are doing so. Meaningful transparency provides access to appropriate levels of information based on the stage of the AI lifecycle and tailored to the role or knowledge of AI actors or individuals interacting with or using the AI system. By promoting higher levels of understanding, transparency increases confidence in the AI system.

This characteristic's scope spans from design decisions and training data to model training, the structure of the model, its intended use cases, and how and when deployment, post-deployment, or end user decisions were made and by whom. Transparency is often necessary for actionable redress related to AI system outputs that are incorrect or otherwise lead to negative impacts. Transparency should consider human-AI interaction: for exam-

ple, how a human operator or user is notified when a potential or actual adverse outcome caused by an AI system is detected. A transparent system is not necessarily an accurate, privacy-enhanced, secure, or fair system. However, it is difficult to determine whether an opaque system possesses such characteristics, and to do so over time as complex systems evolve.

The role of AI actors should be considered when seeking accountability for the outcomes of AI systems. The relationship between risk and accountability associated with AI and technological systems more broadly differs across cultural, legal, sectoral, and societal contexts. When consequences are severe, such as when life and liberty are at stake, AI developers and deployers should consider proportionally and proactively adjusting their transparency and accountability practices. Maintaining organizational practices and governing structures for harm reduction, like risk management, can help lead to more accountable systems.

Measures to enhance transparency and accountability should also consider the impact of these efforts on the implementing entity, including the level of necessary resources and the need to safeguard proprietary information.

Maintaining the provenance of training data and supporting attribution of the AI system's decisions to subsets of training data can assist with both transparency and accountability. Training data may also be subject to copyright and should follow applicable intellectual property rights laws.

As transparency tools for AI systems and related documentation continue to evolve, developers of AI systems are encouraged to test different types of transparency tools in cooperation with AI deployers to ensure that AI systems are used as intended.

3.5 Explainable and Interpretable

Explainability refers to a representation of the mechanisms underlying AI systems' operation, whereas *interpretability* refers to the meaning of AI systems' output in the context of their designed functional purposes. Together, explainability and interpretability assist those operating or overseeing an AI system, as well as users of an AI system, to gain deeper insights into the functionality and trustworthiness of the system, including its outputs. The underlying assumption is that perceptions of negative risk stem from a lack of ability to make sense of, or contextualize, system output appropriately. Explainable and interpretable AI systems offer information that will help end users understand the purposes and potential impact of an AI system.

Risk from lack of explainability may be managed by describing how AI systems function, with descriptions tailored to individual differences such as the user's role, knowledge, and skill level. Explainable systems can be debugged and monitored more easily, and they lend themselves to more thorough documentation, audit, and governance.

Risks to interpretability often can be addressed by communicating a description of why an AI system made a particular prediction or recommendation. (See “Four Principles of Explainable Artificial Intelligence” and “Psychological Foundations of Explainability and Interpretability in Artificial Intelligence” found [here](#).)

Transparency, explainability, and interpretability are distinct characteristics that support each other. Transparency can answer the question of “what happened” in the system. Explainability can answer the question of “how” a decision was made in the system. Interpretability can answer the question of “why” a decision was made by the system and its meaning or context to the user.

3.6 Privacy-Enhanced

Privacy refers generally to the norms and practices that help to safeguard human autonomy, identity, and dignity. These norms and practices typically address freedom from intrusion, limiting observation, or individuals’ agency to consent to disclosure or control of facets of their identities (e.g., body, data, reputation). (See [The NIST Privacy Framework: A Tool for Improving Privacy through Enterprise Risk Management](#).)

Privacy values such as anonymity, confidentiality, and control generally should guide choices for AI system design, development, and deployment. Privacy-related risks may influence security, bias, and transparency and come with tradeoffs with these other characteristics. Like safety and security, specific technical features of an AI system may promote or reduce privacy. AI systems can also present new risks to privacy by allowing inference to identify individuals or previously private information about individuals.

Privacy-enhancing technologies (“PETs”) for AI, as well as data minimizing methods such as de-identification and aggregation for certain model outputs, can support design for privacy-enhanced AI systems. Under certain conditions such as data sparsity, privacy-enhancing techniques can result in a loss in accuracy, affecting decisions about fairness and other values in certain domains.

3.7 Fair – with Harmful Bias Managed

Fairness in AI includes concerns for equality and equity by addressing issues such as harmful bias and discrimination. Standards of fairness can be complex and difficult to define because perceptions of fairness differ among cultures and may shift depending on application. Organizations’ risk management efforts will be enhanced by recognizing and considering these differences. Systems in which harmful biases are mitigated are not necessarily fair. For example, systems in which predictions are somewhat balanced across demographic groups may still be inaccessible to individuals with disabilities or affected by the digital divide or may exacerbate existing disparities or systemic biases.

Bias is broader than demographic balance and data representativeness. NIST has identified three major categories of AI bias to be considered and managed: systemic, computational and statistical, and human-cognitive. Each of these can occur in the absence of prejudice, partiality, or discriminatory intent. Systemic bias can be present in AI datasets, the organizational norms, practices, and processes across the AI lifecycle, and the broader society that uses AI systems. Computational and statistical biases can be present in AI datasets and algorithmic processes, and often stem from systematic errors due to non-representative samples. Human-cognitive biases relate to how an individual or group perceives AI system information to make a decision or fill in missing information, or how humans think about purposes and functions of an AI system. Human-cognitive biases are omnipresent in decision-making processes across the AI lifecycle and system use, including the design, implementation, operation, and maintenance of AI.

Bias exists in many forms and can become ingrained in the automated systems that help make decisions about our lives. While bias is not always a negative phenomenon, AI systems can potentially increase the speed and scale of biases and perpetuate and amplify harms to individuals, groups, communities, organizations, and society. Bias is tightly associated with the concepts of transparency as well as fairness in society. (For more information about bias, including the three categories, see NIST Special Publication 1270, [Towards a Standard for Identifying and Managing Bias in Artificial Intelligence](#).)

4. Effectiveness of the AI RMF

Evaluations of AI RMF effectiveness – including ways to measure bottom-line improvements in the trustworthiness of AI systems – will be part of future NIST activities, in conjunction with the AI community.

Organizations and other users of the Framework are encouraged to periodically evaluate whether the AI RMF has improved their ability to manage AI risks, including but not limited to their policies, processes, practices, implementation plans, indicators, measurements, and expected outcomes. NIST intends to work collaboratively with others to develop metrics, methodologies, and goals for evaluating the AI RMF's effectiveness, and to broadly share results and supporting information. Framework users are expected to benefit from:

- enhanced processes for governing, mapping, measuring, and managing AI risk, and clearly documenting outcomes;
- improved awareness of the relationships and tradeoffs among trustworthiness characteristics, socio-technical approaches, and AI risks;
- explicit processes for making go/no-go system commissioning and deployment decisions;
- established policies, processes, practices, and procedures for improving organizational accountability efforts related to AI system risks;
- enhanced organizational culture which prioritizes the identification and management of AI system risks and potential impacts to individuals, communities, organizations, and society;
- better information sharing within and across organizations about risks, decision-making processes, responsibilities, common pitfalls, TEVV practices, and approaches for continuous improvement;
- greater contextual knowledge for increased awareness of downstream risks;
- strengthened engagement with interested parties and relevant AI actors; and
- augmented capacity for TEVV of AI systems and associated risks.

Part 2: Core and Profiles

5. AI RMF Core

The AI RMF Core provides outcomes and actions that enable dialogue, understanding, and activities to manage AI risks and responsibly develop trustworthy AI systems. As illustrated in Figure 5, the Core is composed of four functions: **GOVERN**, **MAP**, **MEASURE**, and **MANAGE**. Each of these high-level functions is broken down into categories and sub-categories. Categories and subcategories are subdivided into specific actions and outcomes. Actions do not constitute a checklist, nor are they necessarily an ordered set of steps.



Fig. 5. Functions organize AI risk management activities at their highest level to govern, map, measure, and manage AI risks. Governance is designed to be a cross-cutting function to inform and be infused throughout the other three functions.

Risk management should be continuous, timely, and performed throughout the AI system lifecycle dimensions. AI RMF Core functions should be carried out in a way that reflects diverse and multidisciplinary perspectives, potentially including the views of AI actors outside the organization. Having a diverse team contributes to more open sharing of ideas and assumptions about purposes and functions of the technology being designed, developed,

deployed, or evaluated – which can create opportunities to surface problems and identify existing and emergent risks.

An online companion resource to the AI RMF, the NIST AI RMF Playbook, is available to help organizations navigate the AI RMF and achieve its outcomes through suggested tactical actions they can apply within their own contexts. Like the AI RMF, the Playbook is voluntary and organizations can utilize the suggestions according to their needs and interests. Playbook users can create tailored guidance selected from suggested material for their own use and contribute their suggestions for sharing with the broader community. Along with the AI RMF, the Playbook is part of the NIST Trustworthy and Responsible AI Resource Center.

Framework users may apply these functions as best suits their needs for managing AI risks based on their resources and capabilities. Some organizations may choose to select from among the categories and subcategories; others may choose and have the capacity to apply all categories and subcategories. Assuming a governance structure is in place, functions may be performed in any order across the AI lifecycle as deemed to add value by a user of the framework. After instituting the outcomes in **GOVERN**, most users of the AI RMF would start with the **MAP** function and continue to **MEASURE** or **MANAGE**. However users integrate the functions, the process should be iterative, with cross-referencing between functions as necessary. Similarly, there are categories and subcategories with elements that apply to multiple functions, or that logically should take place before certain subcategory decisions.

5.1 Govern

The **GOVERN** function:

- cultivates and implements a culture of risk management within organizations designing, developing, deploying, evaluating, or acquiring AI systems;
- outlines processes, documents, and organizational schemes that anticipate, identify, and manage the risks a system can pose, including to users and others across society – and procedures to achieve those outcomes;
- incorporates processes to assess potential impacts;
- provides a structure by which AI risk management functions can align with organizational principles, policies, and strategic priorities;
- connects technical aspects of AI system design and development to organizational values and principles, and enables organizational practices and competencies for the individuals involved in acquiring, training, deploying, and monitoring such systems; and
- addresses full product lifecycle and associated processes, including legal and other issues concerning use of third-party software or hardware systems and data.

GOVERN is a cross-cutting function that is infused throughout AI risk management and enables the other functions of the process. Aspects of **GOVERN**, especially those related to compliance or evaluation, should be integrated into each of the other functions. Attention to governance is a continual and intrinsic requirement for effective AI risk management over an AI system's lifespan and the organization's hierarchy.

Strong governance can drive and enhance internal practices and norms to facilitate organizational risk culture. Governing authorities can determine the overarching policies that direct an organization's mission, goals, values, culture, and risk tolerance. Senior leadership sets the tone for risk management within an organization, and with it, organizational culture. Management aligns the technical aspects of AI risk management to policies and operations. Documentation can enhance transparency, improve human review processes, and bolster accountability in AI system teams.

After putting in place the structures, systems, processes, and teams described in the **GOVERN** function, organizations should benefit from a purpose-driven culture focused on risk understanding and management. It is incumbent on Framework users to continue to execute the **GOVERN** function as knowledge, cultures, and needs or expectations from AI actors evolve over time.

Practices related to governing AI risks are described in the NIST AI RMF Playbook. Table 1 lists the **GOVERN** function's categories and subcategories.

Table 1: Categories and subcategories for the **GOVERN** function.

Categories	Subcategories
GOVERN 1: Policies, processes, procedures, and practices across the organization related to the mapping, measuring, and managing of AI risks are in place, transparent, and implemented effectively.	GOVERN 1.1: Legal and regulatory requirements involving AI are understood, managed, and documented. GOVERN 1.2: The characteristics of trustworthy AI are integrated into organizational policies, processes, procedures, and practices. GOVERN 1.3: Processes, procedures, and practices are in place to determine the needed level of risk management activities based on the organization's risk tolerance. GOVERN 1.4: The risk management process and its outcomes are established through transparent policies, procedures, and other controls based on organizational risk priorities.

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Table 1: Categories and subcategories for the **GOVERN** function. (Continued)

Categories	Subcategories
	GOVERN 1.5: Ongoing monitoring and periodic review of the risk management process and its outcomes are planned and organizational roles and responsibilities clearly defined, including determining the frequency of periodic review. GOVERN 1.6: Mechanisms are in place to inventory AI systems and are resourced according to organizational risk priorities. GOVERN 1.7: Processes and procedures are in place for decommissioning and phasing out AI systems safely and in a manner that does not increase risks or decrease the organization's trustworthiness.
GOVERN 2: Accountability structures are in place so that the appropriate teams and individuals are empowered, responsible, and trained for mapping, measuring, and managing AI risks.	GOVERN 2.1: Roles and responsibilities and lines of communication related to mapping, measuring, and managing AI risks are documented and are clear to individuals and teams throughout the organization. GOVERN 2.2: The organization's personnel and partners receive AI risk management training to enable them to perform their duties and responsibilities consistent with related policies, procedures, and agreements. GOVERN 2.3: Executive leadership of the organization takes responsibility for decisions about risks associated with AI system development and deployment.
GOVERN 3: Workforce diversity, equity, inclusion, and accessibility processes are prioritized in the mapping, measuring, and managing of AI risks throughout the lifecycle.	GOVERN 3.1: Decision-making related to mapping, measuring, and managing AI risks throughout the lifecycle is informed by a diverse team (e.g., diversity of demographics, disciplines, experience, expertise, and backgrounds). GOVERN 3.2: Policies and procedures are in place to define and differentiate roles and responsibilities for human-AI configurations and oversight of AI systems.
GOVERN 4: Organizational teams are committed to a culture	GOVERN 4.1: Organizational policies and practices are in place to foster a critical thinking and safety-first mindset in the design, development, deployment, and uses of AI systems to minimize potential negative impacts.

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Table 1: Categories and subcategories for the **GOVERN** function. (Continued)

Categories	Subcategories
that considers and communicates AI risk.	GOVERN 4.2: Organizational teams document the risks and potential impacts of the AI technology they design, develop, deploy, evaluate, and use, and they communicate about the impacts more broadly. GOVERN 4.3: Organizational practices are in place to enable AI testing, identification of incidents, and information sharing.
GOVERN 5: Processes are in place for robust engagement with relevant AI actors.	GOVERN 5.1: Organizational policies and practices are in place to collect, consider, prioritize, and integrate feedback from those external to the team that developed or deployed the AI system regarding the potential individual and societal impacts related to AI risks. GOVERN 5.2: Mechanisms are established to enable the team that developed or deployed AI systems to regularly incorporate adjudicated feedback from relevant AI actors into system design and implementation.
GOVERN 6: Policies and procedures are in place to address AI risks and benefits arising from third-party software and data and other supply chain issues.	GOVERN 6.1: Policies and procedures are in place that address AI risks associated with third-party entities, including risks of infringement of a third-party's intellectual property or other rights. GOVERN 6.2: Contingency processes are in place to handle failures or incidents in third-party data or AI systems deemed to be high-risk.

5.2 Map

The **MAP** function establishes the context to frame risks related to an AI system. The AI lifecycle consists of many interdependent activities involving a diverse set of actors (See Figure 3). In practice, AI actors in charge of one part of the process often do not have full visibility or control over other parts and their associated contexts. The interdependencies between these activities, and among the relevant AI actors, can make it difficult to reliably anticipate impacts of AI systems. For example, early decisions in identifying purposes and objectives of an AI system can alter its behavior and capabilities, and the dynamics of deployment setting (such as end users or impacted individuals) can shape the impacts of AI system decisions. As a result, the best intentions within one dimension of the AI lifecycle can be undermined via interactions with decisions and conditions in other, later activities.

This complexity and varying levels of visibility can introduce uncertainty into risk management practices. Anticipating, assessing, and otherwise addressing potential sources of negative risk can mitigate this uncertainty and enhance the integrity of the decision process.

The information gathered while carrying out the **MAP** function enables negative risk prevention and informs decisions for processes such as model management, as well as an initial decision about appropriateness or the need for an AI solution. Outcomes in the **MAP** function are the basis for the **MEASURE** and **MANAGE** functions. Without contextual knowledge, and awareness of risks within the identified contexts, risk management is difficult to perform. The **MAP** function is intended to enhance an organization's ability to identify risks and broader contributing factors.

Implementation of this function is enhanced by incorporating perspectives from a diverse internal team and engagement with those external to the team that developed or deployed the AI system. Engagement with external collaborators, end users, potentially impacted communities, and others may vary based on the risk level of a particular AI system, the makeup of the internal team, and organizational policies. Gathering such broad perspectives can help organizations proactively prevent negative risks and develop more trustworthy AI systems by:

- improving their capacity for understanding contexts;
- checking their assumptions about context of use;
- enabling recognition of when systems are not functional within or out of their intended context;
- identifying positive and beneficial uses of their existing AI systems;
- improving understanding of limitations in AI and ML processes;
- identifying constraints in real-world applications that may lead to negative impacts;
- identifying known and foreseeable negative impacts related to intended use of AI systems; and
- anticipating risks of the use of AI systems beyond intended use.

After completing the **MAP** function, Framework users should have sufficient contextual knowledge about AI system impacts to inform an initial go/no-go decision about whether to design, develop, or deploy an AI system. If a decision is made to proceed, organizations should utilize the **MEASURE** and **MANAGE** functions along with policies and procedures put into place in the **GOVERN** function to assist in AI risk management efforts. It is incumbent on Framework users to continue applying the **MAP** function to AI systems as context, capabilities, risks, benefits, and potential impacts evolve over time.

Practices related to mapping AI risks are described in the NIST AI RMF Playbook. Table 2 lists the **MAP** function's categories and subcategories.

Table 2: Categories and subcategories for the MAP function.

Categories	Subcategories
MAP 1: Context is established and understood.	MAP 1.1: Intended purposes, potentially beneficial uses, context-specific laws, norms and expectations, and prospective settings in which the AI system will be deployed are understood and documented. Considerations include: the specific set or types of users along with their expectations; potential positive and negative impacts of system uses to individuals, communities, organizations, society, and the planet; assumptions and related limitations about AI system purposes, uses, and risks across the development or product AI lifecycle; and related TEVV and system metrics. MAP 1.2: Interdisciplinary AI actors, competencies, skills, and capacities for establishing context reflect demographic diversity and broad domain and user experience expertise, and their participation is documented. Opportunities for interdisciplinary collaboration are prioritized. MAP 1.3: The organization’s mission and relevant goals for AI technology are understood and documented. MAP 1.4: The business value or context of business use has been clearly defined or – in the case of assessing existing AI systems – re-evaluated. MAP 1.5: Organizational risk tolerances are determined and documented. MAP 1.6: System requirements (e.g., “the system shall respect the privacy of its users”) are elicited from and understood by relevant AI actors. Design decisions take socio-technical implications into account to address AI risks.
MAP 2: Categorization of the AI system is performed.	MAP 2.1: The specific tasks and methods used to implement the tasks that the AI system will support are defined (e.g., classifiers, generative models, recommenders). MAP 2.2: Information about the AI system’s knowledge limits and how system output may be utilized and overseen by humans is documented. Documentation provides sufficient information to assist relevant AI actors when making decisions and taking subsequent actions.

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Table 2: Categories and subcategories for the **MAP** function. (Continued)

Categories	Subcategories
	MAP 2.3: Scientific integrity and TEVV considerations are identified and documented, including those related to experimental design, data collection and selection (e.g., availability, representativeness, suitability), system trustworthiness, and construct validation.
MAP 3: AI capabilities, targeted usage, goals, and expected benefits and costs compared with appropriate benchmarks are understood.	MAP 3.1: Potential benefits of intended AI system functionality and performance are examined and documented. MAP 3.2: Potential costs, including non-monetary costs, which result from expected or realized AI errors or system functionality and trustworthiness – as connected to organizational risk tolerance – are examined and documented. MAP 3.3: Targeted application scope is specified and documented based on the system’s capability, established context, and AI system categorization. MAP 3.4: Processes for operator and practitioner proficiency with AI system performance and trustworthiness – and relevant technical standards and certifications – are defined, assessed, and documented. MAP 3.5: Processes for human oversight are defined, assessed, and documented in accordance with organizational policies from the GOVERN function.
MAP 4: Risks and benefits are mapped for all components of the AI system including third-party software and data.	MAP 4.1: Approaches for mapping AI technology and legal risks of its components – including the use of third-party data or software – are in place, followed, and documented, as are risks of infringement of a third party’s intellectual property or other rights. MAP 4.2: Internal risk controls for components of the AI system, including third-party AI technologies, are identified and documented.
MAP 5: Impacts to individuals, groups, communities, organizations, and society are characterized.	MAP 5.1: Likelihood and magnitude of each identified impact (both potentially beneficial and harmful) based on expected use, past uses of AI systems in similar contexts, public incident reports, feedback from those external to the team that developed or deployed the AI system, or other data are identified and documented.

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Table 2: Categories and subcategories for the **MAP** function. (Continued)

Categories	Subcategories
	MAP 5.2: Practices and personnel for supporting regular engagement with relevant AI actors and integrating feedback about positive, negative, and unanticipated impacts are in place and documented.

5.3 Measure

The **MEASURE** function employs quantitative, qualitative, or mixed-method tools, techniques, and methodologies to analyze, assess, benchmark, and monitor AI risk and related impacts. It uses knowledge relevant to AI risks identified in the **MAP** function and informs the **MANAGE** function. AI systems should be tested before their deployment and regularly while in operation. AI risk measurements include documenting aspects of systems' functionality and trustworthiness.

Measuring AI risks includes tracking metrics for trustworthy characteristics, social impact, and human-AI configurations. Processes developed or adopted in the **MEASURE** function should include rigorous software testing and performance assessment methodologies with associated measures of uncertainty, comparisons to performance benchmarks, and formalized reporting and documentation of results. Processes for independent review can improve the effectiveness of testing and can mitigate internal biases and potential conflicts of interest.

Where tradeoffs among the trustworthy characteristics arise, measurement provides a traceable basis to inform management decisions. Options may include recalibration, impact mitigation, or removal of the system from design, development, production, or use, as well as a range of compensating, detective, deterrent, directive, and recovery controls.

After completing the **MEASURE** function, objective, repeatable, or scalable test, evaluation, verification, and validation (TEVV) processes including metrics, methods, and methodologies are in place, followed, and documented. Metrics and measurement methodologies should adhere to scientific, legal, and ethical norms and be carried out in an open and transparent process. New types of measurement, qualitative and quantitative, may need to be developed. The degree to which each measurement type provides unique and meaningful information to the assessment of AI risks should be considered. Framework users will enhance their capacity to comprehensively evaluate system trustworthiness, identify and track existing and emergent risks, and verify efficacy of the metrics. Measurement outcomes will be utilized in the **MANAGE** function to assist risk monitoring and response efforts. It is incumbent on Framework users to continue applying the **MEASURE** function to AI systems as knowledge, methodologies, risks, and impacts evolve over time.

Practices related to measuring AI risks are described in the NIST AI RMF Playbook. Table 3 lists the **MEASURE** function's categories and subcategories.

Table 3: Categories and subcategories for the **MEASURE** function.

Categories	Subcategories
MEASURE 1: Appropriate methods and metrics are identified and applied.	MEASURE 1.1: Approaches and metrics for measurement of AI risks enumerated during the MAP function are selected for implementation starting with the most significant AI risks. The risks or trustworthiness characteristics that will not – or cannot – be measured are properly documented. MEASURE 1.2: Appropriateness of AI metrics and effectiveness of existing controls are regularly assessed and updated, including reports of errors and potential impacts on affected communities. MEASURE 1.3: Internal experts who did not serve as front-line developers for the system and/or independent assessors are involved in regular assessments and updates. Domain experts, users, AI actors external to the team that developed or deployed the AI system, and affected communities are consulted in support of assessments as necessary per organizational risk tolerance.
MEASURE 2: AI systems are evaluated for trustworthy characteristics.	MEASURE 2.1: Test sets, metrics, and details about the tools used during TEVV are documented. MEASURE 2.2: Evaluations involving human subjects meet applicable requirements (including human subject protection) and are representative of the relevant population. MEASURE 2.3: AI system performance or assurance criteria are measured qualitatively or quantitatively and demonstrated for conditions similar to deployment setting(s). Measures are documented. MEASURE 2.4: The functionality and behavior of the AI system and its components – as identified in the MAP function – are monitored when in production. MEASURE 2.5: The AI system to be deployed is demonstrated to be valid and reliable. Limitations of the generalizability beyond the conditions under which the technology was developed are documented.

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Table 3: Categories and subcategories for the **MEASURE** function. (Continued)

Categories	Subcategories
	MEASURE 2.6: The AI system is evaluated regularly for safety risks – as identified in the MAP function. The AI system to be deployed is demonstrated to be safe, its residual negative risk does not exceed the risk tolerance, and it can fail safely, particularly if made to operate beyond its knowledge limits. Safety metrics reflect system reliability and robustness, real-time monitoring, and response times for AI system failures. MEASURE 2.7: AI system security and resilience – as identified in the MAP function – are evaluated and documented. MEASURE 2.8: Risks associated with transparency and accountability – as identified in the MAP function – are examined and documented. MEASURE 2.9: The AI model is explained, validated, and documented, and AI system output is interpreted within its context – as identified in the MAP function – to inform responsible use and governance. MEASURE 2.10: Privacy risk of the AI system – as identified in the MAP function – is examined and documented. MEASURE 2.11: Fairness and bias – as identified in the MAP function – are evaluated and results are documented. MEASURE 2.12: Environmental impact and sustainability of AI model training and management activities – as identified in the MAP function – are assessed and documented. MEASURE 2.13: Effectiveness of the employed TEVV metrics and processes in the MEASURE function are evaluated and documented.
MEASURE 3: Mechanisms for tracking identified AI risks over time are in place.	MEASURE 3.1: Approaches, personnel, and documentation are in place to regularly identify and track existing, unanticipated, and emergent AI risks based on factors such as intended and actual performance in deployed contexts. MEASURE 3.2: Risk tracking approaches are considered for settings where AI risks are difficult to assess using currently available measurement techniques or where metrics are not yet available.

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Table 3: Categories and subcategories for the **MEASURE** function. (Continued)

Categories	Subcategories
	MEASURE 3.3: Feedback processes for end users and impacted communities to report problems and appeal system outcomes are established and integrated into AI system evaluation metrics.
MEASURE 4: Feedback about efficacy of measurement is gathered and assessed.	MEASURE 4.1: Measurement approaches for identifying AI risks are connected to deployment context(s) and informed through consultation with domain experts and other end users. Approaches are documented. MEASURE 4.2: Measurement results regarding AI system trustworthiness in deployment context(s) and across the AI lifecycle are informed by input from domain experts and relevant AI actors to validate whether the system is performing consistently as intended. Results are documented. MEASURE 4.3: Measurable performance improvements or declines based on consultations with relevant AI actors, including affected communities, and field data about context-relevant risks and trustworthiness characteristics are identified and documented.

5.4 Manage

The **MANAGE** function entails allocating risk resources to mapped and measured risks on a regular basis and as defined by the **GOVERN** function. Risk treatment comprises plans to respond to, recover from, and communicate about incidents or events.

Contextual information gleaned from expert consultation and input from relevant AI actors – established in **GOVERN** and carried out in **MAP** – is utilized in this function to decrease the likelihood of system failures and negative impacts. Systematic documentation practices established in **GOVERN** and utilized in **MAP** and **MEASURE** bolster AI risk management efforts and increase transparency and accountability. Processes for assessing emergent risks are in place, along with mechanisms for continual improvement.

After completing the **MANAGE** function, plans for prioritizing risk and regular monitoring and improvement will be in place. Framework users will have enhanced capacity to manage the risks of deployed AI systems and to allocate risk management resources based on assessed and prioritized risks. It is incumbent on Framework users to continue to apply the **MANAGE** function to deployed AI systems as methods, contexts, risks, and needs or expectations from relevant AI actors evolve over time.

Practices related to managing AI risks are described in the NIST AI RMF Playbook. Table 4 lists the **MANAGE** function's categories and subcategories.

Table 4: Categories and subcategories for the **MANAGE** function.

Categories	Subcategories
MANAGE 1: AI risks based on assessments and other analytical output from the MAP and MEASURE functions are prioritized, responded to, and managed.	MANAGE 1.1: A determination is made as to whether the AI system achieves its intended purposes and stated objectives and whether its development or deployment should proceed. MANAGE 1.2: Treatment of documented AI risks is prioritized based on impact, likelihood, and available resources or methods. MANAGE 1.3: Responses to the AI risks deemed high priority, as identified by the MAP function, are developed, planned, and documented. Risk response options can include mitigating, transferring, avoiding, or accepting. MANAGE 1.4: Negative residual risks (defined as the sum of all unmitigated risks) to both downstream acquirers of AI systems and end users are documented.
MANAGE 2: Strategies to maximize AI benefits and minimize negative impacts are planned, prepared, implemented, documented, and informed by input from relevant AI actors.	MANAGE 2.1: Resources required to manage AI risks are taken into account – along with viable non-AI alternative systems, approaches, or methods – to reduce the magnitude or likelihood of potential impacts. MANAGE 2.2: Mechanisms are in place and applied to sustain the value of deployed AI systems. MANAGE 2.3: Procedures are followed to respond to and recover from a previously unknown risk when it is identified. MANAGE 2.4: Mechanisms are in place and applied, and responsibilities are assigned and understood, to supersede, disengage, or deactivate AI systems that demonstrate performance or outcomes inconsistent with intended use.
MANAGE 3: AI risks and benefits from third-party entities are managed.	MANAGE 3.1: AI risks and benefits from third-party resources are regularly monitored, and risk controls are applied and documented. MANAGE 3.2: Pre-trained models which are used for development are monitored as part of AI system regular monitoring and maintenance.

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Table 4: Categories and subcategories for the **MANAGE** function. (Continued)

Categories	Subcategories
MANAGE 4: Risk treatments, including response and recovery, and communication plans for the identified and measured AI risks are documented and monitored regularly.	MANAGE 4.1: Post-deployment AI system monitoring plans are implemented, including mechanisms for capturing and evaluating input from users and other relevant AI actors, appeal and override, decommissioning, incident response, recovery, and change management. MANAGE 4.2: Measurable activities for continual improvements are integrated into AI system updates and include regular engagement with interested parties, including relevant AI actors. MANAGE 4.3: Incidents and errors are communicated to relevant AI actors, including affected communities. Processes for tracking, responding to, and recovering from incidents and errors are followed and documented.

6. AI RMF Profiles

AI RMF *use-case profiles* are implementations of the AI RMF functions, categories, and subcategories for a specific setting or application based on the requirements, risk tolerance, and resources of the Framework user: for example, an AI RMF *hiring profile* or an AI RMF *fair housing profile*. Profiles may illustrate and offer insights into how risk can be managed at various stages of the AI lifecycle or in specific sector, technology, or end-use applications. AI RMF profiles assist organizations in deciding how they might best manage AI risk that is well-aligned with their goals, considers legal/regulatory requirements and best practices, and reflects risk management priorities.

AI RMF *temporal profiles* are descriptions of either the current state or the desired, target state of specific AI risk management activities within a given sector, industry, organization, or application context. An AI RMF Current Profile indicates how AI is currently being managed and the related risks in terms of current outcomes. A Target Profile indicates the outcomes needed to achieve the desired or target AI risk management goals.

Comparing Current and Target Profiles likely reveals gaps to be addressed to meet AI risk management objectives. Action plans can be developed to address these gaps to fulfill outcomes in a given category or subcategory. Prioritization of gap mitigation is driven by the user's needs and risk management processes. This risk-based approach also enables Framework users to compare their approaches with other approaches and to gauge the resources needed (e.g., staffing, funding) to achieve AI risk management goals in a cost-effective, prioritized manner.

AI RMF *cross-sectoral profiles* cover risks of models or applications that can be used across use cases or sectors. Cross-sectoral profiles can also cover how to govern, map, measure, and manage risks for activities or business processes common across sectors such as the use of large language models, cloud-based services or acquisition.

This Framework does not prescribe profile templates, allowing for flexibility in implementation.

Appendix A:

Descriptions of AI Actor Tasks from Figures 2 and 3

AI Design tasks are performed during the Application Context and Data and Input phases of the AI lifecycle in Figure 2. AI Design actors create the concept and objectives of AI systems and are responsible for the planning, design, and data collection and processing tasks of the AI system so that the AI system is lawful and fit-for-purpose. Tasks include articulating and documenting the system's concept and objectives, underlying assumptions, context, and requirements; gathering and cleaning data; and documenting the metadata and characteristics of the dataset. AI actors in this category include data scientists, domain experts, socio-cultural analysts, experts in the field of diversity, equity, inclusion, and accessibility, members of impacted communities, human factors experts (e.g., UX/UI design), governance experts, data engineers, data providers, system funders, product managers, third-party entities, evaluators, and legal and privacy governance.

AI Development tasks are performed during the AI Model phase of the lifecycle in Figure 2. AI Development actors provide the initial infrastructure of AI systems and are responsible for model building and interpretation tasks, which involve the creation, selection, calibration, training, and/or testing of models or algorithms. AI actors in this category include machine learning experts, data scientists, developers, third-party entities, legal and privacy governance experts, and experts in the socio-cultural and contextual factors associated with the deployment setting.

AI Deployment tasks are performed during the Task and Output phase of the lifecycle in Figure 2. AI Deployment actors are responsible for contextual decisions relating to how the AI system is used to assure deployment of the system into production. Related tasks include piloting the system, checking compatibility with legacy systems, ensuring regulatory compliance, managing organizational change, and evaluating user experience. AI actors in this category include system integrators, software developers, end users, operators and practitioners, evaluators, and domain experts with expertise in human factors, socio-cultural analysis, and governance.

Operation and Monitoring tasks are performed in the Application Context/Operate and Monitor phase of the lifecycle in Figure 2. These tasks are carried out by AI actors who are responsible for operating the AI system and working with others to regularly assess system output and impacts. AI actors in this category include system operators, domain experts, AI designers, users who interpret or incorporate the output of AI systems, product developers, evaluators and auditors, compliance experts, organizational management, and members of the research community.

Test, Evaluation, Verification, and Validation (TEVV) tasks are performed throughout the AI lifecycle. They are carried out by AI actors who examine the AI system or its components, or detect and remediate problems. Ideally, AI actors carrying out verification

and validation tasks are distinct from those who perform test and evaluation actions. Tasks can be incorporated into a phase as early as design, where tests are planned in accordance with the design requirement.

- TEVV tasks for design, planning, and data may center on internal and external validation of assumptions for system design, data collection, and measurements relative to the intended context of deployment or application.
- TEVV tasks for development (i.e., model building) include model validation and assessment.
- TEVV tasks for deployment include system validation and integration in production, with testing, and recalibration for systems and process integration, user experience, and compliance with existing legal, regulatory, and ethical specifications.
- TEVV tasks for operations involve ongoing monitoring for periodic updates, testing, and subject matter expert (SME) recalibration of models, the tracking of incidents or errors reported and their management, the detection of emergent properties and related impacts, and processes for redress and response.

Human Factors tasks and activities are found throughout the dimensions of the AI lifecycle. They include human-centered design practices and methodologies, promoting the active involvement of end users and other interested parties and relevant AI actors, incorporating context-specific norms and values in system design, evaluating and adapting end user experiences, and broad integration of humans and human dynamics in all phases of the AI lifecycle. Human factors professionals provide multidisciplinary skills and perspectives to understand context of use, inform interdisciplinary and demographic diversity, engage in consultative processes, design and evaluate user experience, perform human-centered evaluation and testing, and inform impact assessments.

Domain Expert tasks involve input from multidisciplinary practitioners or scholars who provide knowledge or expertise in – and about – an industry sector, economic sector, context, or application area where an AI system is being used. AI actors who are domain experts can provide essential guidance for AI system design and development, and interpret outputs in support of work performed by TEVV and AI impact assessment teams.

AI Impact Assessment tasks include assessing and evaluating requirements for AI system accountability, combating harmful bias, examining impacts of AI systems, product safety, liability, and security, among others. AI actors such as impact assessors and evaluators provide technical, human factor, socio-cultural, and legal expertise.

Procurement tasks are conducted by AI actors with financial, legal, or policy management authority for acquisition of AI models, products, or services from a third-party developer, vendor, or contractor.

Governance and Oversight tasks are assumed by AI actors with management, fiduciary, and legal authority and responsibility for the organization in which an AI system is de-

signed, developed, and/or deployed. Key AI actors responsible for AI governance include organizational management, senior leadership, and the Board of Directors. These actors are parties that are concerned with the impact and sustainability of the organization as a whole.

Additional AI Actors

Third-party entities include providers, developers, vendors, and evaluators of data, algorithms, models, and/or systems and related services for another organization or the organization's customers or clients. Third-party entities are responsible for AI design and development tasks, in whole or in part. By definition, they are external to the design, development, or deployment team of the organization that acquires its technologies or services. The technologies acquired from third-party entities may be complex or opaque, and risk tolerances may not align with the deploying or operating organization.

End users of an AI system are the individuals or groups that use the system for specific purposes. These individuals or groups interact with an AI system in a specific context. End users can range in competency from AI experts to first-time technology end users.

Affected individuals/communities encompass all individuals, groups, communities, or organizations directly or indirectly affected by AI systems or decisions based on the output of AI systems. These individuals do not necessarily interact with the deployed system or application.

Other AI actors may provide formal or quasi-formal norms or guidance for specifying and managing AI risks. They can include **trade associations, standards developing organizations, advocacy groups, researchers, environmental groups, and civil society organizations**.

The general public is most likely to directly experience positive and negative impacts of AI technologies. They may provide the motivation for actions taken by the AI actors. This group can include individuals, communities, and consumers associated with the context in which an AI system is developed or deployed.

Appendix B:

How AI Risks Differ from Traditional Software Risks

As with traditional software, risks from AI-based technology can be bigger than an enterprise, span organizations, and lead to societal impacts. AI systems also bring a set of risks that are not comprehensively addressed by current risk frameworks and approaches. Some AI system features that present risks also can be beneficial. For example, pre-trained models and transfer learning can advance research and increase accuracy and resilience when compared to other models and approaches. Identifying contextual factors in the **MAP** function will assist AI actors in determining the level of risk and potential management efforts.

Compared to traditional software, AI-specific risks that are new or increased include the following:

- The data used for building an AI system may not be a true or appropriate representation of the context or intended use of the AI system, and the ground truth may either not exist or not be available. Additionally, harmful bias and other data quality issues can affect AI system trustworthiness, which could lead to negative impacts.
- AI system dependency and reliance on data for training tasks, combined with increased volume and complexity typically associated with such data.
- Intentional or unintentional changes during training may fundamentally alter AI system performance.
- Datasets used to train AI systems may become detached from their original and intended context or may become stale or outdated relative to deployment context.
- AI system scale and complexity (many systems contain billions or even trillions of decision points) housed within more traditional software applications.
- Use of pre-trained models that can advance research and improve performance can also increase levels of statistical uncertainty and cause issues with bias management, scientific validity, and reproducibility.
- Higher degree of difficulty in predicting failure modes for emergent properties of large-scale pre-trained models.
- Privacy risk due to enhanced data aggregation capability for AI systems.
- AI systems may require more frequent maintenance and triggers for conducting corrective maintenance due to data, model, or concept drift.
- Increased opacity and concerns about reproducibility.
- Underdeveloped software testing standards and inability to document AI-based practices to the standard expected of traditionally engineered software for all but the simplest of cases.
- Difficulty in performing regular AI-based software testing, or determining what to test, since AI systems are not subject to the same controls as traditional code development.

- Computational costs for developing AI systems and their impact on the environment and planet.
- Inability to predict or detect the side effects of AI-based systems beyond statistical measures.

Privacy and cybersecurity risk management considerations and approaches are applicable in the design, development, deployment, evaluation, and use of AI systems. Privacy and cybersecurity risks are also considered as part of broader enterprise risk management considerations, which may incorporate AI risks. As part of the effort to address AI trustworthiness characteristics such as “Secure and Resilient” and “Privacy-Enhanced,” organizations may consider leveraging available standards and guidance that provide broad guidance to organizations to reduce security and privacy risks, such as, but not limited to, the NIST Cybersecurity Framework, the NIST Privacy Framework, the NIST Risk Management Framework, and the Secure Software Development Framework. These frameworks have some features in common with the AI RMF. Like most risk management approaches, they are outcome-based rather than prescriptive and are often structured around a Core set of functions, categories, and subcategories. While there are significant differences between these frameworks based on the domain addressed – and because AI risk management calls for addressing many other types of risks – frameworks like those mentioned above may inform security and privacy considerations in the **MAP**, **MEASURE**, and **MANAGE** functions of the AI RMF.

At the same time, guidance available before publication of this AI RMF does not comprehensively address many AI system risks. For example, existing frameworks and guidance are unable to:

- adequately manage the problem of harmful bias in AI systems;
- confront the challenging risks related to generative AI;
- comprehensively address security concerns related to evasion, model extraction, membership inference, availability, or other machine learning attacks;
- account for the complex attack surface of AI systems or other security abuses enabled by AI systems; and
- consider risks associated with third-party AI technologies, transfer learning, and off-label use where AI systems may be trained for decision-making outside an organization’s security controls or trained in one domain and then “fine-tuned” for another.

Both AI and traditional software technologies and systems are subject to rapid innovation. Technology advances should be monitored and deployed to take advantage of those developments and work towards a future of AI that is both trustworthy and responsible.

Appendix C:

AI Risk Management and Human-AI Interaction

Organizations that design, develop, or deploy AI systems for use in operational settings may enhance their AI risk management by understanding current limitations of human-AI interaction. The AI RMF provides opportunities to clearly define and differentiate the various human roles and responsibilities when using, interacting with, or managing AI systems.

Many of the data-driven approaches that AI systems rely on attempt to convert or represent individual and social observational and decision-making practices into measurable quantities. Representing complex human phenomena with mathematical models can come at the cost of removing necessary context. This loss of context may in turn make it difficult to understand individual and societal impacts that are key to AI risk management efforts.

Issues that merit further consideration and research include:

1. **Human roles and responsibilities in decision making and overseeing AI systems need to be clearly defined and differentiated.** Human-AI configurations can span from fully autonomous to fully manual. AI systems can autonomously make decisions, defer decision making to a human expert, or be used by a human decision maker as an additional opinion. Some AI systems may not require human oversight, such as models used to improve video compression. Other systems may specifically require human oversight.
2. **Decisions that go into the design, development, deployment, evaluation, and use of AI systems reflect systemic and human cognitive biases.** AI actors bring their cognitive biases, both individual and group, into the process. Biases can stem from end-user decision-making tasks and be introduced across the AI lifecycle via human assumptions, expectations, and decisions during design and modeling tasks. These biases, which are not necessarily always harmful, may be exacerbated by AI system opacity and the resulting lack of transparency. Systemic biases at the organizational level can influence how teams are structured and who controls the decision-making processes throughout the AI lifecycle. These biases can also influence downstream decisions by end users, decision makers, and policy makers and may lead to negative impacts.
3. **Human-AI interaction results vary.** Under certain conditions – for example, in perceptual-based judgment tasks – the AI part of the human-AI interaction can amplify human biases, leading to more biased decisions than the AI or human alone. When these variations are judiciously taken into account in organizing human-AI teams, however, they can result in complementarity and improved overall performance.

4. **Presenting AI system information to humans is complex.** Humans perceive and derive meaning from AI system output and explanations in different ways, reflecting different individual preferences, traits, and skills.

The **GOVERN** function provides organizations with the opportunity to clarify and define the roles and responsibilities for the humans in the Human-AI team configurations and those who are overseeing the AI system performance. The **GOVERN** function also creates mechanisms for organizations to make their decision-making processes more explicit, to help counter systemic biases.

The **MAP** function suggests opportunities to define and document processes for operator and practitioner proficiency with AI system performance and trustworthiness concepts, and to define relevant technical standards and certifications. Implementing **MAP** function categories and subcategories may help organizations improve their internal competency for analyzing context, identifying procedural and system limitations, exploring and examining impacts of AI-based systems in the real world, and evaluating decision-making processes throughout the AI lifecycle.

The **GOVERN** and **MAP** functions describe the importance of interdisciplinarity and demographically diverse teams and utilizing feedback from potentially impacted individuals and communities. AI actors called out in the AI RMF who perform human factors tasks and activities can assist technical teams by anchoring in design and development practices to user intentions and representatives of the broader AI community, and societal values. These actors further help to incorporate context-specific norms and values in system design and evaluate end user experiences – in conjunction with AI systems.

AI risk management approaches for human-AI configurations will be augmented by ongoing research and evaluation. For example, the degree to which humans are empowered and incentivized to challenge AI system output requires further studies. Data about the frequency and rationale with which humans overrule AI system output in deployed systems may be useful to collect and analyze.

Appendix D:

Attributes of the AI RMF

NIST described several key attributes of the AI RMF when work on the Framework first began. These attributes have remained intact and were used to guide the AI RMF's development. They are provided here as a reference.

The AI RMF strives to:

1. Be risk-based, resource-efficient, pro-innovation, and voluntary.
2. Be consensus-driven and developed and regularly updated through an open, transparent process. All stakeholders should have the opportunity to contribute to the AI RMF's development.
3. Use clear and plain language that is understandable by a broad audience, including senior executives, government officials, non-governmental organization leadership, and those who are not AI professionals – while still of sufficient technical depth to be useful to practitioners. The AI RMF should allow for communication of AI risks across an organization, between organizations, with customers, and to the public at large.
4. Provide common language and understanding to manage AI risks. The AI RMF should offer taxonomy, terminology, definitions, metrics, and characterizations for AI risk.
5. Be easily usable and fit well with other aspects of risk management. Use of the Framework should be intuitive and readily adaptable as part of an organization's broader risk management strategy and processes. It should be consistent or aligned with other approaches to managing AI risks.
6. Be useful to a wide range of perspectives, sectors, and technology domains. The AI RMF should be universally applicable to any AI technology and to context-specific use cases.
7. Be outcome-focused and non-prescriptive. The Framework should provide a catalog of outcomes and approaches rather than prescribe one-size-fits-all requirements.
8. Take advantage of and foster greater awareness of existing standards, guidelines, best practices, methodologies, and tools for managing AI risks – as well as illustrate the need for additional, improved resources.
9. Be law- and regulation-agnostic. The Framework should support organizations' abilities to operate under applicable domestic and international legal or regulatory regimes.
10. Be a living document. The AI RMF should be readily updated as technology, understanding, and approaches to AI trustworthiness and uses of AI change and as stakeholders learn from implementing AI risk management generally and this framework in particular.

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Synthesis Report

Summary for Policymakers

A Report of the Intergovernmental Panel on Climate Change



CLIMATE CHANGE 2023

Synthesis Report

Summary for Policymakers

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Sources cited in this Synthesis Report

References for material contained in this report are given in curly brackets {} at the end of each paragraph.

In the Summary for Policymakers, the references refer to the numbers of the sections, figures, tables and boxes in the underlying Introduction and Topics of this Synthesis Report.

In the Introduction and Sections of the longer report, the references refer to the contributions of the Working Groups I, II and III (WGI, WGII, WGIII) to the Sixth Assessment Report and other IPCC Reports (in italicized curly brackets), or to other sections of the Synthesis Report itself (in round brackets).

The following abbreviations have been used:

SPM: Summary for Policymakers

TS: Technical Summary

ES: Executive Summary of a chapter

Numbers denote specific chapters and sections of a report.

Other IPCC reports cited in this Synthesis Report:

SR1.5: Global Warming of 1.5°C

SRCCCL: Climate Change and Land

SROCC: The Ocean and Cryosphere in a Changing Climate

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Introduction

This Synthesis Report (SYR) of the IPCC Sixth Assessment Report (AR6) summarises the state of knowledge of climate change, its widespread impacts and risks, and climate change mitigation and adaptation. It integrates the main findings of the Sixth Assessment Report (AR6) based on contributions from the three Working Groups¹, and the three Special Reports². The summary for Policymakers (SPM) is structured in three parts: SPM.A Current Status and Trends, SPM.B Future Climate Change, Risks, and Long-Term Responses, and SPM.C Responses in the Near Term³.

This report recognizes the interdependence of climate, ecosystems and biodiversity, and human societies; the value of diverse forms of knowledge; and the close linkages between climate change adaptation, mitigation, ecosystem health, human well-being and sustainable development, and reflects the increasing diversity of actors involved in climate action.

Based on scientific understanding, key findings can be formulated as statements of fact or associated with an assessed level of confidence using the IPCC calibrated language⁴.

¹ The three Working Group contributions to AR6 are: AR6 Climate Change 2021: The Physical Science Basis; AR6 Climate Change 2022: Impacts, Adaptation and Vulnerability; and AR6 Climate Change 2022: Mitigation of Climate Change. Their assessments cover scientific literature accepted for publication respectively by 31 January 2021, 1 September 2021 and 11 October 2021.

² The three Special Reports are: Global Warming of 1.5°C (2018): an IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty (SR1.5); Climate Change and Land (2019): an IPCC Special Report on climate change, desertification, land degradation, sustainable land management, food security, and greenhouse gas fluxes in terrestrial ecosystems (SRCCL); and The Ocean and Cryosphere in a Changing Climate (2019) (SROCC). The Special Reports cover scientific literature accepted for publication respectively by 15 May 2018, 7 April 2019 and 15 May 2019.

³ In this report, the near term is defined as the period until 2040. The long term is defined as the period beyond 2040.

⁴ Each finding is grounded in an evaluation of underlying evidence and agreement. The IPCC calibrated language uses five qualifiers to express a level of confidence: very low, low, medium, high and very high, and typeset in italics, for example, *medium confidence*. The following terms are used to indicate the assessed likelihood of an outcome or a result: virtually certain 99–100% probability, very likely 90–100%, likely 66–100%, more likely than not >50–100%, about as likely as not 33–66%, unlikely 0–33%, very unlikely 0–10%, exceptionally unlikely 0–1%. Additional terms (extremely likely 95–100%; and extremely unlikely 0–5%) are also used when appropriate. Assessed likelihood is typeset in italics, e.g., *very likely*. This is consistent with AR5 and the other AR6 Reports.

A. Current Status and Trends

Observed Warming and its Causes

A.1 Human activities, principally through emissions of greenhouse gases, have unequivocally caused global warming, with global surface temperature reaching 1.1°C above 1850–1900 in 2011–2020. Global greenhouse gas emissions have continued to increase, with unequal historical and ongoing contributions arising from unsustainable energy use, land use and land-use change, lifestyles and patterns of consumption and production across regions, between and within countries, and among individuals (*high confidence*). {2.1, Figure 2.1, Figure 2.2}

- A.1.1 Global surface temperature was 1.09 [0.95 to 1.20]°C⁵ higher in 2011–2020 than 1850–1900⁶, with larger increases over land (1.59 [1.34 to 1.83]°C) than over the ocean (0.88 [0.68 to 1.01]°C). Global surface temperature in the first two decades of the 21st century (2001–2020) was 0.99 [0.84 to 1.10]°C higher than 1850–1900. Global surface temperature has increased faster since 1970 than in any other 50-year period over at least the last 2000 years (*high confidence*). {2.1.1, Figure 2.1}
- A.1.2 The *likely* range of total human-caused global surface temperature increase from 1850–1900 to 2010–2019⁷ is 0.8°C to 1.3°C, with a best estimate of 1.07°C. Over this period, it is *likely* that well-mixed greenhouse gases (GHGs) contributed a warming of 1.0°C to 2.0°C⁸, and other human drivers (principally aerosols) contributed a cooling of 0.0°C to 0.8°C, natural (solar and volcanic) drivers changed global surface temperature by –0.1°C to +0.1°C, and internal variability changed it by –0.2°C to +0.2°C. {2.1.1, Figure 2.1}
- A.1.3 Observed increases in well-mixed GHG concentrations since around 1750 are unequivocally caused by GHG emissions from human activities over this period. Historical cumulative net CO₂ emissions from 1850 to 2019 were 2400 ± 240 GtCO₂ of which more than half (58%) occurred between 1850 and 1989, and about 42% occurred between 1990 and 2019 (*high confidence*). In 2019, atmospheric CO₂ concentrations (410 parts per million) were higher than at any time in at least 2 million years (*high confidence*), and concentrations of methane (1866 parts per billion) and nitrous oxide (332 parts per billion) were higher than at any time in at least 800,000 years (*very high confidence*). {2.1.1, Figure 2.1}
- A.1.4 Global net anthropogenic GHG emissions have been estimated to be 59 ± 6.6 GtCO₂-eq⁹ in 2019, about 12% (6.5 GtCO₂-eq) higher than in 2010 and 54% (21 GtCO₂-eq) higher than in 1990, with the largest share and growth in gross GHG emissions occurring in CO₂ from fossil fuels combustion and industrial processes (CO₂-FFI) followed by methane, whereas the highest relative growth occurred in fluorinated gases (F-gases), starting from low levels in 1990. Average annual GHG emissions during 2010–2019 were higher than in any previous decade on record, while the rate of growth between 2010 and 2019 (1.3% yr⁻¹) was lower than that between 2000 and 2009 (2.1% yr⁻¹). In 2019, approximately 79% of global GHG

⁵ Ranges given throughout the SPM represent *very likely* ranges (5–95% range) unless otherwise stated.

⁶ The estimated increase in global surface temperature since AR5 is principally due to further warming since 2003–2012 (0.19 [0.16 to 0.22] °C). Additionally, methodological advances and new datasets have provided a more complete spatial representation of changes in surface temperature, including in the Arctic. These and other improvements have also increased the estimate of global surface temperature change by approximately 0.1°C, but this increase does not represent additional physical warming since AR5.

⁷ The period distinction with A.1.1 arises because the attribution studies consider this slightly earlier period. The observed warming to 2010–2019 is 1.06 [0.88 to 1.21]°C.

⁸ Contributions from emissions to the 2010–2019 warming relative to 1850–1900 assessed from radiative forcing studies are: CO₂ 0.8 [0.5 to 1.2]°C; methane 0.5 [0.3 to 0.8]°C; nitrous oxide 0.1 [0.0 to 0.2]°C and fluorinated gases 0.1 [0.0 to 0.2]°C. {2.1.1}

⁹ GHG emission metrics are used to express emissions of different greenhouse gases in a common unit. Aggregated GHG emissions in this report are stated in CO₂-equivalents (CO₂-eq) using the Global Warming Potential with a time horizon of 100 years (GWP100) with values based on the contribution of Working Group I to the AR6. The AR6 WGI and WGIII reports contain updated emission metric values, evaluations of different metrics with regard to mitigation objectives, and assess new approaches to aggregating gases. The choice of metric depends on the purpose of the analysis and all GHG emission metrics have limitations and uncertainties, given that they simplify the complexity of the physical climate system and its response to past and future GHG emissions. {2.1.1}

emissions came from the sectors of energy, industry, transport, and buildings together and 22%¹⁰ from agriculture, forestry and other land use (AFOLU). Emissions reductions in CO₂-FFI due to improvements in energy intensity of GDP and carbon intensity of energy, have been less than emissions increases from rising global activity levels in industry, energy supply, transport, agriculture and buildings. (*high confidence*) {2.1.1}

- A.1.5 Historical contributions of CO₂ emissions vary substantially across regions in terms of total magnitude, but also in terms of contributions to CO₂-FFI and net CO₂ emissions from land use, land-use change and forestry (CO₂-LULUCF). In 2019, around 35% of the global population live in countries emitting more than 9 tCO₂-eq per capita¹¹ (excluding CO₂-LULUCF) while 41% live in countries emitting less than 3 tCO₂-eq per capita; of the latter a substantial share lacks access to modern energy services. Least Developed Countries (LDCs) and Small Island Developing States (SIDS) have much lower per capita emissions (1.7 tCO₂-eq and 4.6 tCO₂-eq, respectively) than the global average (6.9 tCO₂-eq), excluding CO₂-LULUCF. The 10% of households with the highest per capita emissions contribute 34–45% of global consumption-based household GHG emissions, while the bottom 50% contribute 13–15%. (*high confidence*) {2.1.1, Figure 2.2}

Observed Changes and Impacts

- A.2 Widespread and rapid changes in the atmosphere, ocean, cryosphere and biosphere have occurred. Human-caused climate change is already affecting many weather and climate extremes in every region across the globe. This has led to widespread adverse impacts and related losses and damages to nature and people (*high confidence*). Vulnerable communities who have historically contributed the least to current climate change are disproportionately affected (*high confidence*). {2.1, Table 2.1, Figure 2.2, Figure 2.3} (Figure SPM.1)**

- A.2.1 It is unequivocal that human influence has warmed the atmosphere, ocean and land. Global mean sea level increased by 0.20 [0.15 to 0.25] m between 1901 and 2018. The average rate of sea level rise was 1.3 [0.6 to 2.1] mm yr⁻¹ between 1901 and 1971, increasing to 1.9 [0.8 to 2.9] mm yr⁻¹ between 1971 and 2006, and further increasing to 3.7 [3.2 to 4.2] mm yr⁻¹ between 2006 and 2018 (*high confidence*). Human influence was *very likely* the main driver of these increases since at least 1971. Evidence of observed changes in extremes such as heatwaves, heavy precipitation, droughts, and tropical cyclones, and, in particular, their attribution to human influence, has further strengthened since AR5. Human influence has *likely* increased the chance of compound extreme events since the 1950s, including increases in the frequency of concurrent heatwaves and droughts (*high confidence*). {2.1.2, Table 2.1, Figure 2.3, Figure 3.4} (Figure SPM.1)
- A.2.2 Approximately 3.3 to 3.6 billion people live in contexts that are highly vulnerable to climate change. Human and ecosystem vulnerability are interdependent. Regions and people with considerable development constraints have high vulnerability to climatic hazards. Increasing weather and climate extreme events have exposed millions of people to acute food insecurity¹² and reduced water security, with the largest adverse impacts observed in many locations and/or communities in Africa, Asia, Central and South America, LDCs, Small Islands and the Arctic, and globally for Indigenous Peoples, small-scale food producers and low-income households. Between 2010 and 2020, human mortality from floods, droughts and storms was 15 times higher in highly vulnerable regions, compared to regions with very low vulnerability. (*high confidence*) {2.1.2, 4.4} (Figure SPM.1)
- A.2.3 Climate change has caused substantial damages, and increasingly irreversible losses, in terrestrial, freshwater, cryospheric, and coastal and open ocean ecosystems (*high confidence*). Hundreds of local losses of species have been driven by increases in the magnitude of heat extremes (*high confidence*) with mass mortality events recorded on land and in the ocean (*very high confidence*). Impacts on some ecosystems are approaching irreversibility such as the impacts of hydrological changes resulting from the retreat of glaciers, or the changes in some mountain (*medium confidence*) and Arctic ecosystems driven by permafrost thaw (*high confidence*). {2.1.2, Figure 2.3} (Figure SPM.1)

¹⁰ GHG emission levels are rounded to two significant digits; as a consequence, small differences in sums due to rounding may occur. {2.1.1}

¹¹ Territorial emissions.

¹² Acute food insecurity can occur at any time with a severity that threatens lives, livelihoods or both, regardless of the causes, context or duration, as a result of shocks risking determinants of food security and nutrition, and is used to assess the need for humanitarian action. {2.1}

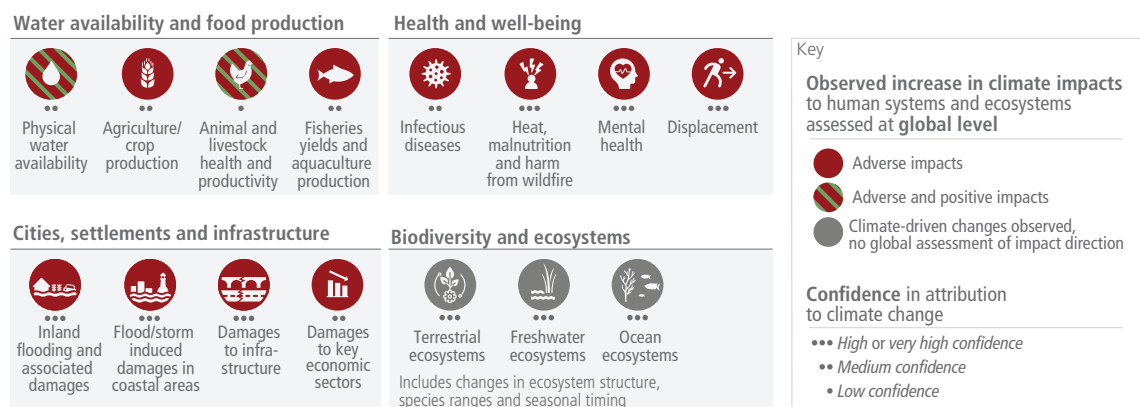
- A.2.4 Climate change has reduced food security and affected water security, hindering efforts to meet Sustainable Development Goals (*high confidence*). Although overall agricultural productivity has increased, climate change has slowed this growth over the past 50 years globally (*medium confidence*), with related negative impacts mainly in mid- and low latitude regions but positive impacts in some high latitude regions (*high confidence*). Ocean warming and ocean acidification have adversely affected food production from fisheries and shellfish aquaculture in some oceanic regions (*high confidence*). Roughly half of the world's population currently experience severe water scarcity for at least part of the year due to a combination of climatic and non-climatic drivers (*medium confidence*). {2.1.2, Figure 2.3} (Figure SPM.1)
- A.2.5 In all regions increases in extreme heat events have resulted in human mortality and morbidity (*very high confidence*). The occurrence of climate-related food-borne and water-borne diseases (*very high confidence*) and the incidence of vector-borne diseases (*high confidence*) have increased. In assessed regions, some mental health challenges are associated with increasing temperatures (*high confidence*), trauma from extreme events (*very high confidence*), and loss of livelihoods and culture (*high confidence*). Climate and weather extremes are increasingly driving displacement in Africa, Asia, North America (*high confidence*), and Central and South America (*medium confidence*), with small island states in the Caribbean and South Pacific being disproportionately affected relative to their small population size (*high confidence*). {2.1.2, Figure 2.3} (Figure SPM.1)
- A.2.6 Climate change has caused widespread adverse impacts and related losses and damages¹³ to nature and people that are unequally distributed across systems, regions and sectors. Economic damages from climate change have been detected in climate-exposed sectors, such as agriculture, forestry, fishery, energy, and tourism. Individual livelihoods have been affected through, for example, destruction of homes and infrastructure, and loss of property and income, human health and food security, with adverse effects on gender and social equity. (*high confidence*) {2.1.2} (Figure SPM.1)
- A.2.7 In urban areas, observed climate change has caused adverse impacts on human health, livelihoods and key infrastructure. Hot extremes have intensified in cities. Urban infrastructure, including transportation, water, sanitation and energy systems have been compromised by extreme and slow-onset events¹⁴, with resulting economic losses, disruptions of services and negative impacts to well-being. Observed adverse impacts are concentrated amongst economically and socially marginalised urban residents. (*high confidence*) {2.1.2}

¹³ In this report, the term 'losses and damages' refers to adverse observed impacts and/or projected risks and can be economic and/or non-economic (see Annex I: Glossary).

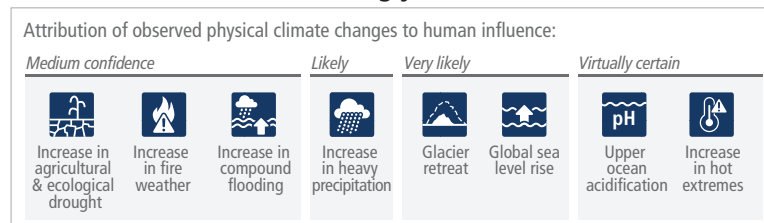
¹⁴ Slow-onset events are described among the climatic-impact drivers of the AR6 WGI and refer to the risks and impacts associated with e.g., increasing temperature means, desertification, decreasing precipitation, loss of biodiversity, land and forest degradation, glacial retreat and related impacts, ocean acidification, sea level rise and salinization. {2.1.2}

Adverse impacts from human-caused climate change will continue to intensify

a) Observed widespread and substantial impacts and related losses and damages attributed to climate change



b) Impacts are driven by changes in multiple physical climate conditions, which are increasingly attributed to human influence



c) The extent to which current and future generations will experience a hotter and different world depends on choices now and in the near term

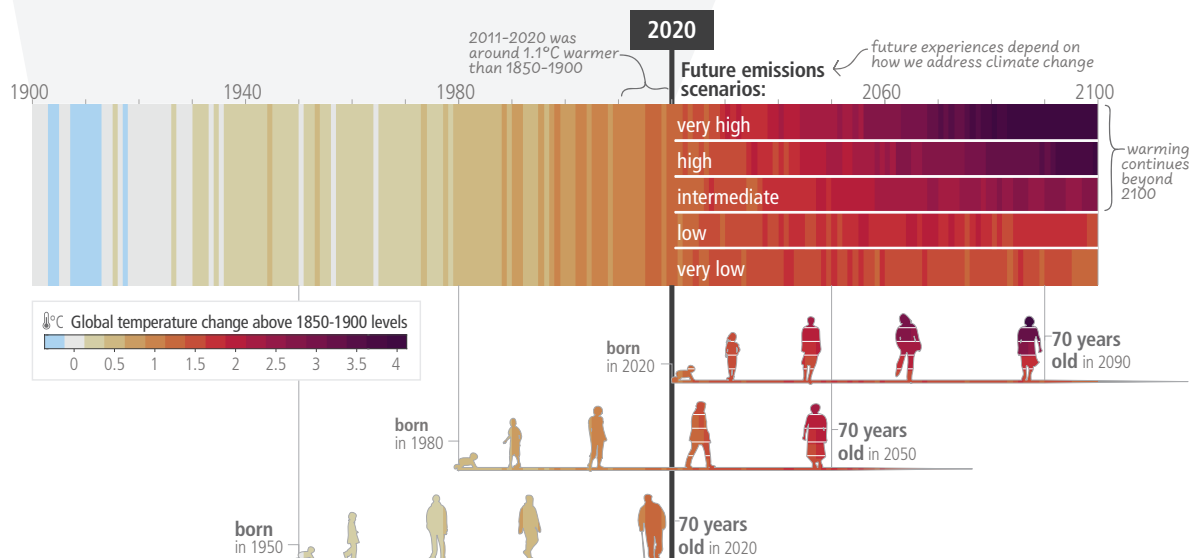


Figure SPM.1: (a) Climate change has already caused widespread impacts and related losses and damages on human systems and altered terrestrial, freshwater and ocean ecosystems worldwide. Physical water availability includes balance of water available from various sources including ground water, water quality and demand for water. Global mental health and displacement assessments reflect only assessed regions. Confidence levels reflect the assessment of attribution of the observed impact to climate change. (b) Observed impacts are connected to physical climate changes including many that have been attributed to human influence such as the selected climatic impact-drivers shown. Confidence and likelihood levels reflect the assessment of attribution of the observed climatic impact-driver to human influence. (c) Observed (1900–2020) and projected (2021–2100) changes in global surface temperature (relative to 1850–1900), which are linked to changes in climate conditions and impacts, illustrate how the climate has already changed and will change along the lifespan of three

representative generations (born in 1950, 1980 and 2020). Future projections (2021–2100) of changes in global surface temperature are shown for very low (SSP1-1.9), low (SSP1-2.6), intermediate (SSP2-4.5), high (SSP3-7.0) and very high (SSP5-8.5) GHG emissions scenarios. Changes in annual global surface temperatures are presented as ‘climate stripes’, with future projections showing the human-caused long-term trends and continuing modulation by natural variability (represented here using observed levels of past natural variability). Colours on the generational icons correspond to the global surface temperature stripes for each year, with segments on future icons differentiating possible future experiences. {2.1, 2.1.2, Figure 2.1, Table 2.1, Figure 2.3, Cross-Section Box.2, 3.1, Figure 3.3, 4.1, 4.3} (Box SPM.1)

Current Progress in Adaptation and Gaps and Challenges

A.3 Adaptation planning and implementation has progressed across all sectors and regions, with documented benefits and varying effectiveness. Despite progress, adaptation gaps exist, and will continue to grow at current rates of implementation. Hard and soft limits to adaptation have been reached in some ecosystems and regions. Maladaptation is happening in some sectors and regions. Current global financial flows for adaptation are insufficient for, and constrain implementation of, adaptation options, especially in developing countries (*high confidence*). {2.2, 2.3}

- A.3.1 Progress in adaptation planning and implementation has been observed across all sectors and regions, generating multiple benefits (*very high confidence*). Growing public and political awareness of climate impacts and risks has resulted in at least 170 countries and many cities including adaptation in their climate policies and planning processes (*high confidence*). {2.2.3}
- A.3.2 Effectiveness¹⁵ of adaptation in reducing climate risks¹⁶ is documented for specific contexts, sectors and regions (*high confidence*). Examples of effective adaptation options include: cultivar improvements, on-farm water management and storage, soil moisture conservation, irrigation, agroforestry, community-based adaptation, farm and landscape level diversification in agriculture, sustainable land management approaches, use of agroecological principles and practices and other approaches that work with natural processes (*high confidence*). Ecosystem-based adaptation¹⁷ approaches such as urban greening, restoration of wetlands and upstream forest ecosystems have been effective in reducing flood risks and urban heat (*high confidence*). Combinations of non-structural measures like early warning systems and structural measures like levees have reduced loss of lives in case of inland flooding (*medium confidence*). Adaptation options such as disaster risk management, early warning systems, climate services and social safety nets have broad applicability across multiple sectors (*high confidence*). {2.2.3}
- A.3.3 Most observed adaptation responses are fragmented, incremental¹⁸, sector-specific and unequally distributed across regions. Despite progress, adaptation gaps exist across sectors and regions, and will continue to grow under current levels of implementation, with the largest adaptation gaps among lower income groups. (*high confidence*) {2.3.2}
- A.3.4 There is increased evidence of maladaptation in various sectors and regions. Maladaptation especially affects marginalised and vulnerable groups adversely. (*high confidence*) {2.3.2}
- A.3.5 Soft limits to adaptation are currently being experienced by small-scale farmers and households along some low-lying coastal areas (*medium confidence*) resulting from financial, governance, institutional and policy constraints (*high confidence*). Some tropical, coastal, polar and mountain ecosystems have reached hard adaptation limits (*high confidence*). Adaptation does not prevent all losses and damages, even with effective adaptation and before reaching soft and hard limits (*high confidence*). {2.3.2}

¹⁵ Effectiveness refers here to the extent to which an adaptation option is anticipated or observed to reduce climate-related risk. {2.2.3}

¹⁶ See Annex I: Glossary. {2.2.3}

¹⁷ Ecosystem-based Adaptation (EbA) is recognized internationally under the Convention on Biological Diversity (CBD14/5). A related concept is Nature-based Solutions (NbS), see Annex I: Glossary.

¹⁸ Incremental adaptations to change in climate are understood as extensions of actions and behaviours that already reduce the losses or enhance the benefits of natural variations in extreme weather/climate events. {2.3.2}

- A.3.6 Key barriers to adaptation are limited resources, lack of private sector and citizen engagement, insufficient mobilization of finance (including for research), low climate literacy, lack of political commitment, limited research and/or slow and low uptake of adaptation science, and low sense of urgency. There are widening disparities between the estimated costs of adaptation and the finance allocated to adaptation (*high confidence*). Adaptation finance has come predominantly from public sources, and a small proportion of global tracked climate finance was targeted to adaptation and an overwhelming majority to mitigation (*very high confidence*). Although global tracked climate finance has shown an upward trend since AR5, current global financial flows for adaptation, including from public and private finance sources, are insufficient and constrain implementation of adaptation options, especially in developing countries (*high confidence*). Adverse climate impacts can reduce the availability of financial resources by incurring losses and damages and through impeding national economic growth, thereby further increasing financial constraints for adaptation, particularly for developing and least developed countries (*medium confidence*). {2.3.2, 2.3.3}

Box SPM.1 The use of scenarios and modelled pathways in the AR6 Synthesis Report

Modelled scenarios and pathways¹⁹ are used to explore future emissions, climate change, related impacts and risks, and possible mitigation and adaptation strategies and are based on a range of assumptions, including socio-economic variables and mitigation options. These are quantitative projections and are neither predictions nor forecasts. Global modelled emission pathways, including those based on cost effective approaches contain regionally differentiated assumptions and outcomes, and have to be assessed with the careful recognition of these assumptions. Most do not make explicit assumptions about global equity, environmental justice or intra-regional income distribution. IPCC is neutral with regard to the assumptions underlying the scenarios in the literature assessed in this report, which do not cover all possible futures.²⁰ {Cross-Section Box.2}

WGI assessed the climate response to five illustrative scenarios based on Shared Socio-economic Pathways (SSPs)²¹ that cover the range of possible future development of anthropogenic drivers of climate change found in the literature. High and very high GHG emissions scenarios (SSP3-7.0 and SSP5-8.5²²) have CO₂ emissions that roughly double from current levels by 2100 and 2050, respectively. The intermediate GHG emissions scenario (SSP2-4.5) has CO₂ emissions remaining around current levels until the middle of the century. The very low and low GHG emissions scenarios (SSP1-1.9 and SSP1-2.6) have CO₂ emissions declining to net zero around 2050 and 2070, respectively, followed by varying levels of net negative CO₂ emissions. In addition, Representative Concentration Pathways (RCPs)²³ were used by WGI and WGII to assess regional climate changes, impacts and risks. In WGIII, a large number of global modelled emissions pathways were assessed, of which 1202 pathways were categorised based on their assessed global warming over the 21st century; categories range from pathways that limit warming to 1.5°C with more than 50% likelihood (noted >50% in this report) with no or limited overshoot (C1) to pathways that exceed 4°C (C8). {Cross-Section Box.2} (Box SPM.1, Table 1)

Global warming levels (GWLs) relative to 1850–1900 are used to integrate the assessment of climate change and related impacts and risks since patterns of changes for many variables at a given GWL are common to all scenarios considered and independent of timing when that level is reached. {Cross-Section Box.2}

¹⁹ In the literature, the terms pathways and scenarios are used interchangeably, with the former more frequently used in relation to climate goals. WGI primarily used the term scenarios and WGIII mostly used the term modelled emission and mitigation pathways. The SYR primarily uses scenarios when referring to WGI and modelled emission and mitigation pathways when referring to WGIII.

²⁰ Around half of all modelled global emission pathways assume cost-effective approaches that rely on least-cost mitigation/abatement options globally. The other half looks at existing policies and regionally and sectorally differentiated actions.

²¹ SSP-based scenarios are referred to as SSPx-y, where 'SSPx' refers to the Shared Socioeconomic Pathway describing the socioeconomic trends underlying the scenarios, and 'y' refers to the level of radiative forcing (in watts per square metre, or W m⁻²) resulting from the scenario in the year 2100. {Cross-Section Box.2}

²² Very high emissions scenarios have become *less likely* but cannot be ruled out. Warming levels >4°C may result from very high emissions scenarios, but can also occur from lower emission scenarios if climate sensitivity or carbon cycle feedbacks are higher than the best estimate. {3.1.1}

²³ RCP-based scenarios are referred to as RCPy, where 'y' refers to the level of radiative forcing (in watts per square metre, or W m⁻²) resulting from the scenario in the year 2100. The SSP scenarios cover a broader range of greenhouse gas and air pollutant futures than the RCPs. They are similar but not identical, with differences in concentration trajectories. The overall effective radiative forcing tends to be higher for the SSPs compared to the RCPs with the same label (*medium confidence*). {Cross-Section Box.2}

Box SPM.1, Table 1: Description and relationship of scenarios and modelled pathways considered across AR6 Working Group reports. {Cross-Section Box.2 Figure 1}

Category in WGIII	Category description	GHG emissions scenarios (SSPx-y*) in WGI & WGII	RCPy** in WGI & WGII
C1	limit warming to 1.5°C (>50%) with no or limited overshoot***	Very low (SSP1-1.9)	
C2	return warming to 1.5°C (>50%) after a high overshoot***		
C3	limit warming to 2°C (>67%)	Low (SSP1-2.6)	RCP2.6
C4	limit warming to 2°C (>50%)		
C5	limit warming to 2.5°C (>50%)		
C6	limit warming to 3°C (>50%)	Intermediate (SSP2-4.5)	RCP 4.5
C7	limit warming to 4°C (>50%)	High (SSP3-7.0)	
C8	exceed warming of 4°C (>50%)	Very high (SSP5-8.5)	RCP 8.5

* See footnote 21 for the SSPx-y terminology.

** See footnote 23 for the RCPy terminology.

*** Limited overshoot refers to exceeding 1.5°C global warming by up to about 0.1°C, high overshoot by 0.1°C-0.3°C, in both cases for up to several decades.

Current Mitigation Progress, Gaps and Challenges

A.4 Policies and laws addressing mitigation have consistently expanded since AR5. Global GHG emissions in 2030 implied by nationally determined contributions (NDCs) announced by October 2021 make it *likely* that warming will exceed 1.5°C during the 21st century and make it harder to limit warming below 2°C. There are gaps between projected emissions from implemented policies and those from NDCs and finance flows fall short of the levels needed to meet climate goals across all sectors and regions. (*high confidence*) {2.2, 2.3, Figure 2.5, Table 2.2}

A.4.1 The UNFCCC, Kyoto Protocol, and the Paris Agreement are supporting rising levels of national ambition. The Paris Agreement, adopted under the UNFCCC, with near universal participation, has led to policy development and target-setting at national and sub-national levels, in particular in relation to mitigation, as well as enhanced transparency of climate action and support (*medium confidence*). Many regulatory and economic instruments have already been deployed successfully (*high confidence*). In many countries, policies have enhanced energy efficiency, reduced rates of deforestation and accelerated technology deployment, leading to avoided and in some cases reduced or removed emissions (*high confidence*). Multiple lines of evidence suggest that mitigation policies have led to several²⁴ Gt CO₂-eq yr⁻¹ of avoided global emissions (*medium confidence*). At least 18 countries have sustained absolute production-based GHG and consumption-based CO₂ reductions²⁵ for longer than 10 years. These reductions have only partly offset global emissions growth (*high confidence*). {2.2.1, 2.2.2}

A.4.2 Several mitigation options, notably solar energy, wind energy, electrification of urban systems, urban green infrastructure, energy efficiency, demand-side management, improved forest and crop/grassland management, and reduced food waste and loss, are technically viable, are becoming increasingly cost effective and are generally supported by the

²⁴ At least 1.8 GtCO₂-eq yr⁻¹ can be accounted for by aggregating separate estimates for the effects of economic and regulatory instruments. Growing numbers of laws and executive orders have impacted global emissions and were estimated to result in 5.9 GtCO₂-eq yr⁻¹ less emissions in 2016 than they otherwise would have been. (*medium confidence*) {2.2.2}

²⁵ Reductions were linked to energy supply decarbonisation, energy efficiency gains, and energy demand reduction, which resulted from both policies and changes in economic structure (*high confidence*). {2.2.2}

public. From 2010 to 2019 there have been sustained decreases in the unit costs of solar energy (85%), wind energy (55%), and lithium-ion batteries (85%), and large increases in their deployment, e.g., >10× for solar and >100× for electric vehicles (EVs), varying widely across regions. The mix of policy instruments that reduced costs and stimulated adoption includes public R&D, funding for demonstration and pilot projects, and demand-pull instruments such as deployment subsidies to attain scale. Maintaining emission-intensive systems may, in some regions and sectors, be more expensive than transitioning to low emission systems. (*high confidence*) {2.2.2, Figure 2.4}

- A.4.3 A substantial ‘emissions gap’ exists between global GHG emissions in 2030 associated with the implementation of NDCs announced prior to COP26²⁶ and those associated with modelled mitigation pathways that limit warming to 1.5°C (>50%) with no or limited overshoot or limit warming to 2°C (>67%) assuming immediate action (*high confidence*). This would make it *likely* that warming will exceed 1.5°C during the 21st century (*high confidence*). Global modelled mitigation pathways that limit warming to 1.5°C (>50%) with no or limited overshoot or limit warming to 2°C (>67%) assuming immediate action imply deep global GHG emissions reductions this decade (*high confidence*) (see SPM Box 1, Table 1, B.6)²⁷. Modelled pathways that are consistent with NDCs announced prior to COP26 until 2030 and assume no increase in ambition thereafter have higher emissions, leading to a median global warming of 2.8 [2.1 to 3.4] °C by 2100 (*medium confidence*). Many countries have signalled an intention to achieve net zero GHG or net zero CO₂ by around mid-century but pledges differ across countries in terms of scope and specificity, and limited policies are to date in place to deliver on them. {2.3.1, Table 2.2, Figure 2.5, Table 3.1, 4.1}
- A.4.4 Policy coverage is uneven across sectors (*high confidence*). Policies implemented by the end of 2020 are projected to result in higher global GHG emissions in 2030 than emissions implied by NDCs, indicating an ‘implementation gap’ (*high confidence*). Without a strengthening of policies, global warming of 3.2 [2.2 to 3.5] °C is projected by 2100 (*medium confidence*). {2.2.2, 2.3.1, 3.1.1, Figure 2.5} (Box SPM.1, Figure SPM.5)
- A.4.5 The adoption of low-emission technologies lags in most developing countries, particularly least developed ones, due in part to limited finance, technology development and transfer, and capacity (*medium confidence*). The magnitude of climate finance flows has increased over the last decade and financing channels have broadened but growth has slowed since 2018 (*high confidence*). Financial flows have developed heterogeneously across regions and sectors (*high confidence*). Public and private finance flows for fossil fuels are still greater than those for climate adaptation and mitigation (*high confidence*). The overwhelming majority of tracked climate finance is directed towards mitigation, but nevertheless falls short of the levels needed to limit warming to below 2°C or to 1.5°C across all sectors and regions (see C7.2) (*very high confidence*). In 2018, public and publicly mobilised private climate finance flows from developed to developing countries were below the collective goal under the UNFCCC and Paris Agreement to mobilise USD 100 billion per year by 2020 in the context of meaningful mitigation action and transparency on implementation (*medium confidence*). {2.2.2, 2.3.1, 2.3.3}

²⁶ Due to the literature cutoff date of WGIII, the additional NDCs submitted after 11 October 2021 are not assessed here. {Footnote 32 in the Longer Report}

²⁷ Projected 2030 GHG emissions are 50 (47–55) GtCO₂-eq if all conditional NDC elements are taken into account. Without conditional elements, the global emissions are projected to be approximately similar to modelled 2019 levels at 53 (50–57) GtCO₂-eq. {2.3.1, Table 2.2}

B. Future Climate Change, Risks, and Long-Term Responses

Future Climate Change

B.1 Continued greenhouse gas emissions will lead to increasing global warming, with the best estimate of reaching 1.5°C in the near term in considered scenarios and modelled pathways. Every increment of global warming will intensify multiple and concurrent hazards (*high confidence*). Deep, rapid, and sustained reductions in greenhouse gas emissions would lead to a discernible slowdown in global warming within around two decades, and also to discernible changes in atmospheric composition within a few years (*high confidence*). {Cross-Section Boxes 1 and 2, 3.1, 3.3, Table 3.1, Figure 3.1, 4.3} (Figure SPM.2, Box SPM.1)

- B.1.1** Global warming²⁸ will continue to increase in the near term (2021–2040) mainly due to increased cumulative CO₂ emissions in nearly all considered scenarios and modelled pathways. In the near term, global warming is *more likely than not* to reach 1.5°C even under the very low GHG emission scenario (SSP1-1.9) and *likely* or *very likely* to exceed 1.5°C under higher emissions scenarios. In the considered scenarios and modelled pathways, the best estimates of the time when the level of global warming of 1.5°C is reached lie in the near term²⁹. Global warming declines back to below 1.5°C by the end of the 21st century in some scenarios and modelled pathways (see B.7). The assessed climate response to GHG emissions scenarios results in a best estimate of warming for 2081–2100 that spans a range from 1.4°C for a very low GHG emissions scenario (SSP1-1.9) to 2.7°C for an intermediate GHG emissions scenario (SSP2-4.5) and 4.4°C for a very high GHG emissions scenario (SSP5-8.5)³⁰, with narrower uncertainty ranges³¹ than for corresponding scenarios in AR5. {Cross-Section Boxes 1 and 2, 3.1.1, 3.3.4, Table 3.1, 4.3} (Box SPM.1)
- B.1.2** Discernible differences in trends of global surface temperature between contrasting GHG emissions scenarios (SSP1-1.9 and SSP1-2.6 vs. SSP3-7.0 and SSP5-8.5) would begin to emerge from natural variability³² within around 20 years. Under these contrasting scenarios, discernible effects would emerge within years for GHG concentrations, and sooner for air quality improvements, due to the combined targeted air pollution controls and strong and sustained methane emissions reductions. Targeted reductions of air pollutant emissions lead to more rapid improvements in air quality within years compared to reductions in GHG emissions only, but in the long term, further improvements are projected in scenarios that combine efforts to reduce air pollutants as well as GHG emissions³³. (*high confidence*) {3.1.1} (Box SPM.1)
- B.1.3** Continued emissions will further affect all major climate system components. With every additional increment of global warming, changes in extremes continue to become larger. Continued global warming is projected to further intensify the global water cycle, including its variability, global monsoon precipitation, and very wet and very dry weather and

²⁸ Global warming (see Annex I: Glossary) is here reported as running 20-year averages, unless stated otherwise, relative to 1850–1900. Global surface temperature in any single year can vary above or below the long-term human-caused trend, due to natural variability. The internal variability of global surface temperature in a single year is estimated to be about $\pm 0.25^\circ\text{C}$ (5–95% range, *high confidence*). The occurrence of individual years with global surface temperature change above a certain level does not imply that this global warming level has been reached. {4.3, Cross-Section Box.2}

²⁹ Median five-year interval at which a 1.5°C global warming level is reached (50% probability) in categories of modelled pathways considered in WGIII is 2030–2035. By 2030, global surface temperature in any individual year could exceed 1.5°C relative to 1850–1900 with a probability between 40% and 60%, across the five scenarios assessed in WGI (*medium confidence*). In all scenarios considered in WGI except the very high emissions scenario (SSP5-8.5), the midpoint of the first 20-year running average period during which the assessed average global surface temperature change reaches 1.5°C lies in the first half of the 2030s. In the very high GHG emissions scenario, the midpoint is in the late 2020s. {3.1.1, 3.3.1, 4.3} (Box SPM.1)

³⁰ The best estimates [and *very likely* ranges] for the different scenarios are: 1.4 [1.0 to 1.8]°C (SSP1-1.9); 1.8 [1.3 to 2.4]°C (SSP1-2.6); 2.7 [2.1 to 3.5]°C (SSP2-4.5); 3.6 [2.8 to 4.6]°C (SSP3-7.0); and 4.4 [3.3 to 5.7]°C (SSP5-8.5). {3.1.1} (Box SPM.1)

³¹ Assessed future changes in global surface temperature have been constructed, for the first time, by combining multi-model projections with observational constraints and the assessed equilibrium climate sensitivity and transient climate response. The uncertainty range is narrower than in the AR5 thanks to improved knowledge of climate processes, paleoclimate evidence and model-based emergent constraints. {3.1.1}

³² See Annex I: Glossary. Natural variability includes natural drivers and internal variability. The main internal variability phenomena include El Niño–Southern Oscillation, Pacific Decadal Variability and Atlantic Multi-decadal Variability. {4.3}

³³ Based on additional scenarios.

climate events and seasons (*high confidence*). In scenarios with increasing CO₂ emissions, natural land and ocean carbon sinks are projected to take up a decreasing proportion of these emissions (*high confidence*). Other projected changes include further reduced extents and/or volumes of almost all cryospheric elements³⁴ (*high confidence*), further global mean sea level rise (*virtually certain*), and increased ocean acidification (*virtually certain*) and deoxygenation (*high confidence*). {3.1.1, 3.3.1, Figure 3.4} (Figure SPM.2)

- B.1.4 With further warming, every region is projected to increasingly experience concurrent and multiple changes in climatic impact-drivers. Compound heatwaves and droughts are projected to become more frequent, including concurrent events across multiple locations (*high confidence*). Due to relative sea level rise, current 1-in-100 year extreme sea level events are projected to occur at least annually in more than half of all tide gauge locations by 2100 under all considered scenarios (*high confidence*). Other projected regional changes include intensification of tropical cyclones and/or extratropical storms (*medium confidence*), and increases in aridity and fire weather (*medium to high confidence*). {3.1.1, 3.1.3}
- B.1.5 Natural variability will continue to modulate human-caused climate changes, either attenuating or amplifying projected changes, with little effect on centennial-scale global warming (*high confidence*). These modulations are important to consider in adaptation planning, especially at the regional scale and in the near term. If a large explosive volcanic eruption were to occur³⁵, it would temporarily and partially mask human-caused climate change by reducing global surface temperature and precipitation for one to three years (*medium confidence*). {4.3}

³⁴ Permafrost, seasonal snow cover, glaciers, the Greenland and Antarctic Ice Sheets, and Arctic sea ice.

³⁵ Based on 2500-year reconstructions, eruptions with a radiative forcing more negative than -1 W m^{-2} , related to the radiative effect of volcanic stratospheric aerosols in the literature assessed in this report, occur on average twice per century. {4.3}

With every increment of global warming, regional changes in mean climate and extremes become more widespread and pronounced

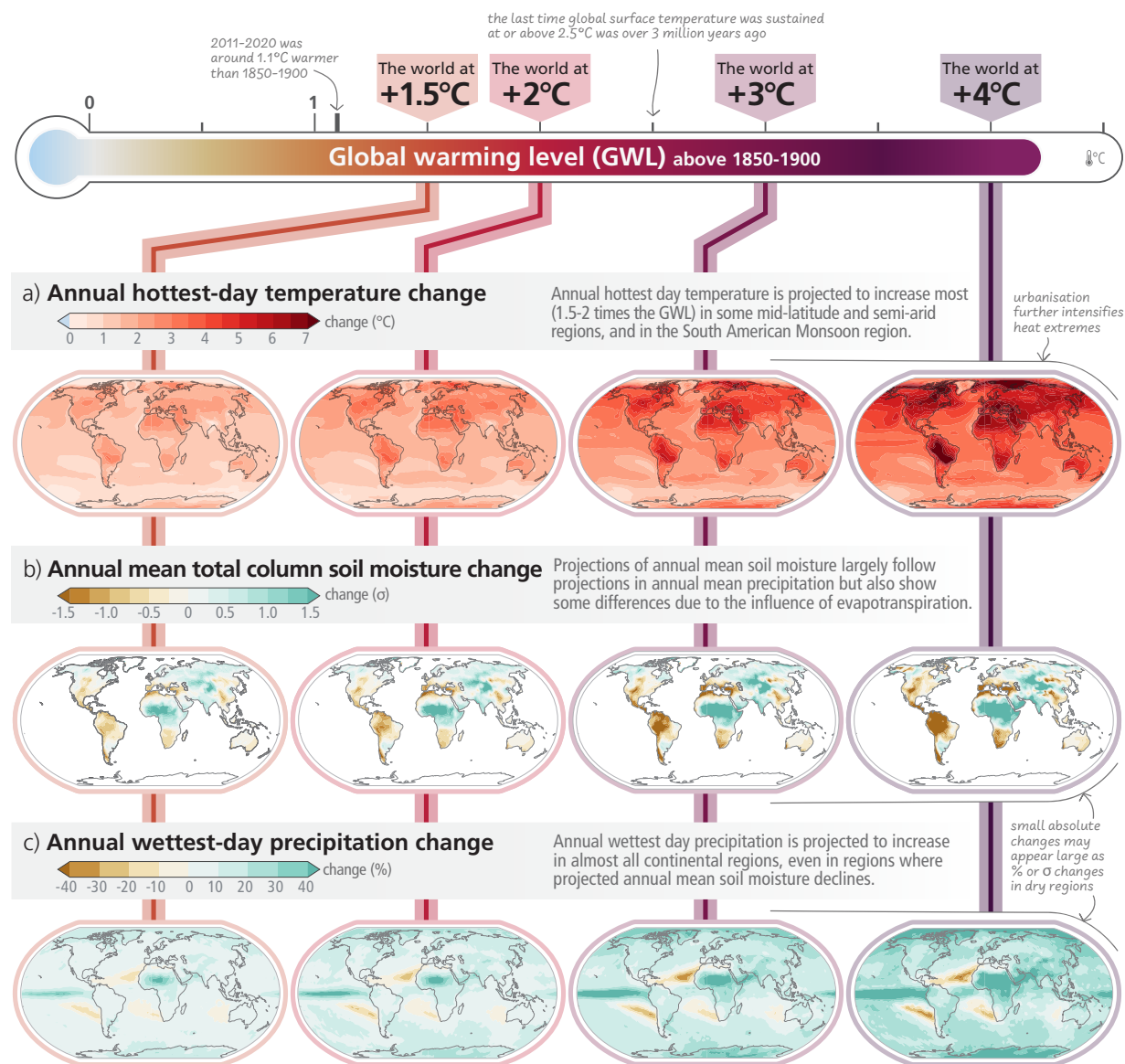


Figure SPM.2: Projected changes of annual maximum daily maximum temperature, annual mean total column soil moisture and annual maximum 1-day precipitation at global warming levels of 1.5°C, 2°C, 3°C, and 4°C relative to 1850–1900. Projected (a) annual maximum daily temperature change (°C), (b) annual mean total column soil moisture change (standard deviation), (c) annual maximum 1-day precipitation change (%). The panels show CMIP6 multi-model median changes. In panels (b) and (c), large positive relative changes in dry regions may correspond to small absolute changes. In panel (b), the unit is the standard deviation of interannual variability in soil moisture during 1850–1900. Standard deviation is a widely used metric in characterising drought severity. A projected reduction in mean soil moisture by one standard deviation corresponds to soil moisture conditions typical of droughts that occurred about once every six years during 1850–1900. The WGI Interactive Atlas (<https://interactive-atlas.ipcc.ch/>) can be used to explore additional changes in the climate system across the range of global warming levels presented in this figure. {Figure 3.1, Cross-Section Box.2}

Climate Change Impacts and Climate-Related Risks

B.2 For any given future warming level, many climate-related risks are higher than assessed in AR5, and projected long-term impacts are up to multiple times higher than currently observed (*high confidence*). Risks and projected adverse impacts and related losses and damages from climate change escalate with every increment of global warming (*very high confidence*). Climatic and non-climatic risks will increasingly interact, creating compound and cascading risks that are more complex and difficult to manage (*high confidence*). {Cross-Section Box.2, 3.1, 4.3, Figure 3.3, Figure 4.3} (Figure SPM.3, Figure SPM.4)

- B.2.1 In the near term, every region in the world is projected to face further increases in climate hazards (*medium to high confidence*, depending on region and hazard), increasing multiple risks to ecosystems and humans (*very high confidence*). Hazards and associated risks expected in the near term include an increase in heat-related human mortality and morbidity (*high confidence*), food-borne, water-borne, and vector-borne diseases (*high confidence*), and mental health challenges³⁶ (*very high confidence*), flooding in coastal and other low-lying cities and regions (*high confidence*), biodiversity loss in land, freshwater and ocean ecosystems (*medium to very high confidence*, depending on ecosystem), and a decrease in food production in some regions (*high confidence*). Cryosphere-related changes in floods, landslides, and water availability have the potential to lead to severe consequences for people, infrastructure and the economy in most mountain regions (*high confidence*). The projected increase in frequency and intensity of heavy precipitation (*high confidence*) will increase rain-generated local flooding (*medium confidence*). {Figure 3.2, Figure 3.3, 4.3, Figure 4.3} (Figure SPM.3, Figure SPM.4)
- B.2.2 Risks and projected adverse impacts and related losses and damages from climate change will escalate with every increment of global warming (*very high confidence*). They are higher for global warming of 1.5°C than at present, and even higher at 2°C (*high confidence*). Compared to the AR5, global aggregated risk levels³⁷ (Reasons for Concern³⁸) are assessed to become high to very high at lower levels of global warming due to recent evidence of observed impacts, improved process understanding, and new knowledge on exposure and vulnerability of human and natural systems, including limits to adaptation (*high confidence*). Due to unavoidable sea level rise (see also B.3), risks for coastal ecosystems, people and infrastructure will continue to increase beyond 2100 (*high confidence*). {3.1.2, 3.1.3, Figure 3.4, Figure 4.3} (Figure SPM.3, Figure SPM.4)
- B.2.3 With further warming, climate change risks will become increasingly complex and more difficult to manage. Multiple climatic and non-climatic risk drivers will interact, resulting in compounding overall risk and risks cascading across sectors and regions. Climate-driven food insecurity and supply instability, for example, are projected to increase with increasing global warming, interacting with non-climatic risk drivers such as competition for land between urban expansion and food production, pandemics and conflict. (*high confidence*) {3.1.2, 4.3, Figure 4.3}
- B.2.4 For any given warming level, the level of risk will also depend on trends in vulnerability and exposure of humans and ecosystems. Future exposure to climatic hazards is increasing globally due to socio-economic development trends including migration, growing inequality and urbanisation. Human vulnerability will concentrate in informal settlements and rapidly growing smaller settlements. In rural areas vulnerability will be heightened by high reliance on climate-sensitive livelihoods. Vulnerability of ecosystems will be strongly influenced by past, present, and future patterns of unsustainable consumption and production, increasing demographic pressures, and persistent unsustainable use and management of land, ocean, and water. Loss of ecosystems and their services has cascading and long-term impacts on people globally, especially for Indigenous Peoples and local communities who are directly dependent on ecosystems to meet basic needs. (*high confidence*) {Cross-Section Box.2 Figure 1c, 3.1.2, 4.3}

³⁶ In all assessed regions.

³⁷ Undetectable risk level indicates no associated impacts are detectable and attributable to climate change; moderate risk indicates associated impacts are both detectable and attributable to climate change with at least *medium confidence*, also accounting for the other specific criteria for key risks; high risk indicates severe and widespread impacts that are judged to be high on one or more criteria for assessing key risks; and very high risk level indicates very high risk of severe impacts and the presence of significant irreversibility or the persistence of climate-related hazards, combined with limited ability to adapt due to the nature of the hazard or impacts/risks. {3.1.2}

³⁸ The Reasons for Concern (RFC) framework communicates scientific understanding about accrual of risk for five broad categories. RFC1: Unique and threatened systems: ecological and human systems that have restricted geographic ranges constrained by climate-related conditions and have high endemism or other distinctive properties. RFC2: Extreme weather events: risks/impacts to human health, livelihoods, assets and ecosystems from extreme weather events. RFC3: Distribution of impacts: risks/impacts that disproportionately affect particular groups due to uneven distribution of physical climate change hazards, exposure or vulnerability. RFC4: Global aggregate impacts: impacts to socio-ecological systems that can be aggregated globally into a single metric. RFC5: Large-scale singular events: relatively large, abrupt and sometimes irreversible changes in systems caused by global warming. See also Annex I: Glossary. {3.1.2, Cross-Section Box.2}

Future climate change is projected to increase the severity of impacts across natural and human systems and will increase regional differences

Examples of impacts without additional adaptation

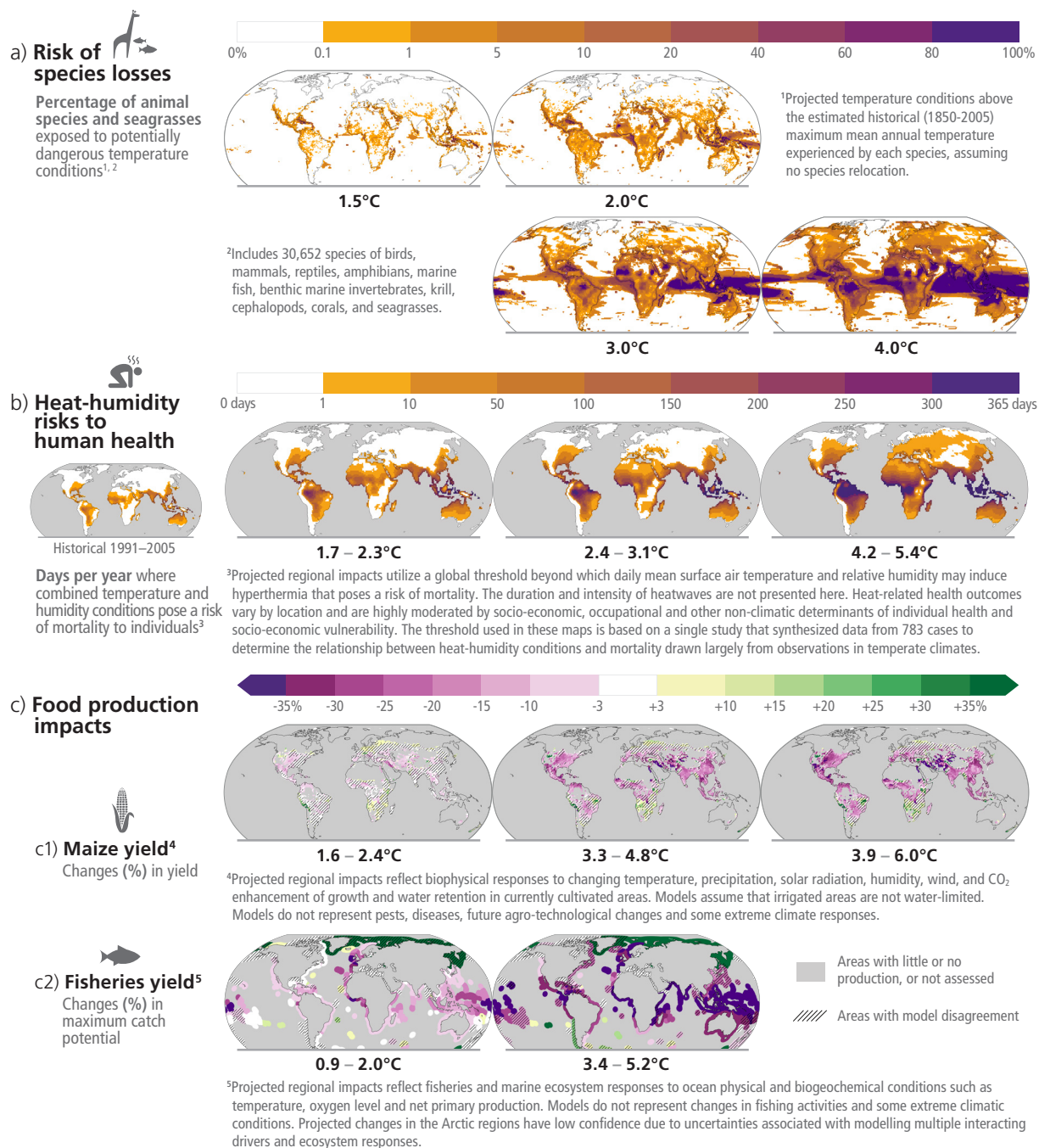
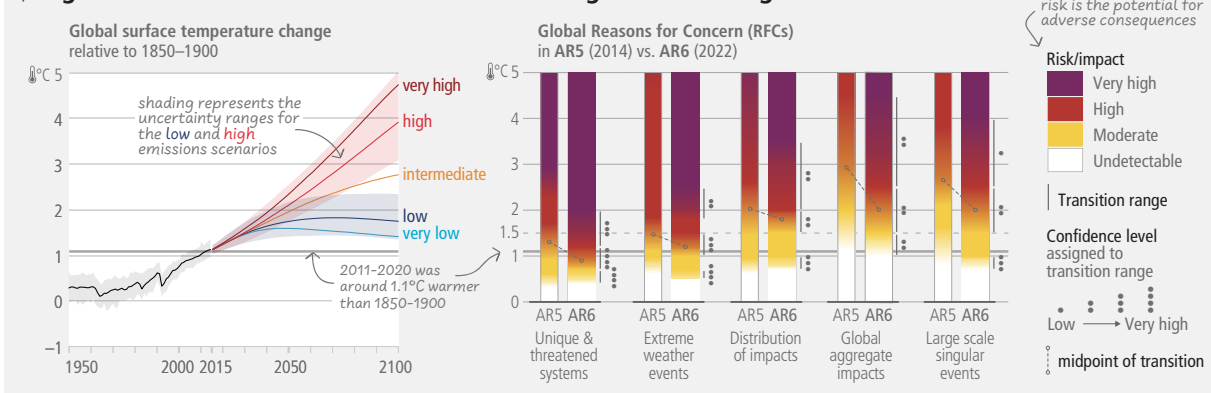


Figure SPM.3: Projected risks and impacts of climate change on natural and human systems at different global warming levels (GWs) relative to 1850-1900 levels. Projected risks and impacts shown on the maps are based on outputs from different subsets of Earth system and impact models that were used to project each impact indicator without additional adaptation. WGII provides further assessment of the impacts on human and natural systems using these projections and additional lines of evidence. **(a)** Risks of species losses as indicated by the percentage of assessed species exposed to potentially dangerous temperature conditions, as defined by conditions beyond the estimated historical (1850–2005) maximum mean annual temperature experienced by each species, at GWs of 1.5°C, 2°C, 3°C and 4°C. Underpinning projections of temperature are from 21 Earth system models and do not consider extreme events impacting ecosystems such as the Arctic. **(b)** Risks to human health as indicated by the days per year of population exposure to hyperthermic conditions that pose a risk of mortality from surface air temperature and humidity conditions for historical period (1991–2005) and at GWs of 1.7°C–2.3°C (mean = 1.9°C; 13 climate models), 2.4°C–3.1°C (2.7°C; 16 climate models) and 4.2°C–5.4°C (4.7°C; 15 climate models). Interquartile ranges of GWs by 2081–2100 under RCP2.6, RCP4.5 and RCP8.5. The presented index is consistent with common features found in many indices included within WGI and WGII assessments. **(c)** Impacts on food production: (c1) Changes in maize yield by 2080–2099 relative to 1986–2005 at projected GWs of 1.6°C–2.4°C (2.0°C), 3.3°C–4.8°C (4.1°C) and 3.9°C–6.0°C (4.9°C). Median yield changes from an ensemble of 12 crop models, each driven by bias-adjusted outputs from 5 Earth system models, from the Agricultural Model Intercomparison and Improvement Project (AgMIP) and the Inter-Sectoral Impact Model Intercomparison Project (ISI-MIP). Maps depict

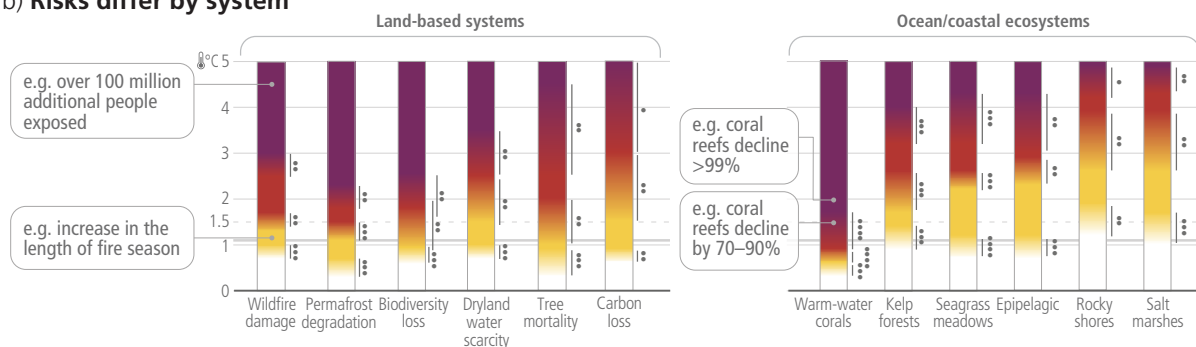
2080–2099 compared to 1986–2005 for current growing regions (>10 ha), with the corresponding range of future global warming levels shown under SSP1-2.6, SSP3-7.0 and SSP5-8.5, respectively. Hatching indicates areas where <70% of the climate-crop model combinations agree on the sign of impact. (c2) Change in maximum fisheries catch potential by 2081–2099 relative to 1986–2005 at projected GWLs of 0.9°C–2.0°C (1.5°C) and 3.4°C–5.2°C (4.3°C). GWLs by 2081–2100 under RCP2.6 and RCP8.5. Hatching indicates where the two climate-fisheries models disagree in the direction of change. Large relative changes in low yielding regions may correspond to small absolute changes. Biodiversity and fisheries in Antarctica were not analysed due to data limitations. Food security is also affected by crop and fishery failures not presented here. {3.1.2, Figure 3.2, Cross-Section Box.2} (Box SPM.1)

Risks are increasing with every increment of warming

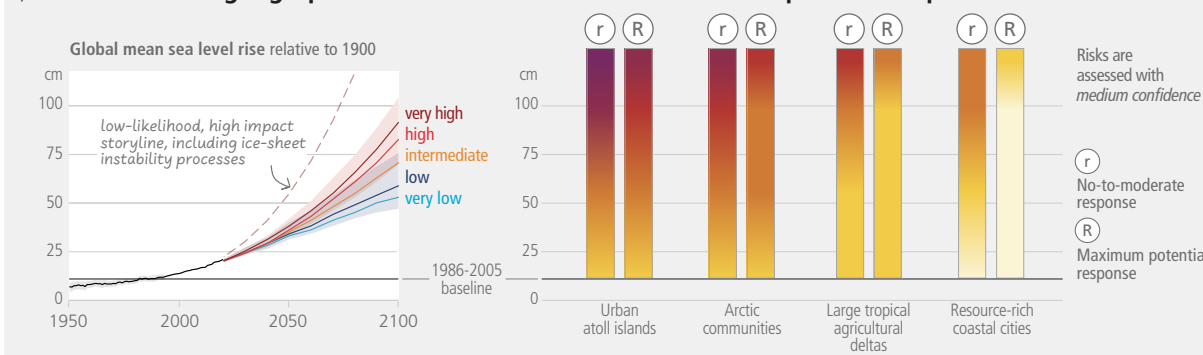
a) High risks are now assessed to occur at lower global warming levels



b) Risks differ by system

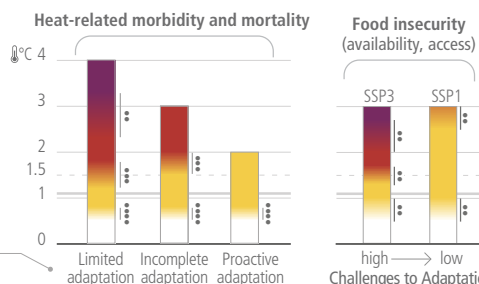


c) Risks to coastal geographies increase with sea level rise and depend on responses



d) Adaptation and socio-economic pathways affect levels of climate related risks

Limited adaptation (failure to proactively adapt; low investment in health systems); incomplete adaptation (incomplete adaptation planning; moderate investment in health systems); proactive adaptation (proactive adaptation management; higher investment in health systems)



The SSP1 pathway illustrates a world with low population growth, high income, and reduced inequalities, food produced in low GHG emission systems, effective land use regulation and high adaptive capacity (i.e., low challenges to adaptation). The SSP3 pathway has the opposite trends.

Figure SPM.4: Subset of assessed climate outcomes and associated global and regional climate risks. The burning embers result from a literature based expert elicitation. **Panel (a): Left** – Global surface temperature changes in °C relative to 1850–1900. These changes were obtained by combining CMIP6 model simulations with observational constraints based on past simulated warming, as well as an updated assessment of equilibrium climate sensitivity. *Very likely* ranges are shown for the low and high GHG emissions scenarios (SSP1-2.6 and SSP3-7.0) (Cross-Section Box.2). **Right** – Global Reasons for Concern (RFC), comparing AR6 (thick embers) and AR5 (thin embers) assessments. Risk transitions have generally shifted towards lower temperatures with updated scientific understanding. Diagrams are shown for each RFC, assuming low to no adaptation. Lines connect the midpoints of the transitions from moderate to high risk across AR5 and AR6. **Panel (b):** Selected global risks for land and ocean ecosystems, illustrating general increase of risk with global warming levels with low to no adaptation. **Panel (c): Left** – Global mean sea level change in centimetres, relative to 1900. The historical changes (black) are observed by tide gauges before 1992 and altimeters afterwards. The future changes to 2100 (coloured lines and shading) are assessed consistently with observational constraints based on emulation of CMIP, ice-sheet, and glacier models, and *likely* ranges are shown for SSP1-2.6 and SSP3-7.0. **Right** – Assessment of the combined risk of coastal flooding, erosion and salinization for four illustrative coastal geographies in 2100, due to changing mean and extreme sea levels, under two response scenarios, with respect to the SROCC baseline period (1986–2005). The assessment does not account for changes in extreme sea level beyond those directly induced by mean sea level rise; risk levels could increase if other changes in extreme sea levels were considered (e.g., due to changes in cyclone intensity). “No-to-moderate response” describes efforts as of today (i.e., no further significant action or new types of actions). “Maximum potential response” represent a combination of responses implemented to their full extent and thus significant additional efforts compared to today, assuming minimal financial, social and political barriers. (In this context, ‘today’ refers to 2019.) The assessment criteria include exposure and vulnerability, coastal hazards, in-situ responses and planned relocation. Planned relocation refers to managed retreat or resettlements. The term response is used here instead of adaptation because some responses, such as retreat, may or may not be considered to be adaptation. **Panel (d):** Selected risks under different socio-economic pathways, illustrating how development strategies and challenges to adaptation influence risk. **Left** – Heat-sensitive human health outcomes under three scenarios of adaptation effectiveness. The diagrams are truncated at the nearest whole °C within the range of temperature change in 2100 under three SSP scenarios. **Right** – Risks associated with food security due to climate change and patterns of socio-economic development. Risks to food security include availability and access to food, including population at risk of hunger, food price increases and increases in disability adjusted life years attributable to childhood underweight. Risks are assessed for two contrasted socio-economic pathways (SSP1 and SSP3) excluding the effects of targeted mitigation and adaptation policies. {Figure 3.3} (Box SPM.1)

Likelihood and Risks of Unavoidable, Irreversible or Abrupt Changes

B.3 Some future changes are unavoidable and/or irreversible but can be limited by deep, rapid, and sustained global greenhouse gas emissions reduction. The likelihood of abrupt and/or irreversible changes increases with higher global warming levels. Similarly, the probability of low-likelihood outcomes associated with potentially very large adverse impacts increases with higher global warming levels. (*high confidence*) {3.1}

- B.3.1** Limiting global surface temperature does not prevent continued changes in climate system components that have multi-decadal or longer timescales of response (*high confidence*). Sea level rise is unavoidable for centuries to millennia due to continuing deep ocean warming and ice sheet melt, and sea levels will remain elevated for thousands of years (*high confidence*). However, deep, rapid, and sustained GHG emissions reductions would limit further sea level rise acceleration and projected long-term sea level rise commitment. Relative to 1995–2014, the *likely* global mean sea level rise under the SSP1-1.9 GHG emissions scenario is 0.15–0.23 m by 2050 and 0.28–0.55 m by 2100; while for the SSP5-8.5 GHG emissions scenario it is 0.20–0.29 m by 2050 and 0.63–1.01 m by 2100 (*medium confidence*). Over the next 2000 years, global mean sea level will rise by about 2–3 m if warming is limited to 1.5°C and 2–6 m if limited to 2°C (*low confidence*). {3.1.3, Figure 3.4} (Box SPM.1)
- B.3.2** The likelihood and impacts of abrupt and/or irreversible changes in the climate system, including changes triggered when tipping points are reached, increase with further global warming (*high confidence*). As warming levels increase, so do the risks of species extinction or irreversible loss of biodiversity in ecosystems including forests (*medium confidence*), coral reefs (*very high confidence*) and in Arctic regions (*high confidence*). At sustained warming levels between 2°C and 3°C, the Greenland and West Antarctic ice sheets will be lost almost completely and irreversibly over multiple millennia, causing several metres of sea level rise (*limited evidence*). The probability and rate of ice mass loss increase with higher global surface temperatures (*high confidence*). {3.1.2, 3.1.3}
- B.3.3** The probability of low-likelihood outcomes associated with potentially very large impacts increases with higher global warming levels (*high confidence*). Due to deep uncertainty linked to ice-sheet processes, global mean sea level rise above the *likely* range – approaching 2 m by 2100 and in excess of 15 m by 2300 under the very high GHG emissions scenario (SSP5-8.5) (*low confidence*) – cannot be excluded. There is *medium confidence* that the Atlantic Meridional Overturning Circulation will not collapse abruptly before 2100, but if it were to occur, it would *very likely* cause abrupt shifts in regional weather patterns, and large impacts on ecosystems and human activities. {3.1.3} (Box SPM.1)

Adaptation Options and their Limits in a Warmer World

B.4 Adaptation options that are feasible and effective today will become constrained and less effective with increasing global warming. With increasing global warming, losses and damages will increase and additional human and natural systems will reach adaptation limits. Maladaptation can be avoided by flexible, multi-sectoral, inclusive, long-term planning and implementation of adaptation actions, with co-benefits to many sectors and systems. (*high confidence*) {3.2, 4.1, 4.2, 4.3}

- B.4.1 The effectiveness of adaptation, including ecosystem-based and most water-related options, will decrease with increasing warming. The feasibility and effectiveness of options increase with integrated, multi-sectoral solutions that differentiate responses based on climate risk, cut across systems and address social inequities. As adaptation options often have long implementation times, long-term planning increases their efficiency. (*high confidence*) {3.2, Figure 3.4, 4.1, 4.2}
- B.4.2 With additional global warming, limits to adaptation and losses and damages, strongly concentrated among vulnerable populations, will become increasingly difficult to avoid (*high confidence*). Above 1.5°C of global warming, limited freshwater resources pose potential hard adaptation limits for small islands and for regions dependent on glacier and snow melt (*medium confidence*). Above that level, ecosystems such as some warm-water coral reefs, coastal wetlands, rainforests, and polar and mountain ecosystems will have reached or surpassed hard adaptation limits and as a consequence, some Ecosystem-based Adaptation measures will also lose their effectiveness (*high confidence*). {2.3.2, 3.2, 4.3}
- B.4.3 Actions that focus on sectors and risks in isolation and on short-term gains often lead to maladaptation over the long term, creating lock-ins of vulnerability, exposure and risks that are difficult to change. For example, seawalls effectively reduce impacts to people and assets in the short term but can also result in lock-ins and increase exposure to climate risks in the long term unless they are integrated into a long-term adaptive plan. Maladaptive responses can worsen existing inequities especially for Indigenous Peoples and marginalised groups and decrease ecosystem and biodiversity resilience. Maladaptation can be avoided by flexible, multi-sectoral, inclusive, long-term planning and implementation of adaptation actions, with co-benefits to many sectors and systems. (*high confidence*) {2.3.2, 3.2}

Carbon Budgets and Net Zero Emissions

B.5 Limiting human-caused global warming requires net zero CO₂ emissions. Cumulative carbon emissions until the time of reaching net zero CO₂ emissions and the level of greenhouse gas emission reductions this decade largely determine whether warming can be limited to 1.5°C or 2°C (*high confidence*). Projected CO₂ emissions from existing fossil fuel infrastructure without additional abatement would exceed the remaining carbon budget for 1.5°C (50%) (*high confidence*). {2.3, 3.1, 3.3, Table 3.1}

- B.5.1 From a physical science perspective, limiting human-caused global warming to a specific level requires limiting cumulative CO₂ emissions, reaching at least net zero CO₂ emissions, along with strong reductions in other greenhouse gas emissions. Reaching net zero GHG emissions primarily requires deep reductions in CO₂, methane, and other GHG emissions, and implies net negative CO₂ emissions³⁹. Carbon dioxide removal (CDR) will be necessary to achieve net negative CO₂ emissions (see B.6). Net zero GHG emissions, if sustained, are projected to result in a gradual decline in global surface temperatures after an earlier peak. (*high confidence*) {3.1.1, 3.3.1, 3.3.2, 3.3.3, Table 3.1, Cross-Section Box.1}
- B.5.2 For every 1000 GtCO₂ emitted by human activity, global surface temperature rises by 0.45°C (best estimate, with a *likely* range from 0.27°C to 0.63°C). The best estimates of the remaining carbon budgets from the beginning of 2020 are 500 GtCO₂ for a 50% likelihood of limiting global warming to 1.5°C and 1150 GtCO₂ for a 67% likelihood of limiting warming to 2°C⁴⁰. The stronger the reductions in non-CO₂ emissions, the lower the resulting temperatures are for a given remaining carbon budget or the larger remaining carbon budget for the same level of temperature change⁴¹. {3.3.1}

³⁹ Net zero GHG emissions defined by the 100-year global warming potential. See footnote 9.

⁴⁰ Global databases make different choices about which emissions and removals occurring on land are considered anthropogenic. Most countries report their anthropogenic land CO₂ fluxes including fluxes due to human-caused environmental change (e.g., CO₂ fertilisation) on 'managed' land in their national GHG inventories. Using emissions estimates based on these inventories, the remaining carbon budgets must be correspondingly reduced. {3.3.1}

⁴¹ For example, remaining carbon budgets could be 300 or 600 GtCO₂ for 1.5°C (50%), respectively for high and low non-CO₂ emissions, compared to 500 GtCO₂ in the central case. {3.3.1}

- B.5.3 If the annual CO₂ emissions between 2020–2030 stayed, on average, at the same level as 2019, the resulting cumulative emissions would almost exhaust the remaining carbon budget for 1.5°C (50%), and deplete more than a third of the remaining carbon budget for 2°C (67%). Estimates of future CO₂ emissions from existing fossil fuel infrastructures without additional abatement⁴² already exceed the remaining carbon budget for limiting warming to 1.5°C (50%) (*high confidence*). Projected cumulative future CO₂ emissions over the lifetime of existing and planned fossil fuel infrastructure, if historical operating patterns are maintained and without additional abatement⁴³, are approximately equal to the remaining carbon budget for limiting warming to 2°C with a likelihood of 83%⁴⁴ (*high confidence*). {2.3.1, 3.3.1, Figure 3.5}
- B.5.4 Based on central estimates only, historical cumulative net CO₂ emissions between 1850 and 2019 amount to about four fifths⁴⁵ of the total carbon budget for a 50% probability of limiting global warming to 1.5°C (central estimate about 2900 GtCO₂), and to about two thirds⁴⁶ of the total carbon budget for a 67% probability to limit global warming to 2°C (central estimate about 3550 GtCO₂). {3.3.1, Figure 3.5}

Mitigation Pathways

B.6 All global modelled pathways that limit warming to 1.5°C (>50%) with no or limited overshoot, and those that limit warming to 2°C (>67%), involve rapid and deep and, in most cases, immediate greenhouse gas emissions reductions in all sectors this decade. Global net zero CO₂ emissions are reached for these pathway categories, in the early 2050s and around the early 2070s, respectively. (*high confidence*) {3.3, 3.4, 4.1, 4.5, Table 3.1} (Figure SPM.5, Box SPM.1)

- B.6.1 Global modelled pathways provide information on limiting warming to different levels; these pathways, particularly their sectoral and regional aspects, depend on the assumptions described in Box SPM.1. Global modelled pathways that limit warming to 1.5°C (>50%) with no or limited overshoot or limit warming to 2°C (>67%) are characterized by deep, rapid, and, in most cases, immediate GHG emissions reductions. Pathways that limit warming to 1.5 °C (>50%) with no or limited overshoot reach net zero CO₂ in the early 2050s, followed by net negative CO₂ emissions. Those pathways that reach net zero GHG emissions do so around the 2070s. Pathways that limit warming to 2 °C (>67%) reach net zero CO₂ emissions in the early 2070s. Global GHG emissions are projected to peak between 2020 and at the latest before 2025 in global modelled pathways that limit warming to 1.5°C (>50%) with no or limited overshoot and in those that limit warming to 2°C (>67%) and assume immediate action. (*high confidence*) {3.3.2, 3.3.4, 4.1, Table 3.1, Figure 3.6} (Table SPM.1)

⁴² Abatement here refers to human interventions that reduce the amount of greenhouse gases that are released from fossil fuel infrastructure to the atmosphere.

⁴³ Ibid.

⁴⁴ WGI provides carbon budgets that are in line with limiting global warming to temperature limits with different likelihoods, such as 50%, 67% or 83%. {3.3.1}

⁴⁵ Uncertainties for total carbon budgets have not been assessed and could affect the specific calculated fractions.

⁴⁶ Ibid.

Table SPM.1: Greenhouse gas and CO₂ emission reductions from 2019, median and 5-95 percentiles. {3.3.1, 4.1, Table 3.1, Figure 2.5, Box SPM.1}

	Reductions from 2019 emission levels (%)				
		2030	2035	2040	2050
Limit warming to 1.5°C (>50%) with no or limited overshoot	GHG	43 [34-60]	60 [49-77]	69 [58-90]	84 [73-98]
	CO ₂	48 [36-69]	65 [50-96]	80 [61-109]	99 [79-119]
Limit warming to 2°C (>67%)	GHG	21 [1-42]	35 [22-55]	46 [34-63]	64 [53-77]
	CO ₂	22 [1-44]	37 [21-59]	51 [36-70]	73 [55-90]

- B.6.2 Reaching net zero CO₂ or GHG emissions primarily requires deep and rapid reductions in gross emissions of CO₂, as well as substantial reductions of non-CO₂ GHG emissions (*high confidence*). For example, in modelled pathways that limit warming to 1.5°C (>50%) with no or limited overshoot, global methane emissions are reduced by 34 [21–57] % by 2030 relative to 2019. However, some hard-to-abate residual GHG emissions (e.g., some emissions from agriculture, aviation, shipping, and industrial processes) remain and would need to be counterbalanced by deployment of CDR methods to achieve net zero CO₂ or GHG emissions (*high confidence*). As a result, net zero CO₂ is reached earlier than net zero GHGs (*high confidence*). {3.3.2, 3.3.3, Table 3.1, Figure 3.5} (Figure SPM.5)
- B.6.3 Global modelled mitigation pathways reaching net zero CO₂ and GHG emissions include transitioning from fossil fuels without carbon capture and storage (CCS) to very low- or zero-carbon energy sources, such as renewables or fossil fuels with CCS, demand-side measures and improving efficiency, reducing non-CO₂ GHG emissions, and CDR⁴⁷. In most global modelled pathways, land-use change and forestry (via reforestation and reduced deforestation) and the energy supply sector reach net zero CO₂ emissions earlier than the buildings, industry and transport sectors. (*high confidence*) {3.3.3, 4.1, 4.5, Figure 4.1} (Figure SPM.5, Box SPM.1)
- B.6.4 Mitigation options often have synergies with other aspects of sustainable development, but some options can also have trade-offs. There are potential synergies between sustainable development and, for instance, energy efficiency and renewable energy. Similarly, depending on the context⁴⁸, biological CDR methods like reforestation, improved forest management, soil carbon sequestration, peatland restoration and coastal blue carbon management can enhance biodiversity and ecosystem functions, employment and local livelihoods. However, afforestation or production of biomass crops can have adverse socio-economic and environmental impacts, including on biodiversity, food and water security, local livelihoods and the rights of Indigenous Peoples, especially if implemented at large scales and where land tenure is insecure. Modelled pathways that assume using resources more efficiently or that shift global development towards sustainability include fewer challenges, such as less dependence on CDR and pressure on land and biodiversity. (*high confidence*) {3.4.1}

⁴⁷ CCS is an option to reduce emissions from large-scale fossil-based energy and industry sources provided geological storage is available. When CO₂ is captured directly from the atmosphere (DACCS), or from biomass (BECCS), CCS provides the storage component of these CDR methods. CO₂ capture and subsurface injection is a mature technology for gas processing and enhanced oil recovery. In contrast to the oil and gas sector, CCS is less mature in the power sector, as well as in cement and chemicals production, where it is a critical mitigation option. The technical geological storage capacity is estimated to be on the order of 1000 GtCO₂, which is more than the CO₂ storage requirements through 2100 to limit global warming to 1.5°C, although the regional availability of geological storage could be a limiting factor. If the geological storage site is appropriately selected and managed, it is estimated that the CO₂ can be permanently isolated from the atmosphere. Implementation of CCS currently faces technological, economic, institutional, ecological-environmental and socio-cultural barriers. Currently, global rates of CCS deployment are far below those in modelled pathways limiting global warming to 1.5°C to 2°C. Enabling conditions such as policy instruments, greater public support and technological innovation could reduce these barriers. (*high confidence*) {3.3.3}

⁴⁸ The impacts, risks, and co-benefits of CDR deployment for ecosystems, biodiversity and people will be highly variable depending on the method, site-specific context, implementation and scale (*high confidence*).

Limiting warming to 1.5°C and 2°C involves rapid, deep and in most cases immediate greenhouse gas emission reductions

Net zero CO₂ and net zero GHG emissions can be achieved through strong reductions across all sectors

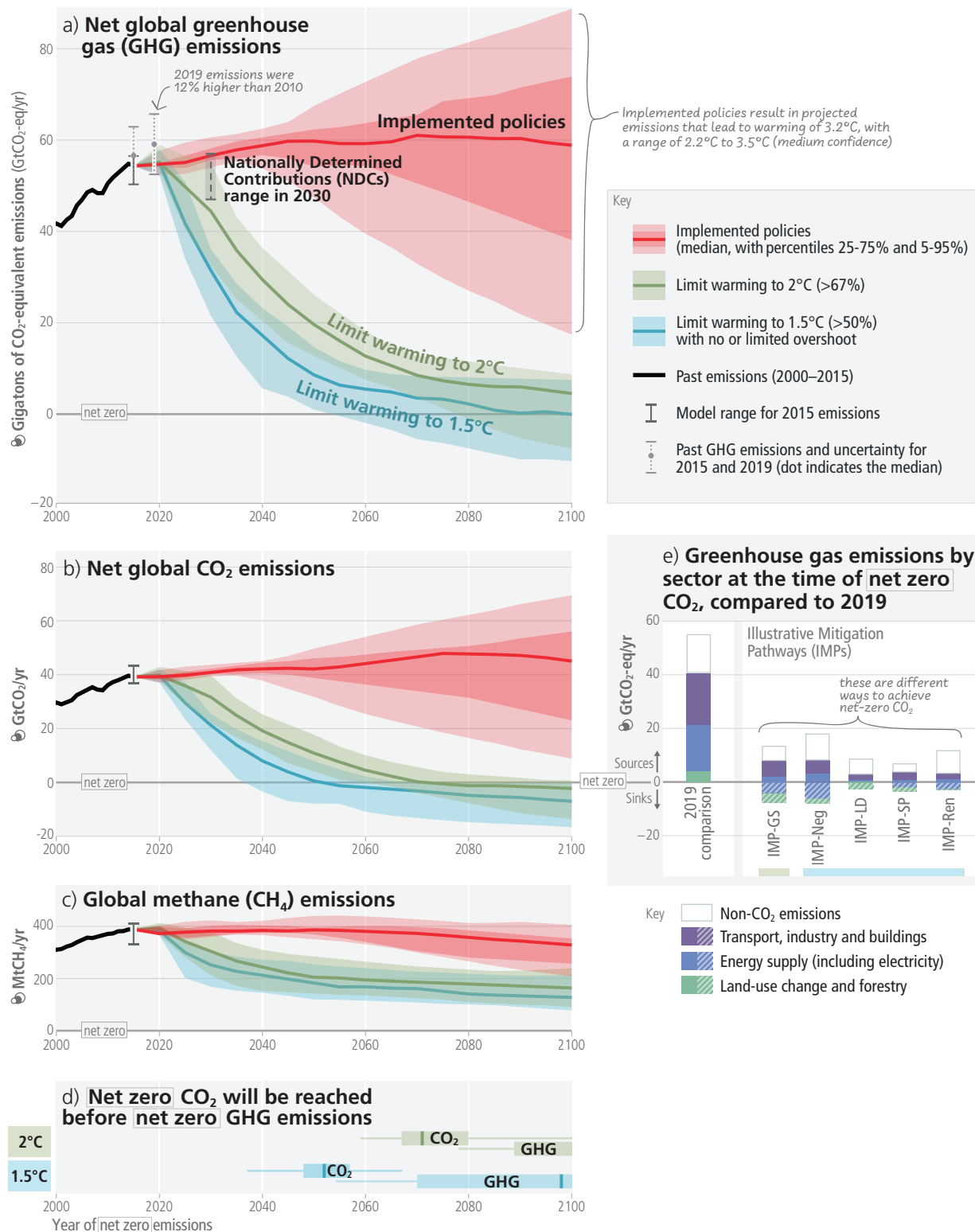


Figure SPM.5: Global emissions pathways consistent with implemented policies and mitigation strategies. Panels (a), (b) and (c) show the development of global GHG, CO₂ and methane emissions in modelled pathways, while panel (d) shows the associated timing of when GHG and CO₂ emissions reach net zero. Coloured ranges denote the 5th to 95th percentile across the global modelled pathways falling within a given category as described in Box SPM.1. The red ranges depict emissions pathways assuming policies that were implemented by the end of 2020. Ranges of modelled pathways that limit warming to 1.5°C (>50%) with no or limited overshoot are shown in light blue (category C1) and pathways that limit warming to 2°C (>67%) are shown in green (category C3). Global emission pathways that would limit warming to 1.5°C (>50%) with no or limited overshoot and also reach net zero GHG in the second half of the century do so between 2070–2075. Panel (e) shows the sectoral contributions of CO₂ and non-CO₂ emissions sources and sinks at the time when net zero CO₂ emissions are reached in illustrative mitigation pathways (IMPs) consistent with limiting warming to 1.5°C with a high reliance on net negative emissions (IMP-Neg) (“high overshoot”), high resource efficiency (IMP-LD), a focus on sustainable development (IMP-SP), renewables (IMP-Ren) and limiting warming to 2°C with less rapid mitigation initially followed by a gradual strengthening (IMP-GS). Positive and negative emissions for different IMPs are compared to GHG emissions from the year 2019. Energy supply (including electricity) includes bioenergy with carbon dioxide capture and storage and direct air carbon dioxide capture and storage. CO₂ emissions from land-use change and forestry can only be shown as a net number as many models do not report emissions and sinks of this category separately. {Figure 3.6, 4.1} (Box SPM.1)

Overshoot: Exceeding a Warming Level and Returning

B.7 If warming exceeds a specified level such as 1.5°C, it could gradually be reduced again by achieving and sustaining net negative global CO₂ emissions. This would require additional deployment of carbon dioxide removal, compared to pathways without overshoot, leading to greater feasibility and sustainability concerns. Overshoot entails adverse impacts, some irreversible, and additional risks for human and natural systems, all growing with the magnitude and duration of overshoot. (*high confidence*) {3.1, 3.3, 3.4, Table 3.1, Figure 3.6}

- B.7.1** Only a small number of the most ambitious global modelled pathways limit global warming to 1.5°C (>50%) by 2100 without exceeding this level temporarily. Achieving and sustaining net negative global CO₂ emissions, with annual rates of CDR greater than residual CO₂ emissions, would gradually reduce the warming level again (*high confidence*). Adverse impacts that occur during this period of overshoot and cause additional warming via feedback mechanisms, such as increased wildfires, mass mortality of trees, drying of peatlands, and permafrost thawing, weakening natural land carbon sinks and increasing releases of GHGs would make the return more challenging (*medium confidence*). {3.3.2, 3.3.4, Table 3.1, Figure 3.6} (Box SPM.1)
- B.7.2** The higher the magnitude and the longer the duration of overshoot, the more ecosystems and societies are exposed to greater and more widespread changes in climatic impact-drivers, increasing risks for many natural and human systems. Compared to pathways without overshoot, societies would face higher risks to infrastructure, low-lying coastal settlements, and associated livelihoods. Overshooting 1.5°C will result in irreversible adverse impacts on certain ecosystems with low resilience, such as polar, mountain, and coastal ecosystems, impacted by ice-sheet melt, glacier melt, or by accelerating and higher committed sea level rise. (*high confidence*) {3.1.2, 3.3.4}
- B.7.3** The larger the overshoot, the more net negative CO₂ emissions would be needed to return to 1.5°C by 2100. Transitioning towards net zero CO₂ emissions faster and reducing non-CO₂ emissions such as methane more rapidly would limit peak warming levels and reduce the requirement for net negative CO₂ emissions, thereby reducing feasibility and sustainability concerns, and social and environmental risks associated with CDR deployment at large scales. (*high confidence*) {3.3.3, 3.3.4, 3.4.1, Table 3.1}

C. Responses in the Near Term

Urgency of Near-Term Integrated Climate Action

- C.1 Climate change is a threat to human well-being and planetary health (*very high confidence*). There is a rapidly closing window of opportunity to secure a liveable and sustainable future for all (*very high confidence*). Climate resilient development integrates adaptation and mitigation to advance sustainable development for all, and is enabled by increased international cooperation including improved access to adequate financial resources, particularly for vulnerable regions, sectors and groups, and inclusive governance and coordinated policies (*high confidence*). The choices and actions implemented in this decade will have impacts now and for thousands of years (*high confidence*). {3.1, 3.3, 4.1, 4.2, 4.3, 4.4, 4.7, 4.8, 4.9, Figure 3.1, Figure 3.3, Figure 4.2} (Figure SPM.1, Figure SPM.6)**
- C.1.1 Evidence of observed adverse impacts and related losses and damages, projected risks, levels and trends in vulnerability and adaptation limits, demonstrate that worldwide climate resilient development action is more urgent than previously assessed in AR5. Climate resilient development integrates adaptation and GHG mitigation to advance sustainable development for all. Climate resilient development pathways have been constrained by past development, emissions and climate change and are progressively constrained by every increment of warming, in particular beyond 1.5°C. (*very high confidence*) {3.4, 3.4.2, 4.1}
- C.1.2 Government actions at sub-national, national and international levels, with civil society and the private sector, play a crucial role in enabling and accelerating shifts in development pathways towards sustainability and climate resilient development (*very high confidence*). Climate resilient development is enabled when governments, civil society and the private sector make inclusive development choices that prioritize risk reduction, equity and justice, and when decision-making processes, finance and actions are integrated across governance levels, sectors, and timeframes (*very high confidence*). Enabling conditions are differentiated by national, regional and local circumstances and geographies, according to capabilities, and include: political commitment and follow-through, coordinated policies, social and international cooperation, ecosystem stewardship, inclusive governance, knowledge diversity, technological innovation, monitoring and evaluation, and improved access to adequate financial resources, especially for vulnerable regions, sectors and communities (*high confidence*). {3.4, 4.2, 4.4, 4.5, 4.7, 4.8} (Figure SPM.6)
- C.1.3 Continued emissions will further affect all major climate system components, and many changes will be irreversible on centennial to millennial time scales and become larger with increasing global warming. Without urgent, effective, and equitable mitigation and adaptation actions, climate change increasingly threatens ecosystems, biodiversity, and the livelihoods, health and well-being of current and future generations. (*high confidence*) {3.1.3, 3.3.3, 3.4.1, Figure 3.4, 4.1, 4.2, 4.3, 4.4} (Figure SPM.1, Figure SPM.6)

There is a rapidly narrowing window of opportunity to enable climate resilient development

Multiple interacting choices and actions can shift development pathways towards sustainability

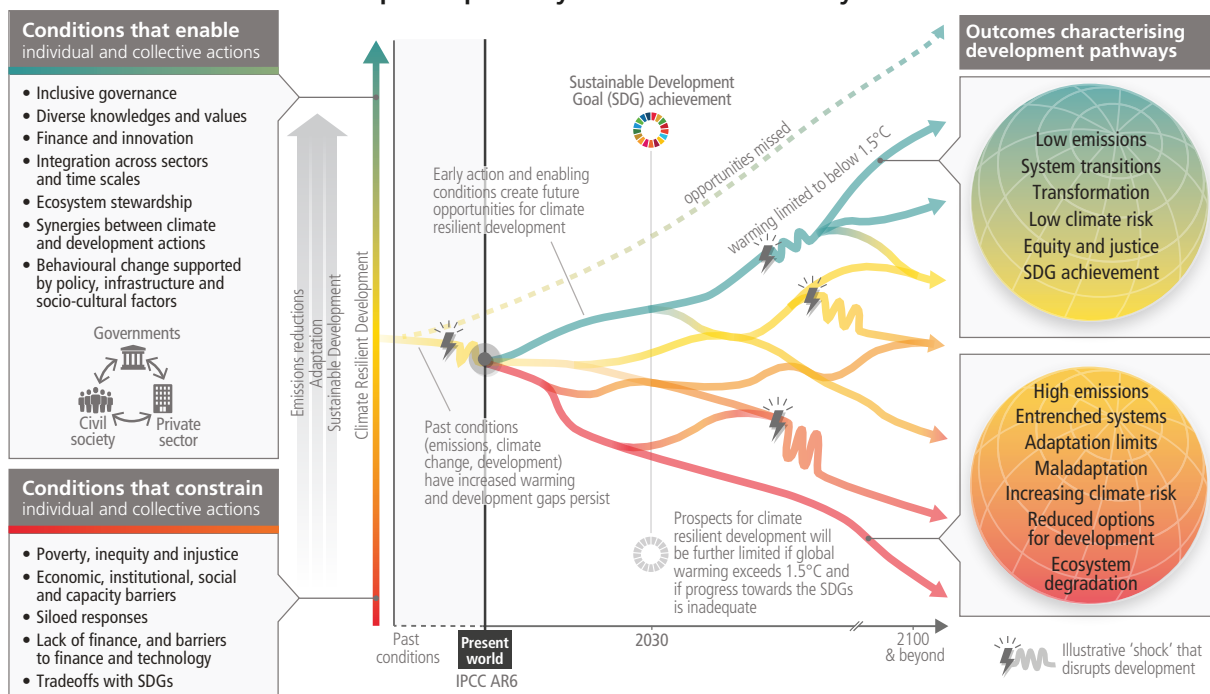


Figure SPM.6: The illustrative development pathways (red to green) and associated outcomes (right panel) show that there is a rapidly narrowing window of opportunity to secure a liveable and sustainable future for all. Climate resilient development is the process of implementing greenhouse gas mitigation and adaptation measures to support sustainable development. Diverging pathways illustrate that interacting choices and actions made by diverse government, private sector and civil society actors can advance climate resilient development, shift pathways towards sustainability, and enable lower emissions and adaptation. Diverse knowledge and values include cultural values, Indigenous Knowledge, local knowledge, and scientific knowledge. Climatic and non-climatic events, such as droughts, floods or pandemics, pose more severe shocks to pathways with lower climate resilient development (red to yellow) than to pathways with higher climate resilient development (green). There are limits to adaptation and adaptive capacity for some human and natural systems at global warming of 1.5°C, and with every increment of warming, losses and damages will increase. The development pathways taken by countries at all stages of economic development impact GHG emissions and mitigation challenges and opportunities, which vary across countries and regions. Pathways and opportunities for action are shaped by previous actions (or inactions and opportunities missed; dashed pathway) and enabling and constraining conditions (left panel), and take place in the context of climate risks, adaptation limits and development gaps. The longer emissions reductions are delayed, the fewer effective adaptation options. [Figure 4.2, 3.1, 3.2, 3.4, 4.2, 4.4, 4.5, 4.6, 4.9]

The Benefits of Near-Term Action

C.2 Deep, rapid, and sustained mitigation and accelerated implementation of adaptation actions in this decade would reduce projected losses and damages for humans and ecosystems (*very high confidence*), and deliver many co-benefits, especially for air quality and health (*high confidence*). Delayed mitigation and adaptation action would lock in high-emissions infrastructure, raise risks of stranded assets and cost-escalation, reduce feasibility, and increase losses and damages (*high confidence*). Near-term actions involve high up-front investments and potentially disruptive changes that can be lessened by a range of enabling policies (*high confidence*). {2.1, 2.2, 3.1, 3.2, 3.3, 3.4, 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8}

C.2.1 Deep, rapid, and sustained mitigation and accelerated implementation of adaptation actions in this decade would reduce future losses and damages related to climate change for humans and ecosystems (*very high confidence*). As adaptation options often have long implementation times, accelerated implementation of adaptation in this decade is important to close adaptation gaps (*high confidence*). Comprehensive, effective, and innovative responses integrating adaptation and mitigation can harness synergies and reduce trade-offs between adaptation and mitigation (*high confidence*). {4.1, 4.2, 4.3}

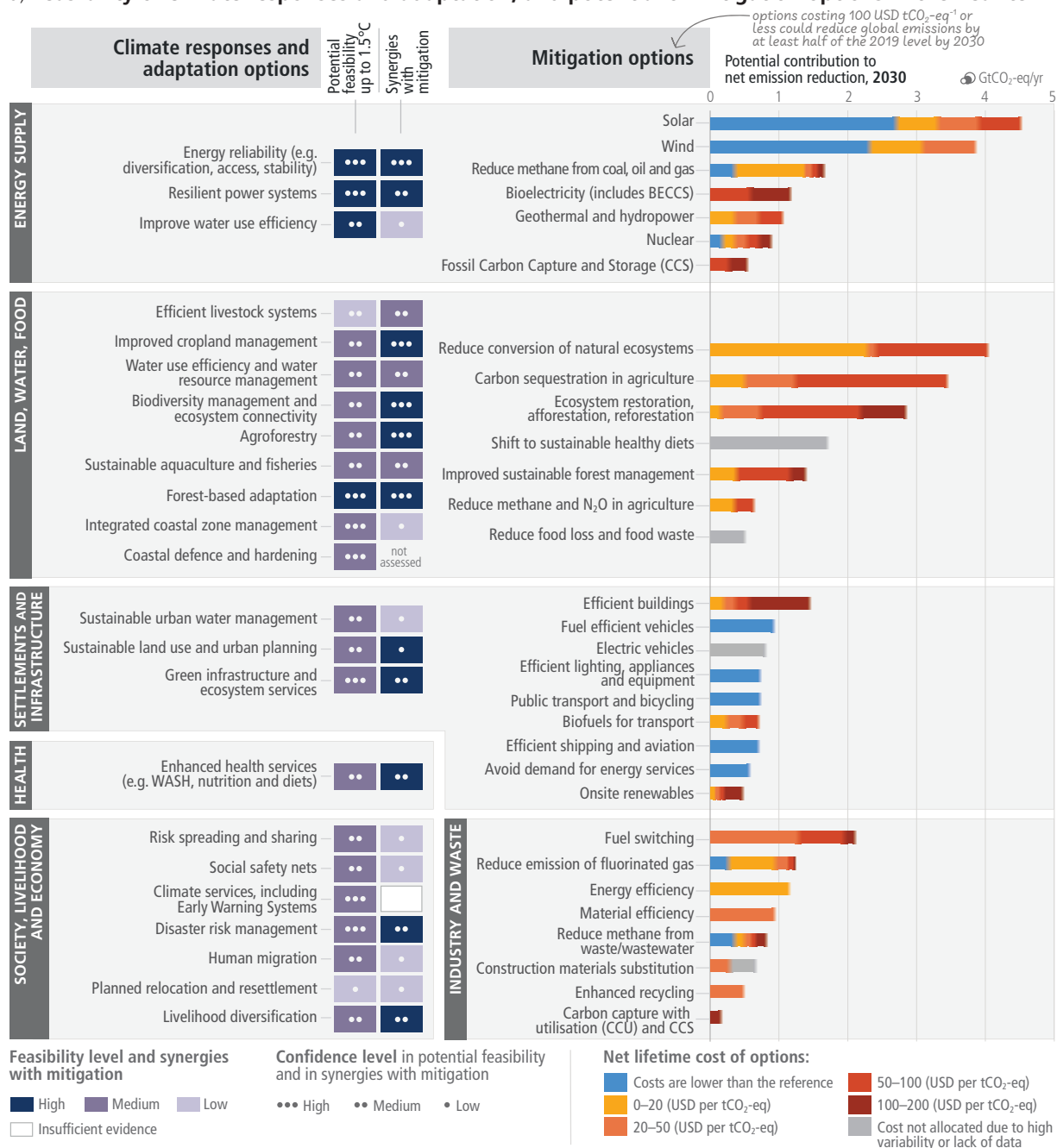
- C.2.2 Delayed mitigation action will further increase global warming and losses and damages will rise and additional human and natural systems will reach adaptation limits. Challenges from delayed adaptation and mitigation actions include the risk of cost escalation, lock-in of infrastructure, stranded assets, and reduced feasibility and effectiveness of adaptation and mitigation options. Without rapid, deep and sustained mitigation and accelerated adaptation actions, losses and damages will continue to increase, including projected adverse impacts in Africa, LDCs, SIDS, Central and South America⁴⁹, Asia and the Arctic, and will disproportionately affect the most vulnerable populations. (*high confidence*) {2.1.2, 3.1.2, 3.2, 3.3.1, 3.3.3, 4.1, 4.2, 4.3} (Figure SPM.3, Figure SPM.4)
- C.2.3 Accelerated climate action can also provide co-benefits (see also C.4) (*high confidence*). Many mitigation actions would have benefits for health through lower air pollution, active mobility (e.g., walking, cycling), and shifts to sustainable healthy diets (*high confidence*). Strong, rapid and sustained reductions in methane emissions can limit near-term warming and improve air quality by reducing global surface ozone (*high confidence*). Adaptation can generate multiple additional benefits such as improving agricultural productivity, innovation, health and well-being, food security, livelihood, and biodiversity conservation (*very high confidence*). {4.2, 4.5.4, 4.5.5, 4.6}
- C.2.4 Cost-benefit analysis remains limited in its ability to represent all avoided damages from climate change (*high confidence*). The economic benefits for human health from air quality improvement arising from mitigation action can be of the same order of magnitude as mitigation costs, and potentially even larger (*medium confidence*). Even without accounting for all the benefits of avoiding potential damages, the global economic and social benefit of limiting global warming to 2°C exceeds the cost of mitigation in most of the assessed literature (*medium confidence*)⁵⁰. More rapid climate change mitigation, with emissions peaking earlier, increases co-benefits and reduces feasibility risks and costs in the long-term, but requires higher up-front investments (*high confidence*). {3.4.1, 4.2}
- C.2.5 Ambitious mitigation pathways imply large and sometimes disruptive changes in existing economic structures, with significant distributional consequences within and between countries. To accelerate climate action, the adverse consequences of these changes can be moderated by fiscal, financial, institutional and regulatory reforms and by integrating climate actions with macroeconomic policies through (i) economy-wide packages, consistent with national circumstances, supporting sustainable low-emission growth paths; (ii) climate resilient safety nets and social protection; and (iii) improved access to finance for low-emissions infrastructure and technologies, especially in developing countries. (*high confidence*) {4.2, 4.4, 4.7, 4.8.1}

⁴⁹ The southern part of Mexico is included in the climatic subregion South Central America (SCA) for WGI. Mexico is assessed as part of North America for WGII. The climate change literature for the SCA region occasionally includes Mexico, and in those cases WGII assessment makes reference to Latin America. Mexico is considered part of Latin America and the Caribbean for WGIII.

⁵⁰ The evidence is too limited to make a similar robust conclusion for limiting warming to 1.5°C. Limiting global warming to 1.5°C instead of 2°C would increase the costs of mitigation, but also increase the benefits in terms of reduced impacts and related risks, and reduced adaptation needs (*high confidence*).

There are multiple opportunities for scaling up climate action

a) Feasibility of climate responses and adaptation, and potential of mitigation options in the near term



b) Potential of demand-side mitigation options by 2050

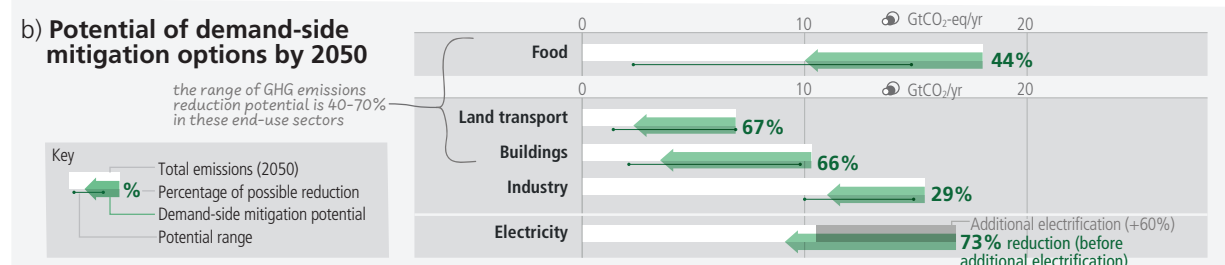


Figure SPM.7: Multiple Opportunities for scaling up climate action. Panel (a) presents selected mitigation and adaptation options across different systems. The left-hand side of panel a shows climate responses and adaptation options assessed for their multidimensional feasibility at global scale, in the near term and up to 1.5°C global warming. As literature above 1.5°C is limited, feasibility at higher levels of warming may change, which is currently not possible to assess robustly. The term response is used here in addition to adaptation because some responses, such as migration, relocation and resettlement may or may not be considered to be adaptation. Forest based adaptation includes sustainable forest management, forest conservation and restoration, reforestation

and afforestation. WASH refers to water, sanitation and hygiene. Six feasibility dimensions (economic, technological, institutional, social, environmental and geophysical) were used to calculate the potential feasibility of climate responses and adaptation options, along with their synergies with mitigation. For potential feasibility and feasibility dimensions, the figure shows high, medium, or low feasibility. Synergies with mitigation are identified as high, medium, and low. The right-hand side of Panel a provides an overview of selected mitigation options and their estimated costs and potentials in 2030. Costs are net lifetime discounted monetary costs of avoided GHG emissions calculated relative to a reference technology. Relative potentials and costs will vary by place, context and time and in the longer term compared to 2030. The potential (horizontal axis) is the net GHG emission reduction (sum of reduced emissions and/or enhanced sinks) broken down into cost categories (coloured bar segments) relative to an emission baseline consisting of current policy (around 2019) reference scenarios from the AR6 scenarios database. The potentials are assessed independently for each option and are not additive. Health system mitigation options are included mostly in settlement and infrastructure (e.g., efficient healthcare buildings) and cannot be identified separately. Fuel switching in industry refers to switching to electricity, hydrogen, bioenergy and natural gas. Gradual colour transitions indicate uncertain breakdown into cost categories due to uncertainty or heavy context dependency. The uncertainty in the total potential is typically 25–50%. **Panel (b)** displays the indicative potential of demand-side mitigation options for 2050. Potentials are estimated based on approximately 500 bottom-up studies representing all global regions. The baseline (white bar) is provided by the sectoral mean GHG emissions in 2050 of the two scenarios (IEA-STEPS and IP_ModAct) consistent with policies announced by national governments until 2020. The green arrow represents the demand-side emissions reductions potentials. The range in potential is shown by a line connecting dots displaying the highest and the lowest potentials reported in the literature. Food shows demand-side potential of socio-cultural factors and infrastructure use, and changes in land-use patterns enabled by change in food demand. Demand-side measures and new ways of end-use service provision can reduce global GHG emissions in end-use sectors (buildings, land transport, food) by 40–70% by 2050 compared to baseline scenarios, while some regions and socioeconomic groups require additional energy and resources. The last row shows how demand-side mitigation options in other sectors can influence overall electricity demand. The dark grey bar shows the projected increase in electricity demand above the 2050 baseline due to increasing electrification in the other sectors. Based on a bottom-up assessment, this projected increase in electricity demand can be avoided through demand-side mitigation options in the domains of infrastructure use and socio-cultural factors that influence electricity usage in industry, land transport, and buildings (green arrow). (Figure 4.4)

Mitigation and Adaptation Options across Systems

C.3 Rapid and far-reaching transitions across all sectors and systems are necessary to achieve deep and sustained emissions reductions and secure a liveable and sustainable future for all. These system transitions involve a significant upscaling of a wide portfolio of mitigation and adaptation options. Feasible, effective, and low-cost options for mitigation and adaptation are already available, with differences across systems and regions. (high confidence) {4.1, 4.5, 4.6} (Figure SPM.7)

C.3.1 The systemic change required to achieve rapid and deep emissions reductions and transformative adaptation to climate change is unprecedented in terms of scale, but not necessarily in terms of speed (*medium confidence*). Systems transitions include: deployment of low- or zero-emission technologies; reducing and changing demand through infrastructure design and access, socio-cultural and behavioural changes, and increased technological efficiency and adoption; social protection, climate services or other services; and protecting and restoring ecosystems (*high confidence*). Feasible, effective, and low-cost options for mitigation and adaptation are already available (*high confidence*). The availability, feasibility and potential of mitigation and adaptation options in the near term differs across systems and regions (*very high confidence*). {4.1, 4.5.1 to 4.5.6} (Figure SPM.7)

Energy Systems

C.3.2 Net zero CO₂ energy systems entail: a substantial reduction in overall fossil fuel use, minimal use of unabated fossil fuels⁵¹, and use of carbon capture and storage in the remaining fossil fuel systems; electricity systems that emit no net CO₂; widespread electrification; alternative energy carriers in applications less amenable to electrification; energy conservation and efficiency; and greater integration across the energy system (*high confidence*). Large contributions to emissions reductions with costs less than USD 20 tCO₂-eq⁻¹ come from solar and wind energy, energy efficiency improvements, and methane emissions reductions (coal mining, oil and gas, waste) (*medium confidence*). There are feasible adaptation options that support infrastructure resilience, reliable power systems and efficient water use for existing and new energy generation systems (*very high confidence*). Energy generation diversification (e.g., via wind, solar, small scale hydropower) and demand-side management (e.g., storage and energy efficiency improvements) can increase energy reliability and reduce vulnerabilities to climate change (*high confidence*). Climate responsive energy markets, updated design standards on energy assets according to current and projected climate change, smart-grid technologies, robust transmission systems and improved capacity to respond to supply deficits have high feasibility in the medium to long term, with mitigation co-benefits (*very high confidence*). {4.5.1} (Figure SPM.7)

⁵¹ In this context, ‘unabated fossil fuels’ refers to fossil fuels produced and used without interventions that substantially reduce the amount of GHG emitted throughout the life cycle; for example, capturing 90% or more CO₂ from power plants, or 50–80% of fugitive methane emissions from energy supply.

Industry and Transport

C.3.3 Reducing industry GHG emissions entails coordinated action throughout value chains to promote all mitigation options, including demand management, energy and materials efficiency, circular material flows, as well as abatement technologies and transformational changes in production processes (*high confidence*). In transport, sustainable biofuels, low-emissions hydrogen, and derivatives (including ammonia and synthetic fuels) can support mitigation of CO₂ emissions from shipping, aviation, and heavy-duty land transport but require production process improvements and cost reductions (*medium confidence*). Sustainable biofuels can offer additional mitigation benefits in land-based transport in the short and medium term (*medium confidence*). Electric vehicles powered by low-GHG emissions electricity have large potential to reduce land-based transport GHG emissions, on a life cycle basis (*high confidence*). Advances in battery technologies could facilitate the electrification of heavy-duty trucks and complement conventional electric rail systems (*medium confidence*). The environmental footprint of battery production and growing concerns about critical minerals can be addressed by material and supply diversification strategies, energy and material efficiency improvements, and circular material flows (*medium confidence*). {4.5.2, 4.5.3} (Figure SPM.7)

Cities, Settlements and Infrastructure

C.3.4 Urban systems are critical for achieving deep emissions reductions and advancing climate resilient development (*high confidence*). Key adaptation and mitigation elements in cities include considering climate change impacts and risks (e.g., through climate services) in the design and planning of settlements and infrastructure; land use planning to achieve compact urban form, co-location of jobs and housing; supporting public transport and active mobility (e.g., walking and cycling); the efficient design, construction, retrofit, and use of buildings; reducing and changing energy and material consumption; sufficiency⁵²; material substitution; and electrification in combination with low emissions sources (*high confidence*). Urban transitions that offer benefits for mitigation, adaptation, human health and well-being, ecosystem services, and vulnerability reduction for low-income communities are fostered by inclusive long-term planning that takes an integrated approach to physical, natural and social infrastructure (*high confidence*). Green/natural and blue infrastructure supports carbon uptake and storage and either singly or when combined with grey infrastructure can reduce energy use and risk from extreme events such as heatwaves, flooding, heavy precipitation and droughts, while generating co-benefits for health, well-being and livelihoods (*medium confidence*). {4.5.3}

Land, Ocean, Food, and Water

C.3.5 Many agriculture, forestry, and other land use (AFOLU) options provide adaptation and mitigation benefits that could be upscaled in the near term across most regions. Conservation, improved management, and restoration of forests and other ecosystems offer the largest share of economic mitigation potential, with reduced deforestation in tropical regions having the highest total mitigation potential. Ecosystem restoration, reforestation, and afforestation can lead to trade-offs due to competing demands on land. Minimizing trade-offs requires integrated approaches to meet multiple objectives including food security. Demand-side measures (shifting to sustainable healthy diets⁵³ and reducing food loss/waste) and sustainable agricultural intensification can reduce ecosystem conversion, and methane and nitrous oxide emissions, and free up land for reforestation and ecosystem restoration. Sustainably sourced agricultural and forest products, including long-lived wood products, can be used instead of more GHG-intensive products in other sectors. Effective adaptation options include cultivar improvements, agroforestry, community-based adaptation, farm and landscape diversification, and urban agriculture. These AFOLU response options require integration of biophysical, socioeconomic and other enabling factors. Some options, such as conservation of high-carbon ecosystems (e.g., peatlands, wetlands, rangelands, mangroves and forests), deliver immediate benefits, while others, such as restoration of high-carbon ecosystems, take decades to deliver measurable results. (*high confidence*) {4.5.4} (Figure SPM.7)

C.3.6 Maintaining the resilience of biodiversity and ecosystem services at a global scale depends on effective and equitable conservation of approximately 30% to 50% of Earth's land, freshwater and ocean areas, including currently near-natural ecosystems (*high confidence*). Conservation, protection and restoration of terrestrial, freshwater, coastal and

⁵² A set of measures and daily practices that avoid demand for energy, materials, land, and water while delivering human well-being for all within planetary boundaries. {4.5.3}

⁵³ 'Sustainable healthy diets' promote all dimensions of individuals' health and well-being; have low environmental pressure and impact; are accessible, affordable, safe and equitable; and are culturally acceptable, as described in FAO and WHO. The related concept of 'balanced diets' refers to diets that feature plant-based foods, such as those based on coarse grains, legumes, fruits and vegetables, nuts and seeds, and animal-sourced food produced in resilient, sustainable and low-GHG emission systems, as described in SRCCL.

ocean ecosystems, together with targeted management to adapt to unavoidable impacts of climate change reduces the vulnerability of biodiversity and ecosystem services to climate change (*high confidence*), reduces coastal erosion and flooding (*high confidence*), and could increase carbon uptake and storage if global warming is limited (*medium confidence*). Rebuilding overexploited or depleted fisheries reduces negative climate change impacts on fisheries (*medium confidence*) and supports food security, biodiversity, human health and well-being (*high confidence*). Land restoration contributes to climate change mitigation and adaptation with synergies via enhanced ecosystem services and with economically positive returns and co-benefits for poverty reduction and improved livelihoods (*high confidence*). Cooperation, and inclusive decision making, with Indigenous Peoples and local communities, as well as recognition of inherent rights of Indigenous Peoples, is integral to successful adaptation and mitigation across forests and other ecosystems (*high confidence*). {4.5.4, 4.6} (Figure SPM.7)

Health and Nutrition

- C.3.7 Human health will benefit from integrated mitigation and adaptation options that mainstream health into food, infrastructure, social protection, and water policies (*very high confidence*). Effective adaptation options exist to help protect human health and well-being, including: strengthening public health programs related to climate-sensitive diseases, increasing health systems resilience, improving ecosystem health, improving access to potable water, reducing exposure of water and sanitation systems to flooding, improving surveillance and early warning systems, vaccine development (*very high confidence*), improving access to mental healthcare, and Heat Health Action Plans that include early warning and response systems (*high confidence*). Adaptation strategies which reduce food loss and waste or support balanced, sustainable healthy diets contribute to nutrition, health, biodiversity and other environmental benefits (*high confidence*). {4.5.5} (Figure SPM.7)

Society, Livelihoods, and Economies

- C.3.8 Policy mixes that include weather and health insurance, social protection and adaptive social safety nets, contingent finance and reserve funds, and universal access to early warning systems combined with effective contingency plans, can reduce vulnerability and exposure of human systems. Disaster risk management, early warning systems, climate services and risk spreading and sharing approaches have broad applicability across sectors. Increasing education including capacity building, climate literacy, and information provided through climate services and community approaches can facilitate heightened risk perception and accelerate behavioural changes and planning. (*high confidence*) {4.5.6}

Synergies and Trade-Offs with Sustainable Development

C.4 Accelerated and equitable action in mitigating and adapting to climate change impacts is critical to sustainable development. Mitigation and adaptation actions have more synergies than trade-offs with Sustainable Development Goals. Synergies and trade-offs depend on context and scale of implementation. (*high confidence*) {3.4, 4.2, 4.4, 4.5, 4.6, 4.9, Figure 4.5}

- C.4.1 Mitigation efforts embedded within the wider development context can increase the pace, depth and breadth of emission reductions (*medium confidence*). Countries at all stages of economic development seek to improve the well-being of people, and their development priorities reflect different starting points and contexts. Different contexts include but are not limited to social, economic, environmental, cultural, political circumstances, resource endowment, capabilities, international environment, and prior development (*high confidence*). In regions with high dependency on fossil fuels for, among other things, revenue and employment generation, mitigating risk for sustainable development requires policies that promote economic and energy sector diversification and considerations of just transitions principles, processes and practices (*high confidence*). Eradicating extreme poverty, energy poverty, and providing decent living standards in low-emitting countries / regions in the context of achieving sustainable development objectives, in the near term, can be achieved without significant global emissions growth (*high confidence*). {4.4, 4.6, Annex I: Glossary}
- C.4.2 Many mitigation and adaptation actions have multiple synergies with Sustainable Development Goals (SDGs) and sustainable development generally, but some actions can also have trade-offs. Potential synergies with SDGs exceed potential trade-offs; synergies and trade-offs depend on the pace and magnitude of change and the development context including inequalities with consideration of climate justice. Trade-offs can be evaluated and minimised by giving emphasis to capacity building, finance, governance, technology transfer, investments, development, context specific gender-based and other social equity considerations with meaningful participation of Indigenous Peoples, local communities and vulnerable populations. (*high confidence*) {3.4.1, 4.6, Figure 4.5, 4.9}

- C.4.3 Implementing both mitigation and adaptation actions together and taking trade-offs into account supports co-benefits and synergies for human health and well-being. For example, improved access to clean energy sources and technologies generates health benefits especially for women and children; electrification combined with low-GHG energy, and shifts to active mobility and public transport can enhance air quality, health, employment, and can elicit energy security and deliver equity. *(high confidence)* {4.2, 4.5.3, 4.5.5, 4.6, 4.9}

Equity and Inclusion

C.5 Prioritising equity, climate justice, social justice, inclusion and just transition processes can enable adaptation and ambitious mitigation actions and climate resilient development. Adaptation outcomes are enhanced by increased support to regions and people with the highest vulnerability to climatic hazards. Integrating climate adaptation into social protection programs improves resilience. Many options are available for reducing emission-intensive consumption, including through behavioural and lifestyle changes, with co-benefits for societal well-being. *(high confidence)* {4.4, 4.5}

- C.5.1 Equity remains a central element in the UN climate regime, notwithstanding shifts in differentiation between states over time and challenges in assessing fair shares. Ambitious mitigation pathways imply large and sometimes disruptive changes in economic structure, with significant distributional consequences, within and between countries. Distributional consequences within and between countries include shifting of income and employment during the transition from high- to low-emissions activities. *(high confidence)* {4.4}
- C.5.2 Adaptation and mitigation actions that prioritise equity, social justice, climate justice, rights-based approaches, and inclusivity, lead to more sustainable outcomes, reduce trade-offs, support transformative change and advance climate resilient development. Redistributive policies across sectors and regions that shield the poor and vulnerable, social safety nets, equity, inclusion and just transitions, at all scales can enable deeper societal ambitions and resolve trade-offs with sustainable development goals. Attention to equity and broad and meaningful participation of all relevant actors in decision making at all scales can build social trust which builds on equitable sharing of benefits and burdens of mitigation that deepen and widen support for transformative changes. *(high confidence)* {4.4}
- C.5.3 Regions and people (3.3 to 3.6 billion in number) with considerable development constraints have high vulnerability to climatic hazards (see A.2.2). Adaptation outcomes for the most vulnerable within and across countries and regions are enhanced through approaches focusing on equity, inclusivity and rights-based approaches. Vulnerability is exacerbated by inequity and marginalisation linked to e.g., gender, ethnicity, low incomes, informal settlements, disability, age, and historical and ongoing patterns of inequity such as colonialism, especially for many Indigenous Peoples and local communities. Integrating climate adaptation into social protection programs, including cash transfers and public works programs, is highly feasible and increases resilience to climate change, especially when supported by basic services and infrastructure. The greatest gains in well-being in urban areas can be achieved by prioritising access to finance to reduce climate risk for low-income and marginalised communities including people living in informal settlements. *(high confidence)* {4.4, 4.5.3, 4.5.5, 4.5.6}
- C.5.4 The design of regulatory instruments and economic instruments and consumption-based approaches, can advance equity. Individuals with high socio-economic status contribute disproportionately to emissions, and have the highest potential for emissions reductions. Many options are available for reducing emission-intensive consumption while improving societal well-being. Socio-cultural options, behaviour and lifestyle changes supported by policies, infrastructure, and technology can help end-users shift to low-emissions-intensive consumption, with multiple co-benefits. A substantial share of the population in low-emitting countries lack access to modern energy services. Technology development, transfer, capacity building and financing can support developing countries / regions leapfrogging or transitioning to low-emissions transport systems thereby providing multiple co-benefits. Climate resilient development is advanced when actors work in equitable, just and inclusive ways to reconcile divergent interests, values and worldviews, toward equitable and just outcomes. *(high confidence)* {2.1, 4.4}

Governance and Policies

C.6 Effective climate action is enabled by political commitment, well-aligned multilevel governance, institutional frameworks, laws, policies and strategies and enhanced access to finance and technology. Clear goals, coordination across multiple policy domains, and inclusive governance processes facilitate effective climate action. Regulatory and economic instruments can support deep emissions reductions and climate resilience if scaled up and applied widely. Climate resilient development benefits from drawing on diverse knowledge. (*high confidence*) {2.2, 4.4, 4.5, 4.7}

- C.6.1 Effective climate governance enables mitigation and adaptation. Effective governance provides overall direction on setting targets and priorities and mainstreaming climate action across policy domains and levels, based on national circumstances and in the context of international cooperation. It enhances monitoring and evaluation and regulatory certainty, prioritising inclusive, transparent and equitable decision-making, and improves access to finance and technology (see C.7). (*high confidence*) {2.2.2, 4.7}
- C.6.2 Effective local, municipal, national and subnational institutions build consensus for climate action among diverse interests, enable coordination and inform strategy setting but require adequate institutional capacity. Policy support is influenced by actors in civil society, including businesses, youth, women, labour, media, Indigenous Peoples, and local communities. Effectiveness is enhanced by political commitment and partnerships between different groups in society. (*high confidence*) {2.2, 4.7}
- C.6.3 Effective multilevel governance for mitigation, adaptation, risk management, and climate resilient development is enabled by inclusive decision processes that prioritise equity and justice in planning and implementation, allocation of appropriate resources, institutional review, and monitoring and evaluation. Vulnerabilities and climate risks are often reduced through carefully designed and implemented laws, policies, participatory processes, and interventions that address context specific inequities such as those based on gender, ethnicity, disability, age, location and income. (*high confidence*) {4.4, 4.7}
- C.6.4 Regulatory and economic instruments could support deep emissions reductions if scaled up and applied more widely (*high confidence*). Scaling up and enhancing the use of regulatory instruments can improve mitigation outcomes in sectoral applications, consistent with national circumstances (*high confidence*). Where implemented, carbon pricing instruments have incentivized low-cost emissions reduction measures but have been less effective, on their own and at prevailing prices during the assessment period, to promote higher-cost measures necessary for further reductions (*medium confidence*). Equity and distributional impacts of such carbon pricing instruments, e.g., carbon taxes and emissions trading, can be addressed by using revenue to support low-income households, among other approaches. Removing fossil fuel subsidies would reduce emissions⁵⁴ and yield benefits such as improved public revenue, macroeconomic and sustainability performance; subsidy removal can have adverse distributional impacts, especially on the most economically vulnerable groups which, in some cases can be mitigated by measures such as redistributing revenue saved, all of which depend on national circumstances (*high confidence*). Economy-wide policy packages, such as public spending commitments and pricing reforms, can meet short-term economic goals while reducing emissions and shifting development pathways towards sustainability (*medium confidence*). Effective policy packages would be comprehensive, consistent, balanced across objectives, and tailored to national circumstances (*high confidence*). {2.2.2, 4.7}
- C.6.5 Drawing on diverse knowledges and cultural values, meaningful participation and inclusive engagement processes—including Indigenous Knowledge, local knowledge, and scientific knowledge—facilitates climate resilient development, builds capacity and allows locally appropriate and socially acceptable solutions. (*high confidence*) {4.4, 4.5.6, 4.7}

⁵⁴ Fossil fuel subsidy removal is projected by various studies to reduce global CO₂ emission by 1 to 4%, and GHG emissions by up to 10% by 2030, varying across regions (*medium confidence*).

Finance, Technology and International Cooperation

C.7 Finance, technology and international cooperation are critical enablers for accelerated climate action. If climate goals are to be achieved, both adaptation and mitigation financing would need to increase many-fold. There is sufficient global capital to close the global investment gaps but there are barriers to redirect capital to climate action. Enhancing technology innovation systems is key to accelerate the widespread adoption of technologies and practices. Enhancing international cooperation is possible through multiple channels. (*high confidence*) {2.3, 4.8}

- C.7.1 Improved availability of and access to finance⁵⁵ would enable accelerated climate action (*very high confidence*). Addressing needs and gaps and broadening equitable access to domestic and international finance, when combined with other supportive actions, can act as a catalyst for accelerating adaptation and mitigation, and enabling climate resilient development (*high confidence*). If climate goals are to be achieved, and to address rising risks and accelerate investments in emissions reductions, both adaptation and mitigation finance would need to increase many-fold (*high confidence*). {4.8.1}
- C.7.2 Increased access to finance can build capacity and address soft limits to adaptation and avert rising risks, especially for developing countries, vulnerable groups, regions and sectors (*high confidence*). Public finance is an important enabler of adaptation and mitigation, and can also leverage private finance (*high confidence*). Average annual modelled mitigation investment requirements for 2020 to 2030 in scenarios that limit warming to 2°C or 1.5°C are a factor of three to six greater than current levels⁵⁶, and total mitigation investments (public, private, domestic and international) would need to increase across all sectors and regions (*medium confidence*). Even if extensive global mitigation efforts are implemented, there will be a need for financial, technical, and human resources for adaptation (*high confidence*). {4.3, 4.8.1}
- C.7.3 There is sufficient global capital and liquidity to close global investment gaps, given the size of the global financial system, but there are barriers to redirect capital to climate action both within and outside the global financial sector and in the context of economic vulnerabilities and indebtedness facing developing countries. Reducing financing barriers for scaling up financial flows would require clear signalling and support by governments, including a stronger alignment of public finances in order to lower real and perceived regulatory, cost and market barriers and risks and improving the risk-return profile of investments. At the same time, depending on national contexts, financial actors, including investors, financial intermediaries, central banks and financial regulators can shift the systemic underpricing of climate-related risks, and reduce sectoral and regional mismatches between available capital and investment needs. (*high confidence*) {4.8.1}
- C.7.4 Tracked financial flows fall short of the levels needed for adaptation and to achieve mitigation goals across all sectors and regions. These gaps create many opportunities and the challenge of closing gaps is largest in developing countries. Accelerated financial support for developing countries from developed countries and other sources is a critical enabler to enhance adaptation and mitigation actions and address inequities in access to finance, including its costs, terms and conditions, and economic vulnerability to climate change for developing countries. Scaled-up public grants for mitigation and adaptation funding for vulnerable regions, especially in Sub-Saharan Africa, would be cost-effective and have high social returns in terms of access to basic energy. Options for scaling up mitigation in developing countries include: increased levels of public finance and publicly mobilised private finance flows from developed to developing countries in the context of the USD 100 billion-a-year goal; increased use of public guarantees to reduce risks and leverage private flows at lower cost; local capital markets development; and building greater trust in international cooperation processes. A coordinated effort to make the post-pandemic recovery sustainable over the longer-term can accelerate climate action, including in developing regions and countries facing high debt costs, debt distress and macroeconomic uncertainty. (*high confidence*) {4.8.1}
- C.7.5 Enhancing technology innovation systems can provide opportunities to lower emissions growth, create social and environmental co-benefits, and achieve other SDGs. Policy packages tailored to national contexts and technological characteristics have been effective in supporting low-emission innovation and technology diffusion. Public policies can

⁵⁵ Finance originates from diverse sources: public or private, local, national or international, bilateral or multilateral, and alternative sources. It can take the form of grants, technical assistance, loans (concessional and non-concessional), bonds, equity, risk insurance and financial guarantees (of different types).

⁵⁶ These estimates rely on scenario assumptions.

support training and R&D, complemented by both regulatory and market-based instruments that create incentives and market opportunities. Technological innovation can have trade-offs such as new and greater environmental impacts, social inequalities, overdependence on foreign knowledge and providers, distributional impacts and rebound effects⁵⁷, requiring appropriate governance and policies to enhance potential and reduce trade-offs. Innovation and adoption of low-emission technologies lags in most developing countries, particularly least developed ones, due in part to weaker enabling conditions, including limited finance, technology development and transfer, and capacity building. (*high confidence*) {4.8.3}

- C.7.6 International cooperation is a critical enabler for achieving ambitious climate change mitigation, adaptation, and climate resilient development (*high confidence*). Climate resilient development is enabled by increased international cooperation including mobilising and enhancing access to finance, particularly for developing countries, vulnerable regions, sectors and groups and aligning finance flows for climate action to be consistent with ambition levels and funding needs (*high confidence*). Enhancing international cooperation on finance, technology and capacity building can enable greater ambition and can act as a catalyst for accelerating mitigation and adaptation, and shifting development pathways towards sustainability (*high confidence*). This includes support to NDCs and accelerating technology development and deployment (*high confidence*). Transnational partnerships can stimulate policy development, technology diffusion, adaptation and mitigation, though uncertainties remain over their costs, feasibility and effectiveness (*medium confidence*). International environmental and sectoral agreements, institutions and initiatives are helping, and in some cases may help, to stimulate low GHG emissions investments and reduce emissions (*medium confidence*). {2.2.2, 4.8.2}

⁵⁷ Leading to lower net emission reductions or even emission increases.

Global Economy Tested Again

The global economy has, to date, withstood a series of shocks, yet another one—this time a military conflict engulfing the Middle East since the end of February—is testing this resilience. This is the latest culmination in a series of events that have been reshaping international relations and raising geopolitical tensions markedly across all regions in recent years (Figure 1.1). The conflict has already inflicted humanitarian costs, damaged critical infrastructure, and severely disrupted maritime and air traffic in the affected region. Economies around the world face repercussions through the direct impact of higher commodity prices, indirect second-order effects on inflation expectations—which tend to be especially sensitive to energy and food prices—and amplification effects coming from risk-off sentiment in financial markets. Commodity-importing emerging market and developing economies are at risk of being hit harder, with a depreciation of their currencies exacerbating the impact of higher energy and food prices. The global economic impact will crucially depend on the conflict’s duration, intensity, and scope, which are inherently unpredictable.

This latest shock comes less than a year since the shift in US trade policies, and the transition to a new international trade system is still ongoing. Following recent court rulings and executive actions, the overall US effective statutory tariff rate is about 5.3 percentage points below the level assumed in the October 2025 *World Economic Outlook* (WEO) (Figure 1.2), and changes in the cases of a few countries are more substantial. The current environment has incentivized a growing number of countries to finalize long-standing trade negotiations or start new partnerships to foster economic ties among themselves, such as the one between the European Union (EU) and MERCOSUR (the Southern Common Market).

Amid these developments, uncertainty, although lower than the peaks it reached in 2025, is still historically high (Figure 1.3). Several inflection points in the coming months may trigger spikes. First and foremost, the situation in the Middle East remains fluid. Odds of a range of outcomes—from ceasefire

to serious escalation of hostilities—shift by the day. On the trade front, an extension, beyond their initial 150 days, of the Section 122 tariffs recently enacted by the US administration requires congressional approval, or similar tariffs would need to be imposed using other legal authorities. The United States–Mexico–Canada Agreement (USMCA) is set for a mandatory joint review at about the same time the extension comes due, in July 2026. Many of the US agreements with other trading partners so far provide only temporary relief and are set to expire by the end of 2026.

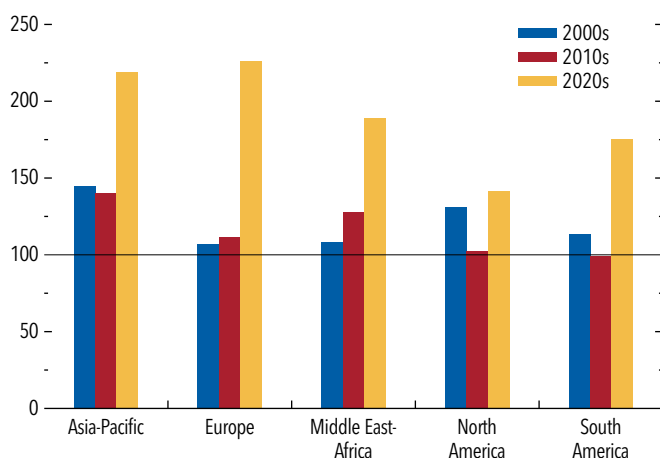
The global economy is facing this next test of resilience as signs of unevenness lie beneath the surface. Activity in the two largest economies, China and the United States, has been stronger than was expected in the October 2025 WEO. But this strength has been uneven. In the case of China, domestic activity—especially in the housing sector—lags behind exports. In the case of the United States, strong activity has been accompanied by low employment growth, amid declining labor force growth.

The unevenness raises downside risks to the outlook, adding to the risks posed by intensifying geopolitical tensions. Medium-term growth prospects remain lackluster, weighed down by geoeconomic fragmentation and structural challenges. That said, it may very well be that current tailwinds, including those from continued fiscal policy support, will last long enough to carry the global economy through the disruptions from the war and to a higher growth path paved by productivity gains from artificial intelligence (AI). Even if they do, however, it will still be crucial to have the right policies in place to make sure that technological transformation leads to broadly balanced growth within and across countries.

Recent Developments: Continued Resilience and Rising Fragility

Before the war, the global economy was performing better than expected, laying the groundwork for upward revisions to forecasts. In aggregate, global growth in the fourth quarter of 2025 increased to

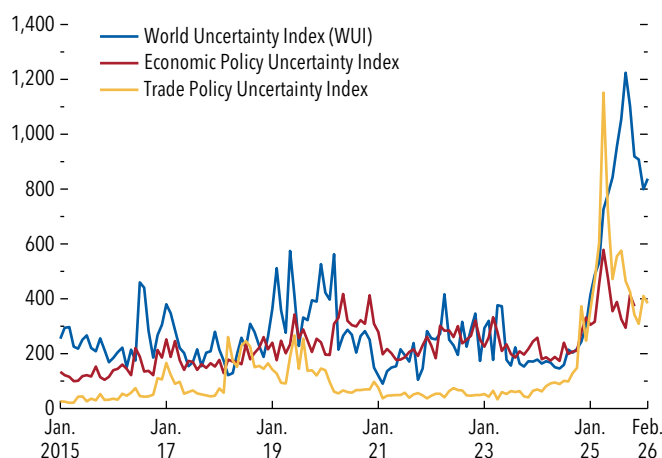
Figure 1.1. Regional Geopolitical Risk
(Index, 1990s = 100)



Sources: Caldara and Iacoviello 2026; and IMF staff calculations.

Note: This figure presents the country-specific geopolitical risk index of Caldara and Iacoviello (2026), a news-based measure of adverse geopolitical events that covers 10 major newspapers in Canada, the United Kingdom, and the United States. The country-level data were downloaded from <https://www.matteoiacoviello.com/gpr.htm> and averaged at the regional and decadal levels and normalized to 100 for the 1990s.

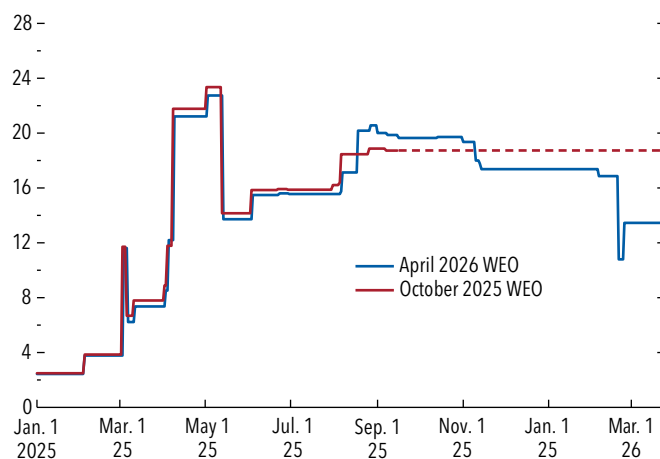
Figure 1.3. Global Uncertainty
(Index)



Sources: Ahir, Bloom, and Furceri 2022 (worlduncertaintyindex.com); Baker, Bloom, and Davis 2016 (policyuncertainty.com); and Caldara and others 2020 (matteoiacoviello.com/tpu.htm).

Note: The uncertainty measures are news- and media-outlets-based indices that quantify media attention to global news related to overall uncertainty (WUI), economic policy uncertainty, and trade policy uncertainty. WUI is divided by 100.

Figure 1.2. US Effective Statutory Tariff Rate
(Percent)



Sources: WTO-IMF Tariff Tracker; and IMF staff calculations.

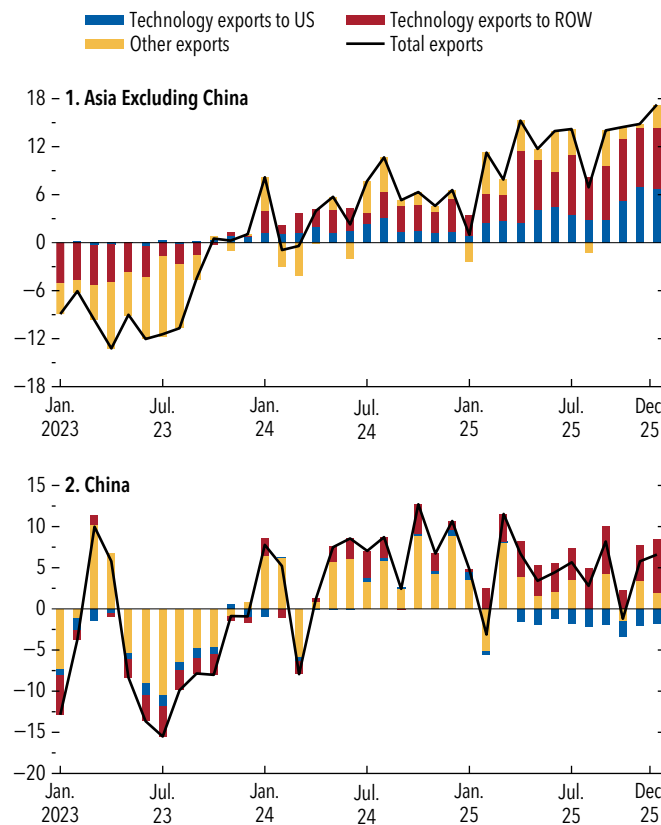
Note: "Effective statutory tariff rate" is a weighted average of announced statutory rates using pretariff (hence, presubstitution) import weights. Calculations include only tariffs that are in effect at the time noted on the x-axis; measures that are not specified and implemented are not included. WEO = *World Economic Outlook*.

3.9 percent on an annualized basis. In China, sequential growth accelerated (per IMF staff seasonal adjustment) to 6.1 percent as strong exports offset weak domestic demand. An increase in fiscal spending fueled stronger activity in Germany, helping growth in the

euro area, excluding Ireland, accelerate to 1.5 percent. Growth in the United States slowed to 0.5 percent, lower than expected in the January 2026 WEO *Update*, as the government shutdown temporarily led to a sharp contraction in public expenditure. Expansion in US technology-related spending remained strong in the fourth quarter, but its effect on GDP was offset by its high import share. Growth in Japan rebounded to 1.3 percent owing to stronger consumption and investment.

Global trade remained robust. Brisk expansion in technology-related exports offset slowing momentum in exports in other product categories (Figure 1.4). This benefited Asian economies in particular, as the main exporters of semiconductors and other equipment sought after by firms raced to invest in digital and AI-related technologies. The rewiring of global supply chains and trading relations continued (Figure 1.5). US imports from China dropped sharply; those from Canada also declined. These dips were offset by increases in imports from Taiwan Province of China, Vietnam, and, to a lesser extent, Mexico. On the other side of the equation, Chinese exports were reoriented from the United States to other Asian economies and, temporarily, to Europe. China's merchandise goods trade surplus hit a record \$1.2 trillion (6 percent of GDP) in 2025.

Figure 1.4. Continued Brisk Growth in Tech-Related Trade Flows
(Percent, year over year)



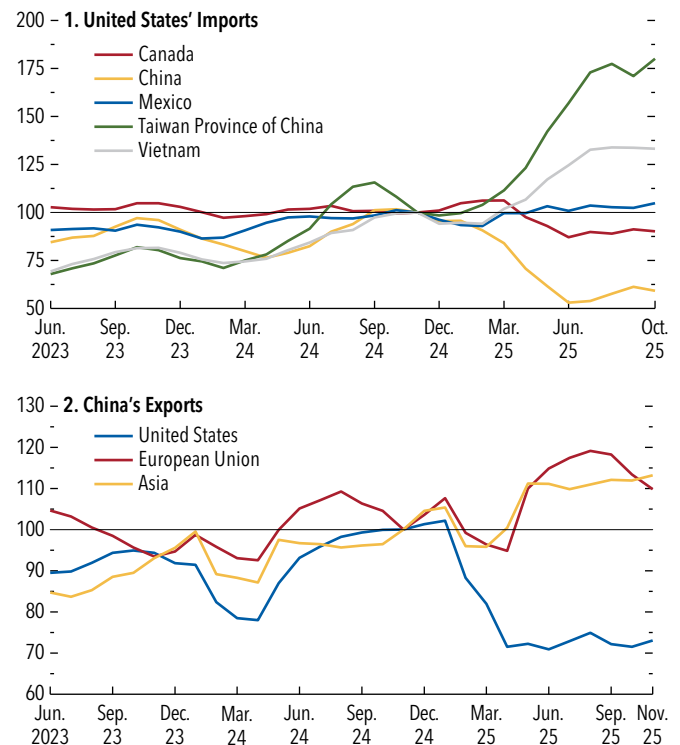
Sources: Haver Analytics; International Trade Center, Trade Map; and IMF staff calculations.

Note: "Technology" exports include those classified under Harmonized System codes 8419, 8470–8473, and 85. "Asia" includes Cambodia, China, Indonesia, Japan, Korea, Malaysia, Singapore, Taiwan Province of China, Thailand, and Vietnam. Data for Vietnam include computers, electronic products, and parts; telephones, mobile phones, and parts; and insulated wires and cables. ROW = rest of the world.

Global inflation has been largely steady. This stability masks some divergence, however. In the United States, above-target inflation persists, with core inflation for personal consumption expenditure maintaining a high year-over-year rate of 3.1 percent in January 2026. To date, evidence indicates that the direct incidence of tariffs has largely fallen on US importers and consumers (Amiti and others 2026; Gimbel 2026; Gopinath and Neiman 2026). In contrast, inflation fell sharply in Japan in January 2026 to below the 2 percent target for the first time since the second quarter of 2022, with the decline largely reflecting the provisional gasoline tax abolition.

Risk-off sentiment following the outbreak of the Middle East conflict has led to a moderate tightening

Figure 1.5. Reorientation of Global Trade
(Index, 2024 = 100)



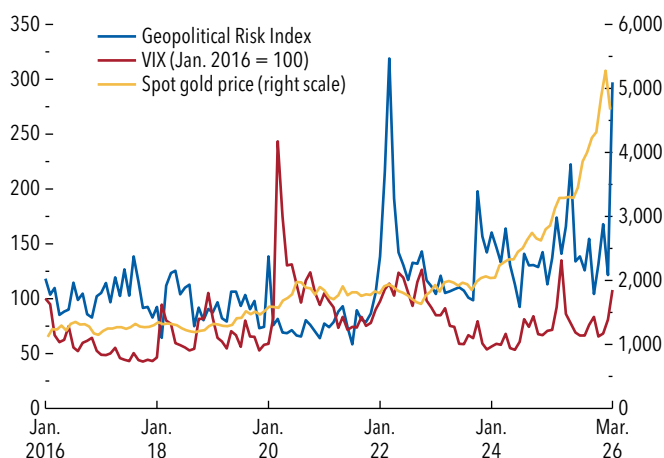
Sources: Antràs and Presbitero 2026; Trade Data Monitor; and IMF staff calculations.

Note: Three-month moving average of non-seasonally adjusted US dollar values, based on cross-border shipments data sourced from Trade Data Monitor. These data can differ from balance of payments data.

of global financial conditions, but they remain accommodative from a historical point of view (see the April 2026 *Global Financial Stability Report*). Concerns about a resurgence of inflation have raised bond yields and driven equity prices down. Emerging markets—especially commodity importers and those with pre-existing vulnerabilities—have been affected the most. The US dollar has strengthened somewhat, reaffirming its safe haven status. Even so, market volatility has been relatively subdued (Figure 1.6). At the same time, geopolitical tensions and other factors have contributed to sharp swings in the gold price (see the Commodity Special Feature).

Fiscal policy remains too loose in many of the largest advanced economies and emerging markets (see the April 2026 *Fiscal Monitor*). The Middle East conflict is putting additional pressure on public finances, both via the direct effects of the conflict and as governments seek ways to protect the most vulnerable from the fallout in commodity markets and may be tempted

Figure 1.6. Global Geopolitical Risk
(Index; US dollars per ounce)



Sources: Caldara and Iacoviello 2022; Chicago Board Options Exchange (CBOE); London Stock Exchange Group, Datastream; and IMF staff calculations.

Note: This figure plots the monthly global geopolitical risk index of Caldara and Iacoviello (2022), a news-based measure of adverse geopolitical events that covers 10 major newspapers in Canada, the United Kingdom, and the United States. Data were downloaded from <https://www.matteoiacoviello.com/gpr.htm> on April 02, 2026. VIX = CBOE Volatility Index.

to offer broad-based fiscal packages, while higher financing costs and weaker activity weigh on revenues. Countries with preexisting fuel subsidies face different fiscal dynamics than those with liberalized energy pricing, while those with links to the Middle East region through remittances confront additional pressure on household incomes and external balances. Monetary policy was becoming more divergent as common global drivers of inflation became less prominent, until the conflict delivered a global negative supply shock.

Growth and Disinflation, Interrupted

Absent the war, sources of recent resilience (Box 1.1) would have been expected to continue to hold global economic activity on a steady path in 2026. Near-term prospects have, however, worsened given the disruptions caused by the closure of the Strait of Hormuz and the attacks on production facilities. Medium-term prospects remain constrained by structural challenges.

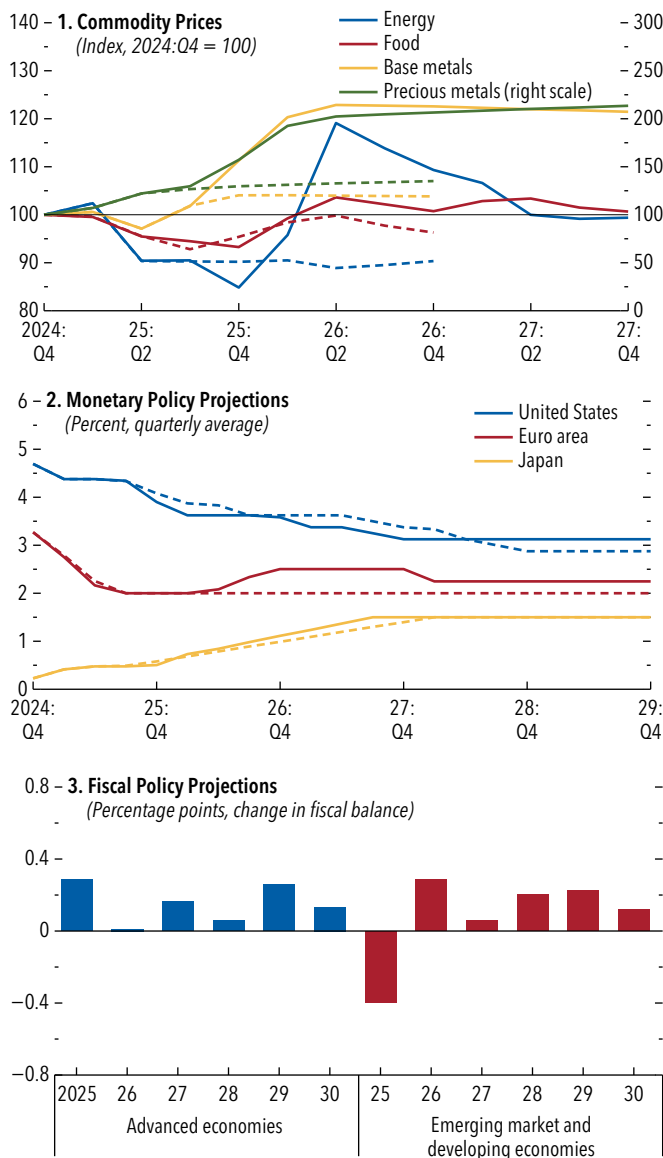
The bottom-up reference forecasts presented in this report are predicated on a relatively short-lived conflict. While more benign scenarios are highly desirable, given the extraordinarily high level of uncertainty, the reference forecasts are complemented with top-down model-based global growth projections under the

assumption of a more prolonged and intense conflict. These scenarios become more likely over time as hostilities and related disruptions continue.

Global Assumptions

The reference forecasts incorporate the impact of the war, based on the assumption that the conflict will last for a few more weeks and a recovery will then gradually take hold, such that the disruptions fade and production and exports from the region normalize by mid-2026. This timeline is broadly captured in the projections for global commodity prices—consistent with futures pricing as of March 10—and interest rates (Figure 1.7). Fiscal and trade policies as they currently stand are assumed to remain in place through the forecast horizon. Uncertainty—encompassing trade and other economic policies as well as geopolitical developments—is assumed to remain elevated through 2027.

- **Commodity price projections:** Prices for energy commodities are expected to rise by 19 percent in 2026, as opposed to the small decline projected in the October 2025 WEO. Oil prices are expected to increase by 21.4 percent on account of disruptions to production and transportation in the Middle East, corresponding to the average petroleum spot price index averaging \$82 per barrel. Natural gas prices are expected to be affected more than oil prices because of the technical complexity of restarting production and the comparatively lower level of reserves to fall back on. Food prices are expected to increase as well, more than projected in October 2025, on account of higher energy and fertilizer prices, disrupted shipping routes, and increased transport costs. Base and precious metal prices are projected to maintain the gains experienced in 2025.
- **Monetary policy projections:** Differentiation of monetary policies in major jurisdictions is expected to continue. In the United States, the federal funds rate is projected to be reduced gradually, reaching its terminal rate of about 3.1 percent by the end of 2027. The policy rate in the euro area is expected to increase by 50 basis points over the course of 2026. In Japan, the policy rate is projected to gradually rise, at a slightly steeper clip than thought in October 2025, toward a neutral setting of about 1.5 percent.
- **Fiscal policy projections:** Fiscal policy in advanced economies, on average, is expected to be neutral in 2026 and tighten in the latter years of the forecast

Figure 1.7. Global Assumptions

Source: IMF staff calculations.

Note: In panels 1 and 2, solid lines denote projections from the April 2026 *World Economic Outlook* (WEO) and dashed lines those from the October 2025 WEO. In panel 3, the fiscal balance used is the general government structural primary balance in percent of potential GDP. The structural primary balance is the cyclically adjusted primary balance excluding net interest payments and corrected for a broader range of noncyclical factors such as changes in asset and commodity prices.

horizon, despite a widening of deficits in major jurisdictions. In the United States, the general government fiscal-balance-to-GDP ratio is expected to deteriorate by 0.7 percentage point in 2026 to 7½ percent, reflecting the impact of the One Big Beautiful Bill Act (OBBBA), partly offset by additional tariff revenues. The fiscal balance is projected

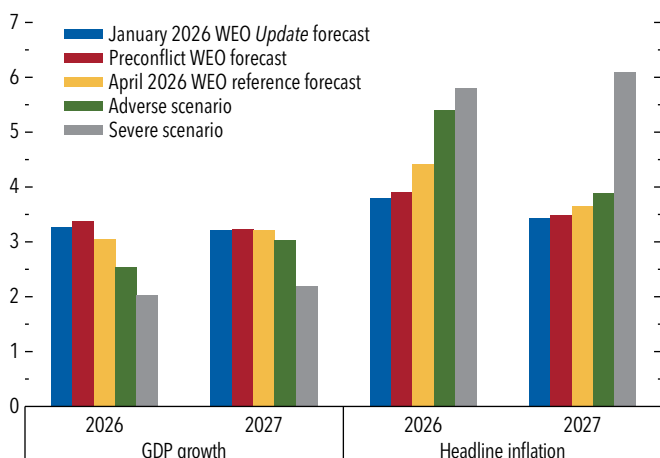
to decline in the euro area, with Germany's deficit registering a widening of over 1 percentage point to 3.8 percent as infrastructure and defense spending ramp up. Japan is also projected to see its deficit widen by 1 percentage point of GDP in 2026, and its fiscal policy is expected to remain moderately expansionary through 2030. Under current policies, US public debt is projected to continue to climb, from 124 percent of GDP in 2025 to 142 percent in 2031. In the euro area, the debt-to-GDP ratio also rises, but by less, from 87 percent in 2025 to 90 percent in 2031. Fiscal policy in emerging market and developing economies, on average, is projected to gradually tighten over the forecast horizon. In China, the deficit is expected to widen by 0.3 percentage point in 2026, before starting to narrow in the medium term. Still, public debt in emerging market and developing economies is projected to rise further, reaching 86 percent of GDP in 2031, from 74 percent in 2025.

- **Trade policy assumptions:** IMF staff projections remain based on real-time current trade policy; that is, they assume that policies as they stood at the end of March are permanent. This is so, even in regard to measures framed as temporary or pending—meaning that US Section 122 tariffs are assumed to be extended or reimposed under different statutes. The US effective statutory tariff rate underlying the projections is 13.5 percent, compared with 18.7 percent in the October 2025 forecast. The corresponding effective tariff rate imposed by the rest of the world on imports from the United States is unchanged at 3.5 percent.

Global Growth Forecast: Fragile with Large Dispersion

- Before the outbreak of the conflict, the bottom-up forecasts would have indicated a stable growth path (“*Preconflict WEO forecast*” in Figure 1.8). Global growth would have been 3.4 percent in 2026 and 3.2 percent in 2027, an upward revision of 0.1 percentage point for 2026 and unchanged for 2027 compared with the forecast in the January 2026 WEO *Update*.
- Under the assumption in the *reference forecast* that the war turns out to be relatively short-lived, global growth is expected to slow down modestly. At 3.1 percent for 2026 and 3.2 percent for 2027, the forecasts mark a deceleration from the estimated

Figure 1.8. Global Growth and Inflation Forecasts
(Percent)

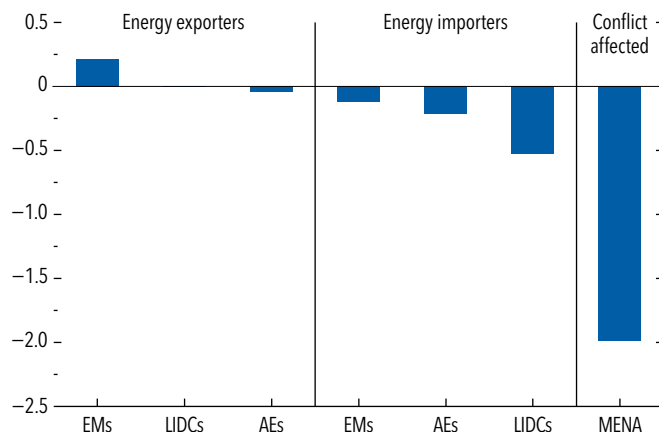


Source: IMF staff estimates.

Note: The three WEO forecasts are constructed bottom-up from individual country projections before and after the outbreak of the Middle East conflict. The two scenarios are model-based top-down estimates. The adverse scenario assumes that oil (gas) prices increase by 80 (160) percent starting in 2026:Q2, relative to the January 2026 WEO Update, with the increase mostly unwinding in 2027, one-year-ahead inflation expectations increase by 50 (90) basis points in advanced economies (emerging markets excluding China), and corporate premiums rise by 50 (100) basis points in advanced economies plus China (emerging markets excluding China), while sovereign spreads in emerging markets excluding China increase by 50 basis points, with the tightening in financial conditions fading in 2027. The severe scenario is calibrated to larger and more persistent shocks. First, oil (gas) prices are assumed to be 100 (200) percent higher than the January 2026 WEO Update, starting in 2026:Q2 and staying at that level in 2027, while food commodity prices increase by 5 (10) percent in 2026 (2027). Second, one-year-ahead inflation expectations increase by 100 (130) basis points in advanced economies (emerging markets excluding China) by 2027. Third, corporate risk premiums rise by 100 (200) basis points in advanced economies plus China (emerging markets excluding China) in 2026–27, while sovereign spreads increase by 100 basis points in emerging markets excluding China over the same period. WEO = *World Economic Outlook*.

3.4 percent achieved in 2025 (Table 1.1). At market exchange rates, world output is projected to grow by 2.6 percent in both 2026 and 2027 (Table 1.2). The relatively modest downward revision to global growth in the reference forecast relative to the January 2026 WEO Update owes to continued tailwinds partially offsetting the negative shocks from the conflict, including lower tariffs, preexisting policy support, and carryover from stronger-than-expected outturns at the end of 2025 and the first quarter of 2026 in some cases. Compared with the preconflict WEO forecasts, growth in the near term is revised downward by 0.2 percentage point. This masks significant variation across countries, with lower-income commodity-importing economies being hit particularly hard through higher

Figure 1.9. GDP Growth Revisions in the Reference Forecast
(Percentage points)



Sources: World Bank, World Development Indicators; and IMF staff calculations.

Note: The figure presents cumulative GDP growth revisions for 2026–27 relative to the January 2026 *World Economic Outlook Update*. Energy exporters and importers are defined using 2022 net energy imports as a share of energy use. Groups are aggregated using purchasing-power-parity weights. EMs and LIDCs exclude MENA. AEs = advanced economies; EMs = emerging markets; LIDCs = low-income developing countries; MENA = Middle East and North Africa.

energy and food prices as well as foreign exchange depreciation (Figure 1.9). Cumulative growth over 2026–27 is revised downward by 0.5 percentage point for low-income net energy-importing economies relative to the January 2026 WEO Update, compared with a downward revision of 0.2 percentage point in energy-importing advanced economies and positive or neutral revisions for net energy-exporting economies.

- Should the conflict become more protracted than assumed in the reference forecast or the resumption of production and transport activities take longer than assumed because of possible scarring from closing of or damage to energy infrastructure, the impact on growth would be larger. To illustrate the potential range of magnitudes, the report considers two top-down model-based downside scenarios: an adverse one and a severe one.
- In the *adverse scenario*, (1) Oil prices are assumed to increase by 80 percent starting in the second quarter of 2026 relative to the January 2026 WEO Update baseline, before falling back to about 20 percent above baseline in 2027, with the increase dissipating in 2028 (corresponding to an average petroleum spot price index of about \$100 per barrel in 2026 and about \$75 in 2027). Gas prices increase for

Table 1.1. Overview of the World Economic Outlook Reference Forecast
(Percent change, unless noted otherwise)

	2025	Projections		Difference from January 2026 WEO Update ¹		Difference from October 2025 WEO ¹	
		2026	2027	2026	2027	2026	2027
World Output	3.4	3.1	3.2	-0.2	0.0	0.0	0.0
Advanced Economies	1.9	1.8	1.7	0.0	0.0	0.2	0.0
United States	2.1	2.3	2.1	-0.1	0.1	0.2	0.0
Euro Area	1.4	1.1	1.2	-0.2	-0.2	-0.1	-0.2
Germany	0.2	0.8	1.2	-0.3	-0.3	-0.1	-0.3
France	0.9	0.9	0.9	-0.1	-0.3	0.0	-0.3
Italy	0.5	0.5	0.5	-0.2	-0.2	-0.3	-0.1
Spain	2.8	2.1	1.8	-0.2	-0.1	0.1	0.1
Japan	1.2	0.7	0.6	0.0	0.0	0.1	0.0
United Kingdom	1.3	0.8	1.3	-0.5	-0.2	-0.5	-0.2
Canada	1.7	1.5	1.9	-0.1	0.0	0.0	0.0
Other Advanced Economies ²	3.0	2.6	2.2	0.6	0.1	0.6	0.1
Emerging Market and Developing Economies	4.4	3.9	4.2	-0.3	0.1	-0.1	0.0
Emerging and Developing Asia	5.5	4.9	4.8	-0.1	0.0	0.2	0.0
China	5.0	4.4	4.0	-0.1	0.0	0.2	-0.2
India ³	7.6	6.5	6.5	0.1	0.1	0.3	0.1
Emerging and Developing Europe	2.0	2.0	2.1	-0.3	-0.3	-0.2	-0.3
Russia	1.0	1.1	1.1	0.3	0.1	0.1	0.0
Latin America and the Caribbean	2.4	2.3	2.7	0.1	0.0	0.0	0.1
Brazil	2.3	1.9	2.0	0.3	-0.3	0.0	-0.2
Mexico	0.6	1.6	2.2	0.1	0.1	0.1	0.2
Middle East and Central Asia	3.6	1.9	4.6	-2.0	0.6	-1.9	0.8
Saudi Arabia	4.5	3.1	4.5	-1.4	0.9	-0.9	1.3
Sub-Saharan Africa	4.5	4.3	4.4	-0.3	-0.2	-0.1	-0.1
Nigeria	4.0	4.1	4.3	-0.3	0.2	-0.1	0.3
South Africa	1.1	1.0	1.3	-0.4	-0.2	-0.2	-0.2
<i>Memorandum</i>							
World Growth Based on Market Exchange Rates	2.9	2.6	2.6	-0.2	0.0	0.0	-0.1
European Union	1.6	1.3	1.4	-0.2	-0.2	-0.1	-0.2
ASEAN-5 ⁴	4.5	4.1	4.4	-0.1	0.0	0.0	0.1
Middle East and North Africa	3.2	1.1	4.8	-2.8	0.8	-2.6	1.1
Emerging Market and Middle-Income Economies	4.4	3.8	4.1	-0.3	0.0	-0.1	0.0
Low-Income Developing Countries	4.8	4.8	4.9	-0.3	-0.2	-0.2	-0.4
World Trade Volume (goods and services)	5.1	2.8	3.8	0.2	0.7	0.5	0.7
Imports							
Advanced Economies	4.7	2.6	3.1	0.8	0.8	1.3	0.9
Emerging Market and Developing Economies	5.7	2.7	4.9	-0.9	0.4	-1.3	0.2
Exports							
Advanced Economies	3.7	2.5	2.7	0.4	0.3	0.8	0.4
Emerging Market and Developing Economies	7.4	3.5	5.4	-0.1	1.2	0.2	1.3
Commodity Prices							
Oil ⁵	-14.4	21.4	-7.6	29.9	-7.7	25.9	-7.4
Nonfuel (average based on world commodity import weights)	9.6	21.7	1.9	14.2	1.0	17.6	1.3
World Consumer Prices⁶	4.1	4.4	3.7	0.6	0.3	0.7	0.3
Advanced Economies ⁷	2.5	2.8	2.2	0.6	0.1	0.6	0.1
Emerging Market and Developing Economies ⁶	5.2	5.5	4.6	0.7	0.3	0.8	0.4

Source: IMF staff estimates.

Note: Real effective exchange rates are assumed to remain constant at the levels prevailing from February 10, 2026, to March 10, 2026. Economies are listed on the basis of economic size. The aggregated quarterly data are seasonally adjusted. WEO = *World Economic Outlook*.

¹ Difference based on rounded figures for the current, January 2026 WEO Update, and October 2025 WEO forecasts.

² Excludes the Group of Seven (Canada, France, Germany, Italy, Japan, United Kingdom, United States) and euro area countries.

³ For India, data and forecasts are presented on a fiscal year basis, and GDP from 2022 onward is based on GDP at market prices with fiscal year 2022/23 as a base year.

⁴ Indonesia, Malaysia, the Philippines, Singapore, and Thailand.

⁵ Simple average of prices of UK Brent, Dubai Fateh, and West Texas Intermediate crude oil. The average price of oil in US dollars a barrel was \$67.74 in 2025; the assumed price, based on futures markets, is \$82.22 in 2026 and \$75.97 in 2027.

⁶ Excludes Venezuela. See the country-specific note for Venezuela in the "Country Notes" section of the Statistical Appendix.

⁷ The assumed inflation rates for 2026 and 2027, respectively, are as follows: 2.6 percent and 2.2 percent for the euro area, 2.2 percent and 2.3 percent for Japan, and 3.2 percent and 2.1 percent for the United States.

Table 1.1. Overview of the World Economic Outlook Reference Forecast (continued)

(Percent change, unless noted otherwise)

	2025	Q4 over Q4 ⁸					
		Projections		Difference from January 2026 WEO Update ¹		Difference from October 2025 WEO ¹	
		2026	2027	2026	2027	2026	2027
World Output	3.3	2.9	3.5	-0.3	0.3	-0.4	...
Advanced Economies	1.9	1.5	1.9	-0.4	0.2	-0.3	...
United States	2.0	2.2	2.1	0.1	0.0	0.2	...
Euro Area	1.2	1.2	1.5	-0.5	0.2	-0.6	...
Germany	0.4	0.7	1.3	-0.3	-0.2	-0.3	...
France	1.2	0.6	1.0	-0.3	-0.3	-0.4	...
Italy	0.8	-0.2	1.8	-0.9	1.1	-0.3	...
Spain	2.6	1.6	2.2	-0.4	0.3	-0.2	...
Japan	0.5	1.0	0.7	-0.5	0.3	-0.1	...
United Kingdom	1.0	1.0	1.9	-0.4	0.0	-0.4	...
Canada	0.7	2.3	1.5	0.0	0.0	0.0	...
Other Advanced Economies ²	4.1	1.5	2.5	-1.3	...	-1.3	...
Emerging Market and Developing Economies	4.4	4.0	4.6	-0.3	0.4	-0.4	...
Emerging and Developing Asia	5.3	4.7	5.1	-0.3	0.3	-0.6	...
China	4.4	3.9	4.6	-0.6	0.5	-1.1	...
India ³	7.5	7.1	6.5	0.6	0.0	0.9	...
Emerging and Developing Europe	1.9	1.8	2.2	-0.3	-0.2	-0.5	...
Russia	1.0	0.6	0.6	0.0	-0.5	0.1	...
Latin America and the Caribbean	2.1	2.9	2.3	0.1	-0.1	0.3	...
Brazil	2.0	2.6	1.9	0.3	-0.3	0.3	...
Mexico	1.8	1.5	2.4	-0.7	0.3	-0.2	...
Middle East and Central Asia	...	...	...	...	...	...	...
Saudi Arabia	5.0	3.1	4.5	-1.4	0.9	-0.9	...
Sub-Saharan Africa	...	...	...	...	...	...	...
Nigeria	3.9	3.8	4.5	-0.5	-2.0	-0.5	...
South Africa	1.5	0.7	1.5	-0.5	-0.2	-0.3	...
<i>Memorandum</i>							
World Growth Based on Market Exchange Rates	2.7	2.4	2.7	-0.3	0.1	-0.4	...
European Union	1.4	1.1	1.8	-0.4	0.2	-0.6	...
ASEAN-5 ⁴	4.9	3.7	4.7	-0.5	0.1	-0.8	...
Middle East and North Africa	...	...	...	...	...	...	...
Emerging Market and Middle-Income Economies	4.4	3.9	4.3	-0.3	0.2	-0.5	...
Low-Income Developing Countries	...	...	...	...	...	...	...
Commodity Prices (US dollars)							
Oil ⁵	-15.1	29.2	-6.7	30.6	-7.6	31.4	...
Nonfuel (average based on world commodity import weights)	14.2	14.6	1.1	13.8	0.5	13.4	...
World Consumer Prices⁶	3.5	3.9	3.0	0.8	0.0	0.9	...
Advanced Economies ⁷	2.5	2.9	2.0	0.8	-0.1	0.9	...
Emerging Market and Developing Economies ⁶	4.2	4.6	3.7	0.7	0.1	0.9	...

⁸For world output, the quarterly estimates and projections account for approximately 90 percent of annual world output at purchasing-power-parity weights. For emerging market and developing economies, the quarterly estimates and projections account for approximately 85 percent of annual emerging market and developing economies' output at purchasing-power-parity weights.

Europe and Asia by 160 percent in the second quarter relative to baseline, before also mostly unwinding in 2027, and food commodity prices increase by 2.5 percent. (2) One-year-ahead inflation expectations increase by as much as 50 basis points by 2027 in advanced economies and as much as 90 basis points in emerging markets excluding China. Inflation expectations are unchanged in China, as current low inflation makes this less of a risk than for other countries. (3) A risk-off episode increases corporate

premiums in advanced economies and China by 50 basis points, while emerging markets excluding China experience a 100 basis point increase as well as a 50 basis point increase in sovereign spreads. The tightening in financial conditions fades in 2027. Given the large impact on inflation expectations, the monetary policy response assigns less weight to output stabilization than usually assumed.

- In the *severe scenario*, (1) The shock to commodity prices is more severe and persistent, with oil prices

Table 1.2. Overview of the World Economic Outlook Reference Forecast at Market Exchange Rate Weights
(Percent change)

	2025	Projections		Difference from January 2026 WEO Update ¹		Difference from October 2025 WEO ¹	
		2026	2027	2026	2027	2026	2027
World Output	2.9	2.6	2.6	-0.2	0.0	0.0	-0.1
Advanced Economies	1.9	1.8	1.7	-0.1	0.0	0.1	-0.1
Emerging Market and Developing Economies	4.3	3.7	3.9	-0.3	0.0	-0.1	0.0
Emerging and Developing Asia	5.3	4.7	4.5	-0.1	0.0	0.2	-0.1
Emerging and Developing Europe	2.0	2.1	2.2	-0.3	-0.3	-0.2	-0.2
Latin America and the Caribbean	2.2	2.3	2.6	0.2	0.0	0.1	0.1
Middle East and Central Asia	3.8	1.6	4.8	-2.5	0.8	-2.4	0.9
Sub-Saharan Africa	4.3	4.1	4.2	-0.3	-0.2	-0.1	-0.1
<i>Memorandum</i>							
European Union	1.5	1.3	1.4	-0.2	-0.1	0.0	-0.1
Middle East and North Africa	3.5	1.0	4.9	-3.1	1.0	-2.9	1.1
Emerging Market and Middle-Income Economies	4.2	3.6	3.9	-0.3	0.1	-0.2	0.0
Low-Income Developing Countries	4.9	4.8	4.9	-0.3	-0.3	-0.2	-0.4

Source: IMF staff estimates.

Note: The aggregate growth rates are calculated as a weighted average, in which a moving average of nominal GDP in US dollars for the preceding three years is used as the weight. WEO = *World Economic Outlook*.¹ Difference based on rounded figures for the current, January 2026 WEO Update, and October 2025 WEO forecasts.

- increasing by 100 percent starting in the second quarter of 2026, relative to the January 2026 WEO Update baseline, but also staying at that level in 2027, before dissipating in 2028 (corresponding to an average petroleum spot price index of about \$110 per barrel in 2026 and about \$125 in 2027). Gas prices for Europe and Asia increase by 200 percent over the same period, and food commodity prices increase by 5 percent in 2026 and 10 percent in 2027. (2) One-year-ahead inflation expectations ratchet up by as much as 100 basis points in advanced economies by 2027 and by as much as 130 basis points in emerging markets excluding China, also by 2027. (3) A significant risk-off episode pushes up corporate premiums in advanced economies and in China by 100 basis points in 2026, and they stay at that level in 2027, while emerging markets excluding China experience a widening in sovereign spreads of 100 basis points over the same period, along with an increase in corporate spreads of 200 basis points. As in the adverse scenario, the monetary policy response is geared toward containing inflationary pressures rather than stabilizing output.
- Under the *adverse scenario*, global growth would be reduced by 0.8 percentage point in 2026, dropping to 2.5 percent. There would also be a modest 0.2 percentage point impact on growth in 2027, bringing global growth to 3.0 percent.

Inflation would be 1.5 percentage points higher at 5.4 percent in 2026, and 0.4 percentage point higher at 3.9 percent in 2027. Most of the impact on inflation and over half the impact on growth in 2026 come from higher energy prices. The more persistent effect on growth in 2027, however, is driven by the tightening in financial conditions and rise in inflation expectations, which implies a modest tightening in policy rates of 50 basis points in advanced economies by 2027 and a somewhat larger increase in emerging market economies.

- Under the *severe scenario*, the effects on global growth are substantial and longer lasting. Global growth would be reduced by 1.3 percentage points in 2026. This would mean a close call for a global recession (growth rate below 2 percent), which has happened only four times since 1980, with the latest two occasions corresponding to the global financial crisis and the COVID-19 pandemic. The effects on growth are also more persistent, with global growth reduced by 1.0 percentage point in 2027, to 2.2 percent. Inflation would be 190 basis points higher in 2026, reaching 5.8 percent, and 260 basis points higher in 2027, reaching 6.1 percent. The increase in oil and gas prices has not only a larger, but also a more persistent, impact on growth, subtracting 0.6 percentage point in 2026 and a further 0.5 percentage point in 2027. The amplification through inflation expectations and financial

conditions would also be sizable, reducing growth by 0.7 percentage point in 2026 and 0.5 percentage point in 2027. This in part reflects a more aggressive monetary policy response. The federal funds rate would increase by 50 basis points in 2026 and 100 basis points in 2027, relative to baseline.

- In both scenarios, the impact on emerging markets would again be greater than that on advanced economies. In the *adverse scenario*, growth in 2026 is lower by 1.3 percentage points in emerging markets excluding China, relative to baseline, and by 0.6 percentage point in advanced economies. The *severe scenario* lowers growth in 2026 by 1.9 percentage points in emerging markets excluding China, almost twice the decline in advanced economies. This reflects a combination of a larger exposure to higher commodity prices and disruption to energy production, a larger increase in inflation expectations, and a more pronounced tightening in financial conditions.

Growth Forecast for Advanced Economies

Under the reference forecast, growth in *advanced economies* is projected to be 1.8 percent in 2026 and 1.7 percent in 2027. The overall effect on growth in advanced economies of the conflict in the Middle East is modest, lowering growth by 0.2 percentage point in 2026 relative to the preconflict forecast, thanks to positive terms-of-trade effects in the United States and stronger growth momentum and offsetting government measures in Japan, with a large negative effect expected only in some net energy-importing economies, such as the euro area and the United Kingdom.

- In the *United States*, the economy is projected to expand by 2.3 percent in 2026, with growth supported by fiscal policy and the lagged impact of monetary policy rate cuts in 2025, even as the rise in trade barriers since April 2025 continues to weigh on the level of activity. This 0.1 percentage point downward revision relative to the January 2026 WEO *Update* reflects the balance of a small negative effect from the war—given the net-energy-exporter status of the United States—and offsets from a rebound in activity in the first quarter of 2026 compared with the fourth quarter of 2025 following the end of the 2025 federal government shutdown, stronger-than-previously-assumed productivity growth, and the associated carryover. While the International Emergency Economic Powers Act (IEEPA) ruling may reduce fiscal revenues raised by

tariffs, the impact on the fiscal balance and activity is expected to be small and spread over the forecast horizon. Growth is projected to remain solid at 2.1 percent in 2027, with a near-term fiscal boost from tax incentives, including those for corporate investment under the OBBBA. Technology-driven momentum is expected to moderate but still provide some offset to lower immigration and moderating consumption. Strong productivity growth is projected to gradually fade and converge to historical norms.

- In the *euro area*, growth is expected to decline from 1.4 percent in 2025 to 1.1 percent in 2026 and to 1.2 percent in 2027. The forecast is revised downward by 0.2 percentage point in each year compared with the January 2026 WEO *Update*, with the effect of better-than-expected growth at the end of 2025 giving way to the negative impact of the Middle East conflict over time. The latter will add to the lingering effects of the persistent rise in energy prices since Russia's invasion of Ukraine, dragging on manufacturing, with additional pressure from the real appreciation of the euro relative to currencies of countries exporting similar products. The impact of the planned increase in defense spending for most countries is expected to materialize only in subsequent years, given commitments to reach target levels gradually by 2035 (see Chapter 2 for estimates of defense spending multipliers). In *Japan*, growth is projected to drop from 1.2 percent in 2025 to 0.7 percent in 2026 and to 0.6 percent in 2027. This marks an upward revision for 2026 relative to the October 2025 figure, reflecting the fiscal stimulus package announced by the new government last November, stronger domestic-demand-driven growth carryover from 2025, and government measures to limit the effects of higher energy prices, partly offset by weaker external demand and the Middle East conflict. In the *United Kingdom*, the war and a slower pace of monetary easing mean that growth is projected to decline from 1.3 percent in 2025 to 0.8 percent in 2026, a downward revision of 0.5 percentage point relative to the October 2025 forecast. Growth is projected to recover to 1.3 percent in 2027, slower than expected before the war as the impact of higher energy prices lingers. In *Canada*, growth is projected to slow from 1.7 percent in 2025 to 1.5 percent in 2026 before recovering to 1.9 percent in 2027. The softer near-term profile reflects weaker momentum at the

end of 2025 and slower population growth, while earlier monetary easing and supportive fiscal policy help sustain domestic demand. This is broadly unchanged from the October 2025 WEO forecast, with the positive terms-of-trade shock of higher oil prices offsetting the other effects of the war in the Middle East.

Growth Forecast for Emerging Market and Developing Economies

In *emerging market and developing economies*, growth is expected to fall to 3.9 percent in 2026 and recover to 4.2 percent in 2027. The conflict in the Middle East has a varied impact on growth given differential exposure—through geographic proximity, financial flows, remittances, and energy dependencies. Overall, it has a larger net impact on growth in emerging market and developing economies compared with advanced economies, lowering growth in 2026 for the former group by 0.3 percentage point relative to the preconflict forecast.

- Growth in *emerging and developing Asia* is expected to decline from 5.5 percent in 2025 to 4.9 percent in 2026 and to 4.8 percent in 2027. Growth in *China* for 2026 is revised upward by 0.2 percentage point, relative to October (a 0.1 percentage point downward revision from January), to 4.4 percent, reflecting the lower US effective tariff rates on Chinese goods, and stimulus measures offset the negative impact of the shock induced by the Middle East conflict. The economy's growth rate is expected to decelerate to 4.0 percent in 2027 as structural headwinds—including those from a grinding slowdown in the housing sector, a declining labor force, decreasing returns on investment, and slower productivity growth—assert themselves. In *India*, growth for 2025 is revised upward by 1.0 percentage point relative to October, to 7.6 percent, reflecting the better-than-expected outturn in the second and third quarters of the fiscal year and sustained strong momentum in the fourth quarter. For 2026, growth is revised upward moderately by 0.3 percentage point (0.1 percentage point relative to January) to 6.5 percent, led by positive contributions from the carryover of the strong 2025 outturn and the decline in additional US tariffs on Indian goods from 50 to 10 percent, which outweigh the adverse impact of the Middle East conflict. Growth is projected to stay at 6.5 percent in 2027. In several South and Southeast Asian economies, disruptions

in the Middle East are expected to reduce tourism and remittance inflows, thereby weakening domestic demand. Growth in the *Philippines* is revised downward by 1.5 percentage points for 2026, relative to January, with the war shock compounding the negative base effects from a weaker-than-expected 2025 outturn related to a sharp decline in public investment and confidence.

- In the *Middle East and Central Asia*, growth is projected to decline from 3.6 percent in 2025 to 1.9 percent in 2026 and recover to 4.6 percent in 2027 as the region experiences the most direct impact of the conflict and the expected subsequent rebound. For commodity exporters directly affected by the conflict, diminished production and exports imply a severe downward revision of GDP growth projections for 2026, depending on the degree of damage suffered in energy and transportation infrastructure as well as the dependence on the Strait of Hormuz and availability of alternative export routes. The contraction of GDP growth for 2026 is therefore more pronounced for *Bahrain*, *Iran*, *Iraq*, *Kuwait*, and *Qatar* and less significant for *Oman*, *Saudi Arabia*, and the *United Arab Emirates*. For all these economies, growth in 2027 is expected to rebound, based on the assumption that energy production and transportation are normalized over the next few months—an assumption that may need to be revised if the duration of the conflict extends and the degree of damage suffered gets reassessed. Growth in *Iran* in 2026 is revised downward by 7.2 percentage points, relative to January, to –6.1 percent, while that for 2027 is revised upward by 1.6 percentage points to 3.2 percent. In *Saudi Arabia*, the growth forecast for 2026 is revised downward by 1.4 percentage point relative to January, to 3.1 percent, and that for 2027 is revised upward by 0.9 percentage point, to 4.5 percent. For commodity importers in the Middle East and North Africa, the terms-of-trade shock from higher commodity prices contributes to a somewhat modest downward revision of growth projections in 2026 and 2027, with some differentiation as a result of varying exposures to imports of energy, energy derivatives, and food items, as well as different economic trajectories before the conflict erupted. In *Egypt*, growth is projected to slow to 4.2 percent in 2026 and recover to 4.8 percent in 2027, a cumulative downward revision of 1.1 percentage points. For Caucasus and Central Asia countries, the growth

momentum experienced over the past few years is expected to continue, with aggregate GDP growth for the group revised upward in 2026 and 2027, by a cumulative 0.3 percentage point.

- Growth in *sub-Saharan Africa* is expected to be relatively stable at 4.3 percent in 2026 and 4.4 percent in 2027. This masks variation across countries, with some in the region—particularly oil-importing non-resource-intensive countries—adversely affected by the Middle East conflict. Key economies continue to benefit from past macroeconomic stabilization and reform efforts. In *Nigeria*, growth momentum is sustained at 4.1 percent in 2026, supported by improved macroeconomic stability and positive terms-of-trade effects, while higher goods and transport costs are headwinds. Growth is expected to strengthen in 2027 to 4.3 percent as these headwinds ease. In *South Africa*, the disruptions from the Middle East conflict are projected to slow growth slightly to 1.0 percent in 2026. The economy is expected to bounce back in 2027, growing at 1.3 percent, supported by a gradual resumption of structural-reform-driven private investment as disruptions from the conflict subside. Growth in other countries in the region as a whole is expected to decline from 5.6 percent in 2025 to 5.2 percent in both 2026 and 2027, revised downward relative to January by a cumulative 0.6 percentage point.
- In *Latin America and the Caribbean*, growth is projected to remain broadly stable at 2.3 percent in 2026 and pick up to 2.7 percent in 2027. The impact from the conflict in the Middle East within the region is heterogeneous, with smaller economies affected more negatively. In *Brazil*, growth is projected to moderate to 1.9 percent in 2026, unchanged from October, and 2.0 percent in 2027. The war is expected to have a small net positive effect in 2026, as a result of the country being a net energy exporter, boosting growth by about 0.2 percentage point. In 2027, slowing global demand, higher input costs (including of fertilizers), and tighter financial conditions are expected to dominate, reducing growth by approximately 0.3 percentage point, compared with the projection in January. Adequate international reserves, low reliance on foreign-currency debt, large government cash buffers, and a flexible exchange rate are expected to help the country weather the shock. In *Mexico*, weaker growth in 2025 amid fiscal consolidation, restrictive monetary policy, and headwinds from

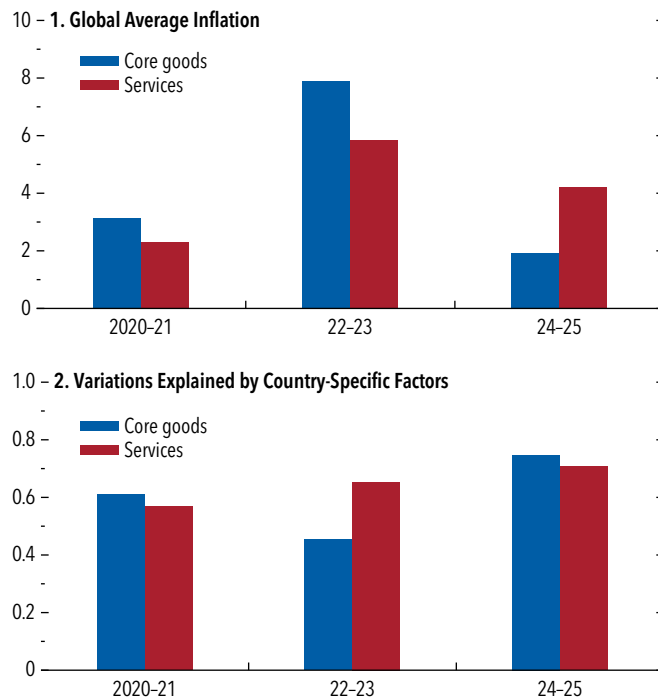
trade tensions is expected to give way to a mild recovery, with the economy expanding at a rate of 1.6 percent in 2026 and 2.2 percent in 2027.

- In *emerging and developing Europe*, a sharp slowdown in 2025 to a growth rate of 2.0 percent is expected to reverse only slightly, with economies in the region expanding at an average rate of 2.0 percent in 2026 and 2.1 percent in 2027. In *Russia*, higher commodity prices are projected to drive the 0.3 percentage point upward revision of 2026 growth relative to January, to 1.1 percent, with the momentum continuing to register another 1.1 percent growth rate in 2027. In *Türkiye*, expected growth is revised downward by 0.8 percentage point for 2026 to 3.4 percent relative to the figure in the January 2026 WEO *Update*, as 2025 growth was weaker than expected and higher oil and gas prices weigh on activity.

Inflation Forecast

Global inflation is projected to pause its decline, with headline inflation increasing from 4.1 percent in 2025 to 4.4 percent in 2026 before falling back to 3.7 percent in 2027. This is a 0.7 percentage point upward revision for 2026 from the figure in the October 2025 WEO, reflecting expected higher energy and food prices. There is divergence across countries, shaped by the stubborn dynamics in services inflation—which tends to have a larger domestic component—and the increasing share of inflation explained by country-specific factors (Figure 1.10). Gradual pass-through from higher tariffs and limited pass-through of higher energy prices along with gradually moderating services inflation amid a broadly balanced labor market mean that US core inflation is projected to return to the country's 2 percent target during 2027. Sustained strong productivity growth slowly converging back to historical norms will provide support for supply-driven disinflation. In the United Kingdom, inflation, which in 2025 increased partly because of one-off changes in regulated prices, is expected to pick up again temporarily toward 4 percent before returning to target by the end of 2027 as the effects of higher energy prices fade and a weakening labor market continues to exert downward pressure on wage growth. In Japan, inflation is expected to moderate in 2026, relative to the outturn in 2025, and converge toward the country's target by the end of 2027 as food and commodity prices ease. In the euro

Figure 1.10. Global Inflation
(Percent)



Sources: Haver Analytics; and IMF staff calculations.

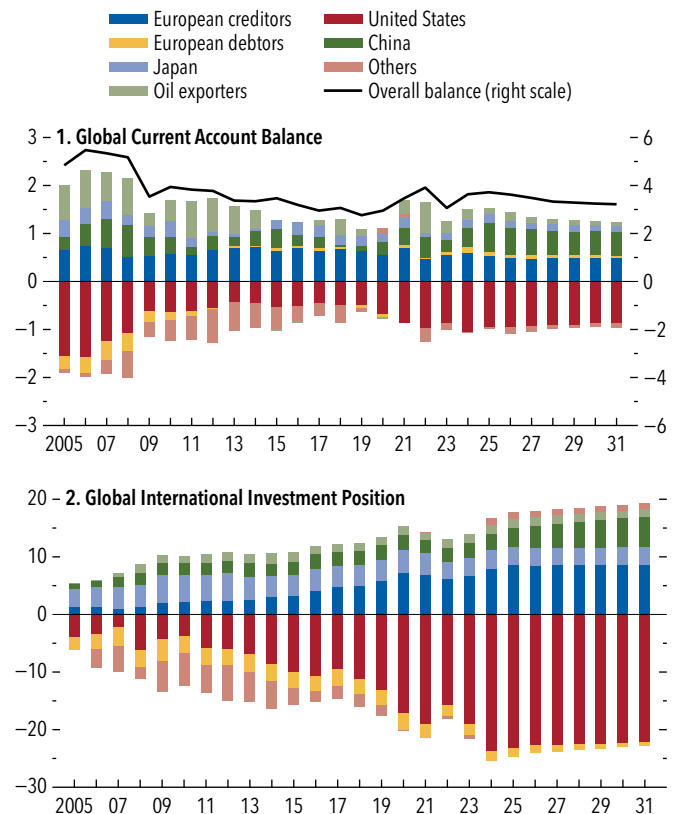
Note: Panel 1 shows the average inflation across 27 countries for which data are available. All numbers are simple averages. Panel 2 shows the share of country-level inflation variation not explained by global inflation, proxied by the first principal component of inflation across 27 countries, computed in a rolling two-year window.

area, headline inflation is projected to increase temporarily to above 2 percent in 2026 and remain above target in 2027. Core inflation is expected to increase more modestly but stay above 2 percent until 2028. Inflation in China is projected to start rising from low levels, whereas inflation in India is expected to return to near target levels after subdued food prices drove a marked decline in 2025.

World Trade Outlook and Global Imbalances

World trade volume growth is expected to decline from 5.1 percent in 2025 to 2.8 percent in 2026 and increase to 3.8 percent in 2027. These dynamics reflect front-loading early on and the impact of tariffs mitigated by adjustments in trade linkages and production chains as time goes by. Exports of both goods and services are projected to decline in percent of world GDP over the forecast horizon, with the decline in services trade being much less pronounced. This reflects the stronger underlying trend growth and greater resilience

Figure 1.11. Current Account and International Investment Positions
(Percent of global GDP)



Source: IMF staff calculations.

Note: "European creditors" are Austria, Belgium, Denmark, Finland, Germany, Italy, Luxembourg, The Netherlands, Norway, Slovenia, Sweden, and Switzerland. "European debtors" are Cyprus, Greece, Ireland, Portugal, and Spain. "Oil exporters" are Algeria, Azerbaijan, Iran, Kazakhstan, Kuwait, Nigeria, Oman, Qatar, Russia, Saudi Arabia, United Arab Emirates, and Venezuela.

to rising risks in services trade compared with that in goods trade (see Box 1.2).

Over the medium term, global imbalances are expected to decline only modestly. Expansionary fiscal packages in some economies with current account surpluses are expected to contribute to this cyclical decline (Figure 1.11). Countering this is a technology-driven business investment surge, which is expected to continue to attract capital flows to the United States even as investment in technology moderates. Stronger productivity growth in the United States could enhance US competitiveness in technology-related services and improve the country's trade balance. But positive wealth effects that boost domestic demand, together with sustained capital inflows driven by higher returns, would dominate and keep the US current account

deficit wider than that observed during the decade preceding the COVID-19 pandemic (also see the discussion of scenario A in Box 1.3). Sustained large fiscal deficits in the United States and China's continued reliance on export-led growth and limited rebalancing to domestic consumption contribute to external imbalances in these two countries.

Medium-Term Outlook

Many countries are facing challenges in lifting medium-term growth prospects, compounded by geoeconomic fragmentation and rising geopolitical risks. Absent decisive policy actions or technological breakthroughs, growth forecasts over the five-year WEO horizon remain mediocre. The global economy is projected to expand at an average annual pace of 3.1 percent in 2028–31, a persistently lackluster performance compared with the prepandemic (2000–19) historical average of 3.7 percent. This pace of growth reflects primarily the slowdown in China's growth, but average annual growth is also expected to slow in several other major Asian economies, the Middle East and Central Asia, and sub-Saharan Africa, as well as in North America and Europe.

Policy-driven reversal of global economic integration and a more volatile international economic environment are expected to hurt the potential of the world economy through multiple interconnected channels (Aiyar and others 2023). First, the catalyst role that international trade played for cross-country income convergence—helping lift large numbers of people out of poverty in the developing world—is expected to be reduced. Second, many tangible benefits that international migration brought to source and destination countries alike are likely to decline with more limited migration flows. This would have particularly adverse implications for poverty reduction and macroeconomic stability in remittance-dependent countries. Third, capital flows, including foreign direct investment, may decrease as firms weigh the costs and benefits under greater uncertainty. All in all, increased barriers to movement of goods and services, investment, and people may reduce technological diffusion, the flow of ideas, and innovation. Estimates of long-term global GDP losses from trade fragmentation alone range between 0.3 percent and 7.0 percent after 10 years (Bolhuis, Chen, and Kett 2023). Crucially, some of these effects may take longer to manifest than the WEO horizon of five years.

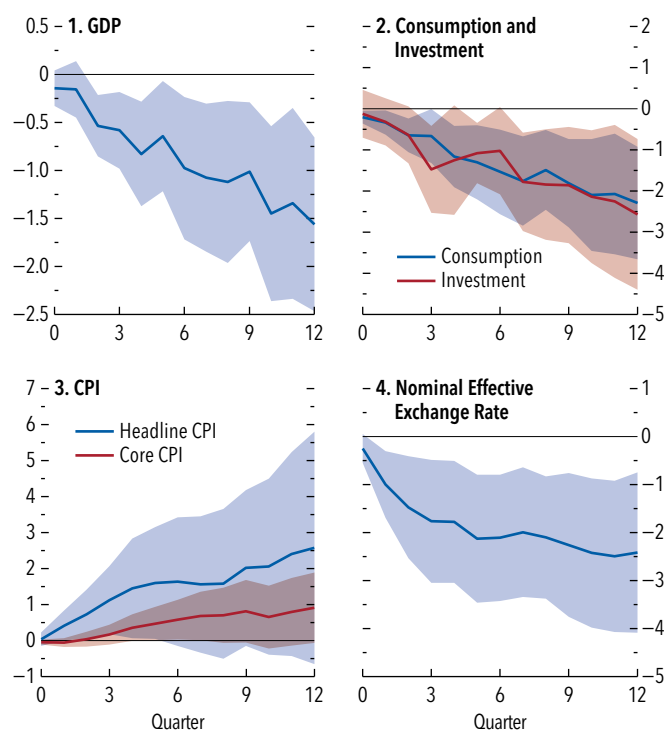
Risks to the Outlook: Downside Dominates

Risks are firmly on the downside, with some adverse risks gaining prominence since the January 2026 WEO *Update*, most notably those related to a more protracted conflict in the Middle East. While recent momentum might prove to be stronger than projected if recent tailwinds such as AI-driven activity moderate less than envisioned or financial conditions remain accommodative, such support may also prove short-lived and is likely to be dominated by downside risks from the conflict in the Middle East. Medium-term risks are also more firmly on the downside, including a potential misallocation of resources resulting from AI exuberance or a continuation of policies that give rise to real, fiscal, and financial vulnerabilities, potentially amplified through financial market mechanisms.

Downside Risks

Further intensification of conflicts and eruption of domestic political tensions. The current war and geopolitical tensions could intensify further. The effects that are already being felt through volatile commodity prices, disrupted supply chains, and exchange rate depreciation could get worse. Food security could be threatened, with disruptions to fertilizer markets ahead of the planting season leading to substantial food price inflation. Erosion of real incomes and increasing poverty in commodity-importing countries could exacerbate those countries' external imbalances and put those with limited reserves at risk of balance of payments distress and social unrest. Political instability could ensue, with the economic toll of the ongoing conflicts imposing knock-on effects on preexisting domestic political tensions. Sub-Saharan African economies may be especially susceptible to such dynamics. An increase in risk aversion or increased frictions in cross-border financial transactions could lead to capital flow reversals and abrupt asset price adjustments, particularly in emerging market economies with weaker policy frameworks as well as smaller fiscal and external buffers (Barrett and others 2021; April 2023 and April 2025 *Global Financial Stability Report*). Surges in military spending could boost economic activity in the short term, but, even when not followed by conflict and war, may distort resource allocation and involve nontrivial macroeconomic trade-offs (see Chapter 2). If conflict breaks out, direct losses—human casualties and

Figure 1.12. Estimated Impact of Geopolitical Risks
(Percent)



Sources: Caldara and others 2026; and IMF staff calculations.

Note: The analysis uses local projections, controlling for lagged country-specific geopolitical risk and changes in macroeconomic variables, as well as country and time fixed effects. Geopolitical risk is measured based on newspaper archives. See Caldara and others (2026) for details. The country-level data were downloaded from <https://www.matteoiacoviello.com/gpr.htm>. The lines denote point estimates, and the shaded areas denote 90 percent confidence intervals. CPI = consumer price index.

destruction of physical capital—are likely to dominate other channels (see Chapter 3).

The severe scenario discussed earlier in the chapter indicates that a more pronounced conflict could result in a major energy crisis, with a significant effect on global output. More generally, the direct macroeconomic effects of country-specific geopolitical risk on the domestic economy can be sizable, even though episodes of elevated risk tend to be relatively short-lived, with a half-life of about two quarters. A one-standard-deviation increase in geopolitical risk is associated with a decline in real GDP of about 0.8 percent one year after the initial shock, with the decrease driven by weaker private consumption and investment (Figure 1.12). To give a sense of magnitude, the average geopolitical risk in Europe (calculated across all country-year observations) rose by about 1.2 standard deviations in 2022 following Russia's

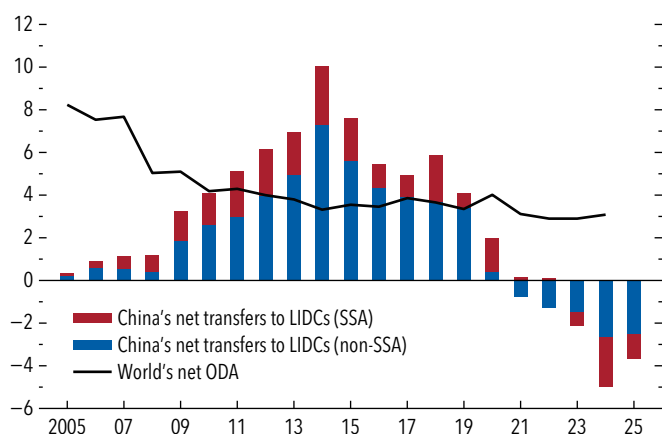
invasion of Ukraine and remained elevated at about 0.5 standard deviation in 2025. Roughly 10 percent of the estimated GDP impact can be attributed to the direct effect of higher oil prices, implying that other transmission channels play a much larger role. Increased geopolitical risk is, on average, associated with a level of prices about 2.5 percent higher relative to that in the no-shock baseline three years after the shock. Core inflation rises as well, although more modestly, suggesting that the inflationary impact comes mostly through the impact on commodity prices, as a result of disruptions to global food and energy supply chains. The nominal exchange rate depreciates by about 1.8 percent one year after the shock, which could compound the upward price pressures.

Reevaluation of productivity gains from new technology. Should AI-driven profitability projections turn out to be overly optimistic, real investment in technology sectors could drop sharply. In addition, equity markets—particularly those with high concentrations in technology sectors—could be vulnerable to a sharp repricing (April 2026 *Global Financial Stability Report*). A stock market correction—on top of the adjustment that has already happened in the stock prices of some technology firms—might lead to a slowdown in private consumption growth through negative wealth effects. Spillovers would spread, directly through trade flows, to export-oriented economies specializing in technology products and through a reversal in capital flows and impacts on cross-border portfolio holdings. An associated tightening of global financial conditions would weigh on global activity more broadly (see scenario B in Box 1.3).

Disruption of the fragile balance of current trade policies. More countries could adopt a protectionist posture, in particular if trade diversion and rerouting become disruptive, breaking the pattern of limited retaliation followed so far. Additional tariffs would further weigh on global growth, while sector-specific tariffs—especially if imposed on upstream industries in supply chains—could create supply bottlenecks and have an outsize impact on economic activity and prices. Nontariff measures targeting critical inputs such as rare earth minerals might also disrupt global supply chains (see the Commodity Special Feature). The effects would be amplified if actions trigger retaliatory tariff and nontariff measures.

Repricing of borrowing costs triggered by fiscal vulnerabilities. Public debt is elevated in several major economies, especially those whose currencies and securities

Figure 1.13. International Flows to LIDCs
(Billions of US dollars)



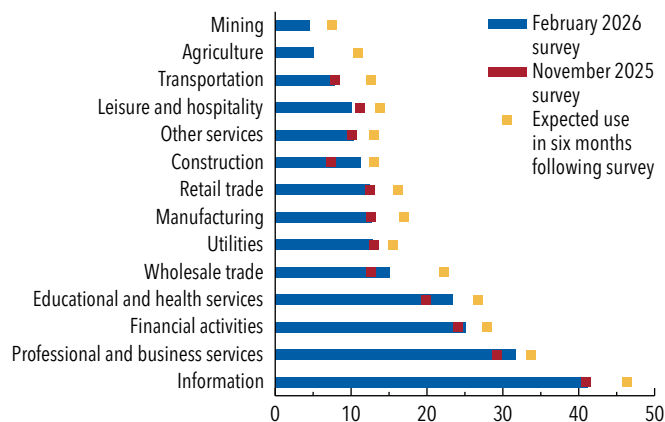
Sources: Organisation for Economic Co-operation and Development; World Bank, World Development Indicators; and IMF staff calculations.

Note: Net official development assistance (ODA) is weighted by and shown as percentages of gross national income. LIDCs = low-income developing countries; SSA = sub-Saharan Africa.

are important in international financial markets (see the April 2026 *Fiscal Monitor*). Fiscal sustainability worries in those economies—potentially triggered by the crystallization of other shocks—could not only put pressure on those economies' own borrowing costs but also tighten broader financial conditions and amplify financial market volatility (see scenario B in Box 1.3) and refinancing risk in some high-debt developing economies. Elections are often associated with fiscal slippage (Shi and Svensson 2006), which, as well as raising borrowing costs, risks triggering boom-bust growth dynamics. For low-income countries, planned reductions in official development assistance (ODA) pose an additional challenge through, among other things, their effects on health, education, and social protection outcomes. Many recipients in sub-Saharan Africa experienced a gradual decline in ODA as a share of GDP after the global financial crisis. Over the same period, Chinese lending expanded significantly, reflecting multiple factors, yet it has weakened in recent years, with net transfers recently turning negative (Figure 1.13; see also Chapter 2 of the April 2026 *Regional Economic Outlook: Sub-Saharan Africa*).

Erosion of confidence in economic institutions. Intensification of political pressure on independent central banks and other policy institutions can erode hard-won public confidence in their ability to fulfill their mandates and lift inflation expectations (Chapter 2 of the October 2025 WEO). Reanchoring expectations—

Figure 1.14. Use of Artificial Intelligence by US Firms by Sector
(Percent)



Sources: US Census Bureau, Business Trends and Outlook Survey; and IMF staff calculations.

Note: Sectors are aggregated using gross-value-added shares. Blue bars and red markers show responses to the survey question: "In the last six months, did this business use Artificial Intelligence (AI) in any of the following business functions?" The yellow markers show responses from the February 2026 survey to the question: "During the next six months, do you think this business will be using AI in any of its business functions?" Survey responses for Construction and Mining sectors were not available from the November 2025 survey.

when it occurs—usually requires a prolonged period of tight monetary policy, with longer-term yields rising because of widening term premiums and higher inflation expectations, ultimately lowering economic growth. Countries may experience persistently higher funding costs and capital outflows if international investors divert investment toward assets that are perceived as safer or more stable.

Upside Risks

Sooner materialization of productivity gains from artificial intelligence. The reference forecast does not include direct effects of AI on productivity, with adoption rates for AI still modest in many sectors (Figure 1.14). But the recent surge in AI-related investment and acceleration in the adoption of AI could substantially increase productivity and boost medium-term growth sooner rather than later, as possibly foretold by the above-trend US productivity growth since 2020. This could lift global growth by as much as 0.3 percentage point in the near term and by 0.1–0.8 percentage point in the medium term (see Box 1.3 of the October 2025 WEO). The benefits could be shared across the economy, provided there

are complementary policies to contain the potential impact on energy prices by relaxing power supply constraints, initiatives to scale up the necessary critical intermediate inputs, and labor market programs to manage workforce transitions. In low-income countries, realizing these benefits might require additional efforts to close gaps in energy and digital infrastructure and reduce the concentration of labor in sectors such as agriculture and mining, in which AI-driven productivity gains are likely to be limited.

Structural reforms as countries seize the moment.

Current challenges and the possibility of transformational technology changes could open a window for structural reform efforts to gain momentum. Accelerated implementation of reforms that upskill the existing labor force, reduce barriers to labor mobility, streamline and rationalize business regulations, reduce internal trade barriers, enhance competition, and promote innovation would make it possible to lift the growth potential of economies in a lasting manner while enhancing their resilience and capacity to adapt. Scenario C in Box 1.3 suggests that a package of reforms across major economies would lift near-term global growth by more than half a percentage point.

Progress in trade talks and enhanced policy predictability. Tangible progress in trade talks could lower tariffs and support global activity. It could also enhance policy predictability, allowing businesses to plan better and unlock investment plans. The gains to investment and activity could be even larger if cooperation extends beyond tariffs to services trade, foreign direct investment, and international taxation. In this context, efforts to develop or complete new regional trade agreements might help reduce trade costs and facilitate adjustment to recent policy changes, such as the EU's recent trade agreements with India and MERCOSUR. A broad reduction in US tariffs and reduction in uncertainty could lift global growth by 0.6 percentage point (see scenario C in Box 1.3). Incidentally, should US Section 122 tariffs fail to be extended and if no new tariffs are imposed under different statutes, the effective tariff rate would turn out to be lower than is assumed in the reference forecast.

Policies: Addressing the Current Shock, Preparing for the Next

The current hostilities in the Middle East pose immediate policy trade-offs: between fighting inflation and preserving growth and between supporting those

affected by the rising cost of living and rebuilding fiscal buffers. Amid frequent global shocks, countries need to calibrate policies to ensure that they not only step up to the moment but also stand up to the next test. There are actions countries can take on their own, some of which can also help increase global stability. Then there are actions that require collective action to reinstate stability in international economic relations.

Getting One's Own House in Order

Preserving Price and Financial Stability

Responding to the commodity price shock. Central banks should be ready to act decisively in line with their mandates. Monetary policy should preserve price stability and be carefully attuned to spillovers from actual inflation to inflation expectations, especially over the medium- to long-term horizon. The transmission of the current war-infused shock to inflation will differ across countries, reflecting varying exposure to commodity markets and the region in which countries are located, the strength of the anchoring of inflation expectations, and the extent of exchange rate depreciation. With the memories of the postpandemic inflation surge still fresh, second-round effects could possibly be larger than they were in 2021–22. At the same time, tightening prematurely could be destabilizing, if financial conditions tighten further—as in the severe scenario presented earlier in the chapter—or consumer and business confidence declines. Reacting strongly to flexible commodity prices, when supply constraints are present only in the related sectors, brings down inflation fast but risks a recession later (Chapter 2 of the October 2024 WEO). Where monetary policy was already properly calibrated before the shock, policymakers may have the option to wait to discern the duration and impact of the shock, also considering that inflation expectations may have shifted by the time the persistence of the shock becomes clear. Where negative demand shocks emerge and activity falls below potential, a reduction in policy rates may be appropriate, but only if risks to price stability remain contained. In non-inflation-targeting economies, such as those with fixed exchange rate regimes, policymakers may need to rely more heavily on other tools, such as fiscal policy, to manage shocks.

Communicating with impact. Clear, timely, and consistent central bank communication is essential at a time of heightened uncertainty and renewed fear of inflationary pressures. Central banks should articulate

their commitment to their mandates, including the resolve not to allow inflation expectations to de-anchor and their determination to tighten policy if incoming data and the evolving balance of risks make it necessary.

Safeguarding independence and credibility. Central bank independence, both legal and operational, is crucial for monetary policy credibility. It helps anchor inflation expectations and protect against fiscal dominance. Credibility that has been built over decades was a crucial factor in bringing inflation down without substantial output losses during the postpandemic inflation surge. But that episode of prolonged above-target inflation has also left the policy credibility of some central banks blemished. That puts an additional premium on avoiding pressures to subordinate monetary policy to fiscal or financial considerations or political objectives. Central banks should be able to maintain a prolonged period of restrictive policy if necessary.

Maintaining a flexible but orderly exchange rate. Exchange rates should generally move flexibly in response to market forces to facilitate macroeconomic adjustment. With the conflict in the Middle East triggering volatility and threatening an abrupt tightening in global financial conditions, foreign exchange movements may become excessive or disorderly, amplifying the impact of higher commodity prices on inflation and financial stability. The IMF's Integrated Policy Framework provides guidance for tailoring country-specific policy responses in instances in which such risk is imminent. In select cases, temporary foreign exchange intervention or targeted capital flow management measures may be warranted, alongside appropriate monetary and fiscal policy stances.

Enhancing prudential oversight. Heightened uncertainty, rising geopolitical risks, and fragilities in asset valuations underscore the need for strong prudential oversight to protect financial stability. In the context of the Middle East conflict and its adverse effects on sovereign risk premiums, financial market volatility, and the stability of financial institutions, countries with significant exposure to the war should devote adequate resources to identifying, quantifying, and managing these risks. Policymakers should be ready to deploy contingency plans against a wide range of shocks and outcomes, including by conducting scenario analysis that contemplates different paths for the hostilities in the Middle East. Macroprudential policies and oversight of nonbank financial institutions should curb risk

taking and fiscal-financial linkages. Preserving sufficient monetary and fiscal policy space and maintaining adequate liquidity, capital, and international reserve buffers remain essential.

Upholding Debt Sustainability

Protecting the vulnerable while staying disciplined. The increase in salient prices—such as those for energy and food—has renewed calls for supporting households and firms, as many countries did in the aftermath of Russia's invasion of Ukraine. Fiscal responses to the Middle East conflict should heed the lessons learned from that episode and ideally adhere to first principles of limiting distortion of price signals and keeping a fiscal and monetary policy mix consistent with price stability. In principle, and under an exceptional set of conditions—that the commodity price shock is temporary, pass-through from headline inflation to core is strong, economic overheating is low, spillovers to global commodity markets are small, and there is available fiscal space—temporary fiscal measures in the form of subsidies, tax cuts, and price caps can help prevent the amplification of large cost-push shocks and smooth inflation. In practice, however, those conditions are difficult to ascertain in real time and, even when they are satisfied, such measures are often regressive, fiscally costly, and politically difficult to roll back. Hence, discretionary fiscal support should typically be avoided. If the cost-of-living squeeze is drastic and some support is unavoidable, as it could very well be if the severe scenario materializes, it should be timely, explicitly temporary, and channeled through tightly targeted transfers to the most vulnerable, with clear sunset clauses and identified offsets through reductions in nonpriority spending or through new revenue measures, particularly where fiscal space is limited. For economies experiencing increased fiscal room—for example, because of windfalls from recent swings in commodity prices—maintaining fiscal discipline is essential to ensure that the gains are used prudently and consistently within a coherent medium-term fiscal framework with debt sustainability at its core.

Replenishing buffers. Rebuilding fiscal buffers is crucial given high public debt levels, eroded fiscal space following a sequence of global shocks, uncertainty surrounding the outcome of the latest conflict, and pressing spending needs. Credible medium-term fiscal consolidation—supported by fiscal transparency and clear communication (April 2026 *Fiscal Monitor*)—should be grounded in realistic assessments

of long-term spending pressures while maintaining a focus on growth-friendly adjustment (October 2025 *Fiscal Monitor*). Dependence on financial repression, monetary financing, or benign market sentiment would carry significant macrofinancial risks and should be avoided. Governments should strengthen revenues through base broadening and improved tax administration, enhance spending efficiency, and reorient expenditures toward high-multiplier areas such as infrastructure, skills development, and well-targeted social protection while crowding in private investment. Strong fiscal frameworks, credible fiscal rules, independent fiscal institutions, and prudent debt management practices are central to supporting these efforts. The mix of expenditure rationalization and revenue mobilization should be calibrated to country circumstances. High-debt countries, in particular, might need to limit state-financed services, better target social spending, and explicitly integrate interest payment risks into fiscal planning. In high-debt low-income countries facing refinancing and rollover risks, international cooperation, timely concessional financing, and debt resolution might be needed.

Promoting Medium-Term Growth

Mobilizing labor. Policymakers should seek to raise labor utilization and job creation, which would also help ease macroeconomic trade-offs and support fiscal sustainability. Labor market institutions should promote mobility and increase matching efficiency, supported by measures that help workers reallocate and stay skill-ready for a job market reshaped by AI. Portable benefits across jobs and contract types, together with affordable childcare and parental leave, can raise labor market participation—particularly among women—and smooth income risks during transitions. Pension and retirement systems should support participation and well-being among older workers through flexibility and actuarially fair incentives, including voluntary part-time work and gradual retirement options (see Chapter 2 of the April 2025 WEO). Migration policies aligned with domestic skill shortages can help alleviate bottlenecks while safeguarding domestic workers (see Chapter 3 of the April 2025 WEO).

Implementing smarter regulation. Reduction of inefficient regulations and constraints through well-targeted and carefully sequenced deregulation can lift impediments to entrepreneurship, investment, and innovation. Efforts should focus on promoting competition, broadening access to finance, and increasing the

efficiency of capital allocation to stimulate risk sharing and productivity growth (see Chapter 3 of the April 2024 WEO). These measures, however, must not come at the expense of prudential standards or macrofinancial stability, as premature or uncoordinated reforms could heighten vulnerabilities and trigger destabilizing boom-bust cycles.

Harnessing technological progress. Digitalization and AI can accelerate productivity growth and expand potential output. To fully capture the associated gains, complementary measures are needed, including investments in skills, energy, and digital infrastructure; competitive markets; and robust frameworks for data governance and cybersecurity (Gopinath 2023). Policies that encourage the diffusion and adoption of new technologies should accompany traditional support for research and development, while competition and product market reforms can facilitate the reallocation of resources toward more productive firms. Where trade or technological shocks are concentrated, targeted and time-bound adjustment assistance, including training, relocation support, and wage insurance, is preferable to open-ended protectionist measures.

Fostering energy transition. Adoption of renewable and energy-efficient systems can deliver multiple benefits, supporting economies in facing both the current challenges and longer-term ones. Actions to accelerate energy transition can help contain the impact on energy prices and enhance resilience to future shocks by improving energy security. They can also bring climate change mitigation goals within closer reach and prepare countries for increased risks from climate-change-related extreme weather events.

Addressing domestic imbalances. Taking actions to correct domestic imbalances helps lay a foundation for sustainable growth at home. Incidentally, it also helps on the external front by reducing global imbalances (see the 2025 *External Sector Report*; IMF 2026a). Hence, it is one of those rare cases in which policy actions can deliver on two fronts rather than impose trade-offs. In China, continued progress toward a more consumption-led growth model would help narrow external surpluses. Near-term fiscal support should focus on boosting household consumption and stabilizing the property sector, while medium-term sustainability will require addressing the overhang of local government debt. In the United States, credible fiscal consolidation would help moderate demand pressures and limit associated global spillovers. In the EU, further deepening of the single market, alongside

growth-enhancing reforms to stimulate private investment, would support more resilient and sustainable growth.

Restoring Predictable International Economic Rules

Advancing efforts to resolve tensions and promote trade. Clear, transparent, and coherent trade policy frameworks are essential to reduce uncertainty, limit volatility, and anchor expectations. Pragmatic cooperation can be critical for containing adjustment costs and lowering distortive barriers to trade and investment flows. This includes safeguarding key global commons and updating international rules to reflect structural shifts in the global economy—including a growing share of services—and to align with policies that bolster job growth, the green transition, or supply-chain resilience. Modernization of trade rules should be targeted and proportionate, focusing on clearly identified cross-border spillovers and respecting legitimate prudential objectives. Negotiations at bilateral, regional, and plurilateral levels should aim to reduce frictions, remain open to new participants, and avoid discriminatory provisions that raise barriers to third parties. They should also steer clear of distortive arrangements—such as purchase commitments and

quantitative restrictions—that are unlikely to address external imbalances stemming from domestic saving and investment dynamics. Policymakers should also avoid export controls and barriers to cross-border trade that would exacerbate supply disruptions, including those associated with the war in the Middle East.

Promoting effective international cooperation. International cooperation will be essential in addressing both the immediate threats from the Middle East conflict and longer-term challenges. In countries facing refugee inflows, integration support should be adequately funded with strong international contributions rather than be left to host countries, which may not have the fiscal space to support it. Access to emergency liquidity, including that provided through IMF facilities, is a crucial backstop against international financial spillovers. For some countries at risk of debt distress, liquidity support will not suffice, and a timely and orderly debt resolution process is the best way to contain the economic fallout. Continued progress in operationalizing international sovereign debt resolution mechanisms—including the Group of Twenty (G20) Common Framework—and greater convergence of practices through the Global Sovereign Debt Roundtable can make necessary restructuring more predictable and less costly.

Box 1.1. Explaining the Resilience of Global Growth in 2025

Higher tariffs and policy uncertainty were projected to have a significant impact on global activity in 2025. In the April 2025 *World Economic Outlook* (WEO) reference forecast—the first WEO projections to incorporate these two factors—global growth was projected to slow by 0.5 percentage point to 2.8 percent in 2025. Eventually, global GDP growth was 3.4 percent in 2025, 0.6 percentage point stronger than expected.

The resilience of global growth reflects not only forecast errors and other unobserved variables, but also a combination of factors that included lower tariffs than announced in early April 2025 and tailwinds such as fiscal policy, financial conditions, and artificial intelligence (AI) investment. While the WEO global growth forecast is the outcome of a bottom-up exercise, simulations conducted using the IMF's Global Integrated Monetary and Fiscal (GIMF) model provide a top-down perspective on what mattered and by how much.

Headwinds as Projected in April 2025

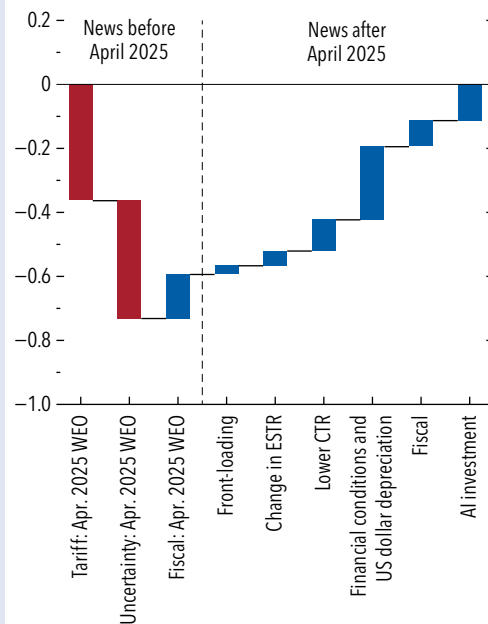
The April 2025 WEO reference forecast was based on information available up to April 4, 2025. It incorporated a rise in the statutory US tariff rate to about 25 percent, alongside countermeasures from Canada and China. Trade policy uncertainty had also increased markedly, contributing to weaker investment incentives and tighter financial conditions. These were only partly offset by fiscal stimulus announced in China and Germany. Taken together, these factors reduced global growth by 0.6 percentage point in 2025, close to the downward revision in the April 2025 WEO (Figure 1.1.1).

Tailwinds since April 2025

Reduced impact of tariffs. US firms and households *front-loading* foreign goods purchases in early 2025 in anticipation of higher tariffs and prices provided a small temporary boost to activity in some economies. The *effective statutory tariff rate* was reduced to about 18 percent by the end of the year, reflecting bilateral trade agreements and the introduction of exemptions. Countermeasures by trading partners were also scaled back. The actual *collected tariff rate* has been persistently below the effective statutory tariff rate (Gopinath and Neiman 2026) and was about half the announced rate in December 2025. While the gap may partly reflect shipment lags, which ought to

The author of this box is Chris Jackson.

Figure 1.1.1. News in 2025 and World GDP Growth (Percent)

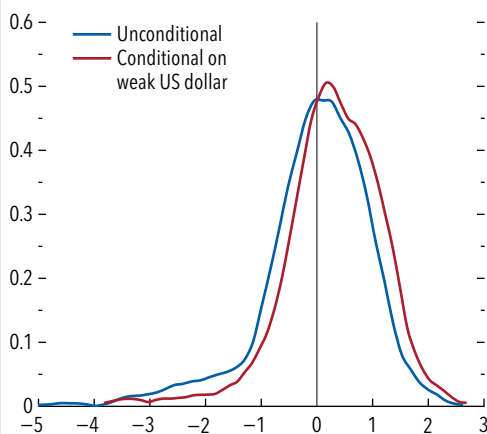


Source: IMF staff calculations.

Note: AI = artificial intelligence; CTR = collected tariff rate; ESTR = effective statutory tariff rate; WEO = *World Economic Outlook*.

dissipate, other factors such as greater use of preferential trade agreements and shifts in the composition of imports may prove more persistent and reflect the adaptability of the private sector.

Accommodative financial conditions. Despite an initial tightening in April, financial conditions eased over the remainder of 2025: Equity prices rose while sovereign spreads narrowed in many emerging markets. This may have reflected some abatement of the uncertainty shock and improved risk sentiment, including that around AI. The US dollar depreciated by 6 percent between April 1 and the end of December, in contrast to the appreciation typically expected after a persistent increase in US tariffs. While the sources of this depreciation—including an increased risk premium on the US dollar and hedging demand—may have reflected medium-term risks, it provided near-term monetary policy space and strengthened balance sheets in emerging market and developing economies (Juselius, Wooldridge, and Xia 2025) and

Box 1.1 (continued)**Figure 1.1.2. Correlation between US Dollar and Emerging Market Growth (Density)**

Sources: Bloomberg Finance L.P.; and IMF staff calculations.
 Note: "Weak US dollar" is defined as a depreciation of at least 5 percent in Bloomberg's US dollar index (USDIX) during 2020, with GDP growth measured as the one-year-ahead outcome following the depreciation. Both unconditional and conditional series are estimated over 1990–2025. The density functions are computed using the Epanechnikov kernel method.

boosted competitiveness in economies with currencies pegged or closely linked to the US dollar. Historically, a weaker US dollar is associated with stronger GDP growth in emerging markets (Figure 1.1.2).

Fiscal support. In addition to the major fiscal packages incorporated in the April 2025 WEO, the United States passed the One Big Beautiful Bill Act in July 2025, which also renewed expiring provisions in the 2017 Tax Cuts and Jobs Act, providing support to activity in 2025.

Investment in artificial intelligence. Technology investment, related to AI, added an estimated 0.5 percentage point to US GDP growth in 2025. The import-intensive nature of this investment implied large spillovers to some parts of the world, notably Asia. The estimates of the boost from AI since April 2025 do not include an impact on general productivity from the adoption of AI by other businesses.

Tallying the Net Effect

Together, these factors added an estimated 0.6 percentage point to global growth in 2025, close to the upward revision in the latest forecasts compared with those in the April 2025 WEO. About a quarter

of the gain came from a lower-than-expected impact of tariffs, while the remainder came from other tailwinds. These factors offset the headwinds incorporated in the April 2025 WEO, with the total impact close to zero. This is somewhat below the upward revision of 0.2 percentage point to the growth forecast compared with that in the October 2024 WEO, suggesting that, alongside these factors (and the usual errors and other unobserved variables that typically explain the divergence of actual outcomes from forecasts), there may be other country-specific factors at play.

A Revival in US Productivity?

In the United States, acceleration in productivity growth since the pandemic is an additional factor that could explain the positive growth surprises. Growth in output per hour worked has shown strength both relative to that in other economies (Chapter 1 of the April 2025 WEO) and relative to its own history (Jefferson 2026), averaging 2.2 percent every year since 2020 compared with a 1.5 percent annual pace over the previous business cycle (2009–19).

While some of this strength may reflect one-off factors, other drivers may signal longer-lasting, structural shifts (Cline, Kahn, and Rich 2025). Investment in labor-saving technologies during pandemic-era labor shortages, as well as reallocation of labor across sectors and increased flexibility to work remotely, may have brought efficiency gains. The surge in new business formation during the pandemic may have changed the composition of businesses toward more innovative and more productive types and bolstered competition (Dao and Platzer 2024). Fiscal policy over the past few years has supported investment in infrastructure and increased tax incentives for private investment. Measurement challenges associated with a concept that is not directly observable aside, compositional issues such as removal of lower-wage workers from the workforce—along with temporary cyclical effects such as increased capacity utilization—could generate noise and significant revisions to past data (Fernald and Li 2024; Gimbel 2026). It may be too early, though, for the benefits of AI adoption to show up materially in the data. While AI adoption indeed correlates with faster productivity growth across sectors, it explains little of the aggregate gain in productivity (Çakır Melek and Miller 2026). AI may not be the major force behind stronger productivity growth yet—and is not part of the reference forecast—but may become so in the future as adoption of AI among sectors broadens.

Box 1.2. Services Trade: An Emerging Engine for Global Growth

The growth of trade in services has outstripped that of trade in goods during recent decades. Between 1985 and 2024, services exports as a share of world GDP expanded by 150 percent, compared with 60 percent for exports of goods, defying the post-2008 “slowbalization” in goods trade (Figure 1.2.1). While goods still account for more than 70 percent of global trade, structural forces could push services trade higher in the years ahead.

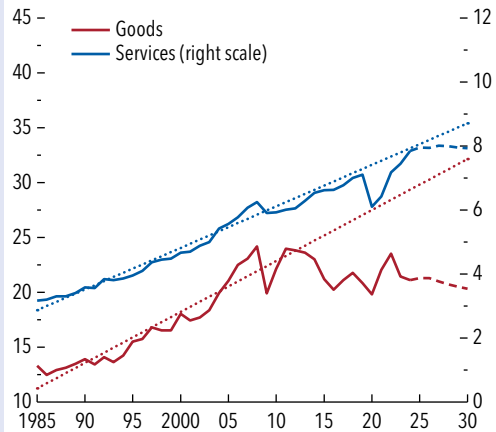
For one, geography is becoming less of a barrier to services trade. Despite improvements in transportation technologies, distance remains a powerful force in shaping goods trade (Disdier and Head 2008). A simple gravity model—regressing bilateral trade on market size and distance—explains about 63 percent of the variation in goods trade patterns in 2019, a percentage comparable to that for the early 2000s. While services trade remains harder to measure than goods trade, evidence from a new research dataset tells a different story for services (Figure 1.2.2). In the early 2000s, a 1 percent increase in geographic distance was associated with a 0.63 percent decline in bilateral trade—the same as for goods. This had fallen to 0.52 percent in 2022–23.

Two forces are driving this change. First, the composition of services trade has shifted dramatically. Transportation and travel once dominated—both tightly bound to the physical movement of goods and people—but their share fell from about 70 percent in 2000 to less than 40 percent in 2023. Most recent growth has come from “modern services”—primarily financial, information technology (IT), and business services in broadly equal measure. These modern services are less constrained by distance. Second, advancements in information and communications technology, accelerated by the pandemic, have made previously nontradable services tradable (Baldwin 2016). Digital platforms, cloud computing, and remote delivery have reduced the need for proximity of suppliers and consumers and fundamentally altered what can be supplied across borders.

The authors of this box are Nan Li and Robert Zymek.

Figure 1.2.1. World Exports

(Percent of world GDP; dotted = pre-GFC trend; dashed = April 2026 WEO reference forecast)

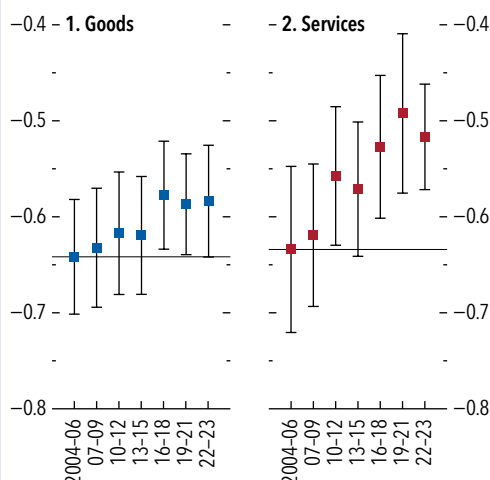


Source: IMF staff calculations.

Note: Dotted lines represent respective linear trends based on 1985–2008 data prior to the global financial crisis (GFC). Dashed lines 2025 and on represent April 2026 WEO reference forecasts. WEO = World Economic Outlook.

Services trade has also proved resilient to rising geopolitical tensions. Geopolitical distance between countries—measured by the degree of their disagreement in UN General Assembly voting—adversely affects their bilateral goods trade intensity, and this effect has intensified since 2016 (Figure 1.2.3). There is no similar trend for services trade. While some modern services—most notably IT and intellectual property licensing—show some sensitivity to geopolitical alignment, aggregate services trade remains largely unaffected, possibly because it is harder to target services trade with traditional protection tools such as tariffs, quotas, and import bans.

The untapped potential is particularly large for emerging markets. Although services have accounted for two-thirds of GDP growth in these economies over the past three decades, this growth has been domestic rather than export led (Kose and Ohnsorge 2024). The structure of global trade networks illustrates this pattern. While goods trade is dominated by the world’s largest economies—the United States, China, Germany, and Japan—with

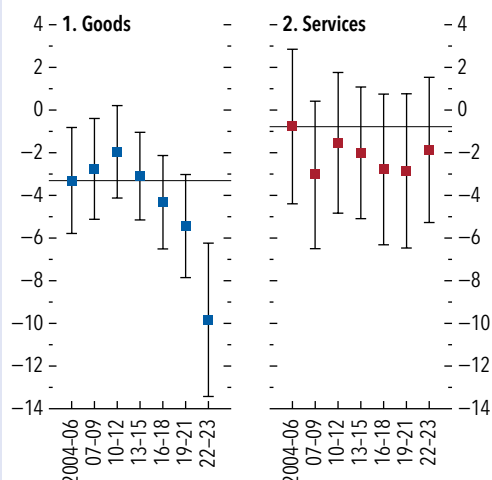
Box 1.2 (continued)**Figure 1.2.2. Impact of Geographic Distance on Bilateral Trade**
(Percent)

Sources: Centre d'Études Prospectives et d'Informations Internationales; and IMF staff calculations.

Note: Based on repeated cross-sectional estimations of a structural gravity model with standard controls and importer and exporter fixed effects. Horizontal lines reflect the values in 2004-06 for each panel. Whiskers reflect 95 percent confidence bands. "Geographic distance" is the weighted average distance between trade partners' most populous cities. See Li and others (2025) for details.

strong geographic clustering, services trade is more concentrated among advanced economies. The United States remains central, but smaller advanced economies such as Ireland and The Netherlands play a larger role than would be expected given their economic size. Meanwhile, emerging markets that are major players in goods trade, such as China, occupy a much smaller position in services trade. This presents both an opportunity and a challenge: Economic size and geography matter less, providing an opening for smaller, distant economies, yet despite some recent growth, many emerging markets remain peripheral to global services flows (Li and others 2025).

The potential gains from services trade can be substantial. Services exporters are more productive and pay higher wages than nonexporters, offering better jobs (Breinlich and Criscuolo 2011). Firms that export both goods and services outperform

Figure 1.2.3. Impact of Geopolitical Distance on Bilateral Trade
(Percent)

Sources: Centre d'Études Prospectives et d'Informations Internationales; and IMF staff calculations.

Note: Based on a panel estimation of a structural gravity model with standard controls; importer-time, exporter-time, and importer-exporter fixed effects; and a time-varying coefficient on geopolitical distance. Horizontal lines reflect the values in 2004-06 for each panel. Whiskers reflect 95 percent confidence bands. "Geopolitical distance" is measured as the ideal-point distance between trade partners. See Li and others (2025) for details.

those focusing on goods alone (Ariu, Mayneris, and Parenti 2020; Berlingieri, Marcolin, and Ornelas 2025), suggesting that services exports can reinforce manufacturing development. Digital transformation creates leapfrogging opportunities by reducing reliance on traditional physical infrastructure.

Realizing these benefits requires addressing trade barriers. Unlike goods trade, in which tariffs are a major obstacle, services trade faces a range of barriers relating to infrastructure, skills, and behind-the-border regulatory barriers: restrictions on foreign ownership, licensing requirements, local-presence requirements, and regulatory standards. These regulatory barriers can be eased through multilateral efforts or deeper bilateral and regional integration frameworks. Complementary investments in digital infrastructure, skills development, and regulatory quality can help countries better capture the gains from expanding services trade.

Box 1.3. Risk Assessment Surrounding the Reference Forecast

This box uses the IMF's Group of Twenty (G20) model to derive confidence bands around the *World Economic Outlook* (WEO) reference forecast and the IMF's Global Integrated Monetary and Fiscal (GIMF) model to assess the impact of plausible shocks and policies that are not included in the current reference forecast. Three scenarios are considered: (A) a deepening of the divergent forces currently shaping the outlook; (B) a reversal of the ongoing boom related to artificial intelligence (AI), coupled with a global risk-off episode; and (C) a reform scenario. Scenario A is neutral for activity over the WEO horizon but deepens global imbalances; scenario B is negative for activity but reduces imbalances; and scenario C is positive for activity and reduces imbalances. These scenarios are distinct from and complement the commodity shock scenario presented in the chapter text.

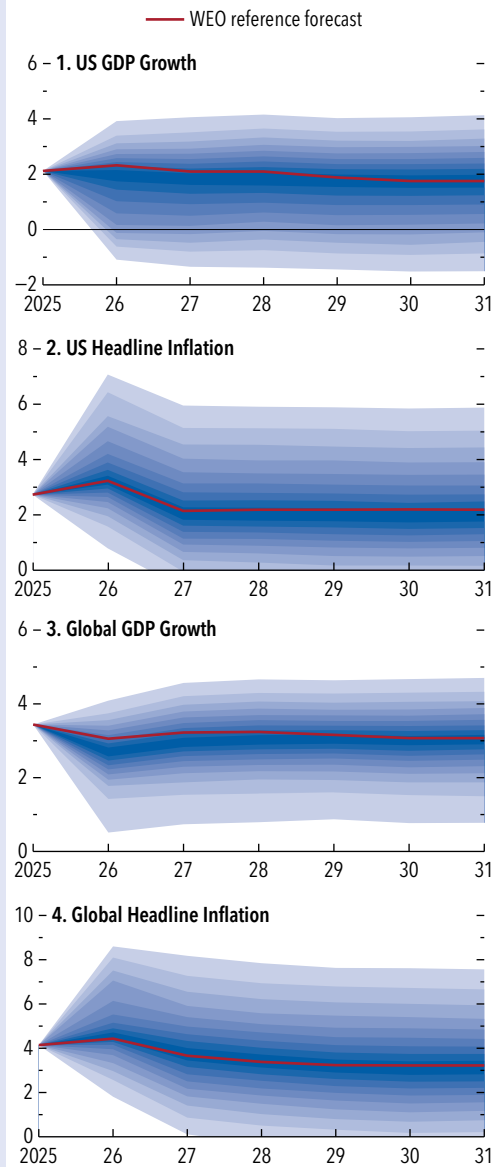
Confidence Bands

The G20 model is used to generate distributions around the reference forecast by drawing on shocks recovered from the historical data (Andrle and Hunt 2020). By oversampling years with negative shocks to growth, the distribution is tilted to align with the growth-at-risk assessment presented in the April 2026 *Global Financial Stability Report*; prominence has been given in this exercise to historical oil price episodes of the 1970s.

Panels 1 and 2 in Figure 1.3.1 show the resulting distributions for US growth and headline inflation. The probability that a recession will occur in 2026—corresponding to an annual growth rate below 1 percent, consistent with a relatively mild recession like that in 2001 and starting in the second quarter—is assessed at about 35 percent, somewhat higher than the recession probability estimated in the October 2025 WEO (at 30 percent). The probability that 2026 US headline inflation will rise above 4 percent is considerably higher, about 33 percent, compared with 17 percent in October 2025.

The authors of this box are Jared Bebee, Benjamin Carton, Chris Jackson, Gene Kindberg-Hanlon, Dirk Muir, Rafael Portillo, Pedro Rodriguez, Philippe Wingender, and Rachel Zhang.

Figure 1.3.1. Forecast Uncertainty around Global Growth and Inflation Projections (Percent)



Source: IMF staff estimates.

Note: Each shade of blue represents a 5 percentage point probability interval. WEO = *World Economic Outlook*.

Box 1.3 (continued)

Panels 3 and 4 in Figure 1.3.1 show the distributions for global growth and headline inflation. While additional quarterly data should have reduced forecast uncertainty relative to the October assessment, the probability that global growth in 2026 will fall below 2 percent has not changed and remains elevated at 25 percent, reflecting increased risks to activity in the remainder of the year. The probability that 2026 global headline inflation will rise above 5 percent is considerably higher at 38 percent (25 percent). In summary, downside risks to growth and upside risks to inflation have increased relative to October 2025.

Scenarios

The scenarios assume that monetary policy responds endogenously, with floating exchange rates in most regions. In scenarios A and B, China's currency adjustment is limited. In scenario C, the renminbi adjusts flexibly. Automatic stabilizers operate on the fiscal side. Layers whose names are identified with an asterisk in the following discussion are similar to layers considered in either April or October 2025.

Scenario A: The Divide Widens

Adoption of AI in the United States deepens, triggering an additional surge in US business investment to capture higher-than-expected productivity from AI technologies. The surge is concentrated in computing and in infrastructure. Investment rises 5 percent above the reference forecast by 2030 and stays 2 percent higher beyond that. Because many AI-related capital goods are sourced from emerging Asia, the import intensity of US investment increases.

The *unbalanced nature of economic activity in China* becomes more pronounced. The scenario considers a stylized policy mix that incentivizes industrial production but hampers consumption. Subsidies for employment in the tradables sector gradually increase by 4 percent of GDP over the WEO horizon, relative to the reference forecast, offset by lower transfers to households. Subsidies result in a gradual loss of productivity in the tradables sector (0.5 percent).

*Lower productivity in Europe** gets entrenched. Starting in 2026, total factor productivity growth in the euro area picks up less than expected and is 0.1 percentage point lower per year over five years, relative to the reference forecast. Investment-specific productivity growth declines by 0.05 percentage point per year over the same period. Productivity growth returns to the level in the reference forecast after 2030, but the effect on productivity is permanent.

Lackluster growth prospects result in a persistent increase in the private saving rate, by 1 percentage point by 2030.

The above layers trigger a *ratcheting up in tariffs* among China, the euro area, and the United States. Starting in mid-2026, these three regions permanently impose additional tariffs of 20 percentage points on all trade among them.

Scenario B: AI Disappoints, Risk Off Ensues

A reappraisal of expected productivity gains from AI leads to a sharp decline in real investment in the technology sector. The *decline is concentrated in the United States*. The fall in total investment, of 3 percent in 2026 relative to the reference forecast, unwinds the recent run-up in technology-related investment. The direct decline in investment elsewhere is more limited, but as in scenario A, the import intensity of investment is higher than usual.

The scenario also includes a layer of *lower asset prices and tighter financial conditions*. Equity prices in the United States fall by 20 percent in 2026, about half the size of the correction after the dot-com bubble burst in the early 2000s. Equities outside the United States fall by 15 percent, given lower exposure to the technology sector. The correction triggers a broader tightening in financial conditions and a loss of risk appetite. Corporate risk premiums temporarily rise by 75 basis points in the United States, 50 basis points in other advanced economies and China, and 75–100 basis points in other emerging markets.

The reduced risk appetite spills over to sovereign bond markets, leading to a *moderate, temporary increase in term premiums* of 30–50 basis points in advanced economies and 75–100 basis points in emerging markets. Finally, the concentration of the shock in the United States manifests as *less appetite for US assets* and puts downward pressure on the US dollar. These shocks are assumed to fade over the WEO horizon.

Scenario C: Reforms Reset Economies

In *China*, the reset involves a short-term fiscal expansion of about 0.5 percent of GDP, including higher social spending that elicits a reduction in the saving rate and measures to support residential investment, followed by a gradual consolidation. Industrial policy support is cut in half and coupled with efforts to increase business dynamism, boosting productivity (IMF 2026b).

In the *euro area*,* public investment increases relative to the reference forecast. It reaches 1 percent

Box 1.3 (continued)

of GDP by 2027, stays at that level until 2030, and remains permanently higher by 0.4 percent after that. Over the WEO horizon, about two-thirds of the surge in spending is financed by higher deficits. From 2030 onward, there is a reallocation of existing spending, such that debt ratios gradually return to their level in the reference forecast. Further progress toward the Capital Market Union results in a permanent reduction in corporate financing costs of 25 basis points.

The *United States** embarks on fiscal reforms to reduce inefficiencies from poorly targeted tax expenditures, shifts from labor to consumption taxes, and contains health care costs. Government consumption is also permanently reduced. Increases in fiscal sustainability lead to a reduction in the term premium of 50 basis points starting in 2026. The overall fiscal deficit gradually decreases by 1 percent of GDP after five years.

*Trade policies** also reset. Tariffs imposed since January 2025 are permanently removed, reducing effective tariff rates on US imports by about 10 percentage points relative to the reference forecast. Trading partners also remove tariffs on US exports, with US exports to China seeing a decrease in effective tariff rates of about 10 percentage points. Greater predictability in trade arrangements reduces uncertainty relative to the reference forecast. The reduction is about the absolute size of the spike observed in 2018–19 in the global economic policy uncertainty measure in Davis (2016).

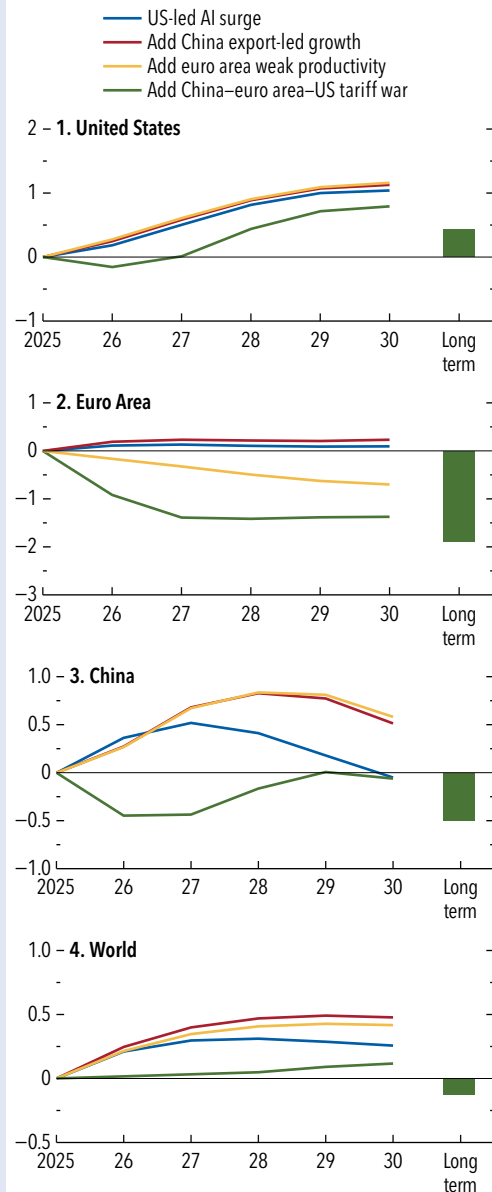
Impact on the World Economy

Figures 1.3.2, 1.3.3, and 1.3.4 present the effects, for scenarios A, B, and C, on the level of GDP during 2026–30 for China, the United States, the euro area, and the world. Figure 1.3.5 shows the effects of the scenarios on the current account balances of the three regions and on China's real effective exchange rate.

Scenario A

The *AI-driven surge* is moderately expansionary for the United States. GDP increases by 0.2 percent relative to the reference forecast in 2026 (0.5 percent in 2027). US investment increases by 2–4 percent in 2026–27, but a large part of the impulse goes to imports (2–3 percent). Higher absorption widens the US current account deficit; the dollar depreciates in real and nominal terms (about 3 percent). US inflation rises modestly, by 10–20 basis points. The increase in the federal funds rate is therefore small, less than 25 basis points by 2027. International spillovers are positive, with global output rising by 0.3 percent

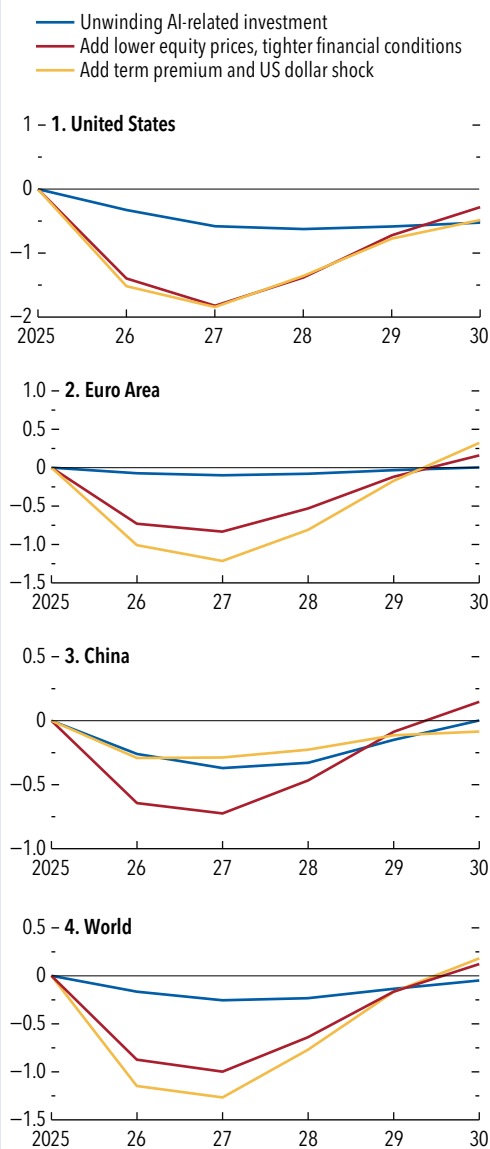
Figure 1.3.2. Impact of Scenario A on GDP
(Percent deviation from reference forecast)



Source: IMF staff estimates.

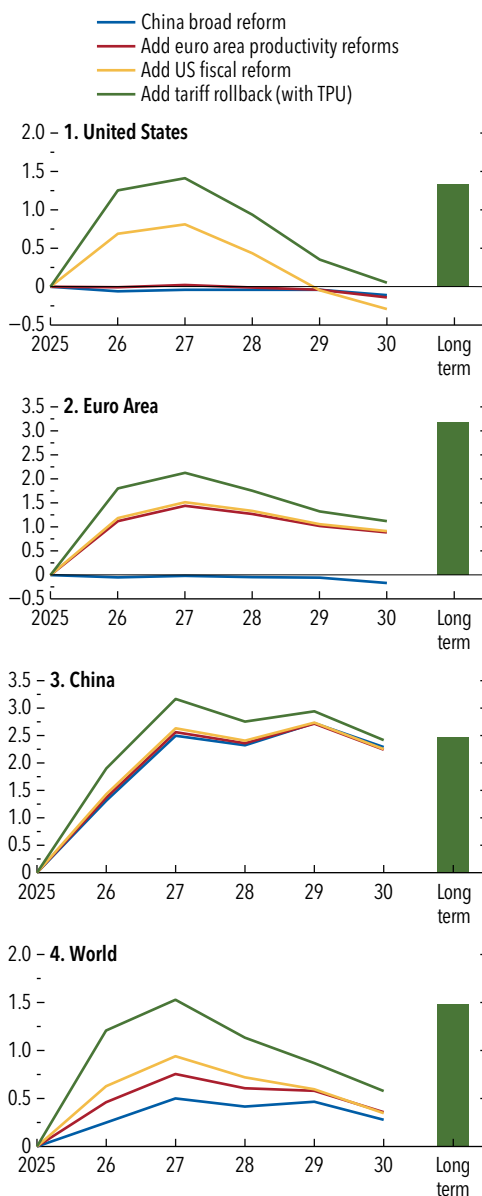
Note: "Long term" is at least 50 years ahead. AI = artificial intelligence.

in 2026–27. Emerging Asia, as the direct recipient of much of the surge in US imports, experiences an increase in output of 0.3–0.4 percent. China also benefits in the short term, through supply chain linkages with the rest of Asia and because the management of

Box 1.3 (continued)**Figure 1.3.3. Impact of Scenario B on GDP**
(Percent deviation from reference forecast)

Source: IMF staff estimates.

Note: AI = artificial intelligence.

Figure 1.3.4. Impact of Scenario C on GDP
(Percent deviation from reference forecast)

Source: IMF staff estimates.

Note: "Long term" is at least 50 years ahead. TPU = trade policy uncertainty.

Box 1.3 (continued)

the exchange rate with respect to the dollar implies that the renminbi depreciates in real terms.

Unbalanced activity in China translates into modest GDP gains, 0.5 percent by 2030, relative to the reference forecast. The policy mix widens imbalances in the tradables sector and the economy more broadly, as it reduces production costs but also constrains domestic demand. Export prices gradually fall, contributing to a decrease in China's terms of trade of 5 percent by 2030. Inflation declines, by 40 basis points by 2028, and the renminbi depreciates in real terms. Real exports increase, reaching 5.5 percent above the reference forecast by 2030, while imports decrease by 3 percent. The boost to activity is limited by the (small) decline in productivity and by subdued domestic demand. The latter is also reflected in an increase in the private saving rate, by 1 percent of GDP. China's current account surplus increases by a similar amount (from this layer alone). Spillovers to other regions are slightly positive but very small. Some manufacturing sectors outside China are negatively affected, while others benefit from cheaper intermediate goods. Investment in the rest of the world benefits from a small reduction in global real interest rates (less than 10 basis points).

Lower productivity in Europe reduces the region's GDP by 1 percent by 2030. Investment and consumption decline by 1.5 and 3 percent, respectively, over the same period. The impact on inflation and monetary policy is limited since demand falls in lock-step with potential. The euro area's current account surplus increases by 0.7 percent of GDP. International spillovers are limited.

The *tariff ratcheting* layer reduces global activity but does not affect imbalances. The impact on activity is largest for China and the euro area, about 0.75 percent in 2026; US GDP declines by 0.4 percent. Regions not directly tariffed benefit in the short term through trade diversion, but the medium- to long-term impact is uniformly negative across countries. Effects also build over time through lower capital accumulation. Tariffs reduce global output by 0.2 percent in 2026 and 0.7 percent in the long term. The impact on current accounts is negligible. Inflation in tariffed countries increases by 10–20 basis points in 2026–27, but disinflationary pressures dominate after 2027 and are most evident in China.

The *combined effect* from the scenario is clearly negative for the euro area, mildly negative for China, and mildly positive for the United States. While the impact on global activity is broadly neutral, the impact on

imbalances is large. Current account surpluses in the euro area and China widen further. The United States experiences an equally large increase in its current account deficit.

Scenario B

The *AI correction* causes a decrease in US investment of 3–4 percent in 2026–27. Greater import intensity mitigates the direct impact on the United States somewhat, with GDP decreasing by 0.3 percent in 2026 and 0.6 percent in 2027, relative to the reference forecast. Activity in Asia (China, Japan, and emerging Asia) decreases by 0.2–0.4 percent of GDP, and global output decreases by 0.2–0.3 percent, in 2026–27.

The *correction in asset prices and tightening in financial conditions* amplify the shock. Wealth effects from lower asset prices lead to a decrease in US consumption of 1 percent of GDP in 2026. Effects on consumption in other countries are substantial, between 0.5–0.7 percent. The hit to investment is amplified by lower asset prices and higher corporate spreads. Global investment declines by about 1.5 percent, with the United States and emerging markets experiencing larger declines. This layer subtracts 0.6–0.7 percent from global output in 2026–27.

Additional *spillovers to sovereign bond markets and US asset demand* are shown together in Figure 1.3.3 but have different effects across countries. Higher term premiums subtract 0.3 percent from global GDP in 2026. Lower demand for US assets is instead globally neutral. It leads to a depreciation of the US dollar by 6 percent in nominal effective terms, supporting external demand for US exports but also adding to the tightening in US financial conditions. The positive effect of net exports slightly dominates for US GDP. China benefits from the depreciation of the US dollar through the management of its exchange rate. Other regions experience a decrease in GDP from lower external demand.

The *combined effect* from the scenario is a large decrease in US GDP, of 1.5 percent in 2026, relative to the reference forecast. Activity in the euro area declines by 1 percent, while the impact on China is milder (0.3 percent). Global activity is 1.2 percent lower. US inflation and policy rates are lower than those in the reference forecast, notwithstanding the depreciation of the US dollar. Finally, global imbalances are reduced over the WEO horizon. As the US experiences a larger shock than other regions, and demand for US assets decreases, the US current

Box 1.3 (continued)

account balance increases (its deficit decreases) by about 1 percent of GDP. The current account surpluses in China and the euro area decrease; in China, the immediate impact is mitigated by the temporary real depreciation of the renminbi (Figure 1.3.3).

Scenario C

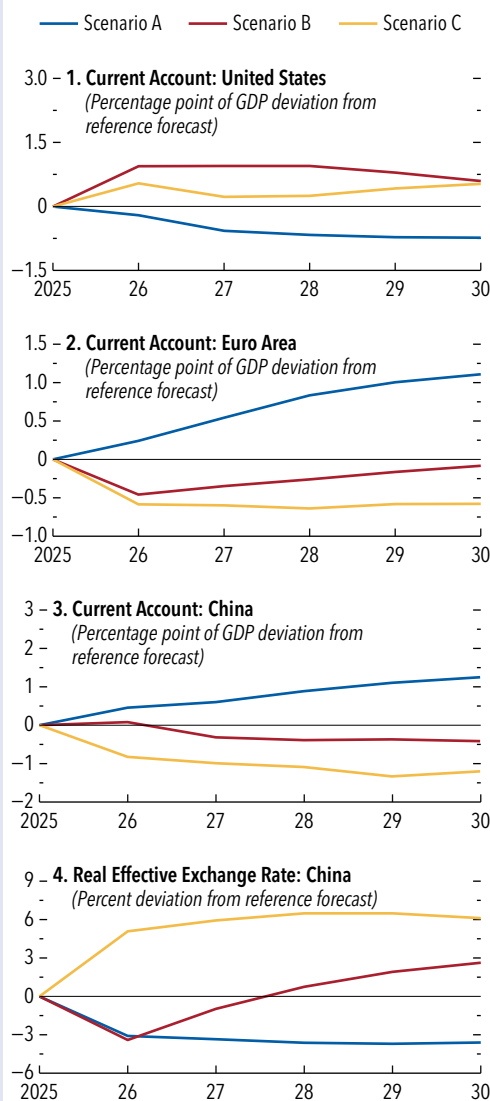
The *reforms in China* lower the private saving rate by 2 percent, boosting private consumption; help rekindle residential investment over 2026–28; and raise long-term productivity and potential output by close to 2 percent. China's GDP increases by 1–2 percent over the WEO horizon, with domestic demand providing a boost to inflation (0.7 percent during 2027–30). The current account surplus decreases by 1.5 percent of GDP from this layer alone. A permanently lower saving rate reduces China's demand for foreign assets and, under exchange rate flexibility, accounts for the real appreciation of the renminbi in the scenario (Figure 1.3.5).

Euro area reforms raise GDP by 1–1.4 percent in 2026–27. Inflation increases by close to 30 basis points over the WEO horizon, with the policy rate increasing by 60 basis points over the same period. The buildup in public capital raises productivity and potential output. Spillovers to other regions are positive but small, and the region's current account surplus decreases.

US fiscal reforms reduce US public debt by 25 percent of GDP over the long term, increasing fiscal sustainability. The combination of growth-friendly measures and lower premiums lifts GDP by 0.7 percent in 2026–27. Inflation net of tax effects is slightly higher, as are policy rates. Lower fiscal deficits contribute to a decrease in the US current account balance.

Tariff rollback and reduced uncertainty raise global GDP by 0.6 percent in 2026, with effects broadly similar across countries. The *combined effect* of scenario C is an increase in global output of 1.2 percent by 2026 (1.5 percent in the long term) and a reduction in global imbalances.

Figure 1.3.5. Impact of Scenarios on Current Accounts and Exchange Rates in the United States, China, and the Euro Area



Source: IMF staff estimates.

Commodity Special Feature: Market Developments and the Economics of Rare Earths

Compared with the October 2025 World Economic Outlook (WEO), the projections for commodity prices have been revised sharply upward. The military conflict in the Middle East has upended the outlook for energy and food as higher oil and gas prices increase the cost of fuels, petrochemicals, and fertilizers. Uncertainty surrounding the outlook for commodity prices remains very high, with risks tilted to the upside. A longer conflict than assumed in the WEO reference forecast would delay and substantially complicate the restoration of oil and gas production and exports to preconflict levels. At the same time, gold and other precious metal prices have retreated from their preconflict peaks. Despite elevated geopolitical uncertainty, retail investor profit taking—amid upward revisions to interest rate expectations and a stronger dollar—has reversed the near-parabolic gains recorded in the first two months of the year. Supply disruptions have driven base metal prices higher in markets already strained by strong demand and limited supply. Food prices have also increased as cereal prices have rebounded from historical lows, mainly as a result of growing weather concerns in large producing regions. Should the conflict linger on, higher transport and fertilizer prices, together with higher demand for biofuel feedstocks, could drive food prices higher. This Special Feature also examines the global implications of shortages in rare earth element markets, showing that such shortages could impose sizable GDP losses on importing countries. Avoiding trade tensions and restrictions remains the first-best outcome to promote a steady supply; de-risking supply chains through targeted industrial policies is fiscally costly, although less so if pursued simultaneously by various importers.

Commodity Market Developments

Oil prices increased 57.6 percent between August 2025 and March 2026 to \$105.8 per barrel as a result

The contributors to this Special Feature are Christian Bogmans (co-lead), Patricia Gomez-Gonzalez, Jorge Miranda Pinto, Jean-Marc Natal (team lead), and Andrea Paloschi, with research assistance from Francis Cuadros Bloch, Maximiliano Jerez Osses, and Joseph Moussa. The Special Feature is based on Bogmans, Cuadros Bloch, and others (forthcoming) and Bogmans, Jerez-Osses, and others (forthcoming).

of the military conflict in the Middle East. Oil prices spiked in March as oil shipments through the Strait of Hormuz—a critical choke point with few alternatives for rerouting—stopped, curtailing about 8.5 million barrels per day (mb/d) of crude oil exports. Usually, about 20 mb/d of oil, equivalent to 20 percent of daily global oil consumption, flow daily through the strait, of which 15 mb/d is crude oil. Major oil-producing facilities were also shut down as a precaution or as storage ran out or was damaged. Global strategic and commercial inventories, standing at a five-year high of 8 billion barrels, offer only a partial buffer.

After reaching a peak of \$119 on March 10, Brent prices retreated following first communication by the US administration that it expected the conflict to be short-lived and then the announcement of a ceasefire. The situation remains fluid and uncertainty is high. The futures curve is in steep backwardation (higher spot prices than futures prices), indicating supply disruptions and elevated geopolitical risk (Figure 1.SF.1, panel 2). If the conflict drags on, prospects of quickly restoring maritime transit and energy production to prewar levels diminish, with obvious ripple effects on refined product prices.

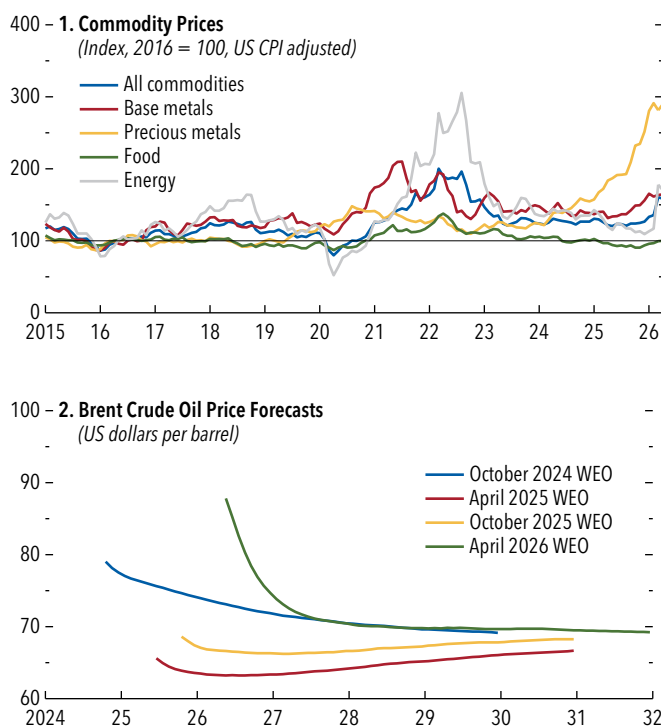
Middle East supply disruptions put upward pressure on European and Asian natural gas prices. Title Transfer Facility (TTF) trading hub prices in Europe rose by 61 percent between August 2025 and March 2026, peaking at \$17.7 per million British thermal units (MMBtu). Asian liquefied natural gas (LNG) prices surged to \$20.8 per MMBtu—an 80.6 percent increase since August 2025—as more than three-quarters of global LNG shipments through the strait (a fifth of global seaborne LNG) are destined for Asia. Prospects for a quick recovery of gas production and exports after the conflict have dimmed dramatically following strikes on the Iranian South Pars gas field, which prompted retaliatory strikes on Persian Gulf energy facilities, including Qatar's Ras Laffan gas field on March 18. At the same time, US Henry Hub prices rose by only 4.9 percent to \$3 per MMBtu, as US LNG exporters are close to capacity, limiting the diversion of domestic gas production to LNG plants. Futures markets for TTF are in steady backwardation,

reaching \$7.5 per MMBtu through 2031, thanks to an expected doubling of US export capacities by 2027. Henry Hub futures prices are expected to hover around \$3.5 per MMBtu through 2031.

Retail traders unwound part of their bets on rising prices for precious metals, while supply disruptions put upward pressure on base metal prices. The IMF's metals price index jumped 36.6 percent between August 2025 and March 2026. Prices for precious metals led the surge, with gold up 44.4 percent and reaching record prices exceeding \$5,000 per ounce as investors sought safe haven assets amid rising geopolitical uncertainty and persistent concerns about the dollar. These dynamics pushed demand for exchange-traded funds (ETFs) for gold to record highs, alongside still-robust central bank purchases. Since the onset of the conflict, however, broad-based profit taking across precious metals—amid upward revisions to interest rate expectations and a stronger dollar—has triggered a sharp correction, bringing gold prices back to their levels at the start of the year. At the same time, supply disruptions have led the increase in the price of base metals. Copper prices surged 29.5 percent following mining accidents in Chile and Indonesia, while aluminum increased 29.8 percent following the shutdown of smelters in Iceland, Mozambique, and the Middle East (which accounts for roughly 9 percent of global aluminum production). Futures markets for base metals suggest further price increases in 2026, indicating resilient demand amid still-fragile supply.

Rising food prices outweighed falling beverage prices, lifting the IMF's food and beverages price index by 4.7 percent between August 2025 and March 2026. Beverage prices plunged by 24.8 percent, led by a 57.4 percent drop in cocoa prices as favorable weather in West Africa boosted supply and inventories while global demand softened. Coffee prices fell by 9.9 percent following a record Brazilian harvest and improving supply conditions in Vietnam. In contrast to beverage prices, food prices are expected to increase by 6.0 percent in 2026. Cereal prices rebounded from historical lows in the first quarter of 2026 owing to growing weather concerns in key producing regions, while futures prices suggest that higher fuel prices are expected to boost demand for biofuel feedstocks such as soybean oil. Should the conflict linger on, higher transport and gas-derived fertilizer prices, together with higher demand for biofuel feedstocks, could drive food prices much higher—particularly those for cereals.

Figure 1.SF.1. Commodity Market Developments



Sources: Bloomberg Finance L.P.; Haver Analytics; IMF, Primary Commodity Price System; International Energy Agency; and IMF staff calculations.

Note: In panel 2, expiration dates are on the x-axis. CPI = consumer price index; WEO = World Economic Outlook.

The Economics of Rare Earths: Global Impact of Shortages and Industrial Policy

Motivation

Since 2020, successive waves of trade restrictions imposed by all major economic blocs have harmed international cooperation and growth. In April 2025, shortly after the US imposed sizable tariffs on most of its trading partners, China—the world's top producer of rare earth elements (REEs)—introduced export licensing requirements for seven REEs and REE-based permanent magnets, causing temporary but serious supply disruptions for manufacturers worldwide.

Like other hard-to-substitute inputs in high-technology manufacturing, produced through geographically concentrated supply chains, REE supply chains are structurally vulnerable. Tensions around these critical inputs have heightened economic security concerns and accelerated efforts to reshore and diversify imports of REE production.

As manufacturing regains strategic importance, understanding the economics of rare earths has become

increasingly urgent. This Special Feature addresses two questions: (1) *What would be the macroeconomic impact of a major REE supply disruption across sectors and countries?* and (2) *What would it cost to sufficiently de-risk the REE supply chain over the next decade?*

Foundations: Rare Earths Market Structure

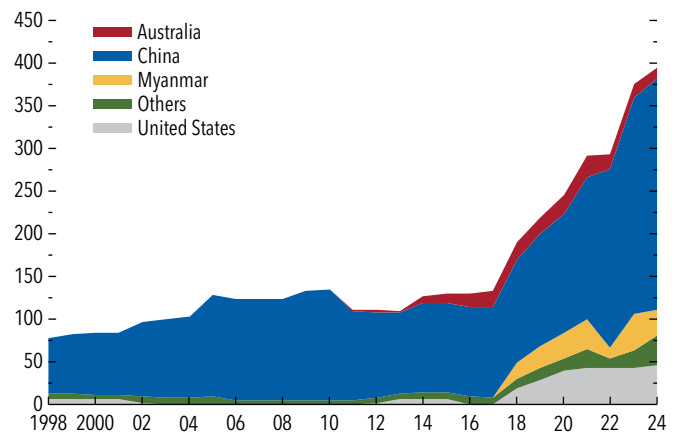
REEs are a group of 17 chemically similar metals, typically divided into two categories based on atomic weight: *light rare earth elements* (LREEs) and *heavy rare earth elements* (HREEs), substantially less abundant in the earth's crust than LREEs. REEs possess unique chemical and physical properties—including exceptional magnetism and catalytic enhancement—that make them valuable inputs across electronic, magnetic, optical, and catalytic applications. They are used in small quantities and are essential for automotive manufacturing (especially electric vehicles), renewable energy, oil refineries, defense systems, semiconductors, and consumer electronics. Arguably the single most important application for rare earths is *permanent magnets*—invented in 1983 by General Motors—in which as many as four REEs are combined with iron and boron to create a highly magnetic alloy that maintains its properties at elevated temperatures. These permanent magnets convert electricity into motion (or vice versa), making them central to both the clean energy transition and advanced manufacturing.

REEs constitute a relatively small commodity market, with rare earth oxides (REOs) valued at about \$6 billion and permanent magnets at approximately \$25 billion in 2024 (Market Data Forecast 2025). As such, the economics of rare earths are increasingly driven by just 4 of the 17 elements: the LREEs neodymium and praseodymium and the HREEs terbium and dysprosium. These “magnet-4” elements jointly comprise 96 percent of the total REO¹ market value despite representing only 23 percent of REO production by weight.²

¹“Total REO” represents the sum of all individual rare earth oxides (REOs) contained in mined material. It provides a consistent measure for comparing rare earth output across mines, deposits, and countries.

²This extreme value concentration reflects both strong demand for permanent magnets—the largest and fastest-growing application, consuming 83 percent of the value of all REOs—and the natural composition of rare earth deposits, which yield lower-demand elements like lanthanum and cerium in far greater proportions than the higher-demand magnet elements neodymium, praseodymium, terbium, and dysprosium.

Figure 1.SF.2. Rare Earth Mining by Country 1998–2024
(Thousands of metric tons per year, in TREO equivalents)



Sources: Nassar and others 2023; National Minerals Information Center, US Geological Survey, “Rare Earths Statistics and Information”; and IMF staff calculations.

Note: TREO = total rare earth oxide.

The rare earth supply chain involves multiple specialized stages: mining, concentration, separation of chemically similar elements through solvent-based extraction, and refining to metals or alloys. The resulting metals or alloys then serve as inputs for downstream manufacturers, including permanent magnet producers. The separation stage is particularly technically demanding, requiring hundreds of sequential processing steps, and it is also a pollution-intensive process. Establishing new capacity for *separation* and *refining*—the main stages of REE processing—requires billions in capital investment, years of regulatory approval, and specialized technical expertise, making rapid diversification of processing capacity extremely difficult.

Technical expertise acquired over decades, established infrastructure, differences in environmental and labor regulations, and subsidies made China the dominant world producer of REEs during the 2000s. Today, China’s dominance varies significantly by rare earth type and supply chain stage. In regard to LREEs, diversification in mining reduced China’s share of global output from 97 percent at its peak in 2010 to 58 percent in 2024 (Figure 1.SF.2), but China maintains 88 percent of the world’s oxide separation capacity and 93 percent of its metal refining. With respect to HREEs, China retains a near monopoly across the entire global supply chain: 98 percent of mining (including mining out of Myanmar; Figure 1.SF.3, panel 2), 97 percent of oxide separation, 95 percent of

metal refining, and 90 percent of permanent magnet production (Bedford 2025).³

This concentration creates potential *choke points* in the supply chain.⁴ While the geographic concentration in HREE mining constitutes an important potential choke point, the *separation* and *refining* stages constitute the most binding bottlenecks, as nearly all rare earth concentrates, regardless of origin, flow through Chinese processing facilities. Permanent magnet manufacturing represents yet another stage in the supply chain in which China holds a dominant market share (90 percent). However, this segment faces lower barriers to capacity expansion: Multiple established magnet producers already operate in Japan, the United States, and Europe, though typically at a smaller scale than in China (Figure 1.SF.3).⁵ In many applications REEs are not truly irreplaceable: Substitutes often exist but come with penalties in terms of efficiency, weight, size, or cost. Research suggests that the possibilities of substituting for HREEs are significantly weaker than those of substituting for the average element, with HREEs scoring 78 out of 100 on a substitutability index (on which 100 indicates no adequate substitute exists), compared with 57 for non-REEs (Graedel and others 2015). While certain rare earths have no substitutes, others may be partially replaced by inferior alternatives given enough time and resources.

Macroeconomic Impact of Supply Disruptions

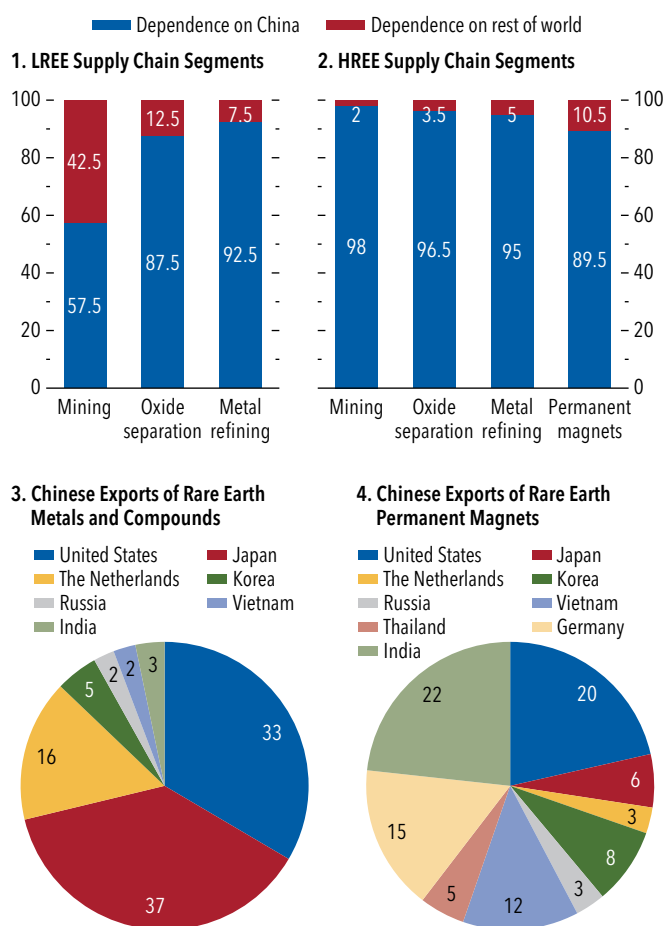
Following a special licensing requirement introduced by China—mainly for HREEs and related products, including permanent magnets—on April 4, 2025, there was a sharp global slowdown in permanent magnet exports between April and June. Exports of these magnets had fallen about 70 percent year over year as of May 2025, indicating a system-wide disruption

³By industry convention, Myanmar's heavy rare earth element (HREE) mining output is often consolidated with China's, reflecting the integration of the two countries' rare earth sectors through upstream investment and downstream processing links.

⁴A choke point emerges when three conditions align: extreme geographic concentration in a single country, potential for disruption, and barriers to rapid diversification.

⁵With China's consumption of REEs, which amounts to 50–60 percent of its production, taken into account, China's market domination is somewhat less apparent. This is especially the case in regard to light rare earth element (LREE) mining, of which China absorbs most of its production at home. For all other supply chain stages, however, dependence on China remains high (74–96 percent of total imports are sourced in China). See Online Annex 1.1, Part I. All online annexes are available at www.imf.org/en/Publications/WEO.

Figure 1.SF.3. Average Import Dependence on China by Rare Earth Supply Chain Segment (Percent)



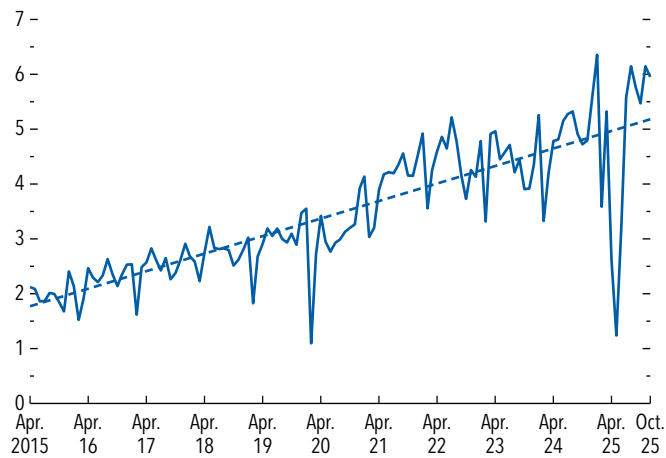
Sources: Bedford 2025; World Bank, World Integrated Trade Solution (WITS); and IMF staff calculations.

Note: Data for panels 1 and 2 are based 2023–24 production, synthesized in Bedford (2025). Data for panels 3 and 4 are from 2024 from WITS. HREE = heavy rare earth element; LREE = light rare earth element.

that extended beyond formally controlled products, but proved short-lived, with monthly Chinese export volumes quickly returning to their positive trend and displaying double-digit year-over-year growth rates (Figure 1.SF.4). In October 2025, China announced further tightening of its REE licensing requirements, which were later suspended in November under a China-US agreement. In January 2026, China restricted HREE exports to Japan. Despite these developments, strong REE export growth continued in January and February 2026. Policymakers in import-dependent countries grew increasingly concerned about the macroeconomic consequences of

Figure 1.SF.4. China's Exports of Rare Earth Permanent Magnets and Rare Earth Magnet Components Intended for Permanent Magnet Production

(Thousands of metric tons; dashed = trend)



Sources: General Administration of Customs of the People's Republic of China (GACC); Hong Kong Trade Development Council, China Customs Statistics; and IMF staff calculations.

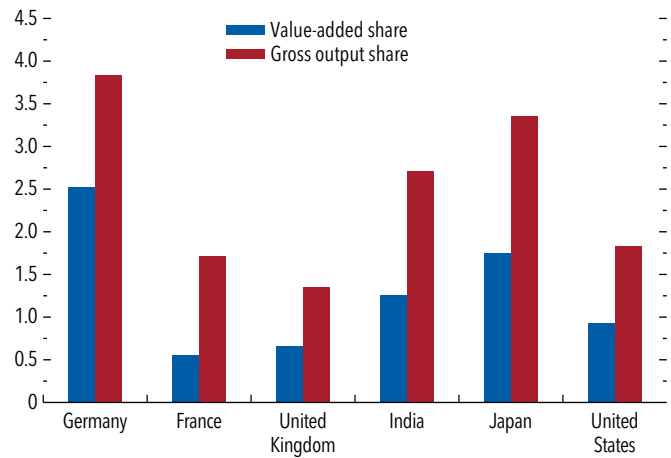
Note: Data are from the series "CN: Export: HS 8: Permanent Magnets or Articles Going to Be Permanent Magnets, of Rare-Earth Metals," Code HS 850511.

potential supply disruptions. Notwithstanding REEs' relatively small market size, their use unlocks trillions of dollars in downstream value creation across sectors globally.

IMF staff analysis of US Geological Survey data shows that rare earths are used as inputs in 34 of the 405 sectors of the US economy (Nassar and others 2025). These sectors jointly added \$233 billion in goods and services value in 2017, equivalent to 0.8 percent of nominal GDP. The estimated share of value added dependent on rare earths is similarly substantial in other advanced economies and major emerging markets: France (0.4 percent), Germany (2.5 percent), India (1.3 percent), Japan (1.7 percent), and the United Kingdom (0.6 percent). These estimates show that REE exposure varies substantially across countries, with the variance driven by differences in sectoral composition and the relative importance of REE-intensive industries such as automotive manufacturing, renewable energy, and electronics manufacturing (Figure 1.SF.5).

This "value added at risk" (VAAR) measure provides a useful first-pass estimate of possible GDP losses from a hypothetical prolonged and severe REE supply disruption. The VAAR measure also reveals the macroeconomic importance of rare earth permanent magnets:

Figure 1.SF.5. Value Added and Gross Output at Risk
(Percent of goods and services using rare earth elements as inputs)



Sources: Organisation for Economic Co-operation and Development (OECD), Input-Output Tables; Nassar and others 2025; and IMF staff calculations.

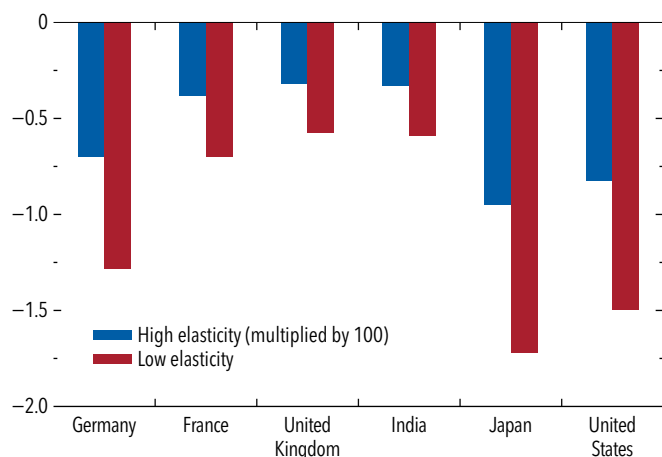
Note: Calculations are based on Nassar and others (2025), which provides industry-specific measures of gross output dependence on rare earths at the Bureau of Economic Analysis 405 × 405 detailed input-output level. These dependence measures are first mapped onto the US input-output structure and then translated to the OECD industry classification input-output with 50 sectors from 2017. Using the OECD input-output framework, country-specific shares of rare-earth-dependent consumption are derived for each OECD-level industry across countries. The value-added share is computed as the value added attributable to rare-earth-dependent activities divided by total GDP. The gross output share is defined as an economy's gross output dependent on rare earths divided by its total gross output.

In the United States, they drive about 70 percent of VAAR.

But the VAAR measure omits important adjustment mechanisms. It likely *overstates* losses by assuming that REEs and magnets cannot be substituted for at all but also *understates* losses by abstracting from cascading input-output (I-O) effects. To quantify potential GDP losses from REE supply disruptions, it is essential to account for both substitutability of other things for REEs and intersectoral linkages between industries directly affected by REE shortages and those that depend on them. For example, if permanent magnets were to become unavailable, electric-vehicle production would be disrupted, with knock-on effects that could ultimately raise transportation costs and ripple through the production of other goods and services. The impact would be larger in industries in which there are no available substitutes for REEs.

To that effect, this Special Feature develops and calibrates a small open economy model with network linkages (Silva and others 2024, extended to incorporate imported REE supply constraints; see Online Annex 1.1, Part II, for more details). The model

Figure 1.SF.6. Output Losses by Country from a Major REE Disruption
(Percent of GDP)



Sources: Organisation for Economic Co-operation and Development; and IMF staff calculations.

Note: The scenarios are based on an 80 percent supply disruption affecting 80 percent of imported goods, specifically rare earths and magnets. The high-elasticity scenario uses a substitution elasticity of 0.8 between the imported goods and their varieties as in Alfaro and others (2025), while the low-elasticity scenario uses an elasticity of 0.015. REE = rare earth element.

analyzes an REE supply shock affecting REE-using sectors—including indirectly through I-O linkages—with implications for prices, real wages, and net foreign assets.⁶ The calibration relies on an REE-augmented I-O table based on US Geological Survey data and is applied to a set of major REE-dependent economies. For each country, the analysis considers a persistent 80 percent reduction in all rare earth inputs—oxides, metals, compounds, and magnets—consistent with the average single-supplier import concentration of each of the advanced economies.

Model simulations show how the magnitude of the supply disruption strongly depends on substitution possibilities. When firms have limited scope to replace rare earths, which is the base case for short horizons of less than a year (Nassar and others 2025), GDP losses may exceed those predicted by the VAAR measure (Figure 1.SF.6). Network amplification means that GDP declines by 1.5 percent in the United States, a number almost twice as large as the VAAR measure. By contrast, GDP declines by about 1.2 percent in

Germany (with a VAAR measure score of 2.5 percent). This difference reflects the stronger forward linkages of US REE-intensive sectors relative to those in Germany, particularly those in motor vehicles, electrical equipment, and computers and electronics. When the substitution elasticity is set higher, reflecting greater opportunities for producers to adjust (typically for horizons longer than five years, similar to those in Alfaro and others 2025), estimated GDP losses are negligible, averaging only 0.006 percent.⁷

Coping with Risks of Supply Disruptions

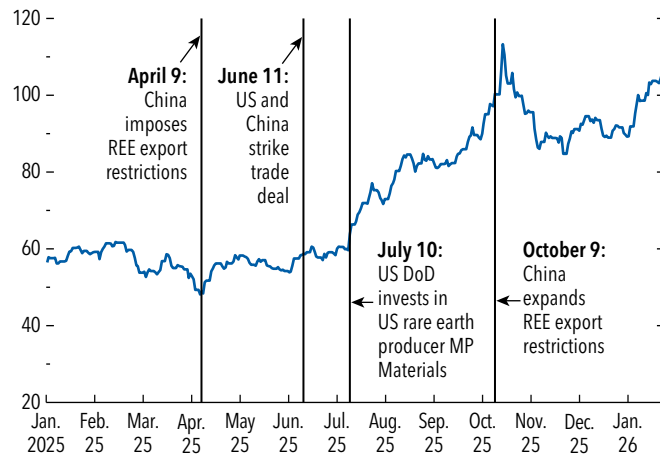
In response to high REE supply concentration, countries and firms are adopting a variety of adaptation strategies. *Stockpiling* provides a short-term buffer against disruptions and may deter coercion, but it does not address underlying structural dependence and may be constrained in practice. *Recycling* holds longer-term promise, but in a rapidly expanding market, it cannot yet serve as a primary supply source. Likewise, the superior performance of permanent magnets makes *large-scale substitution* unlikely in the near term. In this context, *reshoring* and *import diversification* have emerged as the main medium-term responses, despite uncertain viability given the long development timelines for these measures, coordination challenges, and potential shortages of skilled labor.

Following China's introduction of REE export licensing requirements in April 2025, advanced economies accelerated the implementation of industrial policies to reduce reliance on China-centric supply chains. Efforts in this area have focused on three approaches. First, price floors and offtake agreements have been concluded aiming to provide investment certainty in volatile markets, as many REE projects are not commercially viable at current neodymium prices (about \$55 per kilogram; see Online Annex Figure 1.1.1). For example, a July 2025 agreement between the US government and US rare earth producer MP Materials included a price protection mechanism akin to a floor for key REE products, and price floors more broadly featured prominently in discussions at high-level Group of Seven (G7) critical minerals meetings. Second, governments have provided direct financial

⁶The model features domestic general equilibrium adjustment, assuming a common shock across sectors, and limits exports to commodity sectors. REE-intensive sectors, such as motor vehicles, are thus assumed to be fully domestic.

⁷Alfaro and others (2025) estimate industry-specific substitution elasticities between REEs and labor using cross-industry responses to the 2010 REE shock observed over the postshock period 2011–18. Their estimates range from 0.8 to 1.4, reflecting medium-term, innovation-inclusive adjustment.

Figure 1.SF.7. Aggregate Stock Market Value of Publicly Listed Firms in the Rare Earth Industry
(Billions of US dollars)



Sources: S&P Global; S&P Capital IQ Pro; and IMF staff calculations.

Note: Firms are identified using Capital IQ based on keyword searches for rare-earth-related terms in business and long business descriptions. The sample is restricted to operating, publicly listed companies with primary locations outside China. This initial screen yields 315 firms. Each firm is manually reviewed to retain only those with active involvement in rare earth extraction, processing, or project development. Firms with only peripheral references to rare earths are excluded, resulting in a final sample of 89 companies. DoD = US Department of Defense; REE = rare earth element; S&P = Standard and Poor's.

support—through equity stakes, loans, and grants—to supply capital and signal long-term commitment.⁸ Third, agreements—signed in October 2025 between the United States and Australia, Japan, Malaysia, and Thailand—alongside the G7 Critical Minerals Action Plan promote joint financing and coordinated procurement, mobilizing an estimated \$6.4 billion in public and private funding to de-risk REE supply chains. These measures have been effective in improving the financial prospects of publicly listed firms in the industry (Figure 1.SF.7).

Quantifying the Impact of Industrial Policies to De-risk Rare Earth Supply Chains

Because reshoring is costly, policymakers should balance *efficiency losses* from reshoring in normal times against *expected disruption losses* in times of crisis. Optimal de-risking is better viewed as an insurance policy

⁸The July 2025 MP Materials deal included substantial equity and loan components, and the January 2026 agreement between the US Department of Commerce and USA Rare Earth similarly combines government equity, below-market lending, and direct federal funding (grants) to support domestic capacity expansion.

in which the cost of the premium to be paid in normal times (efficiency losses) should be commensurate with the damage expected in crisis times (disruption losses).⁹ Given these considerations, policymakers usually emphasize de-risking rather than decoupling. It should be noted that similar considerations also play a role for other goods and commodities, including energy, food, fertilizers, and semiconductors.

To analyze alternative industrial policies, a calibrated dynamic trade model of the global rare earths market has been developed for this Special Feature. The model features producers in different countries that invest in both extraction and processing capacity to produce raw and processed REEs. The model calibration draws on detailed market, industry, and geological data (see Online Annex 1.1, Part III, for details).

The model is used to assess the effects of two industrial policies—*investment subsidies* and *price floors*—applied to oxide separation (the most crucial processing stage) under two hypothetical implementation scenarios: either *unilaterally*—in the example here by US-based producers—or through *simultaneous* action among all importer countries.¹⁰ To illustrate the economic trade-offs, policy instruments are calibrated to achieve *25 percent self-sufficiency in rare earth processing by 2035 in the US*.¹¹ This is 15 percentage points higher than the comparable figure in the baseline (with unchanged policies) and in line with International Energy Agency projections (IEA 2025).¹² The same benchmark is used in the case of *simultaneous* action in order to highlight the fiscal cost implications of de-risking by various countries instead of unilaterally.

The analysis allows a certain number of conclusions to be drawn (Figure 1.SF.8). First, sizable interventions would be needed to attain the 25 percent self-sufficiency target; for example, in the unilateral scenario, the investment subsidy must cover 77.2 percent of total investment costs for the US to

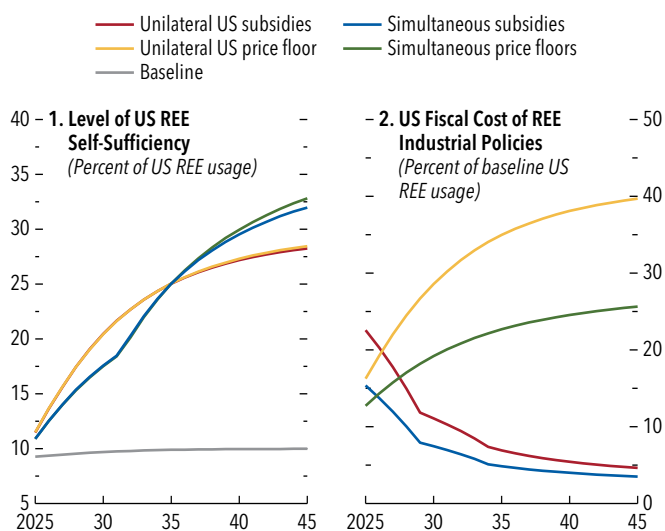
⁹Moreover, security benefits may exhibit diminishing returns, with the first 10–20 percent increase in self-sufficiency delivering the largest reduction in vulnerability. This means that modest self-sufficiency rates can substantially reduce vulnerability to supply disruptions at minimal efficiency costs (Clayton, Maggiori, and Schreger 2024).

¹⁰Price floors are currently being discussed among Group of Seven policymakers, while investment subsidies proxy for a broad class of capital expenditure support measures used in practice, including grants and below-market lending.

¹¹US self-sufficiency is defined here as the share of domestic rare earth consumption supplied either by domestic production or by imports from countries other than China (through friend-shoring).

¹²See Online Annex 1.1, Part III, for an analysis of a more ambitious 50 percent self-sufficiency target.

Figure 1.SF.8. Effectiveness and Fiscal Cost of Alternative Industrial Policies to Achieve 25 Percent REE Self-Sufficiency in the US



Source: IMF staff calculations.

Note: Investment subsidy to US refiners only implemented with a 77.2 percent subsidy; investment subsidy to refiners outside China only implemented with a 77.8 percent subsidy; price floor subsidy to US refiners only implemented with a price floor 2.42 times the period market price; price floor subsidy to refiners outside China implemented with a price floor 2.2 times the period market price. Baseline scenario assumes 4.7 percent global demand growth in 2025–29, 1.42 percent global demand growth in 2030–34. REE = rare earth element.

reach 25 percent self-sufficiency by 2035. The required policy intervention is also sizable under a price floor: 2.4 times the period market price for the unilateral scenario. These large policy interventions reflect the relative efficiency of Chinese producers but also point to important price effects. Because there are already large capacities in both mining and refining of REEs, boosting production in a well-supplied market depresses prices and profits, which reduces private investment incentives and, all else equal, requires more generous government interventions to induce development of additional capacity.

Second, investment subsidies are typically more fiscally efficient than price floors when evaluated in present-value terms. This reflects the fact that investment subsidies are targeted at new capacity, whereas price floors also generate windfall gains for incumbent producers by supporting existing production. As a result, for a given self-sufficiency target, the net present value

of subsidy payments is lower than that of price floor interventions. To achieve the 25 percent target under the unilateral scenario, US fiscal costs associated with the investment subsidy over the first decade amount to 141 percent of the annual US market size—equivalent to about \$1.19 billion (\$0.81 billion).¹³ At the same time, investment subsidies are more costly in the short term (Figure 1.SF.8, panel 2) than price floors as they front-load fiscal outlays, with costs declining in the long term once investment is largely limited to replacing depreciated capital.

Third, *simultaneous* action reduces the fiscal cost of achieving a given self-sufficiency target. For example, when all importing economies incentivize investment in refining, US self-sufficiency is achieved through a less concentrated buildup of capacity, as part of the fiscal cost is outsourced to the incentivizing economies. Under *simultaneous* action, those economies can also leverage the higher US efficiency in REE processing and experience substantial gains in self-sufficiency, at comparatively lower fiscal costs.¹⁴

Conclusion

This Special Feature shows that large disruptions to REE supplies could substantially reduce GDP in many economies, particularly in the short term when substitution options are limited. Avoiding trade tensions and restrictions remains the first-best outcome to promote steady REE supply. Model-based analysis suggests that de-risking supply chains through targeted industrial policies is fiscally costly. Costs are lower if de-risking is pursued by various importers simultaneously and if policy instruments directly target the expansion of new production capacity. IMF research also suggests that industrial policies should be used cautiously (Baquie and others 2025). Beyond industrial policy, governments can promote complementary structural reforms that would lower barriers to entry into REE markets through simpler mining permits, investment in the specialized skills the sector requires—from separation chemistry to metallurgy—and competitive allocation of subsidies.

¹³Global REEs' market size is about \$6 billion. The US share is 14 percent, so roughly \$0.81 billion.

¹⁴See Bogmans, Cuadros Bloch, and others (forthcoming) for more analysis.

Annex Table 1.1.1. European Economies: Real GDP, Consumer Prices, Current Account Balance, and Unemployment
(Annual percent change, unless noted otherwise)

	Real GDP			Consumer Prices ¹			Current Account Balance ²			Unemployment ³		
	2025	Projections		2025	Projections		2025	Projections		2025	Projections	
		2026	2027		2026	2027		2026	2027		2026	2027
Europe	1.6	1.4	1.6	6.2	5.4	4.3	1.6	1.3	1.4	...	...	...
Advanced Europe	1.5	1.2	1.3	2.3	2.6	2.2	2.3	1.9	2.0	5.9	6.0	5.8
Euro Area ^{4,5}	1.4	1.1	1.2	2.1	2.6	2.2	1.6	1.3	1.5	6.3	6.2	6.1
Germany	0.2	0.8	1.2	2.3	2.7	2.3	4.4	3.9	3.8	3.8	3.9	3.5
France	0.9	0.9	0.9	0.9	1.8	1.7	-0.4	-0.3	-0.2	7.6	7.9	7.9
Italy	0.5	0.5	0.5	1.6	2.6	2.4	1.2	0.6	1.1	6.1	6.0	6.1
Spain	2.8	2.1	1.8	2.7	3.0	2.3	2.9	2.2	1.9	10.5	9.8	9.8
The Netherlands	1.9	1.2	1.4	3.0	2.7	2.4	8.8	8.8	8.7	3.9	4.1	4.2
Belgium	1.0	0.7	1.1	3.0	2.8	2.2	-2.1	-2.0	-1.9	6.2	6.2	6.1
Ireland	12.3	2.5	2.4	2.1	3.1	2.4	8.2	8.1	8.1	4.7	4.8	4.8
Austria	0.6	0.7	1.0	3.6	2.5	2.6	1.0	0.4	0.7	5.7	5.7	5.6
Portugal	1.9	1.9	1.8	2.2	3.1	2.3	1.2	0.5	0.7	6.0	5.9	5.9
Greece	2.1	1.8	1.7	2.9	3.5	2.7	-5.7	-6.4	-5.7	8.9	7.4	7.1
Finland	0.2	1.0	1.5	1.8	2.5	2.2	1.3	0.1	0.4	9.7	9.6	9.1
Bulgaria	3.1	2.8	2.5	3.5	3.8	3.7	-5.9	-4.0	-3.2	3.6	3.4	3.3
Slovak Republic	0.8	0.6	1.6	4.2	4.2	3.3	-3.6	-3.0	-1.7	5.4	5.8	5.7
Croatia	3.2	2.6	2.6	4.4	4.4	2.7	-3.2	-3.8	-3.3	4.5	4.6	4.6
Lithuania	2.9	2.9	2.2	3.4	4.0	2.7	1.3	0.2	1.0	6.9	6.5	6.3
Slovenia	1.1	2.0	2.1	2.5	2.9	2.1	3.5	3.2	2.8	3.9	3.9	3.9
Luxembourg	0.6	1.6	1.7	2.5	2.1	2.2	4.7	4.5	4.7	6.0	5.9	6.1
Latvia	2.1	2.2	2.4	3.8	3.0	2.7	-3.4	-3.6	-3.6	6.9	6.7	6.5
Estonia	0.6	1.4	1.9	4.8	3.8	3.0	-0.2	-1.1	-2.0	7.4	7.3	7.0
Cyprus	3.8	3.0	3.0	0.8	2.6	1.6	-8.1	-9.3	-10.3	4.4	4.6	4.7
Malta	4.0	3.7	3.8	2.4	2.5	2.4	5.7	5.8	5.2	3.0	3.0	3.0
United Kingdom	1.3	0.8	1.3	3.4	3.2	2.4	-3.1	-3.4	-3.1	4.9	5.6	5.3
Switzerland	1.3	1.3	1.3	0.2	0.5	0.5	7.1	6.8	7.1	2.8	3.0	2.9
Sweden	1.5	2.0	1.9	2.6	1.5	1.8	6.1	5.3	4.7	8.9	8.6	8.0
Czech Republic	2.5	2.2	2.2	2.5	2.4	2.2	0.7	-1.5	-1.3	2.9	3.0	2.9
Norway	1.1	1.5	1.3	3.0	3.3	2.6	14.1	14.3	13.6	4.5	4.2	4.2
Denmark	2.9	2.0	1.6	1.8	2.0	2.2	12.5	12.3	11.5	2.9	2.9	2.9
Iceland	1.3	1.9	2.1	4.1	4.8	2.8	-3.6	-0.7	-0.1	4.4	4.2	4.3
Liechtenstein	0.0	-0.4	0.4	0.2	0.5	0.7	13.4	12.7	12.7	3.5	3.9	3.8
Andorra	3.9	2.1	1.8	2.4	3.0	2.3	15.9	16.0	16.0	1.1	1.1	1.1
San Marino	1.5	1.3	1.2	2.3	2.8	2.6	17.1	17.1	16.9	4.5	4.4	4.4
Emerging and Developing Europe ⁶	2.0	2.0	2.1	13.5	10.5	8.2	-1.2	-1.2	-1.3	...	...	...
Russia	1.0	1.1	1.1	8.7	5.6	4.3	1.6	2.9	2.3	2.2	2.4	2.6
Türkiye	3.6	3.4	3.5	34.9	28.6	21.4	-1.9	-2.8	-2.5	8.3	8.3	8.7
Poland	3.6	3.3	2.4	3.6	3.3	3.3	-0.7	-1.1	-0.9	3.1	3.4	3.4
Romania	0.7	0.7	2.5	7.3	7.8	3.9	-8.0	-6.8	-6.2	6.1	6.0	5.9
Ukraine ⁷	1.8	2.0	3.5	12.7	6.1	7.7	-15.0	-18.9	-16.6	11.6	10.2	12.0
Hungary	0.4	1.7	2.0	4.4	3.8	3.5	1.3	-0.4	0.4	4.3	4.2	4.0
Belarus	1.3	1.2	1.0	6.6	6.4	6.2	-2.6	-2.7	-3.4	2.9	2.9	2.9
Serbia	2.0	2.8	3.5	3.9	5.2	4.9	-4.9	-5.7	-4.4	8.7	8.8	8.7

Source: IMF staff estimates.

Note: Data for some countries are based on fiscal years. Please refer to Table F in the Statistical Appendix for a list of economies with exceptional reporting periods.

¹ Movements in consumer prices are shown as annual averages. Year-end to year-end changes can be found in Tables A6 and A7 in the Statistical Appendix.

² Percent of GDP.

³ Percent. National definitions of unemployment may differ.

⁴ Current account position corrected for reporting discrepancies in intra-area transactions.

⁵ Based on Eurostat's harmonized index of consumer prices.

⁶ Includes Albania, Bosnia and Herzegovina, Kosovo, Moldova, Montenegro, and North Macedonia.

⁷ See the country-specific note for Ukraine in the "Country Notes" section of the Statistical Appendix.

Annex Table 1.1.2. Asian and Pacific Economies: Real GDP, Consumer Prices, Current Account Balance, and Unemployment
(Annual percent change, unless noted otherwise)

	Real GDP			Consumer Prices ¹			Current Account Balance ²			Unemployment ³		
	2025	Projections		2025	Projections		2025	Projections		2025	Projections	
		2026	2027		2026	2027		2026	2027		2026	2027
Asia	5.0	4.4	4.2	1.4	2.6	2.4	3.5	3.0	2.9	...	...	...
Advanced Asia	2.5	2.0	1.6	2.5	2.4	2.2	5.9	5.3	5.3	2.9	2.9	2.9
Japan	1.2	0.7	0.6	3.2	2.2	2.3	4.8	3.8	3.9	2.5	2.5	2.5
Korea	1.0	1.9	2.1	2.1	2.5	1.9	6.6	5.6	5.4	2.8	2.8	2.9
Australia	2.0	2.0	1.7	2.9	4.0	3.2	-2.6	-2.3	-2.2	4.2	4.2	4.3
Taiwan Province of China	8.7	5.2	3.0	1.7	1.5	1.6	17.4	18.1	18.0	3.4	3.4	3.4
Singapore	5.0	3.5	2.7	0.9	2.3	1.9	16.7	16.6	16.5	2.0	2.0	2.0
Hong Kong SAR	3.5	2.4	2.4	1.4	2.1	1.8	12.2	12.6	12.3	3.6	3.3	3.2
New Zealand	0.2	2.1	2.4	2.8	3.1	2.3	-3.7	-4.0	-3.8	5.3	5.4	5.0
Macao SAR	4.7	3.0	3.1	0.3	1.8	1.9	35.7	35.0	34.2	1.7	1.7	1.7
Emerging and Developing Asia	5.5	4.9	4.8	1.1	2.6	2.5	2.7	2.2	2.0	...	...	...
China	5.0	4.4	4.0	0.0	1.2	1.5	3.7	3.5	3.3	5.1	5.1	5.1
India ⁴	7.6	6.5	6.5	2.1	4.7	4.0	-0.9	-2.0	-1.6	4.9	4.9	4.9
Indonesia	5.1	5.0	5.1	1.9	3.0	2.6	-0.1	-1.1	-0.9	4.9	4.9	4.8
Thailand	2.4	1.5	2.1	-0.1	0.9	1.0	3.1	0.7	1.4	1.0	1.0	1.0
Vietnam	8.0	7.1	6.7	3.3	4.9	4.6	6.7	5.3	4.4	2.2	2.1	2.2
Malaysia	5.2	4.7	4.3	1.4	1.9	2.0	1.6	1.4	1.6	3.0	3.0	3.0
Philippines	4.4	4.1	5.8	1.7	4.3	3.2	-3.3	-4.4	-3.5	4.2	4.7	4.6
Other Emerging and Developing Asia⁵	3.4	4.2	4.0	8.5	8.4	6.4	0.9	-0.3	-1.1	...	...	...
<i>Memorandum</i>												
ASEAN-5 ⁶	4.5	4.1	4.4	1.4	2.6	2.3	3.0	2.2	2.4	...	...	...
Emerging Asia ⁷	5.6	5.0	4.8	0.8	2.4	2.3	2.7	2.2	2.1	...	...	...

Source: IMF staff estimates.

Note: Data for some countries are based on fiscal years. Please refer to Table F in the Statistical Appendix for a list of economies with exceptional reporting periods.

¹ Movements in consumer prices are shown as annual averages. Year-end to year-end changes can be found in Tables A6 and A7 in the Statistical Appendix.

² Percent of GDP.

³ Percent. National definitions of unemployment may differ.

⁴ See the country-specific note for India in the "Country Notes" section of the Statistical Appendix.

⁵ Other Emerging and Developing Asia comprises Bangladesh, Bhutan, Brunei Darussalam, Cambodia, Fiji, Kiribati, Lao P.D.R., Maldives, the Marshall Islands, Micronesia, Mongolia, Myanmar, Nauru, Nepal, Palau, Papua New Guinea, Samoa, the Solomon Islands, Sri Lanka, Timor-Leste, Tonga, Tuvalu, and Vanuatu.

⁶ Indonesia, Malaysia, the Philippines, Singapore, and Thailand.

⁷ Emerging Asia comprises China, India, Indonesia, Malaysia, the Philippines, Thailand, and Vietnam.

Annex Table 1.1.3. Western Hemisphere Economies: Real GDP, Consumer Prices, Current Account Balance, and Unemployment*(Annual percent change, unless noted otherwise)*

	Real GDP			Consumer Prices ¹			Current Account Balance ²			Unemployment ³		
	2025	Projections		2025	Projections		2025	Projections		2025	Projections	
		2026	2027		2026	2027		2026	2027		2026	2027
North America	1.9	2.2	2.1	2.8	3.2	2.3	-3.3	-3.3	-3.2	...	...	...
United States	2.1	2.3	2.1	2.7	3.2	2.1	-3.6	-3.7	-3.6	4.3	4.4	4.2
Mexico	0.6	1.6	2.2	3.8	3.9	3.4	-0.4	-0.4	-0.5	2.6	2.7	2.8
Canada	1.7	1.5	1.9	2.1	2.5	2.1	-0.9	-0.2	-0.3	6.9	6.5	6.3
Puerto Rico ⁴	-0.8	-0.1	1.0	1.4	2.1	2.4	...	...	...	6.4	6.0	5.8
South America⁵	2.7	2.3	2.5	9.7	8.2	5.6	-1.7	-1.3	-1.4	...	...	...
Brazil	2.3	1.9	2.0	5.0	4.0	3.4	-3.0	-2.7	-2.4	6.0	6.8	7.4
Argentina	4.4	3.5	4.0	41.9	30.4	15.7	-1.1	-0.8	-0.6	7.4	7.2	6.9
Colombia	2.6	2.3	2.5	5.1	5.9	5.2	-2.4	-2.5	-2.6	8.0	9.0	10.0
Chile	2.3	2.4	2.6	4.2	2.9	3.3	-2.3	-0.8	-1.8	8.5	8.1	7.6
Peru	3.4	2.8	2.8	1.5	2.5	1.8	3.1	3.4	2.5	5.9	6.3	6.3
Ecuador	3.7	2.5	2.5	0.7	2.9	1.6	5.8	5.2	4.6	3.1	3.1	3.1
Venezuela	1.5	4.0	6.0	252.0	387.4	94.4	2.6	7.1	4.8	...	...	...
Bolivia	-1.2	-3.3	...	19.5	20.7	...	-1.9	1.2	...	3.3	4.5	...
Paraguay	6.0	4.2	3.5	4.1	3.3	3.5	-3.5	-3.0	-2.1	5.2	5.2	5.2
Uruguay	1.8	1.8	2.6	4.7	4.0	4.5	-0.5	-0.8	-0.9	7.5	8.0	8.0
Central America⁶	3.7	3.7	4.0	1.8	2.8	3.4	1.0	-0.3	-0.6	...	...	...
Caribbean⁷	6.2	5.7	8.6	5.9	6.1	6.1	-0.4	0.6	-0.5	...	...	...
<i>Memorandum</i>												
Latin America and the Caribbean ⁸	2.4	2.3	2.7	7.6	6.7	4.9	-1.2	-0.9	-1.1	...	...	...
Eastern Caribbean Currency Union ⁹	2.8	2.4	2.3	1.4	2.2	2.1	-10.6	-10.4	-9.1	...	...	...

Source: IMF staff estimates.

Note: Data for some countries are based on fiscal years. Please refer to Table F in the Statistical Appendix for a list of economies with exceptional reporting periods.

¹ Movements in consumer prices are shown as annual averages. Year-end to year-end changes can be found in Tables A6 and A7 in the Statistical Appendix. Aggregates exclude Venezuela.² Percent of GDP.³ Percent. National definitions of unemployment may differ.⁴ Puerto Rico is a territory of the United States, but its statistical data are maintained on a separate and independent basis.⁵ See the country-specific notes for Argentina and Venezuela in the "Country Notes" section of the Statistical Appendix.⁶ Central America refers to CAPDR (Central America, Panama, and the Dominican Republic) and comprises Costa Rica, the Dominican Republic, El Salvador, Guatemala, Honduras, Nicaragua, and Panama.⁷ The Caribbean comprises Antigua and Barbuda, Aruba, The Bahamas, Barbados, Belize, Dominica, Grenada, Guyana, Haiti, Jamaica, St. Kitts and Nevis, St. Lucia, St. Vincent and the Grenadines, Suriname, and Trinidad and Tobago.⁸ Latin America and the Caribbean comprises Mexico and economies from the Caribbean, Central America, and South America. See the country-specific notes for Argentina and Venezuela in the "Country Notes" section of the Statistical Appendix.⁹ Eastern Caribbean Currency Union comprises Antigua and Barbuda, Dominica, Grenada, St. Kitts and Nevis, St. Lucia, and St. Vincent and the Grenadines, as well as Anguilla and Montserrat, which are not IMF members.

Annex Table 1.1.4. Middle East and Central Asia Economies: Real GDP, Consumer Prices, Current Account Balance, and Unemployment*(Annual percent change, unless noted otherwise)*

	Real GDP			Consumer Prices ¹			Current Account Balance ²			Unemployment ³		
	2025	Projections		2025	Projections		2025	Projections		2025	Projections	
		2026	2027		2026	2027		2026	2027		2026	2027
Middle East and Central Asia	3.6	1.9	4.6	11.2	11.9	9.1	0.7	1.1	0.4	...	...	...
Oil Exporters⁴	3.1	0.5	4.6	10.7	13.4	9.2	2.4	2.8	2.2	...	...	...
Saudi Arabia	4.5	3.1	4.5	2.0	2.3	2.1	-3.0	-1.6	-3.0	...	...	...
Iran	-1.5	-6.1	3.2	50.9	68.9	39.6	0.6	-1.8	-0.2	8.0	9.2	9.0
United Arab Emirates	5.8	3.1	5.3	1.3	2.5	2.0	15.3	11.4	11.4	...	...	...
Kazakhstan	6.5	4.6	4.4	11.4	10.7	10.1	-3.9	-1.0	-2.0	4.6	4.6	4.5
Algeria	3.8	3.8	2.9	1.4	2.9	3.0	-7.7	-1.2	-2.5	...	...	...
Iraq	-0.4	-6.8	11.3	0.3	3.0	3.3	-1.7	-5.3	0.2	...	...	...
Qatar	2.8	-8.6	8.6	0.6	3.9	2.5	14.5	11.0	9.9	...	...	...
Kuwait	3.5	-0.6	2.8	2.4	2.8	2.5	23.3	26.0	21.8	...	...	...
Azerbaijan	1.4	2.2	2.5	5.6	6.0	5.1	5.5	9.7	5.4	5.3	5.3	5.2
Oman	2.4	3.5	3.4	1.0	1.7	1.9	1.3	7.5	5.6	...	...	...
Turkmenistan	3.6	2.6	2.0	2.7	3.9	4.7	3.3	3.7	1.6	...	...	...
Bahrain	3.1	-0.5	4.5	-0.1	2.4	1.2	5.9	3.9	4.8	...	...	...
Oil Importers^{5,6}	4.4	4.1	4.5	12.1	9.8	8.9	-3.8	-3.3	-4.0	...	...	...
Egypt	4.4	4.2	4.8	20.4	13.2	11.1	-4.2	-4.2	-4.6	7.3	7.4	7.1
Pakistan ⁷	3.1	3.6	3.5	4.5	7.2	8.4	0.5	-0.4	-0.9	7.1	6.9	6.5
Morocco	4.9	4.9	4.5	0.8	1.3	1.6	-2.1	-3.1	-2.9	13.0	12.2	11.3
Uzbekistan	7.7	6.5	5.9	8.8	7.0	5.6	-3.9	-1.3	-3.4	4.8	4.3	3.8
Tunisia	2.5	2.1	1.6	5.3	6.5	7.2	-2.8	-4.2	-5.8	15.2	...	...
Sudan ⁷	3.2	0.7	8.1	100.2	75.1	44.6	-8.1	-4.5	-10.3	60.6	61.3	59.3
Jordan	2.7	2.7	3.1	1.8	2.3	2.2	-5.6	-6.2	-5.1	21.3	21.3	21.3
Georgia	7.5	5.3	5.0	3.9	4.4	3.0	-2.6	-5.0	-4.5	13.9	13.9	13.9
Armenia	7.2	5.3	5.5	3.3	3.6	3.4	-6.7	-6.0	-5.2	13.0	12.8	12.7
Tajikistan	8.4	6.0	4.8	3.4	4.0	4.7	16.6	6.3	2.5	...	...	...
Kyrgyz Republic	11.1	6.1	6.1	8.2	10.6	10.0	-23.4	-7.1	-6.1	4.0	4.0	4.0
Mauritania	4.2	4.4	4.5	1.6	4.1	4.1	-5.8	-6.5	-6.4	...	...	...
West Bank and Gaza ⁷	...	...	...	10.5	...	...	...	...	...	...	...	...
<i>Memorandum</i>												
Caucasus and Central Asia	6.2	4.8	4.5	8.4	8.0	7.2	-2.2	0.0	-1.6	...	...	...
Middle East, North Africa, Afghanistan, and Pakistan ⁶	3.2	1.4	4.6	11.7	12.6	9.4	1.1	1.2	0.7	...	...	...
Middle East and North Africa	3.2	1.1	4.8	12.9	13.5	9.6	1.4	1.4	0.9	...	...	...
Israel ⁸	2.9	3.5	4.4	3.0	2.3	2.1	1.5	1.9	2.3	3.0	3.2	3.3

Source: IMF staff estimates.

Note: Data for some countries are based on fiscal years. Please refer to Table F in the Statistical Appendix for a list of economies with exceptional reporting periods.

¹ Movements in consumer prices are shown as annual averages. Year-end to year-end changes can be found in Tables A6 and A7 in the Statistical Appendix.² Percent of GDP.³ Percent. National definitions of unemployment may differ.⁴ Includes Libya.⁵ Includes Djibouti, Lebanon, Somalia, and Yemen. See the country-specific note for Lebanon and Yemen in the "Country Notes" section of the Statistical Appendix.⁶ Excludes Afghanistan and Syria because of a lack of adequate data. See the country-specific notes in the "Country Notes" section of the Statistical Appendix.⁷ See the country-specific notes for Sudan and West Bank and Gaza in the "Country Notes" section of the Statistical Appendix.⁸ Israel, which is not a member of the economic region, is shown for reasons of geography but is not included in the regional aggregates.

Annex Table 1.1.5. Sub-Saharan African Economies: Real GDP, Consumer Prices, Current Account Balance, and Unemployment
(Annual percent change, unless noted otherwise)

	Real GDP			Consumer Prices ¹			Current Account Balance ²			Unemployment ³		
	2025	Projections		2025	Projections		2025	Projections		2025	Projections	
		2026	2027		2026	2027		2026	2027		2026	2027
Sub-Saharan Africa	4.5	4.3	4.4	12.5	8.8	8.8	-0.9	-0.6	-1.1	...	...	...
Oil Exporters ⁴	4.0	3.7	4.0	21.3	14.6	14.5	2.6	3.8	1.8	...	...	...
Nigeria ⁵	4.0	4.1	4.3	23.0	16.0	15.9	5.1	5.8	3.1	...	...	...
Angola	3.1	2.3	2.6	20.2	12.9	12.8	0.4	2.2	1.0	...	...	...
Gabon	2.5	2.7	2.8	2.0	2.5	2.4	-3.1	-4.3	-5.5	...	...	...
Chad	5.6	5.2	4.9	-2.6	0.5	3.9	0.2	0.8	0.3	...	...	...
Equatorial Guinea	-6.4	-2.7	-1.3	2.8	3.2	2.9	-3.0	-1.6	-2.4	...	...	...
Middle-Income Countries ⁶	3.5	3.2	3.4	4.5	4.3	4.3	-0.7	-1.0	-1.4	...	...	...
South Africa	1.1	1.0	1.3	3.2	3.9	3.4	-0.5	-0.9	-1.4	32.4	32.5	32.4
Kenya	4.9	4.5	4.7	4.1	5.9	5.9	-2.7	-4.1	-3.7	...	...	...
Ghana	6.0	4.8	4.9	14.2	5.8	7.8	7.9	10.1	9.5	...	...	...
Côte d'Ivoire	6.5	6.2	6.3	0.1	1.8	2.0	-1.1	-1.1	-3.1	...	...	...
Cameroon	3.1	3.3	3.8	3.4	3.5	3.2	-3.8	-5.1	-5.8	...	...	...
Senegal	7.9	2.2	2.3	1.4	2.5	2.2	-5.6	-6.2	-5.8	...	...	...
Zambia	3.8	4.3	4.7	13.9	9.0	8.0	-3.5	0.9	1.9	...	...	...
Low-Income Countries ⁷	6.5	6.6	6.4	11.1	6.6	6.8	-3.9	-3.9	-2.9	...	...	...
Ethiopia	9.2	9.2	7.9	13.2	11.8	10.7	-0.9	-2.4	-1.9	...	...	...
Tanzania	5.9	5.9	6.1	3.3	4.0	4.3	-2.4	-2.3	-2.1	...	...	...
Democratic Republic of the Congo	5.7	5.9	5.4	7.4	3.3	6.4	-3.7	-2.0	-1.6	...	...	...
Uganda	6.7	7.5	8.2	3.6	4.0	4.9	-6.2	-3.9	-2.7	...	...	...
Mali	4.9	5.5	5.7	2.3	2.2	2.0	-0.8	-1.5	1.6	...	...	...
Burkina Faso	5.0	4.9	4.8	-0.5	1.5	2.1	6.7	8.6	8.4	...	...	...

Source: IMF staff estimates.

Note: Data for some countries are based on fiscal years. Please refer to Table F in the Statistical Appendix for a list of economies with exceptional reporting periods.

¹ Movements in consumer prices are shown as annual averages. Year-end to year-end changes can be found in Tables A6 and A7 in the Statistical Appendix.

² Percent of GDP.

³ Percent. National definitions of unemployment may differ.

⁴ Includes Republic of Congo and South Sudan.

⁵ See the country-specific note for Nigeria in the "Country Notes" section of the Statistical Appendix.

⁶ Includes Benin, Botswana, Cabo Verde, Comoros, Eswatini, Lesotho, Mauritius, Namibia, São Tomé and Príncipe, and Seychelles.

⁷ Includes Burundi, the Central African Republic, Eritrea, The Gambia, Guinea, Guinea-Bissau, Liberia, Madagascar, Malawi, Mozambique, Niger, Rwanda, Sierra Leone, Togo, and Zimbabwe.

Annex Table 1.1.6. Summary of World Real per Capita Output*(Annual percent change; in constant 2021 international dollars at purchasing power parity)*

	Average									Projections	
	2008-17	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
World	1.8	2.5	1.8	-3.9	5.7	2.8	2.3	2.4	2.5	2.7	2.3
Advanced Economies	0.8	1.9	1.5	-4.4	6.0	2.5	0.9	1.2	1.6	1.6	1.5
United States	0.8	2.4	2.1	-2.9	5.8	2.0	2.1	1.9	1.5	2.0	1.8
Euro Area ¹	0.4	1.6	1.4	-6.3	6.5	3.4	-0.1	0.6	1.1	0.9	1.0
Germany	1.1	1.0	0.9	-4.0	4.1	1.1	-1.8	-0.8	0.2	0.8	1.2
France	0.3	1.3	1.7	-7.9	6.4	2.2	1.1	0.8	0.6	0.6	0.6
Italy	-0.8	1.0	0.6	-8.6	9.7	5.2	1.0	0.8	0.6	0.6	0.6
Spain	0.1	1.8	1.1	-11.1	6.5	5.0	1.3	2.5	1.4	0.9	0.8
Japan	0.6	1.0	-0.1	-4.0	3.9	1.7	1.2	0.2	1.7	1.2	1.2
United Kingdom	0.5	1.1	0.7	-10.2	8.2	4.1	-1.0	0.0	1.0	0.3	0.9
Canada	0.5	1.3	0.4	-6.1	5.3	2.9	-0.8	-0.9	0.6	1.6	1.8
Other Advanced Economies ²	1.7	2.1	1.3	-2.1	6.0	2.0	0.6	1.8	2.6	2.2	1.8
Emerging Market and Developing Economies	3.3	3.3	2.5	-3.2	5.9	3.2	3.4	3.4	3.4	3.4	3.1
Emerging and Developing Asia	6.1	5.6	4.5	-1.3	7.1	4.1	5.0	4.9	5.0	4.4	4.3
China	7.7	6.4	5.7	2.2	8.5	3.2	5.5	5.1	5.2	4.6	4.3
India ³	5.3	5.3	2.8	-6.7	8.8	6.8	6.3	6.1	6.7	5.6	5.6
Emerging and Developing Europe	1.8	3.4	2.4	-1.9	7.6	1.7	3.8	4.3	2.4	2.4	2.1
Russia	1.0	2.6	2.1	-2.5	6.3	-1.3	4.2	5.4	1.6	1.5	1.3
Latin America and the Caribbean	0.8	0.2	-0.9	-8.0	6.7	3.6	1.5	1.6	1.6	1.6	2.0
Brazil	0.8	1.1	0.6	-3.9	4.3	2.6	2.8	3.0	1.9	1.6	1.6
Mexico	0.3	1.0	-1.3	-9.1	5.4	2.9	2.2	0.5	-0.3	0.9	1.4
Middle East and Central Asia	1.1	0.8	0.3	-4.4	3.0	4.2	0.5	0.8	1.8	4.4	2.8
Saudi Arabia	0.5	5.9	2.1	-8.3	9.2	7.2	-4.0	-2.0	2.5	1.1	2.4
Sub-Saharan Africa	1.5	0.5	0.3	-5.7	1.4	1.9	1.3	1.5	1.9	1.8	1.9
Nigeria ³	2.1	-0.4	0.0	-8.3	-1.0	2.2	1.2	1.9	1.9	1.9	2.2
South Africa	0.2	0.1	-1.2	-7.4	3.9	1.0	-0.4	-0.7	-0.1	-0.3	-0.1
<i>Memorandum</i>											
European Union	0.7	2.1	1.8	-5.7	6.7	3.4	0.0	0.9	1.4	1.2	1.3
ASEAN-5 ⁴	3.5	3.8	3.2	-5.5	3.4	4.6	3.1	3.9	3.6	3.2	3.5
Middle East and North Africa	0.7	0.3	-0.1	-4.6	3.1	4.5	0.5	0.3	1.6	-0.8	2.9
Emerging Market and Middle-Income Economies	3.6	3.7	2.7	-2.9	6.6	3.5	3.8	3.8	3.7	3.1	3.4
Low-Income Developing Countries	2.7	2.2	2.2	-3.8	1.4	2.7	1.8	1.8	2.5	4.3	2.7

Source: IMF staff estimates.

Note: Data for some countries are based on fiscal years. Please refer to Table F in the Statistical Appendix for a list of economies with exceptional reporting periods.

¹ Data are calculated as the sum of individual euro area countries.² Excludes the Group of Seven (Canada, France, Germany, Italy, Japan, United Kingdom, United States) and euro area countries.³ See the country-specific notes for India and Nigeria in the "Country Notes" section of the Statistical Appendix.⁴ ASEAN-5 comprises Indonesia, Malaysia, the Philippines, Singapore, Thailand.

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World health statistics 2025

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Sustainable Development Goals



**World Health
Organization**

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Foreword

To make progress, we must be able to measure progress. Data about the health of populations are foundational to understanding trends, making policies and directing resources to where they will have the greatest benefit. Without timely, trusted and actionable data, threats remain invisible, systems underperform and opportunities to save lives are lost.

World health statistics is the world's annual health report card. It shows where progress has been made – and where it has stalled. This year's edition highlights the impacts of the COVID-19 pandemic, which reversed many of the gains made in the previous two decades. Between 2000 and 2019, healthy life expectancy rose by more than five years, maternal mortality fell by one third, child mortality more than halved and premature deaths fell – driven by political commitment, investment, innovation and stronger health systems. In 2020 and 2021, COVID-19 killed millions of people, put health systems under severe strain and wiped 1.8 years off healthy life expectancy.

This report also shows that at the current pace, the world will miss the target in the Sustainable Development Goals (SDGs) to reduce premature deaths from noncommunicable diseases (NCDs) by one third. While mortality rates have declined, the number of premature NCD deaths continues to rise, due to population growth and ageing. Tobacco use is declining but not fast enough, while alcohol consumption has decreased in some regions but stagnated in others. Hypertension and diabetes remain inadequately controlled, and air pollution continues to claim millions of lives globally.

Likewise, progress on maternal and child mortality has slowed considerably, and many countries are off-course for the 2030 targets, due to underfunding of primary health care and huge gaps in access to essential services, such as skilled care at birth, immunization and access to health workers. Domestic investment must increase, particularly because where the need is greatest, resources are most limited.

WHO's Thirteenth General Programme of Work translated the health-related targets in the SDGs into the “triple billion” targets for 2018–2025: one billion more people living healthier lives; one billion more people covered by universal health coverage without financial hardship; and one billion more people better protected from health emergencies. By the end of 2024, we estimate that the first target was already exceeded, but the second two are likely to be missed: about 1.4 billion more people were living healthier lives, but only 431 million more people were enjoying access to universal health coverage, and 637 million more people were better protected from health emergencies – encouraging progress, but less than is needed to reach the SDGs.

Nevertheless, it's not 2030 yet. With stronger leadership and scaled-up delivery, the gap can still be closed. Accordingly, the Fourteenth General Programme of Work sets ambitious new targets: to keep 6 billion people healthier, expand access to affordable health services for 5 billion people, and to see 7 billion people better protected from health emergencies between 2025 and 2028.

Health data are central to achieving these targets, which is why WHO is working with countries and partners to modernize information systems and link data to delivery, through the World Health Data Hub, the SCORE package of health data tools, and the Delivery for Impact approach.

Because every life counts.



Dr Tedros Adhanom Ghebreyesus

Director-General
World Health Organization

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Abbreviations

ABR	adolescent birth rate	MNT	maternal and neonatal tetanus
AMR	antimicrobial resistance	MRSA	methicillin-resistant <i>Staphylococcus aureus</i>
APC	alcohol per capita consumption	NCDs	noncommunicable diseases
ARIs	acute respiratory infections	NMR	neonatal mortality rate
ARR	annual rate of reduction	NTDs	neglected tropical diseases
ART	antiretroviral therapy	ODA	official development assistance
ASR	age-standardized death rate	OOP	out-of-pocket (health spending)
CDR	crude death rate	PAB	protection at birth
COVID-19	coronavirus disease	PCV3	pneumococcal-conjugate vaccine third dose
DHS	Demographic and Health Survey(s)	PM2.5	fine particulate matter less than 2.5 µm in diameter
DTP3	diphtheria–tetanus–pertussis vaccine third dose	RMNCH	reproductive, maternal, newborn and child health
GLASS	Global Antimicrobial Resistance and Use Surveillance System	SDG	Sustainable Development Goal
GPW13	Thirteenth General Programme of Work	SPAR	States Parties Self-Assessment Annual Reporting Tool
GPW14	Fourteenth General Programme of Work	SPL	societal poverty line
GTS	Global technical strategy for malaria 2016–2030	TB	tuberculosis
HALE	healthy life expectancy	U5MR	under-five mortality rate
HIC	high-income country	UHC	universal health coverage
IA2030	Immunization Agenda 2030	UI	uncertainty interval
IHRs	international health regulations	UMIC	upper-middle-income country
IQR	interquartile range	UNAIDS	Joint United Nations Programme on HIV/AIDS
LIC	low-income country	USAID	United States Agency for International Development
LMIC	lower-middle-income country	WASH	water, sanitation and hygiene
MCV2	measles-containing vaccine second dose	WHA	World Health Assembly
MIC	middle-income country	WHO	World Health Organization
MICS	Multiple Indicator Cluster Survey(s)		
MMR	maternal mortality ratio		

Introduction

The *World health statistics* report is the annual compilation of health and health-related indicators, which has been published by the World Health Organization (WHO) since 2005.

The 2025 edition consists of four chapters and the accompanying annexes. Chapter 1 presents an in-depth analysis of global and regional estimates of life expectancy, healthy life expectancy and progress in the reduction of premature mortality. Chapter 2 reviews the status of the health-related Sustainable Development Goal (SDG) indicators, covering age- and cause-specific mortality, infectious diseases, risk factors for health, and universal health coverage (UHC) and health systems. A summary of global and regional trends towards the achievement of selected health-related SDG indicators is further presented in Annex 1. Chapter 3 presents updated projections on WHO's Triple Billion targets, based on the most recent available data. It also describes WHO's transition from its thirteenth general programme of

work (GPW13) to the fourteenth general programme of work (GPW14), aligning with emerging global health challenges and priorities. Chapter 4 looks at inequalities in immunization as an important global health issue.

The information presented in *World health statistics 2025* is based on data available from global monitoring as of April 2025. These data have been compiled primarily from publications and databases produced and maintained by WHO, other United Nations entities (such as UNICEF, UNAIDS, and the Population Division in the Department of Economic and Social Affairs of the United Nations), United Nations Inter-Agency bodies of which WHO is a member, and other international organizations. A summary of the methodology for each chapter is described in Annex 2. Tables of health statistics by country and area, WHO region and globally can be accessed via <https://www.who.int/data/gho/publications/world-health-statistics>.

Key messages

The major contributors to the 5.4-year pre-pandemic increase in global healthy life expectancy (HALE) at birth between 2000 (58.1 years) and 2019 (63.5 years) were mortality reduction from communicable and perinatal conditions among ages under 5 years, and from noncommunicable diseases (NCDs) among those 30 years and older. However, worsening morbidity due to diabetes among ages 30 years and older has led to a 0.14-year loss in HALE. Each WHO region exhibits distinctive patterns given their unique compositions of mortality and morbidity.

Between 2019 and 2021, the HALE loss can be almost entirely explained by mortality both directly and indirectly attributable to coronavirus disease (COVID-19) among those aged 30 years and older, both globally and in most WHO regions. The pandemic also caused increased morbidity from anxiety disorders and depressive disorders, each responsible for a 0.06-year (or 3-week) loss in global HALE in 2019–2021, and together effectively wiped out 80% of the positive contribution to HALE (0.15 years, or 8 weeks) by mortality decline from all NCDs combined in that two-year period.

The contributors to the HALE difference between males and females show a mixed pattern. While lower female mortality from injuries among 5–69-year-olds and from NCDs including ischaemic heart disease and stroke among females of 30 years and older gives a lead in HALE compared with males, mortality in maternal conditions and breast cancers, and higher female morbidity from conditions including back and neck pain, gynaecological diseases, migraine, depressive disorders and anxiety disorders have effectively eliminated a considerable share of the female HALE lead.

While lower mortality from communicable and perinatal causes explains a significant amount of the HALE lead that populations in high-income countries (HICs) had against those in low-income countries (LICs) and lower-middle-income countries (LMICs), higher mortality from tracheal, bronchus and lung cancers and drug use disorders, and higher morbidity from falls and back and neck pain has cost HICs some loss in HALE compared with other income groups.

Premature mortality was in decline globally and in all WHO regions from 2000 to the onset of the COVID-19 pandemic. Despite the overall progress prior to the pandemic, stagnation was seen after 2015. Globally, the pace of decline is projected to slow between now and 2050, compared with the rates observed in 2000–2019. Only about one third of countries are projected to have

accelerated progress in 2019–2050 compared with 2000–2019, mostly located in the Region of the Americas and the Western Pacific Region.

As the world moves closer to 2030, overall progress is insufficient to meet the health-related SDG and other global targets. Declines were observed in mortality from causes addressed by the SDG indicators, including maternal mortality, child and neonatal mortality, premature NCD mortality, injury mortality, and mortality attributed to unsafe water, sanitation and hygiene (WASH) and air pollution. However, the progress for all these indicators is either insufficient or stalled, and is currently off-track for achieving their respective global targets.

There has been mixed progress in the global fight against infectious diseases, with declines in HIV and tuberculosis (TB) incidence rates, as well as in the number of people requiring interventions against neglected tropical diseases (NTDs), while malaria incidence rates have increased since 2015 and hepatitis B and antimicrobial resistance (AMR) continue to pose challenges.

Similarly, the world is off-track for meeting most of the global targets in reducing the prevalence or exposure to health risk factors such as malnutrition, tobacco use, unsafe WASH, air pollution and violence against women and girls, despite improvement in some of these areas.

However, the world is on-track to reach the 20% reduction in total per-capita alcohol consumption by 2030, although progress differs by region.

While gains in UHC service coverage index (SDG indicator 3.8.1) were observed globally over the past two decades, progress has slowed in recent years. In 2019, some 344 million people were pushed or further pushed into extreme poverty by out-of-pocket (OOP) health spending, while 13.5% of the global population spent more than 10% of their household budget on OOP payments for health.

The estimated global shortage of health workers of 15.4 million in 2020 decreased to 14.7 million in 2023. The projected 2030 shortage of 11.1 million shows slow progress in closing the gap, with the WHO African and Eastern Mediterranean regions projected to bear nearly 70% of the shortage.

Progress towards achieving the Triple Billion targets remains uneven. The expansion of UHC is insufficient, with only about 431 million additional people gaining access to essential health services without incurring



financial hardship by 2024 compared with the 2018 baseline set in the WHO Thirteenth General Programme of Work, and only a projected 500 million by 2025 – just half of the targeted one billion. Meanwhile, close to 637 million additional individuals are expected to be better protected from health emergencies by 2024. This is expected to increase to only 697 million in 2025 – a substantial gain that nonetheless remains a little over 30% short of the one-billion target. On the plus side, an estimated 1.35 billion more people will experience healthier lives by 2024, rising to 1.5 billion by 2025, thus exceeding the original target of one billion. However, this progress remains insufficient to put the world on track to achieve the health-related SDGs by 2030. Moreover, recent interruptions in international aid threaten to disrupt health services and systems, disproportionately impacting countries and communities with the greatest health-care needs. Safeguarding the gains made

towards the Triple Billion targets remains of paramount importance for the global community in the years ahead.

Within-country inequalities in childhood immunization related to economic status and education level of the mother persist. Positively, inequalities in the last decade have diminished, compared with those in the previous decade, in LICs and LMICs. For instance, economic-related inequality in DTP3 coverage and zero-dose prevalence have reduced by more than half across LICs. Eliminating economic-related inequality across 88 LICs and middle-income countries is associated with improving the national average DTP3 coverage across these countries by 10 percentage points and halving zero-dose prevalence. Achieving equity in immunization coverage requires focused attention on urban poor areas, remote/rural areas, conflict areas, and gender-related inequities and barriers.

1

Change and inequality in healthy longevity, and the contributing causes

Steady increases in life expectancy and healthy life expectancy (HALE) at birth were observed globally in 2000–2019, followed by rapid declines between 2019 and 2021 due to the COVID-19 pandemic. Various factors exhibit diverse epidemiological patterns across different stages in an individual's life course and in different populations, contributing in various degrees to the change in life expectancy and HALE over time or the inequality therein between populations. This chapter reviews the contribution of different causes to the change in HALE before and during the COVID-19 pandemic, and to the inequality in HALE between males and females and between income groups as defined by the World Bank.

Unless otherwise stated, the principal source of data for this chapter is the World Health Organization (WHO) Global health estimates 2021 (1).

The global community enjoyed steady improvements in many aspects of population health between the turn of the century until the onset of the pandemic, from increasing access to clean water and sanitation to declining prevalence of tobacco smoking, from rising proportion of births attended by skilled health personnel (2) to doubling effective coverage for hypertension treatment (3), and from dropping mortality due to HIV/AIDS to improving child survival (4,5). These improvements led to a decline in mortality from a broad spectrum of causes of death throughout the human lifespan.

Both life expectancy and HALE increased between 2000 and 2019. Global life expectancy at birth increased by 6.3 years in this period, from 66.8 years in 2000 to 73.1 years in 2019, with males gaining 6.2 years (from 64.4 to 70.6 years) and females gaining 6.5 years (from 69.2 to 75.7 years) during this period. Simultaneously, global HALE at birth increased by 5.4 years, from

58.1 years in 2000 to 63.5 years in 2019 (from 57.0 to 62.3 years for males and from 59.3 years to 64.6 years females) (Fig. 1.1).

However, this progress faced a major setback as the COVID-19 pandemic wreaked havoc across the globe. With COVID-19 becoming one of the leading causes of deaths globally and in many of the WHO regions in 2020 and 2021 and claiming an excess toll of lives indirectly through other causes, both life expectancy and HALE rolled back by nearly a decade within just two years. Global life expectancy at birth dropped by 0.7 years to 72.5 years in 2020 (back to the level of 2016), and by a further 1.1 years to 71.4 years in 2021 (back to the level of 2012). Similarly, global HALE at birth dropped to 62.8 years in 2020 (back to the level of 2016) and 61.9 years in 2021 (back to the level of 2012) (Fig. 1.1).

This chapter reviews the contribution of different causes to the change in HALE before and during the COVID-19 pandemic, and to the inequality in HALE between males and females and between income groups (6,7) as defined by the World Bank.

Figure 1.1 Life expectancy (LE) and healthy life expectancy at birth (HALE), by WHO region, 2000, 2019 and 2021



Source: WHO (1).

While the overall contribution to the HALE gain from changing morbidity in 2000–2019 was relatively minor and, for the majority of causes, the contribution was positive, it is worth noting that morbidity from diabetes among adults 30 years and older increased and was responsible for a 0.14-year HALE loss in 2000–2019 (Fig. 1.2).

Source: WHO (1).



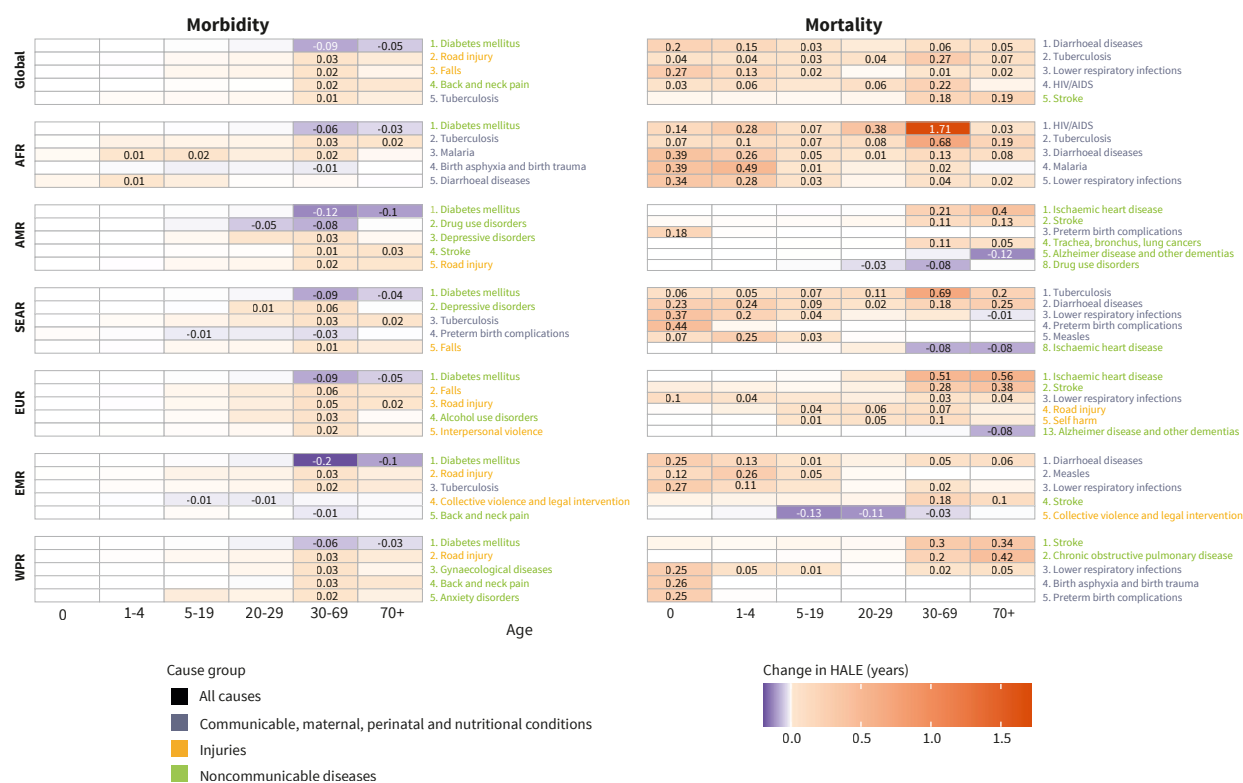
1.1.2 WHO regions

As the cause and the age profile of mortality and morbidity vary across geographical locations, the contributions to HALE by cause and age also exhibit diverse patterns.

In the WHO African Region, the most remarkable contributor to the HALE gain in 2000–2019 was from reduction in HIV/AIDS mortality, resulting in a total

of a 2.61-year gain in HALE, within which a reduction among adults aged 30–69 years was the major source (1.71 years). Decline in mortality from tuberculosis in the same age group also contributed to a large share (0.68 years or 57.6%) of the 1.18-year HALE gain accumulated throughout the entire life course. Mortality decline among children aged under 5 years from malaria, diarrhoeal diseases and lower respiratory infections also contributed considerably to the HALE gain in 2000–2019, at 0.88, 0.65 and 0.62 years, respectively (Fig. 1.3).

Figure 1.3 Leading contributing causes for change in healthy life expectancy (HALE) between 2000 and 2019, by cause, age group and WHO region, for both sexes combined



AFR: African Region; AMR: Region of the Americas; SEAR: South-East Asia Region; EUR: European Region; EMR: Eastern Mediterranean Region; WPR: Western Pacific Region.

Source: WHO (1).

Similarly, in the Eastern Mediterranean Region and the South-East Asia Region, reduction in mortality from communicable diseases was the primary driver of the HALE gain in 2000–2019. Diarrhoeal diseases contributed a total of 1.01 years (0.47 years among those aged under 5 years) in HALE gain in the South-East Asia Region and 0.51 years (0.38 years among those aged under 5 years) in the Eastern Mediterranean Region. A mortality decline in lower respiratory infections among children aged under 5 years led to HALE gains of 0.57 years in the South-East Asia Region and 0.38 years in the Eastern Mediterranean Region. While the Eastern Mediterranean Region enjoyed a 0.26-year gain in HALE due to declining mortality from ischaemic heart disease among adults aged 30 years and older, the South-East Asia Region saw an opposite trend, with rising mortality from the diseases causing 0.16-year HALE loss among the same age groups. Also worth noting is the 0.3-year loss attributed to increasing diabetes morbidity among those aged 30 years and older in the Eastern Mediterranean Region, constituting the largest loss among all WHO regions. The Eastern Mediterranean Region is also the only region where a sizable HALE loss was attributed to mortality increase in collective violence and legal intervention, at 0.28 years, of which 0.24 years was due to rising mortality among those aged 5–29 years (Fig. 1.3).

In contrast, in the Region of the Americas the contribution to the gain in HALE was more heavily concentrated in NCDs, with declining mortality among adults aged 30 years and older from ischaemic heart disease (0.61 years), stroke (0.24 years), and trachea, bronchus and lung cancers (0.16 years) driving the gain (Fig. 1.3).

However, some negative patterns were also observed. Rising mortality due to Alzheimer disease and other dementias among adults aged 70 years and older was responsible for a 0.12-year HALE loss. While improvement in mortality from diabetes contributed to a minor increase (0.03 years) in HALE in 2000–2019, worsening morbidity from the disease caused a 0.22-year loss. Additionally,

drug use disorder in the Region of the Americas contributed negatively through increasing both mortality and morbidity, causing 0.11-year and 0.13-year losses, respectively, a unique phenomenon compared with all other WHO regions (Fig. 1.3).

In the European Region, there was a 4.69-year gain in HALE between 2000 and 2019, with 93.5% (or 4.39 years) of this gain attributed to declining mortality. By far, the leading driver in HALE gain was the decline in mortality from ischaemic heart disease among adults aged 30 years and older, accounting for a 1.07-year HALE gain, or over one fifth of the total gain. This is followed by mortality reduction from stroke among the same age groups, contributing to a further 0.66-year increase in HALE. The European Region also benefitted from reduced mortality from lower respiratory infections, road injuries and suicide across many age groups over the life course, achieving a HALE gain of about 0.2 years for each cause (0.6 years in total). However, increasing mortality from Alzheimer disease and other dementias among adults aged 70 years and older and increasing morbidity from diabetes led to a loss in HALE, at 0.08 and 0.14 years respectively (Fig. 1.3).

In the Western Pacific region, HALE increased by 4.53 years between 2000 and 2019, about 97% (or 4.43 years) of which was achieved by reducing mortality. The most prominent HALE gain was achieved through reducing mortality from stroke and chronic obstructive pulmonary disease among adults aged 30 years and older, accounting for an increase of 0.64 and 0.62 years, respectively. Declining mortality from birth asphyxia and birth trauma, preterm birth complications and lower respiratory infections among children aged under 1 year led to an increase in HALE of around 0.25 years for each of the three diseases (0.76 years in total). As in other regions, diabetes-related morbidity increased among adults aged 30 years and older and caused a 0.1-year loss in HALE (Fig. 1.3).



1.2 Change in HALE during the COVID-19 pandemic in 2019–2021

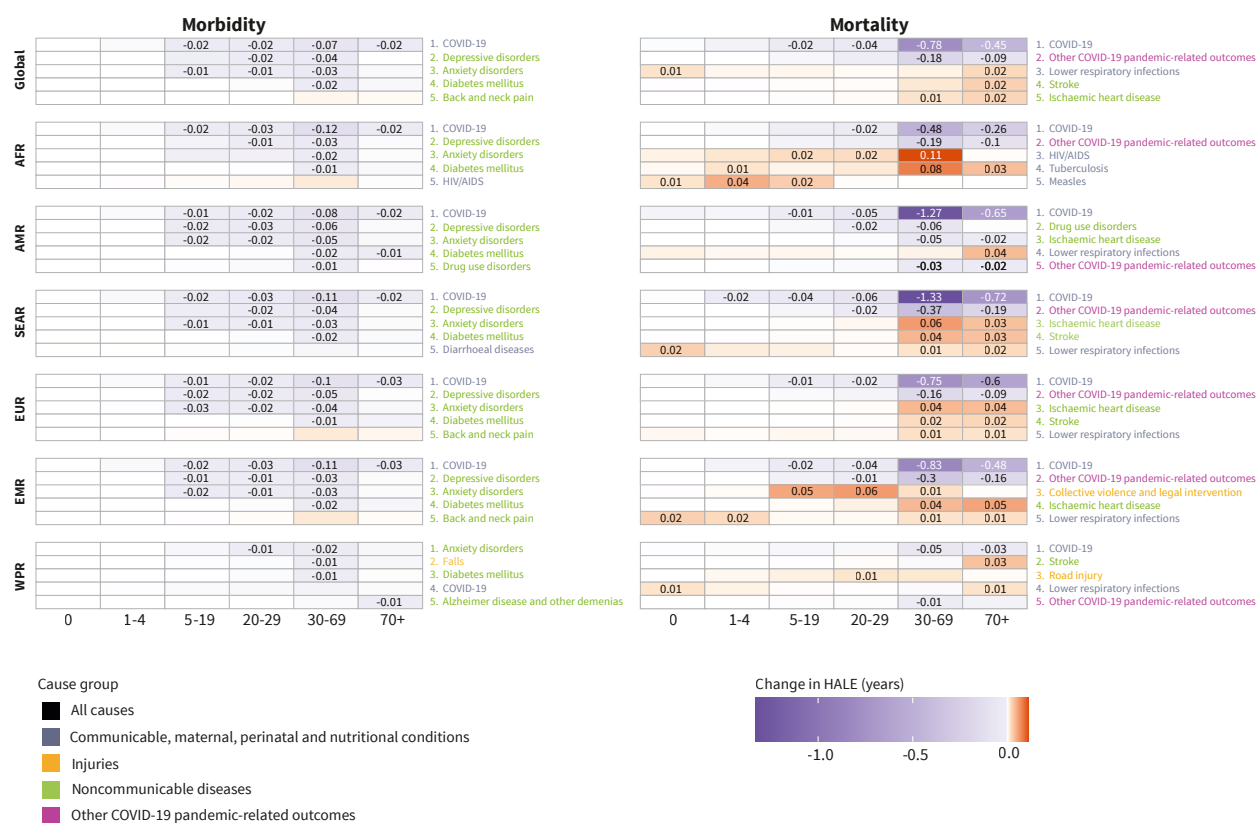
1.2.1 Change attributed to mortality

The Global HALE dropped by 1.54 years between 2019 and 2021. Mortality from the emerging COVID-19 and other outcomes related to the COVID-19 pandemic led to 1.29-year and 0.28-year losses, respectively, in HALE, totalling a staggering 1.57-year loss within just two years and exceeding the total gains in reduced mortality and morbidity from other causes. The majority of the loss was concentrated among those aged 30 years and older, at

1.23 years and 0.27 years, respectively, attributed to the two causes (Fig. 1.4).

While gain due to continued decline in mortality from causes including HIV/AIDS and tuberculosis was observed in 2019–2021 in the African Region, the region suffered from a net loss of 0.66 years in HALE. The loss is predominantly driven by mortality from COVID-19 (0.74 years) and other COVID-19 pandemic-related outcomes (0.29 years) among adults aged 30 years and older (Fig. 1.4).

Figure 1.4 Leading contributing causes for change in healthy life expectancy (HALE) between 2019 and 2021, by cause, age group and WHO region, for both sexes combined



AFR: African Region; AMR: Region of the Americas; SEAR: South-East Asia Region; EUR: European Region; EMR: Eastern Mediterranean Region; WPR: Western Pacific Region.

Source: WHO (1).

Much of the 2.72-year HALE loss in the Region of the Americas can be attributed to COVID-19 mortality among adults aged 30 years and older (1.92 years) and morbidity for all ages (0.15 years). Loss in HALE can also be explained by causes for which the previous pre-pandemic progress in mortality reduction was completely reversed or halted, including ischaemic heart disease, which contributed negatively to the HALE change between 2019 and 2021 (–0.07 years in 2019–2021 vs 0.61 years in 2000–2019) and stroke, which contributed almost no gain between 2019 and 2021 (compared with a 0.24-year gain in 2000–2019) (Fig. 1.4).

Mortality from COVID-19 and other COVID-19 pandemic-related outcomes among adults aged 30 years and older contributed 2.05 and 0.56 years, respectively, to the total 2.5-year HALE loss in the South-East Asia Region between 2019 and 2021. This outweighed HALE gains achieved through mortality reductions from causes including ischaemic heart disease (0.09 years) and stroke (0.08 years) among those aged 30 years and older during this two-year period (Fig. 1.4).

The overall HALE dropped by 1.5 years in 2019–2021 in both the European Region (66.1 to 64.6 years) and the Eastern Mediterranean Region (60.4 to 58.9 years), and mortality from COVID-19 and other COVID-19 pandemic-related outcomes among adults aged 30 years and older accounted for 1.35 and 0.25 years, respectively, of the loss in the European Region and 1.31 and 0.46 years, respectively, in the Eastern Mediterranean Region (Fig. 1.4).

The COVID-19 pandemic had a very limited impact on HALE in the Western Pacific region in 2020 and 2021, resulting in virtually no change in HALE (68.4 years in 2019 and 68.2 years in 2021). Mortality from COVID-19 and other COVID-19 pandemic-related outcomes only accounted for 0.09 and 0.02 years of loss, respectively, in HALE, which were compensated for by minor gains from mortality decline from other causes, including stroke and road injuries (Fig. 1.4).

1.2.2 Change attributed to morbidity

The decline in HALE between 2019 and 2021 at global and regional levels can be partly attributed to increasing morbidity. A shared pattern at both global and regional levels is that the HALE loss in 2019–2021 associated with morbidity increase can largely be attributed to three causes: COVID-19, anxiety disorders and depressive disorders. It is also worth noting that the pre-pandemic HALE gains in 2000–2019 attributable to declining morbidity from anxiety disorders and depressive disorders were completely cancelled out by the subsequent loss associated with these two causes during the first two years of the COVID-19 pandemic (Fig. 1.4).

Globally, morbidity due to COVID-19 led to a 0.13-year HALE loss in 2019–2021. The pandemic also caused increasing morbidity due to anxiety disorders and depressive disorders, each responsible for a 0.06-year (or 3-week) loss in HALE in 2019–2021. This loss (0.12 years, or 6 weeks) effectively wiped out 80% of the positive contribution to HALE (0.15 years, or approximately 8 weeks) by mortality decline from all NCDs combined in that two-year period. By WHO regions, morbidity due to COVID-19 led to HALE loss in 2019–2021 ranging from 0.01 years in the Western Pacific region to 0.18–0.20 years in the South-East Asia Region, the Eastern Mediterranean Region and the African Region. Increasing morbidity due to anxiety disorders was responsible for losses ranging from 0.03 years in the African Region to 0.10 years in the Region of the Americas. Similarly, the contribution to overall HALE loss in 2019–2021 from increasing morbidity due to depressive disorders ranged from 0.01 years in the Western Pacific region to 0.11 years in the Region of the Americas (Fig. 1.4).



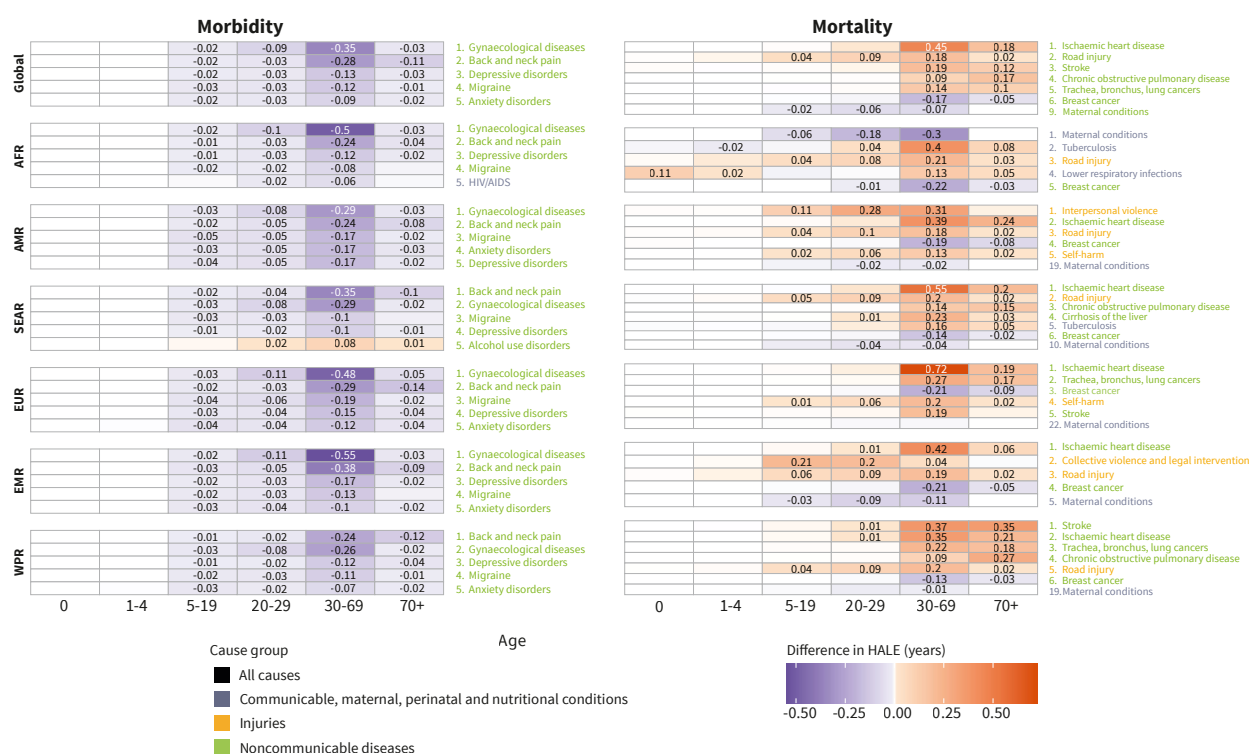
1.3 Gap in HALE between males and females in 2019 and 2021

1.3.1 Global

On average, women live longer than men, as they benefit from lower mortality, especially from causes that are affected by behavioural or genetic factors (8,9,10,11). In 2019, just before the COVID-19 pandemic, the global female life expectancy was about 5.1 years longer than that of males (75.70 versus 70.61 years). However, with higher overall risk of morbidity among females, when disability is accounted for, the gap between the global male and female HALE in 2019 reduced to about 2.3 years (8,10). The lower level of mortality among females would have rendered them a 3.85-year advantage in HALE, yet they also faced higher levels of morbidity than males, offsetting the HALE advantage associated with mortality by 1.56 years (Fig. 1.5).

Lower female mortality from NCDs among adults aged 30 years and older from ischaemic heart disease (accounting for 0.63 years), stroke (0.31 years), chronic obstructive pulmonary disease (0.26 years), and trachea, bronchus and lung cancers (0.24 years) are major contributors to the female advantage in HALE. In addition, lower mortality among those aged 5–69 years old from injuries including road injury (0.32 years), interpersonal violence (0.17 years) and suicide (0.12 years) constitute an important source of the female HALE advantage. However, mortality from sex-specific causes, such as breast cancer and maternal conditions, reduced the HALE advantage for females, at 0.22 and 0.15 years, respectively (Fig. 1.5).

Figure 1.5 Leading contributing causes for the difference in healthy life expectancy (HALE) in 2019 between males and females, by cause and age group



AFR: African Region; AMR: Region of the Americas; SEAR: South-East Asia Region; EUR: European Region; EMR: Eastern Mediterranean Region; WPR: Western Pacific Region.

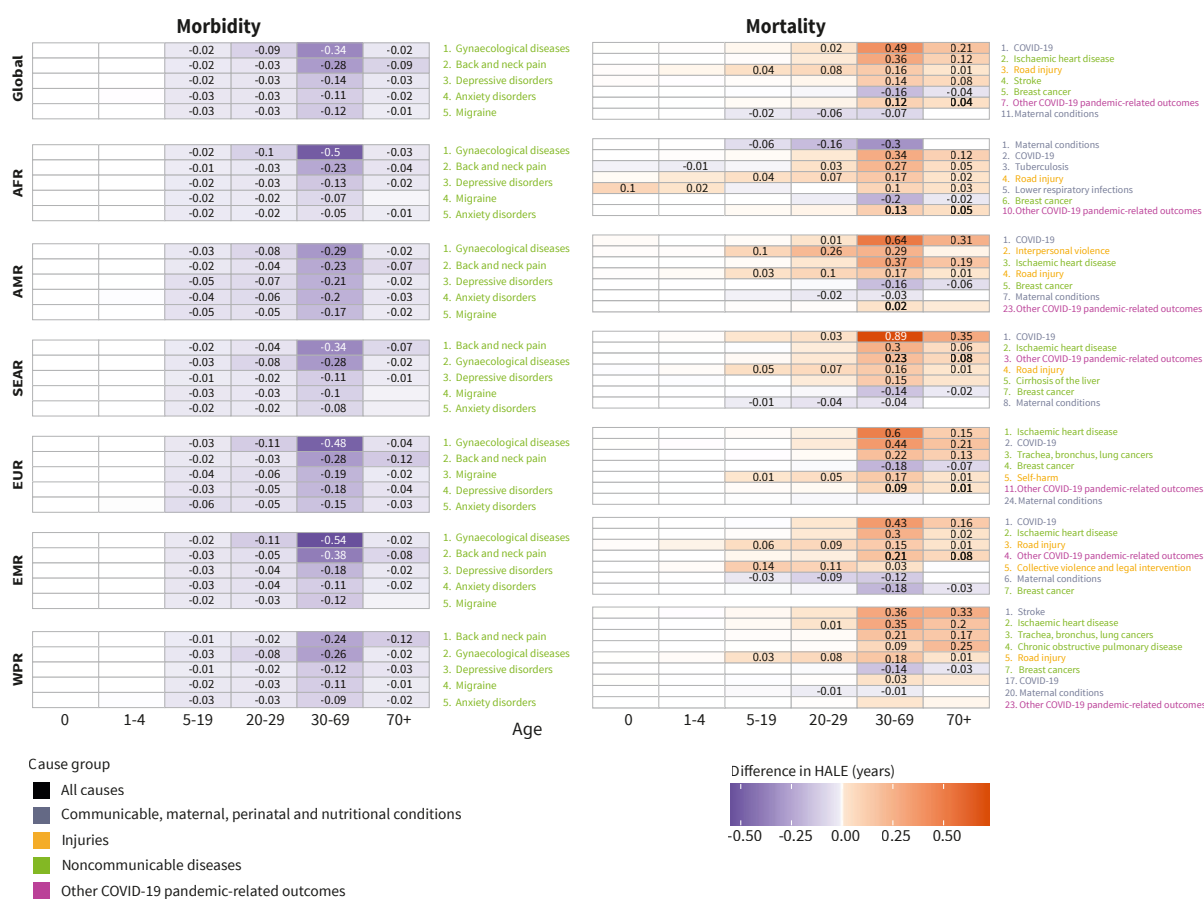
Source: WHO (1).

In addition, a higher rate of morbidity from many conditions has cost more healthy life-years among females. Back and neck pain, gynaecological diseases and migraine among females aged 30 years and older reduced female HALE by 0.39 years, 0.38 years and 0.13 years, respectively, compared with males, and thus effectively eliminated the HALE advantage associated with mortality due to ischaemic heart disease and chronic obstructive pulmonary disease. Furthermore, depressive disorders and anxiety disorders reduced the overall additional HALE for females by 0.20 and 0.16 years, respectively (Fig. 1.5).

In 2021, HALE for females was about 2.2 years higher than for males, representing the net of a 3.8-year advantage

associated with mortality and a 1.7-year disadvantage associated with morbidity. In general, the mortality and morbidity contributors to the female–male gap in HALE remained largely the same, although the size of each contribution diminished to different degrees. What is most noteworthy is that lower female mortality from COVID-19 (0.72 years) and other COVID-19 pandemic-related outcomes (0.17 years) have contributed to a 0.89-year HALE advantage over males globally. This contribution is dominated by those aged 30 years and older (0.70 and 0.16 years, respectively, for COVID-19 and other COVID-19 pandemic-related outcomes), a pattern that also holds true at regional and country levels (Fig. 1.6).

Figure 1.6 Leading contributing causes for the difference in healthy life expectancy (HALE) in 2021 between males and females, by cause and age group



AFR: African Region; AMR: Region of the Americas; SEAR: South-East Asia Region; EUR: European Region; EMR: Eastern Mediterranean Region; WPR: Western Pacific Region.

Source: WHO (1).



1.3.2 WHO regions

Similar patterns were observed across WHO regions, yet given each region's unique epidemiological profile, there are distinctive characteristics in the health inequality between male and females in each region. In the African Region, sex-difference in mortality from communicable diseases was the primary source of the HALE gap (1.35 years) between sexes in 2019. Lower tuberculosis mortality among adults aged 30 years and older brought a 0.48-year HALE advantage to females. Lower mortality from lower respiratory infections among children aged under 5 years and adults aged 30 years and older contributed a further 0.30 years. However, these gains were partially offset by mortality from maternal conditions in reproductive ages, which cost females 0.54 years in HALE (Fig. 1.5). In 2021, lower female mortality from COVID-19 and other COVID-19 pandemic-related outcomes accounted for an additional 0.47 and 0.18 years, respectively, in female HALE over male. This is predominantly contributed by females aged 30 years and older, accounting for 0.46 and 0.18 years, respectively, for the two causes (Fig. 1.6).

A unique contributor to the female HALE lead (2.49 years) in 2019 in the Region of the Americas was mortality due to interpersonal violence, accounting for 0.71 years of the HALE gap and predominantly concentrated in those aged 5–69 years (Fig. 1.5). In 2021, lower female mortality from COVID-19 and other COVID-19 pandemic-related outcomes explained 0.96 and 0.03 years, respectively, of the lead in female HALE (2.96 years in total) (Fig. 1.6).

Stroke is the most significant cause of death that contributed to the female HALE lead (3.43 years in total) in the Western Pacific region in 2019, accounting for 0.73 years (Fig. 1.5). In 2021, mortality from COVID-19 and other COVID-19 pandemic-related outcomes had very limited impact on the HALE gap between males and females, accounting for only 0.03 and 0.01 years, respectively (Fig. 1.6).

The Eastern Mediterranean Region had the narrowest male–female gap in HALE in 2019, at 0.35 years. Mortality from collective violence and legal intervention stands out as a prominent driver for the HALE gap compared with other regions, contributing 0.45 years to the female HALE lead, with fatality among those aged 5–29 years accounting for over 90% of it (0.41 years). The relatively narrow overall HALE gap can be partially explained by large contributions from some conditions for which females faced higher mortality or morbidity, particularly

morbidity from gynaecological diseases (–0.71 years) with women aged 20–69 years accounting for –0.66 years of this (Fig. 1.5). In 2021, the female lead in HALE declined slightly to 0.33 years, despite the addition of major contributions associated with mortality from COVID-19 (0.61 years) and other COVID-19 pandemic-related outcomes (0.3 years). This was due to declining contributions from mortality from causes including collective violence and legal intervention (down from 0.45 years in 2019 to 0.28 years in 2021) and ischaemic heart disease (down from 0.49 years in 2019 to 0.34 years in 2021) (Fig. 1.6).

The South-East Asia Region had the second lowest male–female gap in HALE in 2019 after the Eastern Mediterranean Region, at 0.90 years. Females benefitted from lower mortality from ischaemic heart disease, which accounted for 0.76 years of the female HALE advantage, primarily concentrated in those aged 30 years and older (0.75 years) (Fig. 1.5). The overall HALE gap remained relatively constant in 2021, at 0.91 years. Lower female mortality from COVID-19 and other COVID-19 pandemic-related outcomes accounted for 1.28 and 0.33 years, respectively, of the lead in female HALE. It is worth noting that the contribution associated with ischaemic heart disease halved, to 0.37 years, compared with 2019, indicating that as a competing risk the emerging deaths attributed to the COVID-19 pandemic have narrowed the male–female mortality gap associated with ischaemic heart disease (Fig. 1.6).

Females in the European Region enjoyed a 2.98-year HALE lead in 2019 compared with males. Lower female mortality from ischaemic heart disease among adults aged 30 years and older represents the greatest contributor, accounting for a total of 0.91 years. Lower female mortality from trachea, bronchus and lung cancers in the same age group adds another 0.44 years to the female HALE lead (Fig. 1.5). In 2021, the male–female HALE gap dropped to 2.76 years, with lower female mortality from COVID-19 and other COVID-19 pandemic-related outcomes accounting for 0.65 and 0.11 years, respectively, and the contribution associated with mortality from ischaemic heart disease (down to 0.75 years) and trachea, bronchus and lung cancers declining from the 2019 levels (down to 0.36 years). Contributions from all four causes to the female HALE lead were concentrated in those aged 30 years and older (Fig. 1.6).

1.4 Gap in HALE between high-income and other income groups in 2019 and 2021

Populations living in lower-resourced settings face many socioeconomic challenges and higher health risks than those living in higher-resourced settings (12), resulting in disproportionately higher morbidity and mortality from many causes and poorer overall health outcome as measured by life expectancy and HALE (1).

When looking at HALE in high-income countries compared with other income groups (as classified by the World Bank), the main health issues that explain the differences – and how much they contribute – vary depending on which groups are being compared.

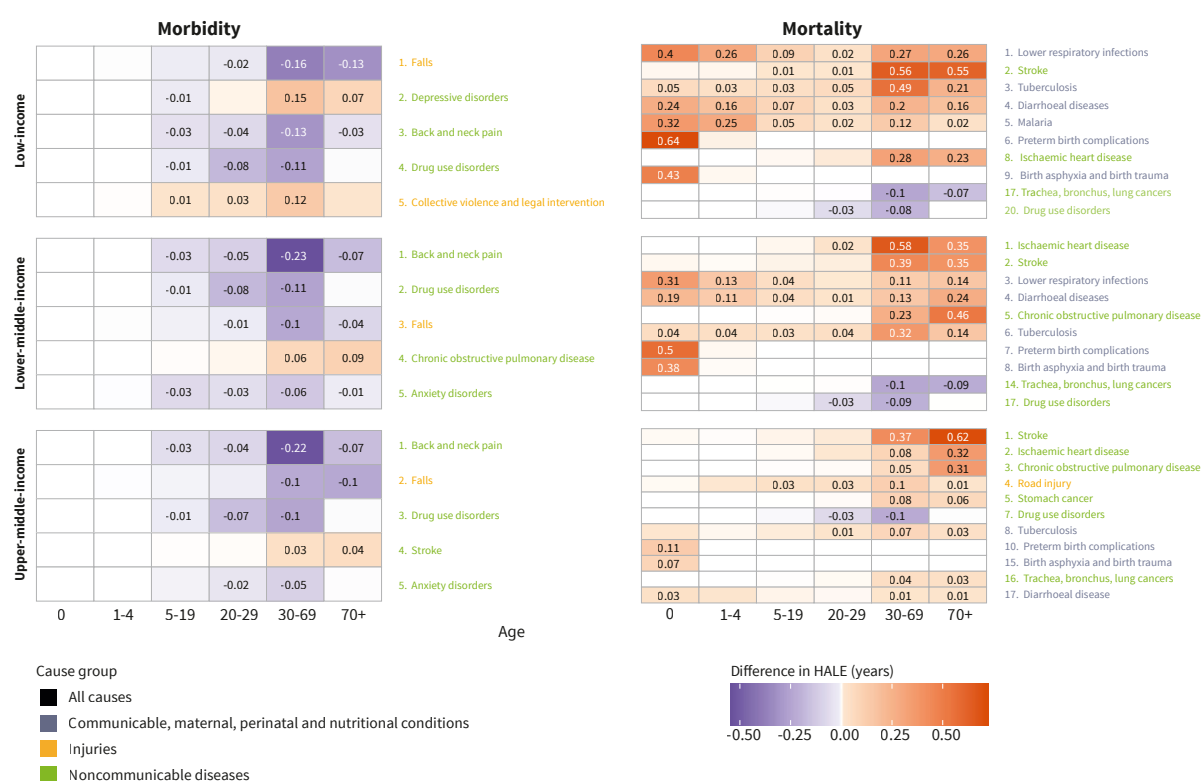
In 2019, people in high-income countries lived longer healthy lives partly because they had lower death rates from certain diseases and conditions, including birth asphyxia and birth trauma, preterm birth complications, lower respiratory infections, diarrhoeal diseases and

tuberculosis. These factors explained 4.08 years of the 12.62-year HALE gap between high- and low-income countries, and 3.08 years of the 8.61-year gap between high- and lower-middle-income countries.

However, these same health issues only explained a small part – about 0.38 years – of the 2.08-year HALE difference between high- and upper-middle-income countries.

Other major health conditions that contributed to the HALE gap were ischaemic heart disease and stroke. Lower death rates from these conditions in high-income countries accounted for: 0.52, 0.95 and 0.4 years of the HALE gap due to ischaemic heart disease, and 1.14, 0.76 and 1.01 years due to stroke, in comparisons with low-, lower-middle- and upper-middle-income countries, respectively (Fig. 1.7).

Figure 1.7 Leading contributing causes for the difference in healthy life expectancy (HALE) in 2019 between high-income and other income groups, by cause and age group



Source: WHO (1).



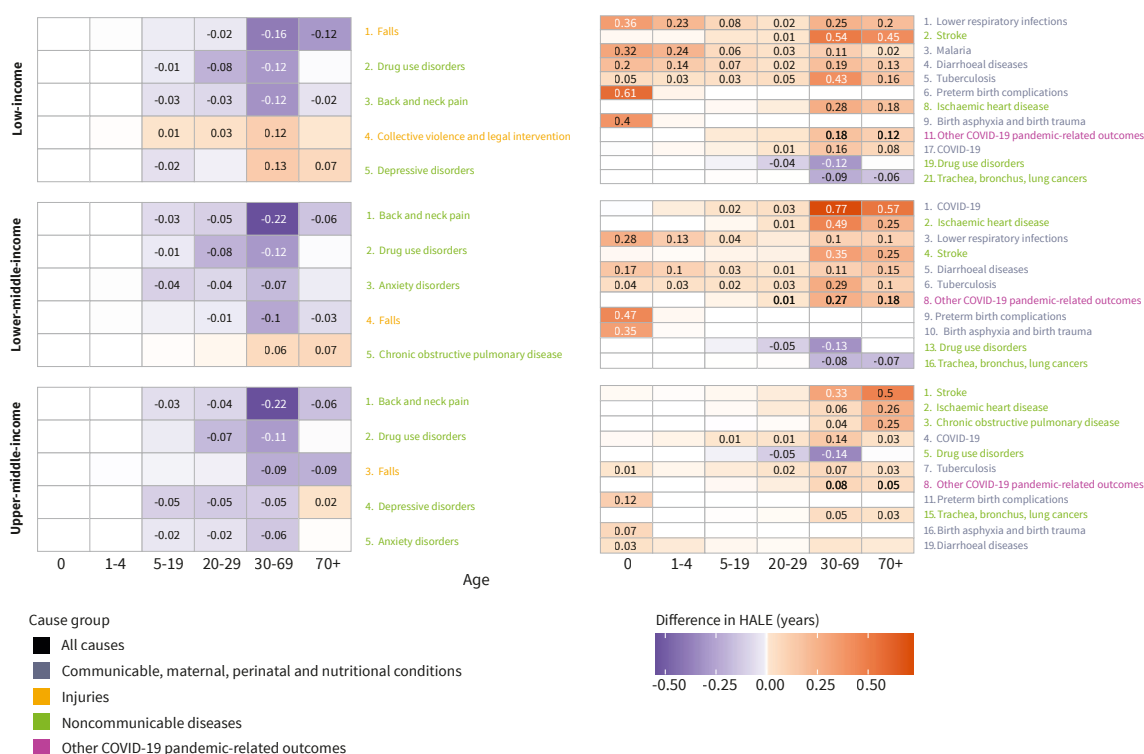
However, high-income countries did not succeed in every aspect. Higher mortality due to trachea, bronchus and lung cancers, particularly among adults aged 30 years and older, led to a negative contribution to the difference in HALE when comparing high-income countries against low-income (–0.17 years) and lower-middle-income countries (–0.2 years). Additionally, populations in high-income countries on average faced higher mortality from drug use disorders in comparison with all the other three income groups, costing high-income countries 0.12, 0.13 and 0.13 years of the HALE lead, respectively, compared with low-income, lower-middle-income and upper-middle-income countries, respectively (Fig. 1.7).

High-income countries also sustained higher rates of morbidity due to some injuries and NCDs. While overall morbidity in high-income countries was still lower than that in low- and lower-middle-income countries in 2019, it was higher than upper-middle-income countries and led to a nearly 1-year negative contribution to the HALE

gap between high- and upper-middle-income countries. The major causes leading to high-income countries' disadvantage include falls (losing 0.32, 0.16 and 0.21 years compared with low-, lower-middle- and upper-middle-income countries, respectively), back and neck pain (–0.21, –0.38 and –0.36 years, respectively) and drug use disorders (–0.20, –0.20 and –0.19 years, respectively) (Fig. 1.7).

These patterns in general continued in 2021. However, the different impacts of the COVID-19 pandemic in individual income groups has influenced the gaps in HALE to various degrees compared with high-income countries. Lower mortality due to COVID-19 and other COVID-19 pandemic-related outcomes in high-income countries accounted for the HALE lead of 0.25 years and 0.32 years, respectively, compared with low-income countries; 1.39 and 0.47 years, respectively, compared with lower-middle-income countries; and 0.21 and 0.14 years, respectively, compared with upper-middle-income countries (Fig. 1.8).

Figure 1.8 Leading contributing causes for the difference in healthy life expectancy (HALE) in 2021 between high-income and other income groups, by cause and age group



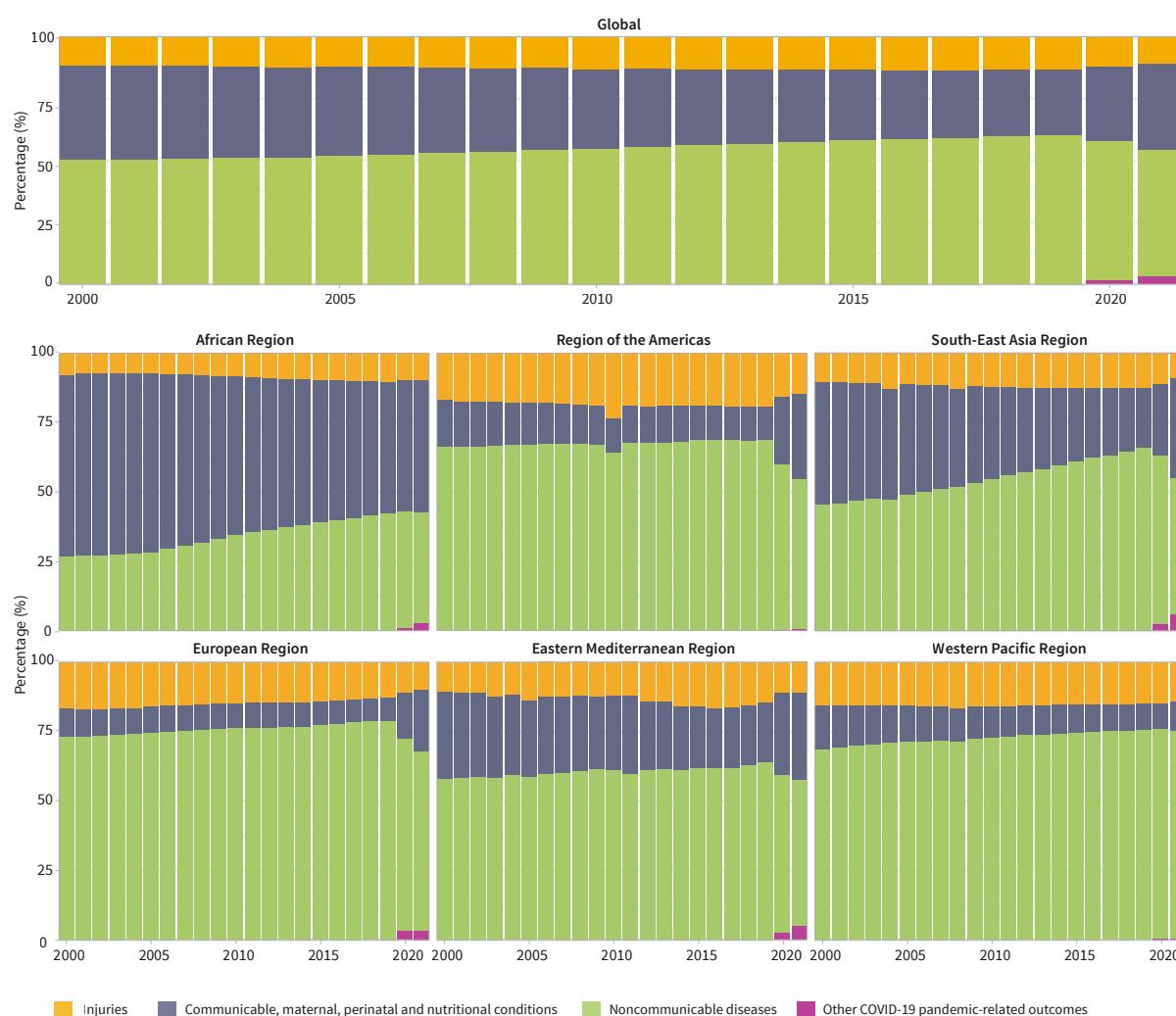
Source: WHO (1).

1.5 Progress in premature mortality reduction

Deaths occurring before the age of 70 years are considered premature and the majority of these deaths are preventable. Taking into account of the changing population structure over time and its difference across regions, the age-standardized death rate (ASR) for ages under 70 years was in steady decline from 2000 until the start of the COVID-19 pandemic in 2019, both globally and at regional level. It is evident that before the COVID-19 pandemic, NCDs were making up an increasing share of the ASRs of all deaths under age of 70 years across all WHO regions, while the share of communicable

diseases was in decline (Fig. 1.9). In 2019, the African Region had the highest ASR of all deaths under age of 70 years (665 per 100 000 population) among all WHO regions, over 80% higher than the global average (366 per 100 000 population) and nearly triple that of the Western Pacific Region (233 per 100 000), the region with the lowest ASRs. The African Region also had the largest share of communicable diseases in the overall ASR under age of 70 years in 2019, at 47.4%, some 5.5 times of that in the European Region (8.6%), which had the lowest share (Fig. 1.9).

Figure 1.9 Composition of causes of death in the age-standardized death rates for ages under 70 years, by WHO region, 2000–2021



Source: WHO (1).



The probability of dying before the age of 70 years is another important indicator for overall population health with a focus on mortality. Premature mortality had been in decline globally and in all WHO regions, from 2000 to the onset of the COVID-19 pandemic. Globally, a newborn had about 40.4% chance of dying before reaching the age of 70 years in 2000, declining to 29.9% in 2019. Health disparities associated with geography and income are also significant. Africa, the WHO region with the highest premature mortality in 2019, had a probability (46.3%) of dying before the age of 70 years that was double that of the European Region (23.2%) and the Western Pacific region (21.5%), which had the lowest probability. Similarly, the probability in low-income countries (46.8%) was also more than double that of high-income countries (19.8%).

The COVID-19 pandemic has had a marked impact on global health, with global premature mortality increasing by 15.0% between 2019 (29.9%) and 2021 (34.3%). The pandemic impacts were unequal across WHO regions and World Bank income groups. The mortality toll due to COVID-19 was heaviest in relative terms in the Region of the Americas and the South-East Asia Region, where premature mortality increased by 28.9% and 23.7%, respectively. In contrast, the pandemic had a limited impact in the African Region and the Western Pacific Region, where premature mortality increased by just 5.7% and 2.0%, respectively, between 2019 and 2021. By World Bank income groups, the premature mortality in low-income countries increased modestly, by just 5%, during the first two years of the pandemic, while all other income groups experienced a 14.3–17.8% increase.

Despite the overall progress pre-pandemic, stagnation was seen after 2015. The global annual rate of reduction (ARR) in premature mortality was just 1.1% between 2015 and 2019, down from 1.7% during 2000–2015. Among WHO regions, the Region of the Americas and the South-East Asia Region experienced the greatest slowdown, with ARR halving between 2015 and 2019 compared with 2000–2015. The Eastern Mediterranean Region was the only region where some acceleration was seen in 2015–2019, while the progress in the African Region slowed only slightly and the region continued to have the most rapid rate of decline in premature mortality, followed by the European Region. Similarly, progress slowed in 2015–2019 across all World Bank income groups, with low-income countries seeing the smallest relative reduction in ARR and remaining the group with the fastest declining premature mortality.

1.5.1 The prospect of premature mortality by 2050

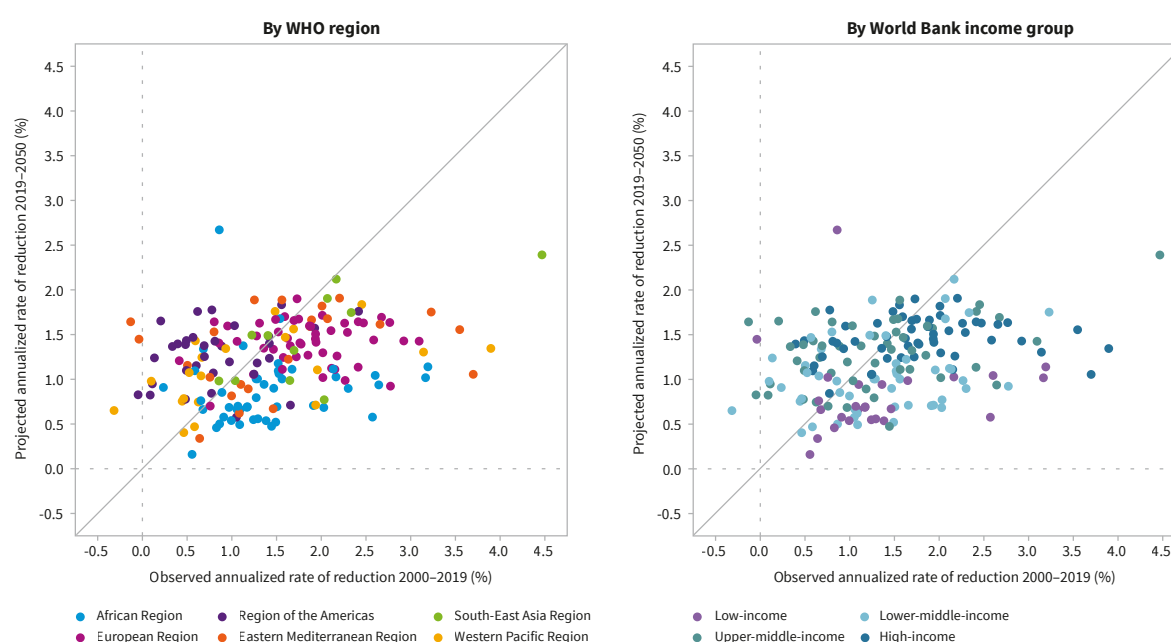
Using the WHO mortality estimates in 2019 as the baseline and incorporating the projected age–sex-specific mortality trends by country from the United Nations *World population prospects 2024* (13), it is anticipated that the reduction in premature mortality during the three decades between 2019 and 2050 would be moderate compared with the observed progress in 2000–2019. Globally, the probability of dying before the age of 70 years is projected to decline by 24.2% between 2019 and 2050, with reductions ranging from 20.2% in the African Region and 24.5% in the Eastern Mediterranean Region to 33.9% in the Western Pacific region and 35.4% in the South-East Asia Region. This represents a global slowdown, given it had taken one fewer decade to achieve a slightly larger reduction (26.1%) in 2000–2019, equivalent to an estimated ARR of 1.6%, compared with only a projected 0.9% in 2019–2050. Overall, only about one third of countries are projected to have accelerated progress in 2019–2050 compared with 2000–2019. This would largely occur in the Region of the Americas and the Western Pacific Region, where at least half of the countries (76% and 52%, respectively) would see an acceleration (Fig. 1.10). Upper-middle-income countries are the only World Bank income group where more than half (52%) of the countries are projected to see accelerated progress in 2019–2050, while only four (or 15%) of the low-income countries will achieve this (Fig. 1.10).

The Region of the Americas is the only WHO region that is expected to see some accelerated progress, with a projected ARR at 1.2% in 2019–2050, up from 0.9% in 2000–2019. While the ARR in the South-East Asia Region is expected to be largely stable, the other four regions are all expected to see considerable deceleration, with ARR dropping by up to 60% in the African Region. Slowed progress is projected for all World Bank income groups. Although a one-third reduction in premature mortality is expected in both upper-middle-income and high-income countries in 2019–2050, the ARRs are considerably lower than that in 2000–2019 (down by 31.4% and 19.5%, respectively). The ARR in low-income countries is expected to be more than halved the level in 2000–2019, leading to only about a one-fifth reduction in premature mortality in 2019–2050.

Should the global community wish to halve premature mortality by 2050 (14), considerable acceleration would be needed in reducing mortality rates in those aged 0–69 years in the next three decades. The global decline would have to be 1.4 times the observed rate in 2000–2019 to cut the premature mortality in 2019 by half by mid-century. The required regional rate of acceleration relative to the ARR in 2000–2019 would range from 1.1 times in the European Region and the Western Pacific region to 2.4 times in the Region of the Americas,

and from 1.2 times in upper-middle-income countries to 1.6 times in lower-middle-income countries. Allowing the mortality trends to continue as projected without interventions would shift a heavier burden of acceleration for meeting the target to future years. At the projected ARR for 2019–2050, a minimum of 2.5 times acceleration is required to halve premature mortality by 2050 globally, and a minimum of about a threefold acceleration in the African Region and low-income countries.

Figure 1.10 Annual rate of reduction in probability of death under 70 years of age, observed (2000–2019) versus projected (2019–2050)



Source: WHO (1).



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¹ Unless otherwise stated, all references accessed on 5 May 2025.

The background of the entire page is a solid blue color. Overlaid on this background is a repeating pattern of stylized circular charts. Each chart consists of a large outer circle and a smaller inner circle. The space between the two circles is divided into several radial segments, resembling a donut chart or a gauge. The charts are arranged in a grid-like fashion, with some overlapping. The number '2' is a large, bold, orange-yellow color. The small circular icon is a multi-colored ring, similar to the UN Sustainable Development Goals logo.

2.

Health-related Sustainable Development Goals

This chapter presents the latest available evidence of trends in health-related SDG indicators, assessing progress towards achieving the global targets. The topics covered in this chapter include child mortality, cause-specific mortality, infectious diseases, risk factors for health, and UHC and health systems.

2.1 Mortality-related Sustainable Development Goal indicators

2.1.1 Maternal and child mortality

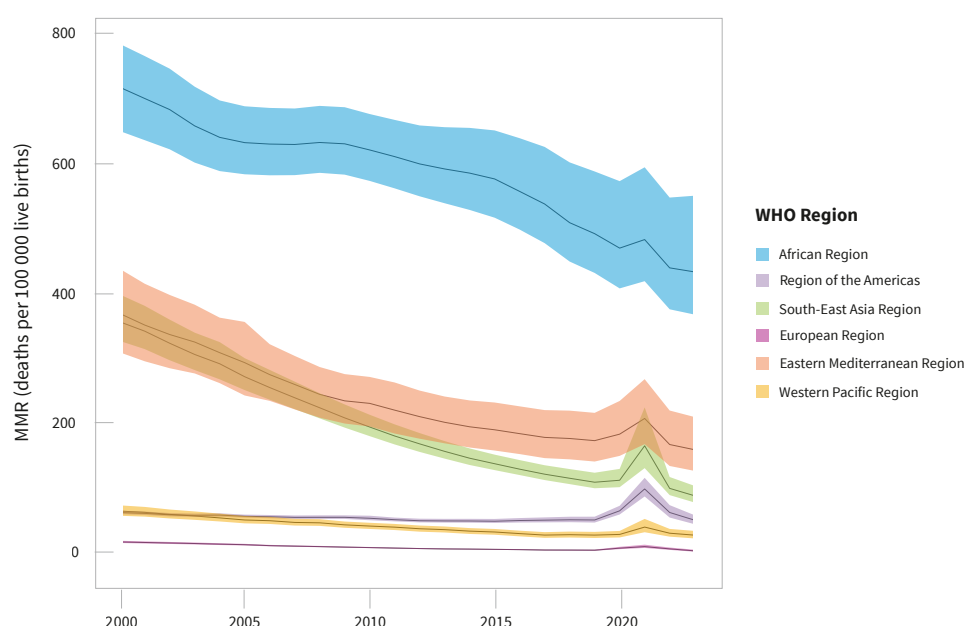
Maternal mortality

Globally, an estimated 260 000 (uncertainty interval [UI]: 230 000–309 000) women died in 2023 due to maternal conditions, equivalent to an overall maternal mortality ratio (MMR; SDG indicator 3.1.1) of 197 (UI: 174–234) maternal deaths per 100 000 live births. This represents a decline in the global MMR by 40% since 2000, when the rate was 328 (UI: 308–352) per 100 000 live births. Between 2000 and 2023, the average ARR in global MMR was 2.2% (UI: 1.4–2.8%). This led to a significant fall in the number of maternal deaths globally, from 444 000 in 2000 to 260 000 in 2023. A closer examination of the first eight years of the SDG era, between the beginning of 2016 and the end of 2023, reveals that the global MMR declined by 10%, from an estimated 220 (UI: 203–242) maternal deaths per 100 000 live births, at a global average ARR of 1.6% (UI: 0.04–2.7%) (1). If this ARR were to continue until

2030, the global MMR would be 177 in 2030 – two and a half times the SDG target (below 70) (1,2). Achieving the SDG target will require an average ARR of 14.8% over the remaining seven years (2024–2030). This represents an unprecedented challenge and would be equivalent to almost 700 000 deaths averted between 2024 and 2030, compared with a scenario where the 2016–2023 global ARR continues.

MMR declined in all WHO regions between 2000 and 2023. The greatest decline was achieved in the South-East Asia Region, where MMR declined by 73.6%, from 365 (UI: 336–407) in 2000 to 96 (UI: 86–112) maternal deaths per 100 000 live births in 2023 (Fig. 2.1). This equates to an average ARR of 5.8% (UI: 5.1–6.5%). MMR also declined by 56.0% in the Eastern Mediterranean Region, 55.2% in the European Region, 51.2% in the Western Pacific Region, and 39.6% in the African Region between 2000 and 2023. The smallest percentage reduction in MMR was in the Region of the Americas at 16.9%, equivalent to an ARR of 0.8% (UI: 0.2–1.3%).

Figure 2.1 Estimates of maternal mortality ratio (per 100 000 live births) 2000–2023, by WHO region



Source: WHO et al. (1).

In 2023, the African Region had a high MMR of 442 (UI: 376–560) per 100 000 live births and alone accounted for approximately 70% of global maternal deaths in 2023. The Eastern Mediterranean and South-East Asia Regions each accounted for about 12% of global maternal deaths, with MMRs of 167 (UI: 135–218) and 96 (UI: 86–112), respectively. The MMR was also low in the Region of the Americas, at 59 (UI: 52–66), and the Western Pacific Region, at 35 (UI: 30–42). The European Region had a very low MMR in 2023, at 11 (UI: 10–13) (Fig. 2.1) (1).

Under-five mortality

Child survival has improved remarkably since 2000, with the global under-five mortality rate (U5MR; SDG indicator 3.2.1) declining by more than half (52%) in the last two decades, from 77 (UI: 76–78) deaths per 1000 live births in 2000 to 37 (UI: 35–41) deaths per 1000 live births in 2023 (Fig. 2.2). Accordingly, the number of global under-five deaths fell to 4.8 million (UI: 4.5–5.3 million) in 2023, a number that is still unacceptably high, despite accounting for less than half of the under-five deaths in 2000 (10.1 million [UI: 9.9–10.2 million]) and almost two thirds less than the estimated 13 million (UI: 12.8–13.2 million) in 1990 (3).

Although the progress is noteworthy at global level, stark inequalities persist across regions and income levels. The WHO African Region remains the region with the highest U5MR, estimated at 67 (UI: 60–80) deaths per 1000 live births in 2023 (Fig. 2.2). The African Region alone accounted for 55% (2.6 million [UI: 2.4–3.1 million]) of the global deaths of children under age 5 in 2023. In 2023, the U5MR in low-income countries (LICs) (62 [UI: 56–75] deaths per 1000 live births) and lower-middle-income countries (LMICs) (43 [UI: 39–50] deaths per 1000 live births) was, respectively, 13 and 9 times the rate in high-income countries (HICs) (5 [UI: 5–5] deaths per 1000 live births) (3).

Globally, infectious diseases, including acute respiratory infections, malaria and diarrhoea, along with pre-term birth complications, birth asphyxia and trauma, and congenital anomalies remain the leading causes of death for children aged under 5. Access to basic lifesaving interventions such as skilled delivery at birth, quality postnatal care, early and continued breastfeeding and adequate nutrition, immunization and treatment for common childhood diseases can save many young lives (3).

Figure 2.2 Child mortality rate (per 1000 live births), by WHO region, 1990, 2000 and 2023



Source: UN IGME (3).



Neonatal mortality

The number of deaths occurring in the first 28 days of life (the newborn period) declined from 4.2 million (UI: 4.0–4.3 million) in 2000 to 2.3 million (UI: 2.1–2.6 million) in 2023. However, the 45% decline in the neonatal mortality rate (NMR; SDG indicator 3.2.2) from 2000 (31 [UI: 30–32] deaths per 1000 live births) to 2023 (17 [UI: 16–19] deaths per 1000 live births), has been slower compared with the 58% decline in the mortality rate of children aged 1–59 months (from 47 [UI: 47–48] deaths per 1000 children aged 28 days in 2000 to 20 [UI: 18–23] deaths per 1000 children aged 28 days in 2023) (Fig. 2.2). This has led to a larger share of global newborn deaths among total under-five deaths, up from 41% in 2000 to 48% in 2023 (3).

The chances of survival through the first 28 days of life vary depending on where a child is born. The African Region had the highest regional NMR in 2023, at 26 (UI: 23–31) deaths per 1000 live births, followed by the Eastern Mediterranean Region with 24 (UI: 21–29) deaths per 1000 live births. The South-East Asia Region had the highest regional NMR in 1990 at 53 (UI: 51–55) deaths per 1000 live births (Fig. 2.2), but has experienced a sharp decline with a ARR of 3.6% (UI: 3.2–4.0%) double that of the African Region (1.7% [UI: 1.1–2.0%]) in 1990–2023. Country-level NMR in 2023 ranged from 0.6 to 40.2 deaths per 1000 live births and the risk of dying before the 28th day of life for a child born in the highest-mortality country was approximately 65 times higher than the lowest-mortality country (3).

Meeting the SDG targets for child survival

If current trends continue, 60 countries will not meet the SDG target for under-five mortality. Of these, 47 countries will need to more than double their current rate of progress or reverse an increasing trend and then accelerate progress to achieve the SDG target by 2030. Even more countries are at risk of missing the SDG target for neonatal mortality. A total of 65 countries will need to accelerate the mortality decline to meet the target on time, with 59 of those countries needing to more than double their current rate of decline or reverse an increasing trend to meet the target by 2030. Reaching this target in all countries will avert 8 million under-5 deaths between 2024 and 2030, some 42% of which would be among neonates (3).

2.1.2 Mortality due to noncommunicable diseases

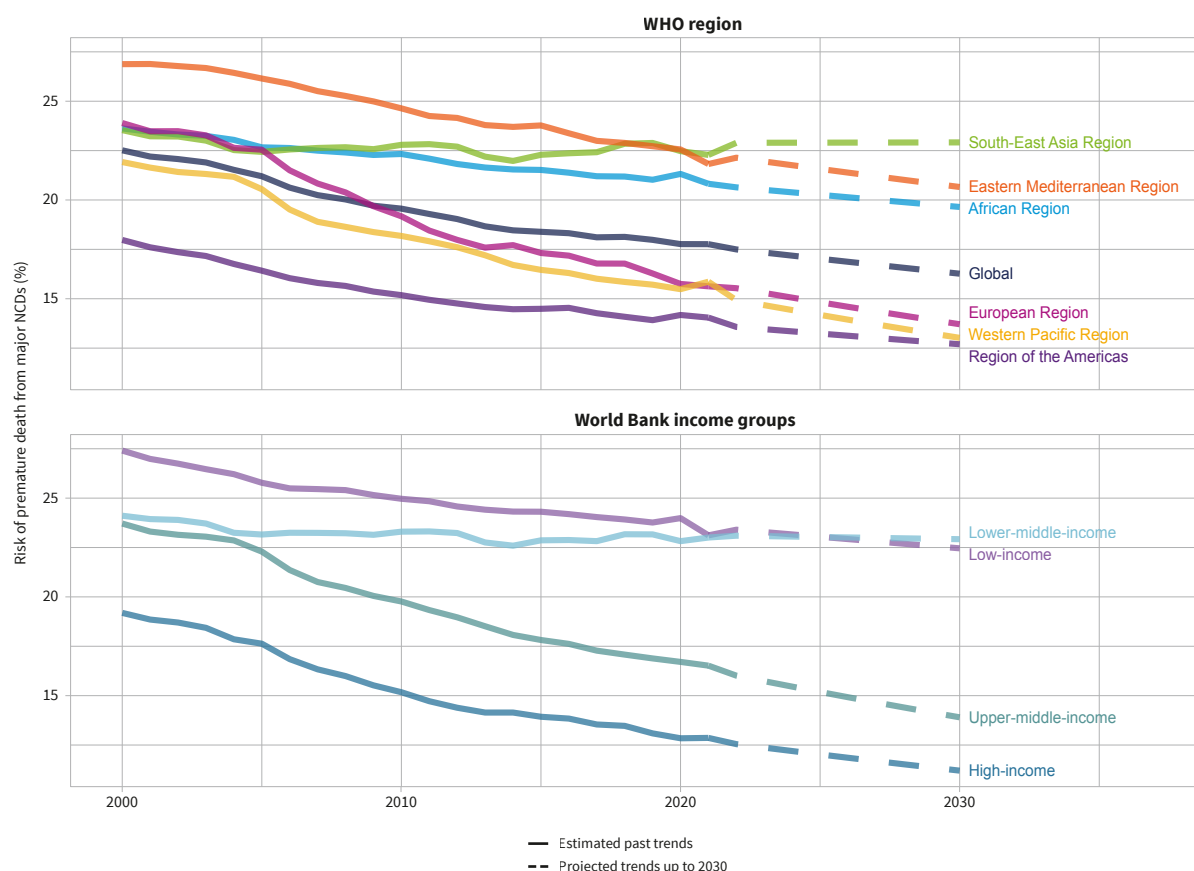
Globally, in 2021, some 18 million people under the age of 70 years died from an NCD, accounting for over half of deaths in that age group. The risk of premature death from NCD, which is measured by SDG indicator 3.4.1,

is the unconditional probability of dying from any of the four main NCDs (cardiovascular disease, cancer, diabetes or chronic respiratory disease) between the ages of 30 and 70 years (2). It is estimated that, in 2000, a 30-year-old had a 22.5% chance of dying from one of the four main NCDs before the age of 70 years. This risk fell to 18.4% in 2015 and further to 18.0% in 2019, just prior to the onset of the COVID-19 pandemic (4). The reliability of the estimated risk during 2020–2021 was potentially affected by disruptions associated with the COVID-19 pandemic as countries' ability to monitor trends in NCD mortality was compromised and may have resulted in misclassification of NCD deaths. Where reliable data are available, the risk of premature death from an NCD plateaued during this period.

The observed pre-pandemic trend translates to an ARR of 1.3% between 2000 and 2015 and 0.5% between 2015 and 2019, indicating a remarkable deceleration in progress. With the required ARR to meet the one-third reduction in premature NCD mortality standing at 2.7%, the pre-pandemic progress since 2015 was equivalent to just one fifth of what is needed to meet the target and calls for coordinated action to prevent and treat NCDs.

The level and pace of progress vary by WHO region. The Eastern Mediterranean Region faced the highest risk of premature NCD mortality in 2000 (26.9%), which progressively declined to 22.7% in 2019, at a level that was only lower than the South-East Asia Region (22.9%) that year. Despite the higher risk compared to most other regions, the Eastern Mediterranean Region is the only region where some acceleration of progress was observed during the SDG era (ARR at 1.2% in 2015–2019 compared with 0.8% in 2000–2015). In contrast, the South-East Asia Region saw overall stagnation in 2000–2019, with a slight uptick in risk between 2015 and 2019. The African Region (23.6% in 2000) experienced a moderate but steady decline, with an ARR at about 0.6% in 2000–2019 and reaching a premature NCD mortality of 21.0% in 2019. The European (23.9% in 2000) and Western Pacific (21.9% in 2000) Regions saw the fastest decline among all regions during both 2000–2015 (ARRs at 2.2% and 1.9%, respectively) and 2015–2019 (ARRs at 1.5% and 1.2%, respectively), reaching premature NCD mortality of 16.3% and 15.7%, respectively, in 2019. The Region of the Americas had the lowest risk of NCD premature mortality in 2000 (18.0%) and remained the lowest in 2019 (13.9%), although the pace of progress slowed, with an ARR at 1.4% in 2000–2015 and 1.1% in 2015–2019.

Based on these results, it is evident that neither the world nor any one region is on track to meet the target of reducing the risk for premature NCD mortality by one third by 2030. The South-East Asia Region, where premature NCD mortality has remained high since 2000, is at greatest risk of missing the target (Fig. 2.3).

Figure 2.3 Observed and projected trends for risk of premature mortality from NCDs (%), by WHO region

Source: WHO (4).

2.1.3 Mortality due to injuries

Road injury

In 2021, an estimated 1.18 million (UI: 1.05–1.32 million) people died due to road injuries globally (SDG indicator 3.6.1). The greatest burden was observed in South-East Asia Region, with around 318 000 (UI: 283 000–358 000) deaths, or 28% of the global burden, and the Western Pacific Region with almost 296 000 (UI: 276 000–316 000) deaths, or 25% of the global burden. Together, these regions account for more than half of the global road traffic deaths. These are followed by the African Region (229 000 [UI: 185 000–274 000] deaths); the Region of the Americas (146 000 (UI: 135 000–158 000) deaths); the Eastern Mediterranean Region (128 000 [UI: 108 000–145 000] deaths); and the European Region (64 000 [UI: 58 000–68 000] deaths) (4,5).

The African Region had the highest crude death rate (CDR) in 2021 (19.5 [UI: 15.7–23.4] per 100 000 population). This was nearly three times that of the European Region, which had the lowest rate of 6.8 (UI: 6.2–7.3) per 100 000 population (Fig. 2.4). By World Bank income levels, the highest death rate from road injuries was observed in LICs, at 21.1 (UI: 16.8–25.2) per 100 000 population, while the lowest was in HICs, at 8.1 (UI: 7.5–8.6) per 100 000 population. Disparity between sexes was also evident. Males were at elevated risk of dying from road injuries compared with females. Globally, the male-to-female ratio of CDRs from road injuries was 3.0, ranging from 2.3 in the African Region to above 3.5 in the South-East Asia Region and the Region of the Americas (4).



Road fatalities declined by 5.5% from 2010–2021, falling from 1.25 million (UI: 1.11–1.39 million) deaths in 2010 (4). This is equivalent to a decline from 17.9 (UI: 15.7–19.9) deaths per 100 000 population to 14.9 (UI: 13.2–16.6) deaths per 100 000 population, and is in spite of the growth in global population and number of vehicles. This indicates that while efforts to enhance road safety are making progress, they remain insufficient to achieve the United Nations Decade of Action for Road Safety 2021–2030 goal of reducing deaths and injuries by half by 2030 (6). There was also a universal decline in the CDRs across all regions in the period. The slowest progress was seen in the Region of the Americas, although rates still declined by 8.9%, from 15.6 (UI: 14.5–16.7) per 100 000 population in 2010 to 14.2 (UI: 13.1–15.4) per 100 000 population in 2021.

Suicide

An estimated 727 000 people died from suicide globally in 2021, corresponding to a crude suicide mortality rates (CDR; SDG indicator 3.4.2) of 9.2 (UI: 6.9–11.6) per 100 000 population. The rate in males (12.4 (UI: 9.6–15.4) per 100 000) was more than double that in females (5.9 (UI: 4.2–7.7) per 100 000). The highest male-to-female ratio was found in the European Region (3.4) and the Region of the Americas (3.7), while the lowest was in the South-East Asia Region (1.4) (4).

Crude suicide mortality rates and their change over time exhibited wide variations across WHO regions. The European Region had the highest rate among all regions in 2000 (21.5 [UI: 19.9–23.0] per 100 000 population), but has seen rapid decline, with the rate falling by over two fifths, to 12.4 (UI: 10.3–14.4) per 100 000 population in 2021. A rapid decline was also seen in the Western Pacific Region, where the rate in 2000 (14.7 [UI: 10.1–16.9] per 100 000 population) fell by 35% to 9.5 (UI: 7.1–12.3) per 100 000 population in 2021. However, a slight increase was seen in the African Region, where the CDR from suicide in 2021 (7.3 [UI: 4.5–10.7] per 100 000 population) was about 5% higher than in 2000 (6.9 [UI: 4.5–10.0] per 100 000). The Region of the Americas has seen a concerning trend, with the rate increasing by nearly 30% from 7.6 (UI: 7.1–8.3) per 100 000 population in 2000 to 9.9 (UI: 8.9–11.0) per 100 000 population in 2021. The South-East Asia Region had the second highest rate after the European Region in 2021 (10.1 [UI: 7.4–12.6] per 100 000), yet experienced a 20.4% decline from the rate in 2000 (12.7 [UI: 9.4–15.1] per 100 000). The Eastern Mediterranean Region continued to have the lowest rate throughout the period, with a 24.3% decline from 4.7 (UI: 2.7–7.4) per 100 000 population in 2000 to 3.6 (UI: 2.0–5.9) per 100 000 population in 2021 (Fig. 2.4) (4).

Homicide

Nearly 484 000 (UI: 360 000–648 000) people were victims of intentional homicide (defined as the unlawful death inflicted upon a person with the intent to cause death or serious injury) in 2021. Despite having a slight increase from 480 000 (UI: 400 000–578 000) in 2000, this corresponded to a decline in CDR of 27.1%, from 7.8 (UI: 6.5–9.4) per 100 000 population (SDG indicator 16.1.1) in 2000 to 6.1 (UI: 4.5–8.2) per 100 000 population in 2021 (Fig. 2.4). About 80% of the victims were men (4).

The WHO Region of the Americas had the highest mortality burden from homicide in 2021, with a total of 198 000 (UI: 169 000–232 000) deaths, equivalent to a CDR of 19.3 (UI: 16.4–22.6) per 100 000 population – triple the global average. This region accounted for 40.9% of the total global homicide deaths but only 12.9% of the global population. In contrast, the Western Pacific Region accounted for about a quarter of the global population, but only 6.1% of global homicide deaths, with a CDR (1.5 (UI: 1.1–2.1) per 100 000 population) that was 8% of that in the Region of the Americas. The African Region had the second highest CDR in homicide mortality in 2021, with a CDR that was only half of that in the Region of the Americas, at 9.7 (UI: 5.9–15.0) per 100 000 population, accounting for 114 000 deaths. Compared with the Region of the Americas and the African Region, the CDR of homicide was also relatively low in the European, South-East Asia and Eastern Mediterranean Regions, at 2.6 (UI: 2.1–3.3) per 100 000 population, 3.7 (UI: 2.7–4.9) per 100 000 population and 5.4 (UI: 3.2–8.5) per 100 000 population, respectively (Fig. 2.4) (4).

Men face disproportionately higher risk of homicide deaths than women and the sex disparity was unevenly distributed across WHO regions. Globally, the male-to-female ratio for CDR for homicide mortality in 2021 was 4.0, ranging from 2.9 in the European Region to 7.7 in the Region of the Americas. A strong age pattern was also observed, with young adults aged 20–29 years having the highest age-specific mortality rate, peaking at over 12 deaths per 100 000 population globally and up to nearly 40 deaths per 100 000 population in the Region of the Americas (4).

Unintentional poisoning

Globally, about 59 000 (UI: 32 000–91 000) people died from unintentional poisoning in 2021, a decline of approximately 4600 from 2000 (64 000 [UI: 47 000–92 000]). This represents a decline of a quarter from a CDR of 1.0 (UI: 0.8–1.5) to 0.7 (UI: 0.4–1.1) per 100 000 population (SDG indicator 3.9.3). The highest CDRs in 2021 were observed in the WHO African Region, at 1.2 (UI: 0.7–2.1) per 100 000 population, closely followed by the Western Pacific Region at just under 1.2 (UI: 0.5–1.7)

per 100 000 population. The greatest decline in CDR was achieved in the European Region, with a decline of nearly two thirds between 2000 and 2021, followed by the South-East Asia and Eastern Mediterranean Regions, with a 36% and 30% decline, respectively. However, the rate stagnated in the African Region and a slight increase was seen in the Region of the Americas and the Western Pacific Region (Fig. 2.4) (4).

Sex and age disparities were observed. Men are at higher risk of dying from unintentional poisoning. Men died at a rate 67% higher than women in 2021, primarily

due to occupational hazards and higher rates of risky behaviours. The Region of the Americas and the European Region saw the highest male-to-female ratios, at 2.2 and 2.6 respectively, whereas the lowest ratio was in the African Region, at 1.4. Additionally, the youngest and oldest populations were at the highest risk of dying from unintentional poisoning, with those aged under 5 years and 65 years and older accounting for just 18% of the global population but 38% of the global deaths from unintentional poisoning in 2021 (4).

Figure 2.4 Crude death rates for injuries (per 100 000 population), by type, sex and WHO region, 2000 and 2021



AFR: African Region; AMR: Region of the Americas; SEAR: South-East Asia Region; EUR: European Region; EMR: Eastern Mediterranean Region; WPR: Western Pacific Region.

Source: WHO (4).

2.1.4 Mortality attributable to environmental risk factors

Air pollution mortality

Almost the entire global population is exposed to outdoor (ambient) air pollution in both cities and rural areas and more than a quarter (2.4 billion) is exposed to dangerous levels of household air pollution. Exposure to air pollution increases the risk for many negative health outcomes (7). Five health outcomes – acute lower respiratory infections,

lung cancer, cerebrovascular diseases, ischemic heart disease and chronic obstructive pulmonary disease – are currently included in SDG indicator 3.9.1 on mortality attributed to air pollution from particulate matter. In 2019, an estimated 6.7 million deaths worldwide were attributable to indoor and outdoor air pollution from particulate matter jointly – corresponding to an age-standardized mortality rate of 104 (UI: 81–130) deaths per 100 000 population (8,9). People living in LICs and middle-income countries (MICs) disproportionately bear the burden of outdoor air pollution (see also section 2.3.3 subsection “Air pollution”).



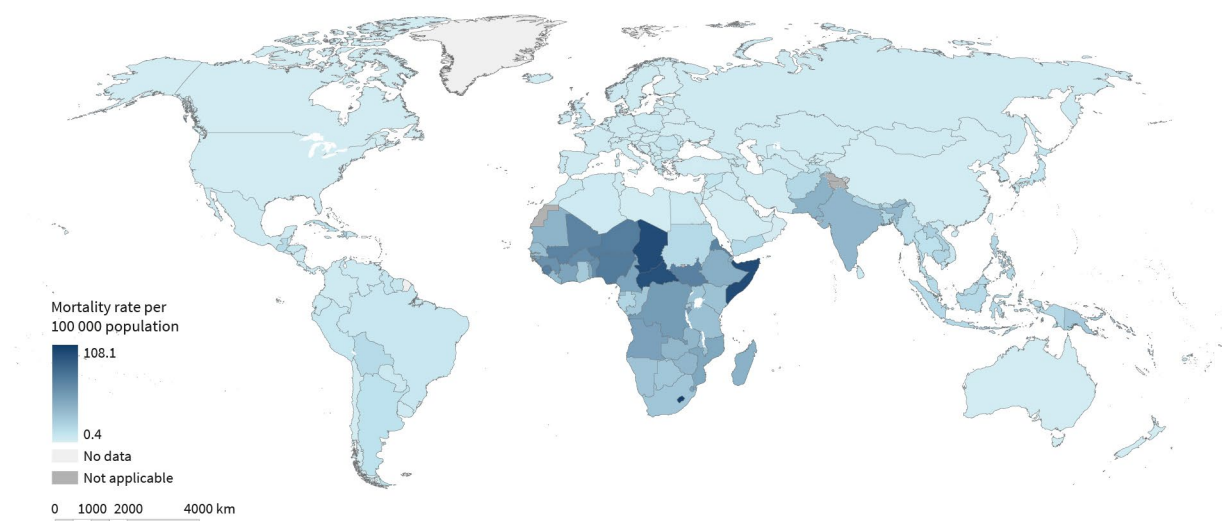
WASH mortality

Millions of people globally lack access to safe water, sanitation and hygiene (WASH) services and consequently suffer from or are exposed to a multitude of preventable illnesses.

Four health outcomes – diarrhoea, acute respiratory infections, undernutrition and soil-transmitted helminthiasis – are included in the SDG indicator 3.9.2 on mortality attributed to unsafe WASH. It is estimated that 1.4 million deaths in 2019 were attributable to unsafe

WASH services, of which the vast majority were attributable to diarrhoea (73.9%) and acute respiratory infections (25.4%) (10). The global WASH-attributable mortality rate was 18.3 deaths per 100 000 population, ranging from 3.7 in HICs up to 41.7 in LICs (10). Among the six WHO regions, the highest mortality rate was found in the African Region (46.7 per 100 000 population) and the South-East Asia Region (29.6 per 100 000 population) in 2019 (11). These two regions alone accounted for 79% of the total global deaths due to unsafe WASH (Fig. 2.5).

Figure 2.5 Mortality rate attributed to exposure to unsafe WASH services (per 100 000 population), 2019



Source: WHO (10).

2.2 Infectious diseases

In the infectious disease section, the indicators on new HIV infections, tuberculosis and malaria incidence, hepatitis B and neglected tropical diseases are monitored towards progress within SDG target 3.3. In addition, this section also presents the status of AMR.

2.2.1 HIV

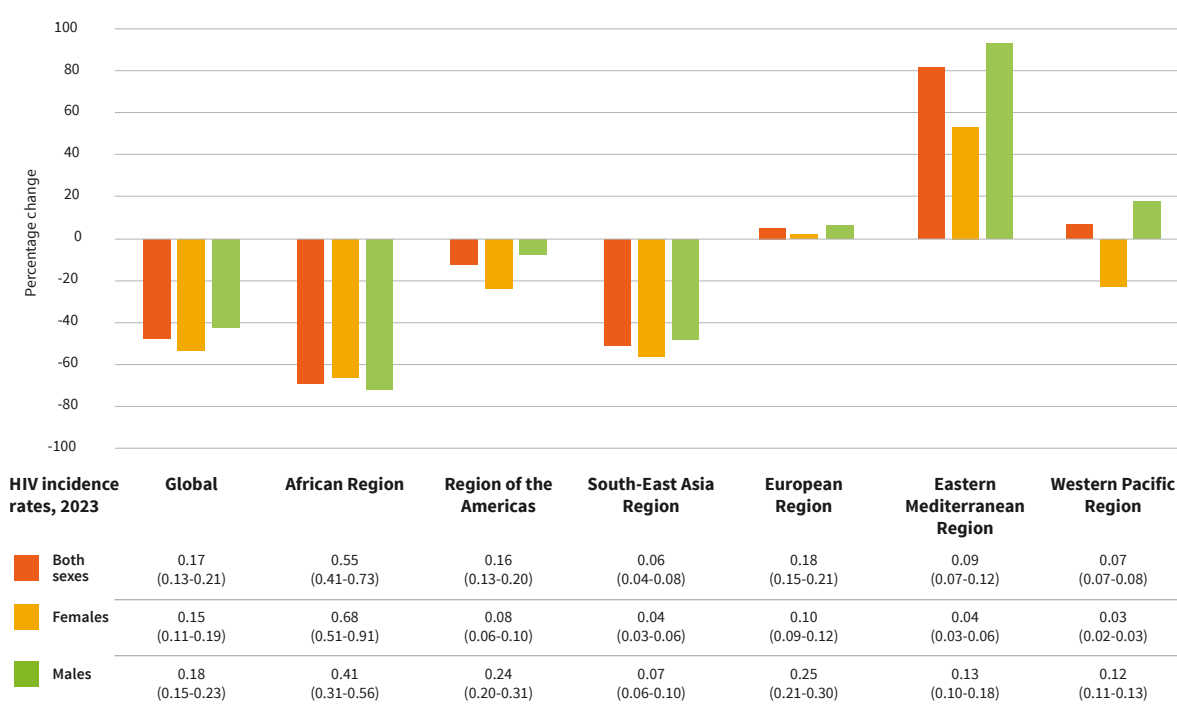
Globally, an estimated 39.9 million (UI: 36.1–44.6 million) people were living with HIV by the end of 2023, of which 30.7 million (UI: 27.0–31.9 million) were receiving lifesaving antiretroviral therapy (12,13).

In 2023, an estimated 1.3 million (UI: 1.0–1.7 million) HIV infections occurred, down from 2.8 million (UI: 2.3–3.6 million) in 2000 and 2.1 million (UI: 1.7–2.7 million) in 2010. Globally, the HIV incidence rate (number of new HIV infections per 1000 uninfected population, SDG indicator 3.3.1) was 0.17 (UI: 0.13–0.21) in 2023, a 48% reduction from 0.32 (UI: 0.25–0.40) in 2010. Regions show different levels of progress since 2010: An 81%

increase in HIV incidence rate was observed in the Eastern Mediterranean Region, while it is notable that this region had the lowest number of new infections across all WHO regions in 2023. The South-East Asian and the African Regions recorded a 51% and 69% decrease in HIV incidence rate in the same time period, respectively, while the latter region continued to bear the heaviest HIV burden in 2023. Further disaggregation by sex show that the biggest reduction in HIV incidence rate between 2010 and 2023 was among males in the African Region (–72%) (Fig. 2.6) (12,14,15).

The HIV pandemic and the global response has made progress over the past decade, with fewer people acquiring HIV in 2023 than at any point since the mid-1990s. However, the number of new infections in 2023 was still more than three times the 2025 target of 370 000 or fewer new infections. If this trend continues, the target of ending AIDS as a public health threat by 2030, which requires a 90% reduction in new infections against the 2010 baseline, will not be reached (16).

Figure 2.6 Percentage change in the number of new HIV infections per 1000 uninfected population, between 2010 and 2023, and HIV incidence rates, 2023, globally and by WHO region



Note: Percentage change in HIV incidence rate was calculated from unrounded incidence rates in the figure and the corresponding text.

Source: UNAIDS/WHO estimates, 2024 (14,15).



2.2.2 Tuberculosis

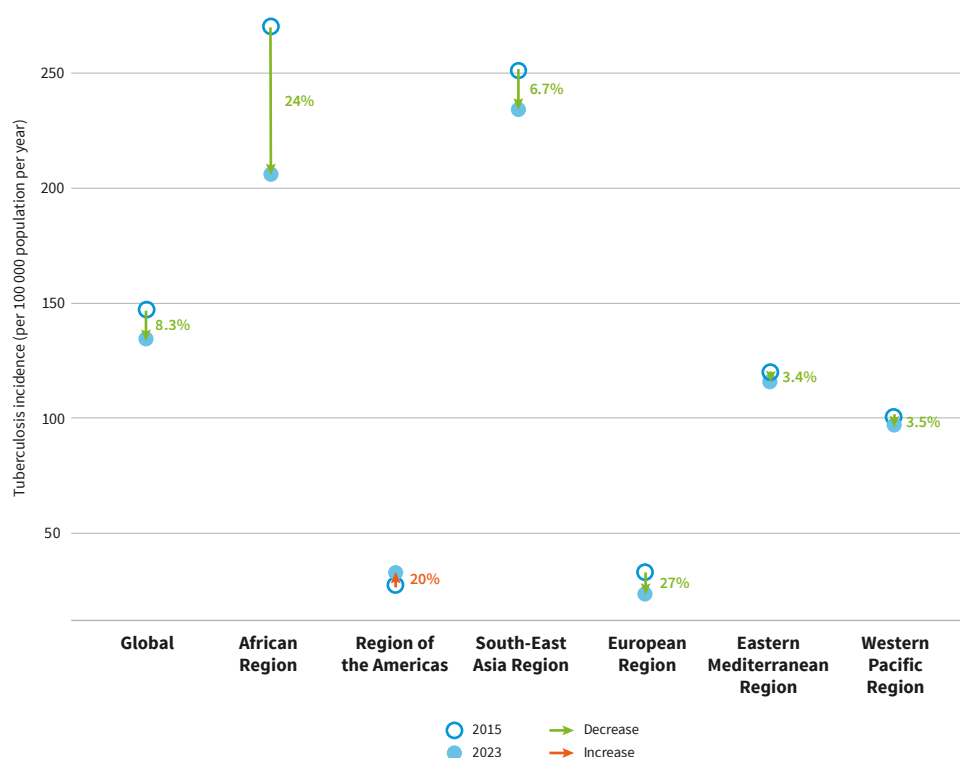
The global increase in new TB cases that has been observed since the start of the COVID-19 pandemic has slowed and is now stabilizing. In 2023, an estimated 10.8 million people (UI: 10.1–11.7 million) developed TB, a slight rise from 10.7 million (UI: 10.0–11.5 million) in 2022. However, this remains higher than the 10.4 million (UI: 9.7–11.1 million) cases in 2021 and 10.1 million (UI: 9.5–10.7 million) in 2020 (18).

Among those affected who developed TB in 2023, some 55% were men, 33% were women and 12% were children and young adolescents aged 0–14 years. Most of the

people who develop TB disease each year are in 30 high TB burden countries, accounting for 87% of the global total in 2023 (18).

The global TB incidence rate (SDG indicator 3.3.2) in 2023 was 134 (UI: 125–145) per 100 000 population, similar to the rate in 2022, and an increase from 129 (UI: 121–136) per 100 000 population in 2020. The net reduction between 2015 and 2023 was 8.3%, falling short of the WHO End TB Strategy target of an 80% reduction by 2030 (17,18). The African and European regions showed the most progress, with decreases of 24% and 27%, respectively (Fig. 2.7) (18).

Figure 2.7 Percentage change in tuberculosis incidence rate between 2015 and 2023, globally and by WHO region



Note: Percentage change in TB incidence rate was calculated from unrounded incidence rates.

Source: Global tuberculosis report 2024 (18).

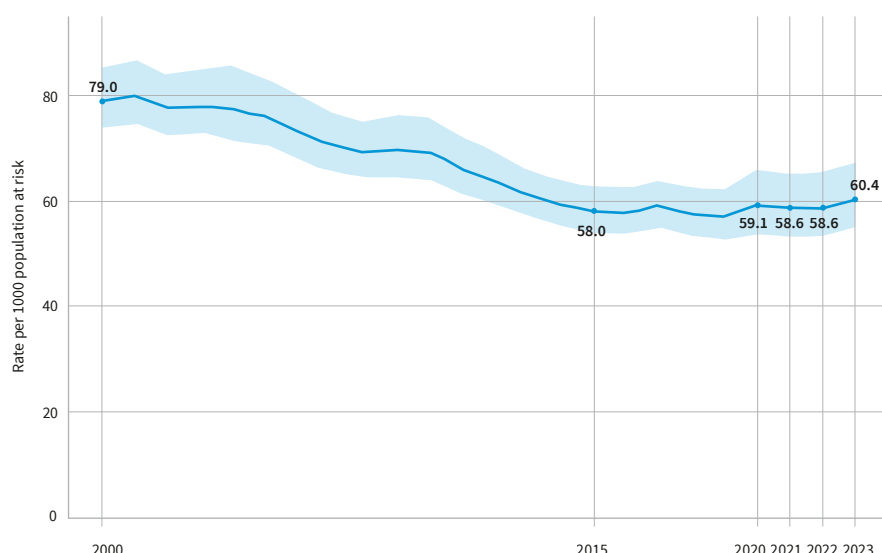
2.2.3 Malaria

In 2023, there were an estimated 263 million (UI: 238–294 million) malaria cases globally, an increase of 11 million cases from the previous year. The global incidence rate (SDG indicator 3.3.3) in 2023 was 60.4 (UI: 54.8–67.5) cases per 1000 population at risk, a rise from 58.6 (UI: 53.1–65.6) cases per 1000 population at risk in 2022, and a 4.1% increase from 58.0 (UI: 53.8–62.9) cases per 1000 population at risk in 2015 (Fig. 2.8). The WHO African Region continues to carry the heaviest burden of the disease, accounting for an estimated 94% of cases and 95% of deaths worldwide in 2023 (19).

The Global technical strategy for malaria 2016–2030 (GTS) target for reducing malaria case incidence by 90% by 2030 from the 2015 baseline is unlikely to be achieved (20). In 2023, global malaria incidence remained nearly three times higher than the level required to meet the target. However, the WHO South-East Asia Region is on track to meet the GTS 2030 target (19).

Despite these challenges, some notable progress towards elimination has been made. The number of malaria-endemic countries declined from 85 in 2022 to 83 in 2023, as two countries maintained zero indigenous cases for three consecutive years. By 2024, a total of 26 countries that were endemic for malaria in 2000 have achieved this milestone (19).

Figure 2.8 Global incidence rate of malaria (per 1000 population at risk), 2000–2023



Source: World malaria report 2024 (19).

2.2.4 Hepatitis B

As a major public health challenge of this decade, viral hepatitis B and C lead to chronic disease for hundreds of millions of people and can both cause deaths from liver cirrhosis and liver cancer.

Globally, 1.3 million people died from viral hepatitis-related causes in 2022, including 1.1 million deaths attributed to hepatitis B alone. This represents an increase of deaths attributed to chronic hepatitis B from the 820 000 in 2019. The Western Pacific Region bears the greatest mortality burden, with 518 000 deaths (or 47.1% of the global total) related to hepatitis B (21). Rapid scale-up of treatment is therefore critical to prevent deaths among people living with chronic hepatitis B. Without this

expansion in access, the world will face increasing cases of liver cancer in the next generation, with associated increasing care costs and hepatitis-related deaths.

In 2022, an estimated 254 million people were living with hepatitis B globally. About 5% of the general population in the WHO African Region and Western Pacific Region are living with hepatitis B. Children under the age of five are at risk of vertical (mother-to-child) transmission of hepatitis B, notably in settings with high hepatitis B prevalence. Notably, the African Region is the only region where the hepatitis B prevalence among children younger than five years (SDG indicator 3.3.4) exceeded 1% and yet only 17% of newborns in the Region receive the hepatitis B birth dose vaccination (22).



Global new infections of viral hepatitis dropped from 3.0 million in 2019 to 2.2 million in 2022, including 1.2 million newly infected with hepatitis B. The African Region accounted for 63% (or 771 000) of all new hepatitis B infections globally in 2022. This underscores the importance of a focus on scaling up access to viral hepatitis services in Africa, including the hepatitis B birth-dose vaccination, expansion of hepatitis B treatment in the general population and specifically among pregnant women as recently recommended in the 2024 WHO guidelines on people with chronic hepatitis B infection (23).

2.2.5 Neglected tropical diseases

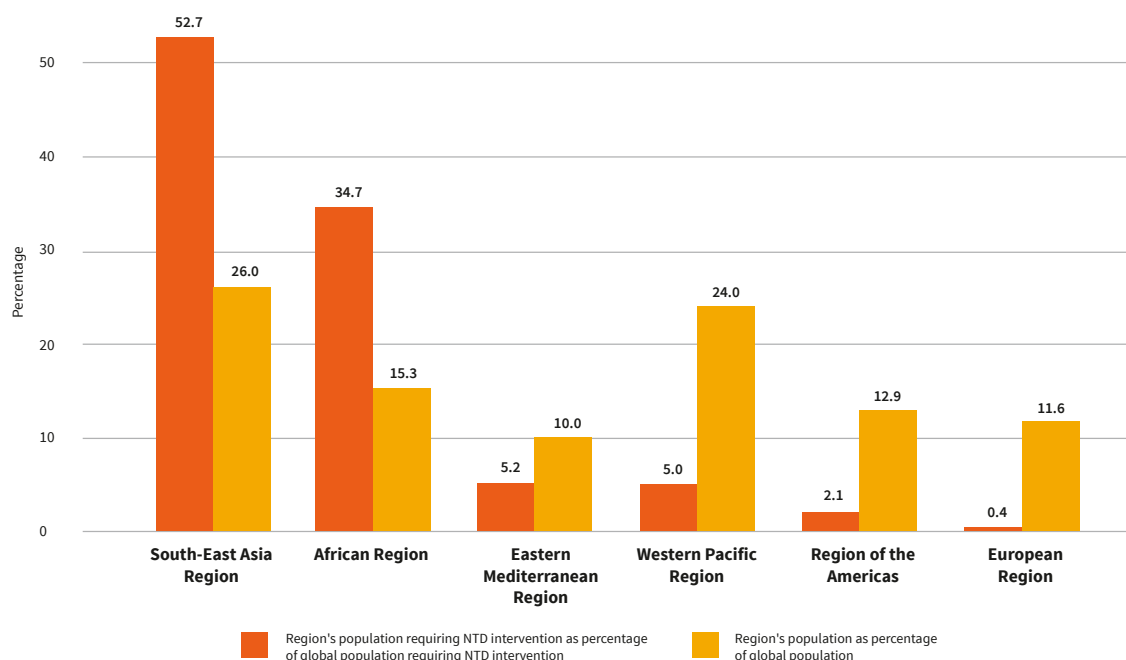
Neglected tropical diseases (NTDs) are a diverse group of conditions with complex epidemiology.¹ They affect some of the world's most vulnerable people, including those living in remote or impoverished communities, creating NTD-related generational cycles of poverty. Addressing the burden of NTDs can break this cycle and

contribute to the health, social and economic well-being of entire communities. Globally, in 2023, 1.495 billion people required interventions against NTDs (SDG indicator 3.3.5), about 122 million people fewer than in 2022. The 2023 number represents a 32% decrease from 2010, falling short of the global target of 90% reduction by 2030. The South-East Asia Region and the African Region, home to 41% of the global population, continue to disproportionately bear the highest burden of NTDs, accounting for 87% of the global number of people needing interventions in 2023 (Fig. 2.9) (24,25).

As of December 2024, 54 countries, areas and territories worldwide have eliminated at least one NTD, as acknowledged by WHO. This is still only about halfway towards the global target of at least 100 countries eliminating at least one NTD by 2030 (24,25).

This slow progress reflects the challenges of addressing NTDs on a global scale. Massive efforts, including through innovative solutions, are needed to accelerate progress in the global fight against NTDs, especially in the context of uncertain and evolving health, political and financial landscapes.

Figure 2.9 Share of people requiring interventions against NTDs and share of population out of global total, by WHO region, 2023



Note: The regions are shown in descending order of the number of people requiring interventions against NTDs.

Source: Global Health Observatory (24).

¹ NTDs include: Buruli ulcer; Chagas disease; dengue and chikungunya; dracunculiasis; echinococcosis; foodborne trematodiasis; human African trypanosomiasis; leishmaniasis; leprosy; lymphatic filariasis; mycetoma, chromoblastomycosis and other deep mycoses; noma; onchocerciasis; rabies; scabies and other ectoparasitoses; schistosomiasis; soil-transmitted helminthiasis; snakebite envenoming; taeniasis/cysticercosis; trachoma; and yaws.

2.2.6 Antimicrobial resistance

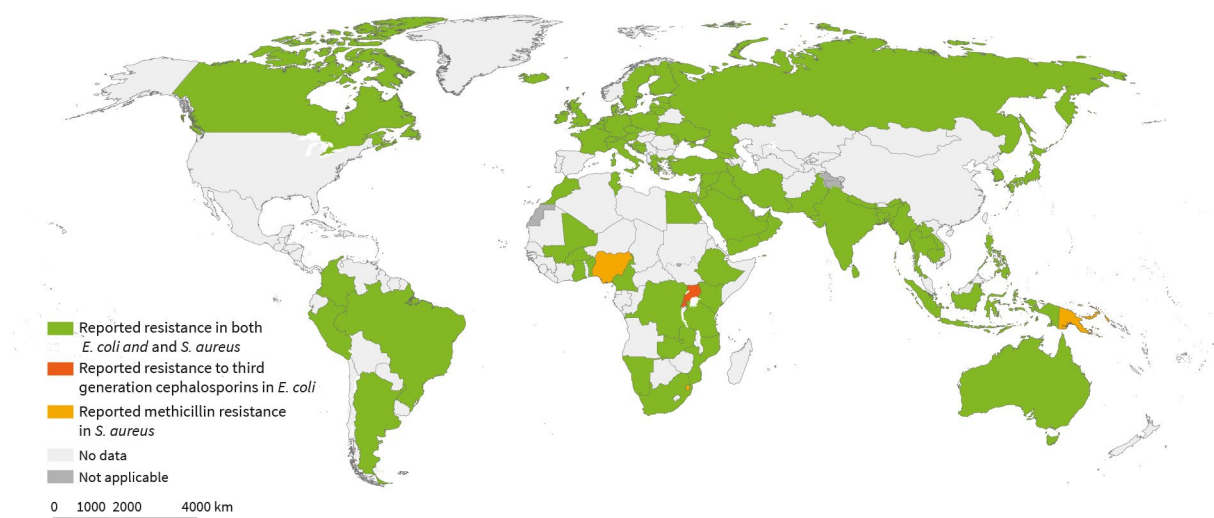
SDG indicator 3.d.2 monitors the proportion of bloodstream infections caused by antimicrobial-resistant organisms, specifically *Escherichia coli* resistant to third-generation cephalosporins and methicillin-resistant *Staphylococcus aureus* (MRSA). In 2022, median resistance proportions were 45.5% (interquartile range [IQR]: 18.2–70.6) for *E. coli* and 32.2% (IQR: 12.5–54.6) for MRSA, up from 26.9% (IQR: 11.1–42.4) and 13.4% (IQR: 3.6–36.7) in 2016 (27).

These figures should be interpreted with caution due to changes in countries reporting data, with progressively more data collected from resource-limited settings over time. In 2016, only 16 countries reported data on *E. coli* resistance and 15 on MRSA, compared to 85 and 86 countries, respectively, in 2022 (Fig. 2.10). Many LICs and MICs have limited testing coverage, often restricted to tertiary hospitals and research facilities, where patients are more likely to have received multiple antibiotics,

leading to higher and potentially biased resistance rates. Countries with better testing coverage reported much lower resistance proportions in 2022: 11.4% (IQR: 9.4–16.4) for *E. coli* and 8.5% (IQR: 3.3–27.9) for MRSA, based on median resistance from 21 and 22 countries with testing coverage above the 75th percentile, respectively (27). The 2024 WHO Bacterial Priority Pathogen List (26) classifies *E. coli* resistant to third-generation cephalosporins as a critical priority due to increasing trends, while MRSA is considered as high priority based on its prevalence and impact in health-care settings, and decreasing trends.

Current global AMR data reflect the challenges of early-stage surveillance systems. Strengthening these systems, particularly in LICs and MICs, is essential for improving data accuracy and representativeness to inform interventions to prevent and reduce AMR. The Global Antimicrobial Resistance and Use Surveillance System (GLASS) dashboard (27) provides more information on the current AMR situation in the human health sector.

Figure 2.10 Countries reporting bloodstream infections caused by *Escherichia coli* resistant to third-generation cephalosporins and methicillin-resistant *Staphylococcus aureus* in 2022



Source: Data from GLASS dashboard (27).



2.3 Risk factors for health

A risk factor for health is any attribute, characteristic or exposure of an individual that increases their likelihood of developing a disease or injury. This section presents progress towards achieving the SDG targets related to nutritional, behavioural and environmental risk factors that may affect individual and population health.

2.3.1 Nutritional risk factors

Multisectoral factors affect the nutritional status of individuals and populations. Mitigating the risks of malnutrition – manifested as undernutrition (wasting, stunting, underweight), micronutrient-related conditions, overweight and obesity – requires multifaceted approaches through nutrition-specific (e.g. food, diet) and nutrition-sensitive strategies (e.g. WASH, education, gender). In fact, undernutrition and food safety are inextricably linked and comprise the essential components of a healthy and sustainable food system. The following sections present the latest available estimates for the global SDG indicators for stunting, wasting and overweight among children under 5 years of age, and anaemia in women aged 15–49 years.

Child malnutrition

The indicators to monitor progress in reducing child malnutrition among children under 5 years of age include the prevalence of stunting (SDG indicator 2.2.1),

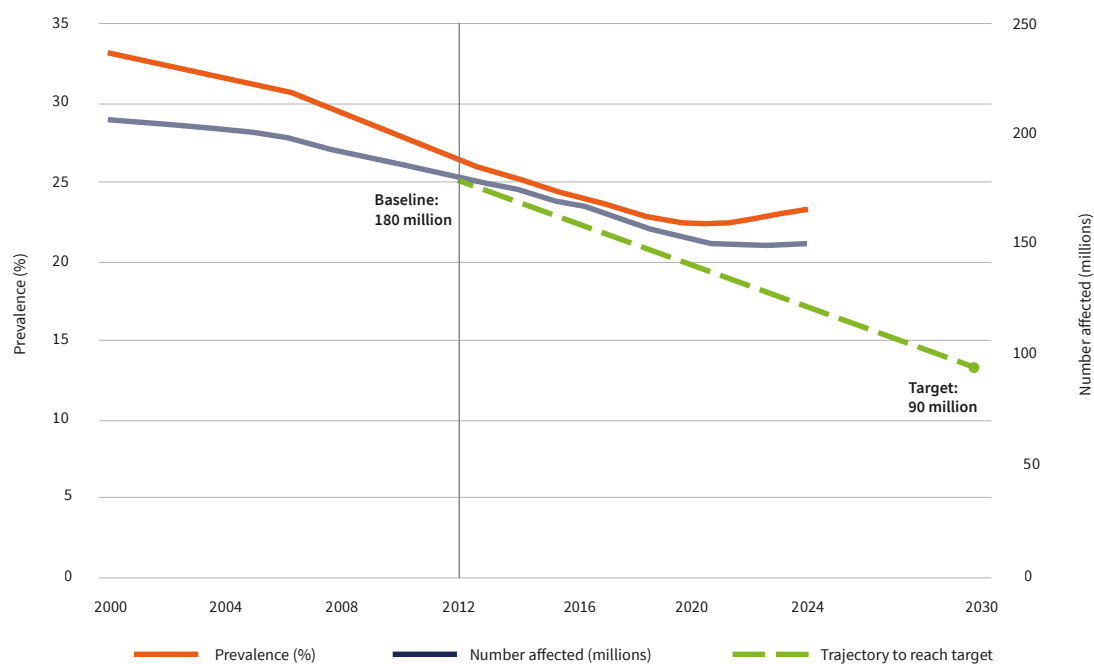
and the prevalence of wasting and overweight (SDG indicator 2.2.2).

The global prevalence of stunting among children under 5 years of age had been steadily declining from 33.1% (UI: 32.7–33.5%) in 2000 to 22.4% (UI: 22.0–22.8%) in 2020 but had slightly increased since to reach 23.2% (UI: 22.6–23.8%) in 2024 (Fig. 2.11) (28).

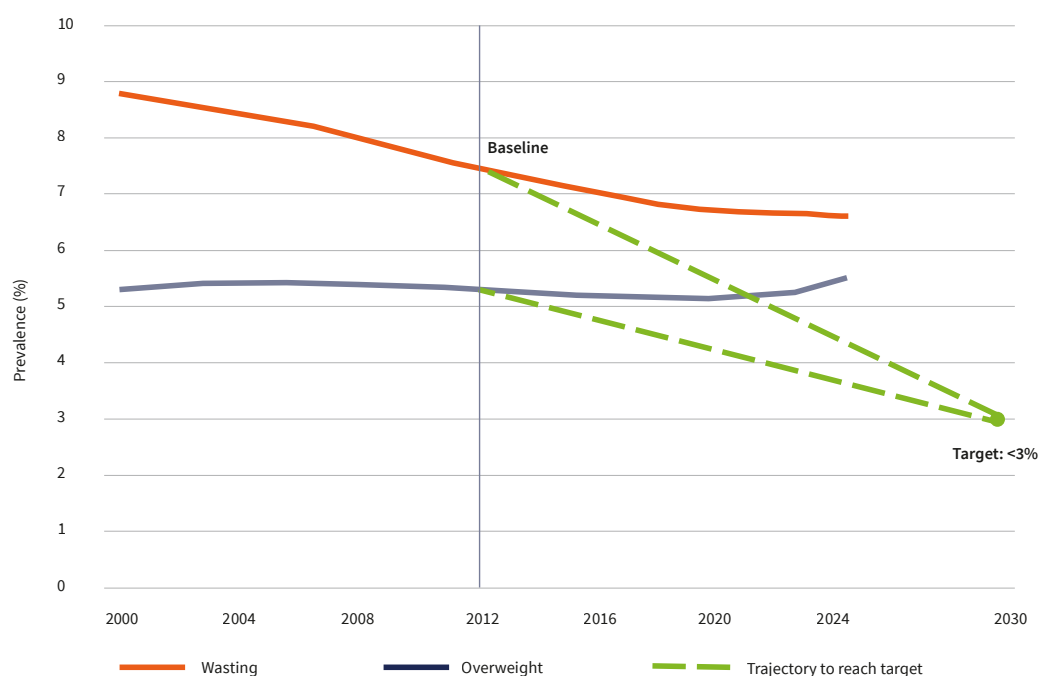
The number of children affected by stunting worldwide fell from 180 million in 2012 to 150 million in 2024 (Fig. 2.11) – a net reduction of 17% over the period (28). This pace of reduction is insufficient to achieve the global target of halving the number of children affected by stunting by 2030 from the 2012 baseline (28,29).

Similarly, the world is off track to reach the target of reducing the prevalence of wasting and overweight to less than 3% each by 2030 (Fig. 2.12) (29). In 2024, the global prevalence of wasting among children under 5 years of age stood at 6.6% (UI: 5.3–8.1%), having declined by 11% since 2012, when the prevalence was 7.4% (UI: 6.3–8.7%) (28).

While there was a sign of slight decline in the global prevalence of overweight among children under 5 years of age in the 2010s, it has slightly increased again since 2020 to reach 5.5% (UI: 4.9–6.1%) in 2024. Between 2012 and 2024, there was a net increase of 3% in the global prevalence of overweight, necessitating a reversal of trend and an acceleration of progress to meet the 2030 target (28).

Figure 2.11 Global prevalence of stunting among children under 5 years of age, 2000–2024, and the 2030 target

Source: UNICEF/WHO/World Bank Group joint child malnutrition estimates, 2025 (28).

Figure 2.12 Global prevalence of wasting and overweight among children under 5 years of age, 2000–2024, and the 2030 target

Source: UNICEF/WHO/World Bank Group joint child malnutrition estimates, 2025 (28).



Anaemia among women

Anaemia is an important global health issue mainly affecting young children, menstruating girls and women, and pregnant and postpartum women. It can affect school performance and productivity in adult life leading to economic losses. During pregnancy, anaemia has been associated with poor maternal and birth outcomes.

The global prevalence of anaemia among all women 15–49 years of age (SDG indicator 2.2.3) declined marginally from 29.3% (UI: 26.4–32.4%) in 2000 to 27.6% (UI: 25.3–29.7%) in 2012 but slightly increased since to reach 30.7% (UI: 26.6–34.9%) in 2023 (Fig. 2.13). At this rate, the world is off track to meet the target of 50% reduction by 2030 (from a 2012 baseline) (29,30).

Among pregnant women globally, the prevalence decreased from 40.3% (UI: 37.7–43.2%) in 2000 to 35.3% (UI: 33.3–37.3%) in 2012, however, it remained stagnant since. In 2023, the prevalence was 35.5% (UI: 32.1–39.1%). By contrast, prevalence among non-pregnant women only decreased slightly from 28.7% (UI: 25.7–31.9%) in 2000 to 27.1% (UI: 24.9–29.4%) in 2012 and increased to 30.5% (UI: 26.2–34.8%) in 2023.

Regional trends in the prevalence of anaemia among all women 15–49 years of age indicate a slight increase in all WHO regions since 2012 except in the Western Pacific Region, where it has remained unchanged (Fig. 2.13). The South-East Asia Region shows the highest level of

anaemia since 2000. In 2023, anaemia affected almost half (46.4%, UI: 35.5–56.1%) of all women aged 15–49 years in the region.

2.3.2 Behavioural risk factors

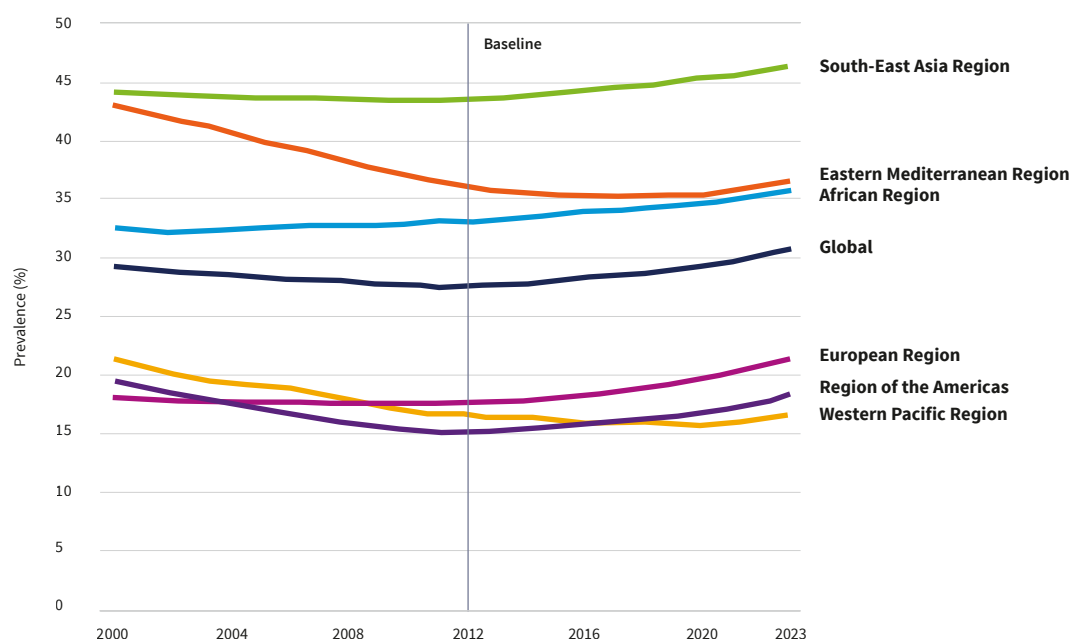
Among the established risk factors associated with NCDs are modifiable behaviours such as tobacco use, alcohol consumption, physical inactivity and unhealthy diet. This section summarizes the global progress towards the reduction of tobacco use and alcohol consumption.

Tobacco use

Tobacco use is a major risk factor for cardiovascular and respiratory diseases, different types of cancer and many other debilitating health conditions. There is no safe level of exposure to tobacco, and all forms of tobacco are harmful.

In 2022, an estimated 1.25 billion persons aged 15 years and older globally were current tobacco users. This corresponds to an age-standardized prevalence (SDG indicator 3.a.1) of 20.9% (UI: 18.8–22.9%), a 21% relative decline from 26.4% in 2010. If this pace of decline is maintained, the global prevalence would be reduced by 25% by 2025, short of the 30% reduction target. Nevertheless, the African Region and the South-East Asia Region are currently on track to achieve the target, and other regions have also achieved reductions in average prevalence (31,32).

Figure 2.13 Trends in the prevalence of anaemia among women aged 15–49 years, globally and by WHO region, 2000–2023



Source: WHO (30).

Prevalence remained much higher among men than women globally. In 2022, there were four times as many male tobacco users as female users among those aged 15 years and older. The prevalence among female tobacco users fell by 34% between 2010 and 2022, exceeding even the 2025 global target. Among male users, prevalence has so far declined by only 17%. However, the fastest reduction among male tobacco users was seen in the “initiation ages” (age group 15–24 years), a 20% relative reduction since 2010, which bodes well for continuing reductions in prevalence of tobacco use among men in the future (31).

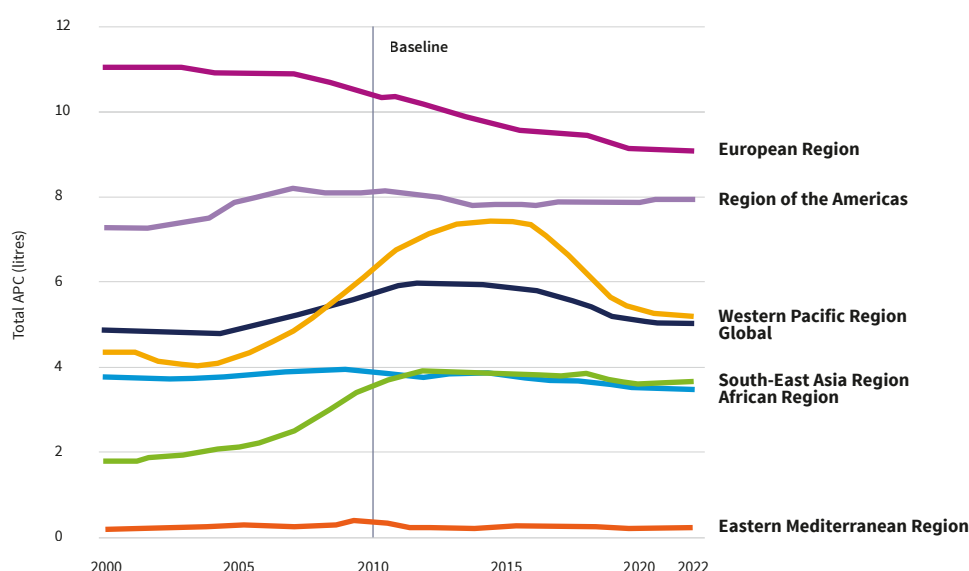
Alcohol consumption

Drinking alcohol is associated with risks of developing NCDs such as liver diseases, cardiovascular diseases and different types of cancers, as well as mental health and behavioural conditions such as depression, anxiety and alcohol use disorders. Alcohol consumption also causes significant harm to others, as it can lead to injuries such as those caused by road traffic crashes. The Global alcohol action plan 2022–2030 includes a target of at least a 20% relative reduction in the harmful use of alcohol by 2030 (in comparison with 2010) (33).

SDG indicator 3.5.2 is defined as total alcohol per capita consumption (APC) in persons aged 15 years and older within a calendar year, measured in litres of pure alcohol. Globally, the total APC declined by 12% from 5.7 litres (UI: 5.4–6.2) in 2010 to 5.0 litres (UI: 4.7–5.4) in 2022. If this rate of reduction is maintained, the global total APC will reach 4.6 litres in 2030, reaching the 20% reduction target. This is remarkable progress, especially given that in the early 2010s the global total APC was increasing, continuing the upward trend since 2000s, driven mainly by the rise in total APC in the South-East Asia Region and the Western Pacific Region (Fig. 2.14) (34).

The South-East Asia Region has curbed the rise and the APC level in 2022 had returned to its 2010 level. The Western Pacific Region has reversed the trend and is now on track to reach 20% reduction in APC by 2030. The Eastern Mediterranean Region has remained the region with the lowest APC across all regions. The European Region continues to have the highest total APC, although it has declined by 12% since 2010 (Fig. 2.14) (34).

Figure 2.14 Trends in total alcohol consumption per capita (APC), globally and by WHO region, 2000–2022



Source: WHO (34).



2.3.3 Environmental risk factors

Water, sanitation and hygiene

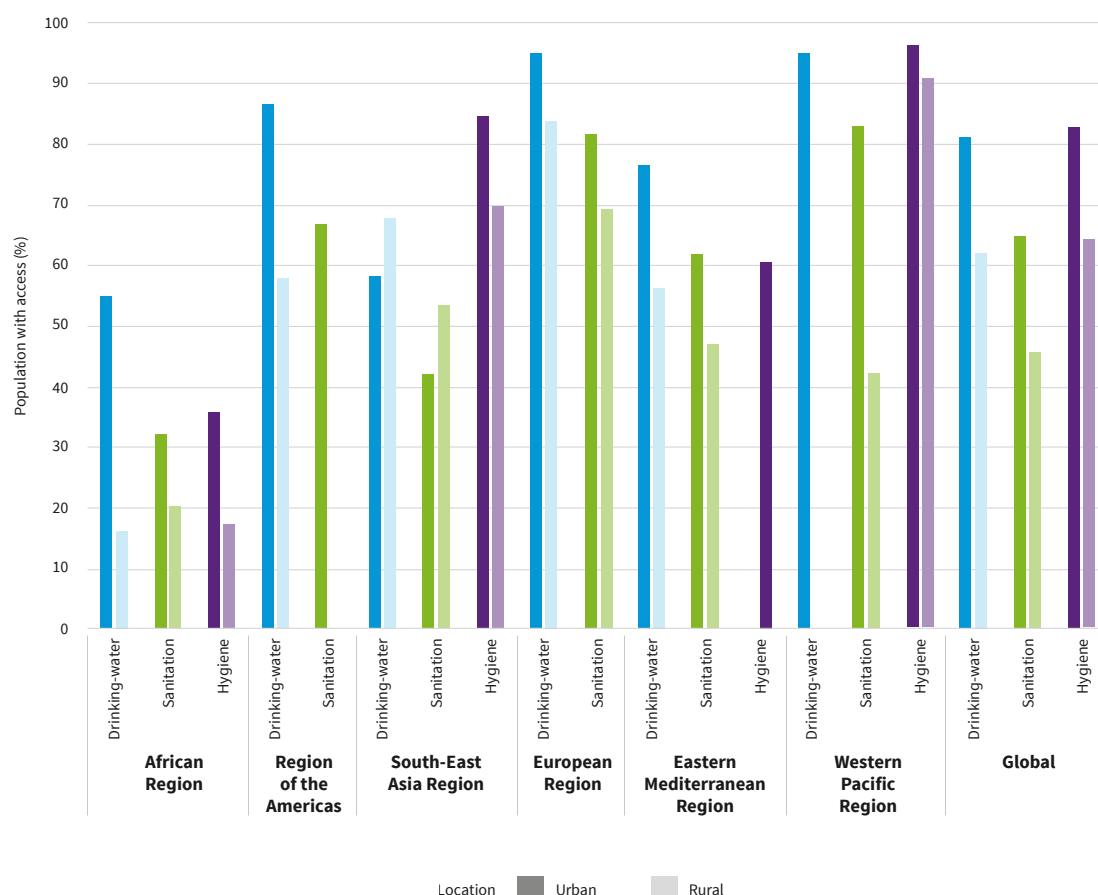
Safe drinking water, sanitation and hygiene (WASH) is crucial to human health and well-being. Contaminated water and poor sanitation are linked to transmission of diseases such as cholera, diarrhoea, dysentery, hepatitis A, typhoid and polio.

The global coverage of safely managed drinking water (SDG indicator 6.1.1) increased from 69% in 2015 to 73%

in 2022. During the same period, the global coverage of safely managed sanitation (SDG indicator 6.2.1) rose from 49% to 57% and that of basic hygiene services (SDG indicator 6.2.1) increased from 67% to 75%. Despite this progress, if current trends continue the world will not achieve universal coverage of safe WASH by 2030 (35).

Globally, and in all WHO regions with aggregate data for 2022, coverage of safe WASH is higher in urban areas than in rural areas, except in the South-East Asia Region, where the coverage of both safely managed drinking water and safely managed sanitation was higher in rural areas (Fig. 2.15) (35).

Figure 2.15 Coverage of safe WASH by location, globally and by WHO region, 2022



Note: Absence of bar for some services and locations means no aggregate statistics are available for 2022.
Source: UNICEF and WHO (35).

SDG indicator 6.3.1 monitors the proportion of total, domestic and industrial wastewater flows that are safely treated. Many countries, however, are at an early stage of developing wastewater monitoring systems and there is currently a lack of reporting, especially from industrial sources. Among the 73 countries (representing 42% of the global population) reporting volumes of total wastewater generated and treated, 76% of total wastewater flows received some level of treatment in 2022. Among the 42 countries (representing 12% of global population) reporting the type of treatment, 60% of total wastewater flows was “safely” treated (i.e. had at least secondary treatment, or treatment complying with relevant standards) (36). An estimated 56% of household wastewater was safely treated in 2024, based on data from 142 countries and areas (representing 88% of all household wastewater flows and 84% of the global population) (37). While current data are insufficient to accurately assess progress since the start of the SDG era (mainly due to the lack of baseline data for 2015), data availability has greatly improved. Still, there is little evidence of improvements in the proportion of wastewater that is safely treated globally since global reporting began in 2021.

Official development assistance (ODA) disbursements to the water sector (SDG indicator 6.a.1) declined slightly from US\$ 8.9 billion in 2022 to US\$ 8.7 billion in 2023,

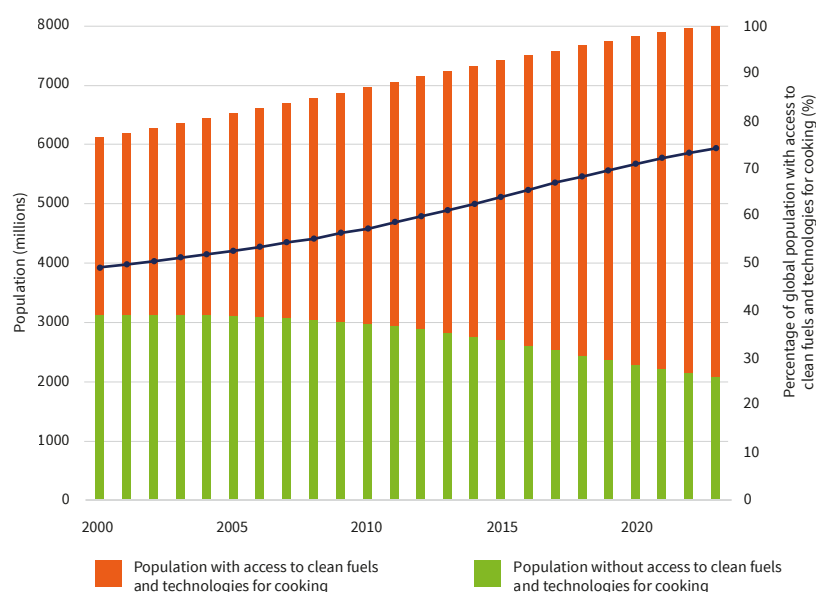
however they remained higher than disbursements in 2020 and 2021, which had the lowest levels since the start of the SDGs in 2015. ODA commitments to the water sector decreased more substantially, from US\$ 11.4 billion in 2022 to US\$ 10.2 billion in 2023 (38).

Air pollution

Air pollution, both indoor and outdoor (ambient), poses major risks to health. The main pathway of exposure to air pollution is through the respiratory tract. However, the effects on health extend beyond respiratory diseases, as some air pollutants can penetrate the bloodstream via the lungs and circulate throughout the entire body leading to systemic inflammation and carcinogenicity.

Simple stoves paired with charcoal, coal, crop waste, dung, kerosene or wood are an important source of household air pollution. The latest estimates reveal that 2.1 billion (UI: 1.9–2.5 billion) people worldwide were still relying on these polluting cooking fuels and technologies in 2023 – equivalent to slightly over a quarter (26%, UI: 23–30%) of the global population. The proportion of the global population with primary reliance on clean cooking fuels and technologies had increased from 49% (UI: 45–53%) in 2000 to 64% (UI: 60–68%) in 2015 and 74% (UI: 70–77%) in 2023 (Fig. 2.16) (39). Under current trends, the world will miss the target of universal access by 2030.

Figure 2.16 Global trends in the access to clean cooking fuels and technologies, 2000–2023



Source: WHO (39).



In terms of ambient air pollution, SDG indicator 11.6.2 tracks the annual average concentration of fine particulate matter in urban areas. Globally, air quality levels for particulate matter are still more than six times the recommended level for protecting public health according to the WHO air quality guidelines, which is set at the annual mean of $5 \mu\text{g}/\text{m}^3$. Nearly all populations in HICs, LICs and MICs are exposed to unsafe levels of air pollution. However, the proportion of population in LICs and MICs exposed to levels above the interim target 4 (annual mean of $10 \mu\text{g}/\text{m}^3$) is twice that in HICs, at 93.5% and 47.1%, respectively. The proportion of population in LICs and MICs exposed to levels exceeding interim target 1 (annual mean of $35 \mu\text{g}/\text{m}^3$) is over 10 times that in HICs (42.0% and 3.7%, respectively). In addition, people living in LICs and MICs are less likely to have air quality standards that can protect them. The recent publication of the WHO air quality standard database shows that only 32% of LICs and MICs (17 countries) have annual fine particulate matter (PM_{2.5}) standards while 76% of HICs (44 countries) have such protective standards (40).

2.3.4 Risks to girls' and women's health

Adolescent pregnancy

Adolescent pregnancy is a worldwide issue with significant health, social and economic impacts. Teenage

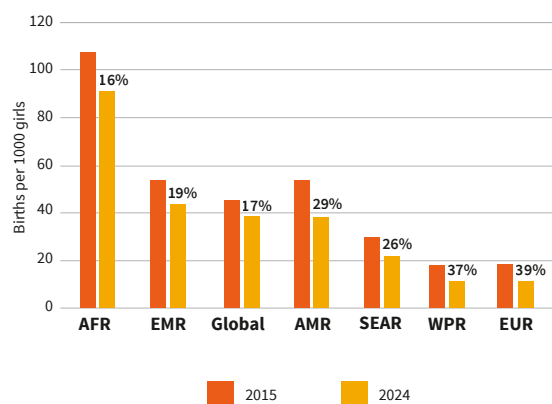
mothers aged 10–19 years face higher risks of eclampsia, puerperal endometritis and systemic infections than women aged 20–24 years, and babies of teenage mothers face higher risks of low birth weight, preterm birth and severe neonatal conditions.

The global adolescent birth rate (ABR; SDG indicator 3.7.2) among girls aged 15–19 years declined by 17%, from 45.9 births per 1000 girls in 2015 to 38.3 births per 1000 girls in 2024. The global ABR among girls aged 10–14 years fell at almost double the rate, with a 31% decline from 1.5 births per 1000 girls in 2015 to 1.0 birth per 1000 girls in 2024. While this decline is encouraging, there were still far too many births among adolescent mothers in 2024, accounting for about 9.4% of all births globally (41).

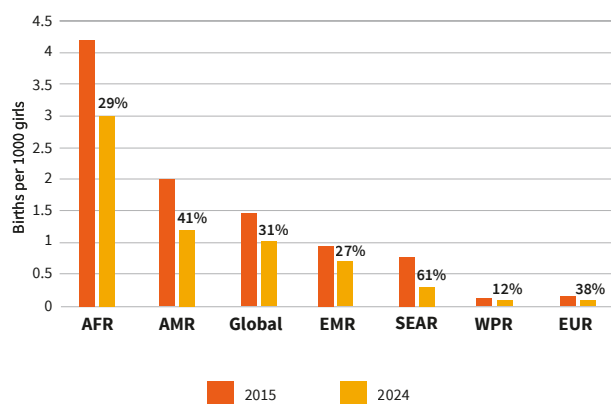
Progress is uneven among regions (Fig. 2.17), as well as among and within countries. The African Region continued to have the highest ABR in both adolescent age groups (10–14 years and 15–19 years). In 2024, the ABR among girls aged 15–19 years was 90.5 births per 1000 girls, over eight times that of the region with the lowest ABR, the Western Pacific Region. The gaps in ABR among girls aged 10–14 years were even larger: at 3.0 births per 1000 girls, it was 35 times that of the regions with the lowest ABR, the European Region. By region and age group, the biggest relative reduction in ABR occurred in the South-East Asia Region, where ABR among girls aged 10–14 years fell by 61%, from 0.8 births per 1000 girls in 2015 to 0.3 births per 1000 girls in 2024 (41).

Figure 2.17 Adolescent birth rate, globally and by WHO region, 2015–2024

15–19 years



10–14 years



AFR: African Region; AMR: Region of the Americas; EMR: Eastern Mediterranean Region; EUR: European region; SEAR: South-East Asia Region; WPR: Western Pacific Region.

Note: The regions are shown in descending order of the adolescent birth rate (ABR) in 2024. Values shown above the 2024 bars denote the relative reduction of ABRs between 2015 and 2024.

Source: World population prospects 2024 (41).

Violence against women

Violence against women and girls – particularly intimate partner violence (IPV) and sexual violence – is a violation of human rights and a major public health problem. Such violence often has devastating impacts on the lives and well-being of the victims, as well as their children and families.

According to the latest available global estimates, in 2018 over a quarter (26%, UI: 22–30%) of ever-partnered women aged 15 years or older had experienced IPV at least once in their lifetime (SDG indicator 5.2.1) and 10% (UI: 8–12%) in the past 12 months (42). There has been insufficient progress in reducing IPV against women and girls since 2000 (43).

In 2018, an estimated 6% (UI: 4–9%) of all women globally had been subjected to non-partner sexual violence in their lifetime since the age of 15 (SDG indicator 5.2.2). The true prevalence is likely to be much higher (42).

Population-based surveys which record survivors' responses provide the best information to measure prevalence of IPV and sexual violence. However, even such surveys often have limitations, such as variations in case definitions and recall periods, stigma and repercussions associated with the disclosure of violence against women, and limited data collection from women aged 50 years and older.

2.4 Universal health coverage and health systems

Universal health coverage aims to ensure that all individuals and communities have access to the quality health services they need without suffering financial hardship. When more people have access to affordable quality health services, it leads to better overall public health and living standards, including, but not limited to, improved life expectancy, lower rates of preventable diseases and lower rates of poverty due to medical costs.

2.4.1 Service delivery

To achieve UHC and improve health outcomes, health systems need strong service delivery mechanisms that ensure comprehensive service coverage, a capable health workforce and adherence to international health regulations (IHRs) to address both routine and emergency health needs.

Service coverage

While gains in UHC service coverage index (UHC SCI; SDG indicator 3.8.1) have been observed globally over the past

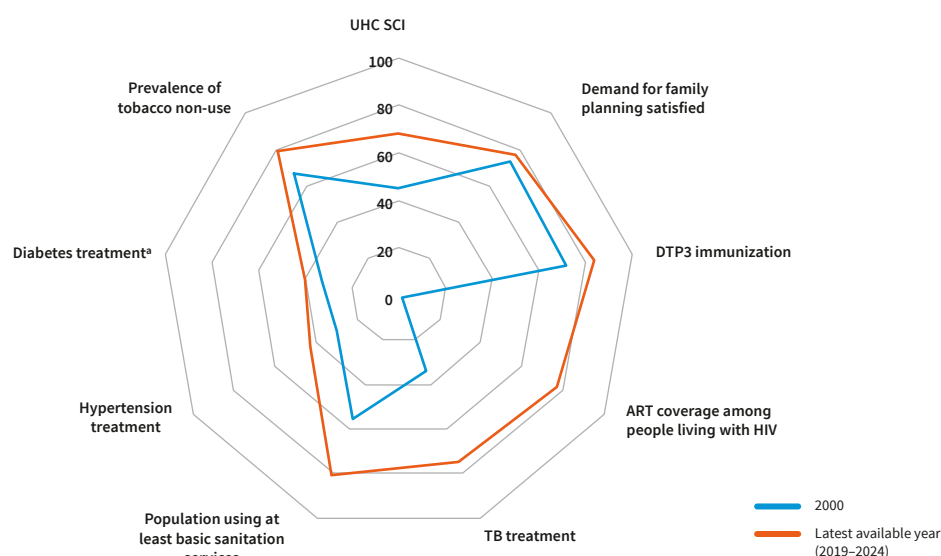
two decades, progress has slowed in recent years (44). The UHC SCI is composed of 14 tracer indicators selected as a manageable subset to represent overall coverage. These indicators are organized into four domains: reproductive, maternal, newborn and child health (RMNCH); infectious diseases; NCDs; and service capacity and access.

The progress made in the UHC SCI has not been even across domains (44). Figure 2.18 shows the change since 2000 in UHC SCI and selected tracers or related health SDG service coverage indicators.² Globally, service coverage tracer indicators improved by a range of 3–75 percentage points from 2000 to the latest available year (2019–2024). There remains significant scope for progress in the area of NCDs, which contribute to the largest proportion of disease burden globally. Hypertension and diabetes treatment coverage (age-standardized) remains below 50% of the target population globally (42% and 40% in 2019 and 2022, respectively) and progress needs to be accelerated (14,15,18,31,35,44–48).

² Service coverage indicators (scale of 0–100) with a global baseline of 2000 and a latest available of 2019 or later.



Figure 2.18 Universal health coverage service coverage index (UHC SCI) and selected tracers globally, 2000 and latest available year (2019–2024)



ART: Antiretroviral therapy; DTP3: diphtheria, tetanus toxoid and pertussis vaccination; TB: tuberculosis.

^a Diabetes treatment will be a UHC SCI tracer in the next round of reporting, see Box 2.2 for more information.

Source: Sanitation (35), UHC SCI (44), ART (14,15), TB (18), hypertension (45), diabetes (46), tobacco (31), family planning (47), DTP3 (48).

In the domain of RMNCH, the proportion of women of reproductive age having their need for family planning met with modern methods (SDG indicator 3.7.1) has seen a modest improvement of an average of 1.2 percentage points in each of the last two decades globally, with no WHO region reaching universal coverage (range 58.1–87.3%) in 2024 (47). The global proportion of births attended by skilled health personnel (SDG indicator 3.1.2) increased by 26 percentage points between 2000 and 2024, to 87%. As of 2024, the WHO Region of the Americas and Western Pacific and European Regions have achieved universal or nearly universal coverage, while only 74% of births are attended by a medical doctor, nurse or midwife in the WHO African Region (49).

Vaccination rates of all four childhood vaccines tracked within SDG indicator 3.b.1³ stagnated or declined between 2020 and 2022. For third dose diphtheria-pertussis-tetanus containing vaccine (DTP3), which is a useful marker of vaccination system functioning, in 2023 vaccination rates had still not fully returned to their 2019 pre-pandemic level, suggesting substantial residual dysfunction in global vaccine delivery systems. Vaccination rates for the other three vaccines increased, although this was partially due to new introductions.

Globally, DTP3 still has the highest coverage of the four vaccines, at 84% in 2023.

The WHO Region of the Americas has the highest coverage of the human papillomavirus (HPV) vaccination among girls aged 9–14 years, reaching 55% within 18 years of introduction of the vaccine. However, globally HPV coverage remains low, meaning millions of girls will remain susceptible to cervical cancer over their lifetimes. Among WHO regions, the European Region has the highest uptake of third dose pneumococcal conjugate vaccine (PCV3), with coverage at 86% in 2023. In 2023, the WHO European and Western Pacific Regions had DTP3 and measles-containing-vaccine second-dose (MCV2) coverage of 90% or higher, and the South-East Asia Region reached 90% for DTP3 (48). However, the 90% global target set by the Immunization Agenda is unlikely to be met for any of these vaccinations if current trends continue (50).

Data collected from about 20 countries between 2015 and 2018 on treatment for alcohol and drug use disorders (SDG indicator 3.5.1) indicate a median contact coverage of 3.2% and 10.0%, respectively. About 40% of the 145 countries who responded to Global Survey 2019 on Progress on SDG health target 3.5 do not collect any data

³ Diphtheria-tetanus-pertussis (DTP3), measles-containing-vaccine second-dose (MCV2), pneumococcal conjugate 3rd dose (PCV3) and human papillomavirus (HPV) vaccine.

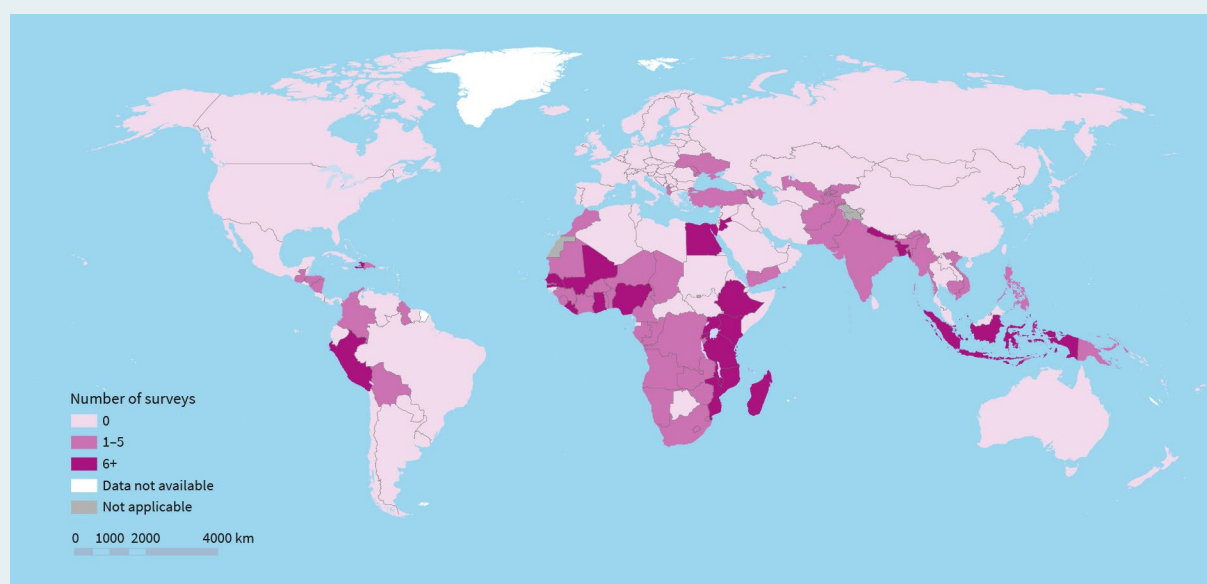
on service provision and service utilization for treatment of alcohol and drug use disorders (51). Alcohol and drug use are risk factors for multiple health conditions, such as cardiovascular diseases, liver diseases, cancer, respiratory diseases, injuries and mental health disorders, all of which contribute to NCD-related morbidity and mortality (52).

Surveys are essential for gathering population health data, including, but not limited to, fertility rates, nutrition, child health and disease prevalence. The data collected by surveys support informing health policy, improving health systems and identifying emerging health issues (see Box 2.1).

Box 2.1 Demographic and Health Surveys in universal health coverage service coverage index (UHC SCI)

Over half of the UHC SCI tracers use surveys such as Demographic and Health Surveys (DHS) as a data source. The DHS programme was initiated in the early 1980s by the United States Agency for International Development (USAID) with the aim of providing countries with reliable and timely data on critical health and demographic indicators. It aimed to fill a gap in data collection, particularly in developing countries where such information was often lacking or outdated. The first survey was conducted in El Salvador in 1985. Since 2000, there have been around 300 DHS conducted (see Fig. 2.19) (53). The suspension of USAID funding delays crucial public health data, resulting in the short-term in an evidence gap to allocate resources and develop policies. Without immediate actions to replace this data collection, the evidence gap will lead to less effective health interventions and widening health disparities. As statistical offices and ministries of health develop a strategy to mitigate the loss of the DHS, other questionnaires such as the WHO World Health Survey Plus (54) can guide comprehensive data collection to monitor progress towards population health targets and health-related SDGs.

Figure 2.19 Number of demographic and health surveys conducted by DHS programme by country, 2000–2024



Note: Includes all survey types with data available as of 27 March 2025.

Source: DHS (53).



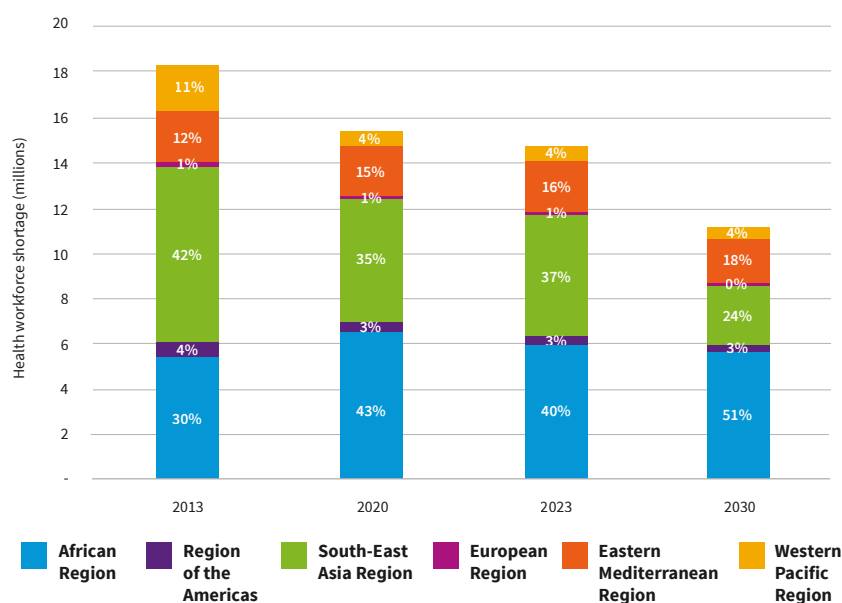
Health workforce

Health workers are crucial for achieving UHC. Their role goes beyond providing medical care: an ample, well-trained and properly distributed workforce ensures that health services are accessible, effective and of good quality for all, regardless of income or place of residence.

The estimated global stock of health workers exceeds 70 million in 2023, with the five occupations included in SDG target 3.c (dentists, medical doctors, midwifery personnel, nursing personnel and pharmacists) accounting for 52 million (55). Data for 2023 show a 26% increase in the density of dentists, medical doctors, midwifery personnel, nursing personnel and pharmacists since 2013, but with significant differences by region and income group (for instance, on average, 1 for every 64 people in HICs versus 1 for every 621 people in LICs) (56).

The estimated global shortage of 15.4 million health workers in 2020 decreased to 14.7 million in 2023. The projected 2030 shortage of 11.1 million shows slow progress in closing the gap (Fig. 2.20), with the WHO African and Eastern Mediterranean Regions projected to bear nearly 70% of the shortage. HICs and upper-middle-income countries (UMICs), which account for 53% of the global population and 80% of workers in the five occupations under SDG target 3.c, have the highest densities of health workers. Nevertheless, these countries will likely need to recruit 8.4 million additional health workers by 2030 to maintain density levels and meet the rising demands of growing and ageing populations, potentially accelerating international migration of health workers.

Figure 2.20 Distribution of the global health workforce shortage by WHO region (for 2013, 2020 and 2023, and projected for 2030)



Note: Percentages represent the relative share of the global health workforce shortage.

Source: NHWA (56).

International health regulations

IHRs are essential to service delivery as they ensure countries have the necessary capacities within their health systems to prevent, prepare, detect in a timely manner and respond to public health emergencies. IHRs therefore contribute to global health resilience and the achievement of sustainable health outcomes for all populations.

IHRs States Parties self-assessment annual reporting tool (SPAR) second edition (57), which assesses 15 capacities,

was launched in 2021 and is in its fourth year of reporting. Globally, the capacities average (SDG target 3.d.1) has wavered around 64% between 2021 and 2024. The three main achievement and challenges capacities measured by SPAR (Fig. 2.21) have changed minimally (1 point) globally, with the exception of challenge C1 (Policy, legal and normative instruments to implement IHR), which improved by 3 points globally and by 11 in the South-East Asia Region. The WHO African Region improved its score (from 2 to 6 points) in all three main challenges and two of the achievements between 2021 and 2024 (58).

Figure 2.21 Global top three challenges and achievements scores in 2024, by WHO region

	Global	African Region	Region of the Americas	South-East Asia Region	European Region	Eastern Mediterranean Region	Western Pacific Region
Top global challenges							
C14 Chemical events	53	35	51	53	69	55	53
C1 Policy, legal and normative instruments to implement IHR	55	38	55	61	64	61	62
C15 Radiation emergencies	57	38	53	55	77	67	47
Top global achievements							
C5 Surveillance	80	73	79	83	85	80	79
C7 Health emergency management	71	60	74	75	75	69	76
C8 Health services provision	71	56	73	78	79	75	73

Scores: dark green: > 80; light green: 61–80; orange: 41–60; yellow: 21–40. Change from 2021: bold: increased by more than 1 point; italic: decreased by more than 1 point; not bold or italic: change is 1 point or less.

Source: eSPAR (58).

2.4.2 Health financing

Sustainable, efficient and effective health financing is critical for achieving UHC. By pooling resources, reducing out-of-pocket payments and ensuring financial protection, health financing allows countries to provide equitable access to quality health care for all populations.

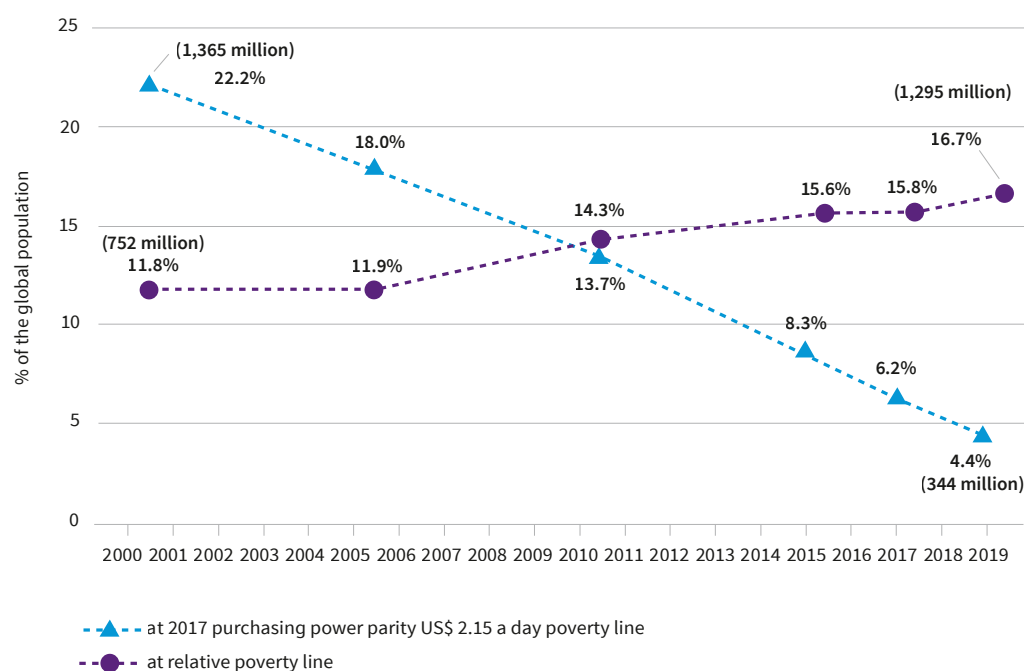
Financial protection

The financial protection dimension of UHC is achieved when there are no financial barriers to accessing needed health services and goods, and out-of-pocket (OOP) health spending is not a source of financial hardship. SDG indicator 3.8.2 measures the proportion of people experiencing catastrophic OOP health spending. The latest data show that the global proportion of the population spending more than 10% of their household budget on health OOP increased from 2000 to 2019, reaching 13.5%, while the global proportion of the population spending more than 25% reached 3.8%.

It is also important to track how many people are impoverished or further impoverished by OOP health spending, as even relatively small OOP payments can threaten the living standards of people living near or in poverty. In 2019, some 344 million people were pushed or further pushed into extreme poverty by OOP health spending, while 13.5% of the global population spent more than 10% of their household budget on OOP payments for health. The proportion of people facing impoverishing OOP spending at the extreme poverty line declined from 22.2% in 2000 to 4.4% in 2019. In contrast, the share at the relative poverty line rose from 11.8% in 2000 to 16.7% in 2019 (Fig. 2.22). These trends suggest that while OOP health spending was less likely to push people into extreme poverty by 2019, it still disproportionately affected those in the lower income brackets (44).



Figure 2.22 Trends in the incidence of impoverishing health spending at the extreme and relative poverty lines globally, 2000–2019



Source: WHO and World Bank (44).

Domestic and international funding

Domestic public funds provide long-term sustainability, equity and a foundation for health systems, while international funds address funding gaps, support global initiatives and bring innovation to health care delivery.

The increase in the prioritization of health within general government spending seen in 2020 and 2021 (the health component of SDG indicator 1.a.2) among middle- and high-income groups was not sustained in 2022, and even declined slightly in LICs, LMICs and UMICs. However, the share of health expenditure within total government spending remained higher in 2022 than in 2019 in all income groups (59). Government funding is essential to subsidize health services and ensure they are affordable for everyone, including low-income or vulnerable populations.

Development assistance to medical research is vital for addressing both emerging health challenges and

ongoing diseases, particularly in LICs and MICs, where the basic health portion addresses fundamental health-care needs that are critical to UHC. In 2023, LICs received the highest weighted average ODA per capita (SDG indicator 3.b.2), at US\$ 6.04, compared with US\$ 0.88 and US\$ 0.43 for the LMICs and UMICs, respectively. However, ODA varies significantly by country; the five countries with the highest per capita ODA are mostly island states in the middle- and high-income groupings (60).

Since the global indicator framework for SDGs was agreed in 2017 (61), challenges including the rising burden of NCDs, the impact of global health crises like the COVID-19 pandemic, workforce shortages and financial sustainability, have increased. Indicators used to monitor UHC need to adapt to the changing world, improvements in data availability and understanding of populations being left behind (Box 2.2).

Box 2.2 Revising SDGs to improve monitoring of progress towards UHC

Together, SDG indicators 3.8.1 and 3.8.2 are essential to monitoring progress towards UHC: 3.8.1 aims to assess whether needed services are available and used by those in need, while 3.8.2 assesses whether people are financially burdened by the cost of care. The two indicators were revised as part of the Inter-Agency and Expert Group on Sustainable Development Goal Indicators 2025 comprehensive review (62) and approved by United Nations Statistical Commission in March 2025 (63).

The revision of SDG indicator 3.8.1 follows the same principals as its original construction (64) but aims to update to indicators with improved data availability for better tracking and alignment with existing SDG indicators. The tracers were selected as a subset of overall coverage and all equally contribute to the UHC SCI. The revision will apply weights based on the population affected by each tracer to better reflect the service burden on a health system; the four domain areas will still equally contribute to the overall SCI.

The revision of SDG indicator 3.8.2 captures financial hardship caused by both substantial and impoverishing OOP health spending. Initially, the budget share definition was retained for SDG indicator 3.8.2, with a 10% threshold used to identify large OOP spending. This was intended to identify hardship in poorer populations. However, global monitoring reports on UHC have shown that many people living in poverty, who spend less than 10% of their household budget on health, are still pushed or further pushed into extreme or relative poverty. The revised indicator is defined as the proportion of the population with OOP household expenditure on health exceeding 40% of the household discretionary budget. A household's discretionary budget is defined as household total consumption expenditure or income net of the societal poverty line (SPL). The SPL is used to assess the cost of basic needs across countries at all income levels. The discretionary budget approach takes into account that poorer households allocate most of their resources to necessities and in effect can spend very little on health OOP without struggling financially, while wealthier households can spend much more before OOP spending leads to financial hardship.

These revisions align with the SDG principle of ensuring no one is left behind and aim to better reflect the realities of service coverage and financial hardship across countries. The new definitions will be reflected in the next *Tracking universal health coverage: global monitoring report* (65) to be published at the end of 2025.

2.5 SDG progress to date

As the world moves closer to 2030, progress is insufficient to meet the health-related SDG and other global targets. While there have been improvements for most of the indicators reviewed in this chapter, only one – reducing alcohol consumption – is on track to meet the 2030 target

at the global level. The assessment of progress in selected health-related SDG indicators by WHO region is presented in Annex 1. Concerted efforts are needed to accelerate progress and reverse regressing trends globally, as well as at regional, country and community levels.



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3.

Progress in achieving the Triple Billion targets

The Triple Billion targets, a cornerstone of WHO's Thirteenth General Programme of Work (GPW13), track the collective progress by WHO and its Member States in promoting, providing and protecting health worldwide. The latest update illustrates the progress made since the start of GPW13 in 2018 and the key challenges particularly in the universal health coverage and health emergencies protection billions.

Anchored in the health-related SDGs, WHO's GPW13 provides a strategic roadmap to enhance health and well-being for all (1). More importantly, GPW13 championed the results framework with impact measurement at its core to reinforce organizational accountability and transparency. The foundation of the impact measurement is the 46 outcome indicators centred on promoting the overall health and quality of life for the people, providing essential health services, and protecting people from health emergencies. The Triple Billion targets give strategic clarity to the impact measurement of GPW13. By summarizing the outcome indicators into three ambitious, but easy to understand and communicate goals, the Triple Billion targets effectively convey global health priorities, motivate collective action by the global

community, facilitate transparency and accountability, and complement the strategic priorities set by WHO's GPW13 to address current and emerging global health priorities.

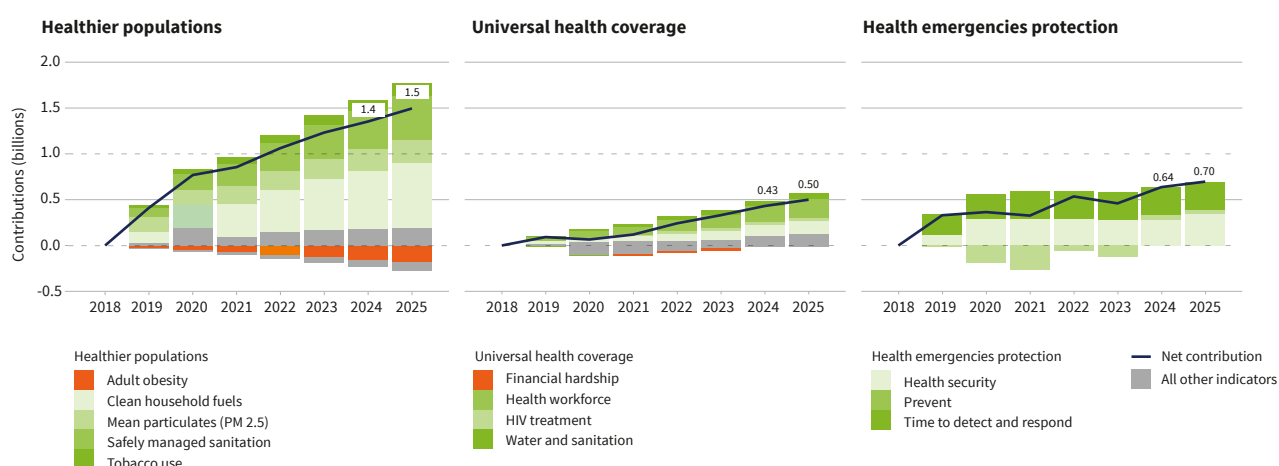
For GPW13, the WHO Member States aim to enable one billion more people achieve better health and overall well-being, to provide one billion more people with access to essential health services without incurring financial hardship, and to better protect one billion more people from health emergencies worldwide by the end of the GPW13, extended from 2023 to 2025. This chapter provides a summary of the progress made in achieving the Triple Billion targets with projections to 2025, incorporating the latest available data from the outcome indicators at the country level.

3.1 Triple Billion targets progress

Progress towards achieving the Triple Billion targets remains uneven (Fig. 3.1). Growth in providing essential health services is insufficient, with only 431 million more people gaining affordable health-care access by 2024 relative to the GPW13 baseline year of 2018. This is projected to rise to 500 million by 2025 – half the targeted one billion. Furthermore, close to 637 million more individuals will be better protected from health emergencies by 2024. Despite significant progress, this is expected only to increase to 697 million people by 2025, around 30% short of the one-billion target. Remarkably,

an estimated 1.4 billion more people will experience healthier lives by 2024, with 1.5 billion expected in 2025, surpassing the original target of one billion. Yet, such progress is not enough to put the world on track to achieve the health-related SDGs by 2030 (2,3). Moreover, recent interruptions in international aid threaten to disrupt health services and systems, disproportionately impacting countries and communities with the greatest health-care needs. Safeguarding the gains made in achieving the Triple Billion targets is a paramount task for the global community in the years ahead.

Figure 3.1 Progress in achieving the Triple Billion targets and estimated contributions by indicator



3.2 Progress in healthier populations

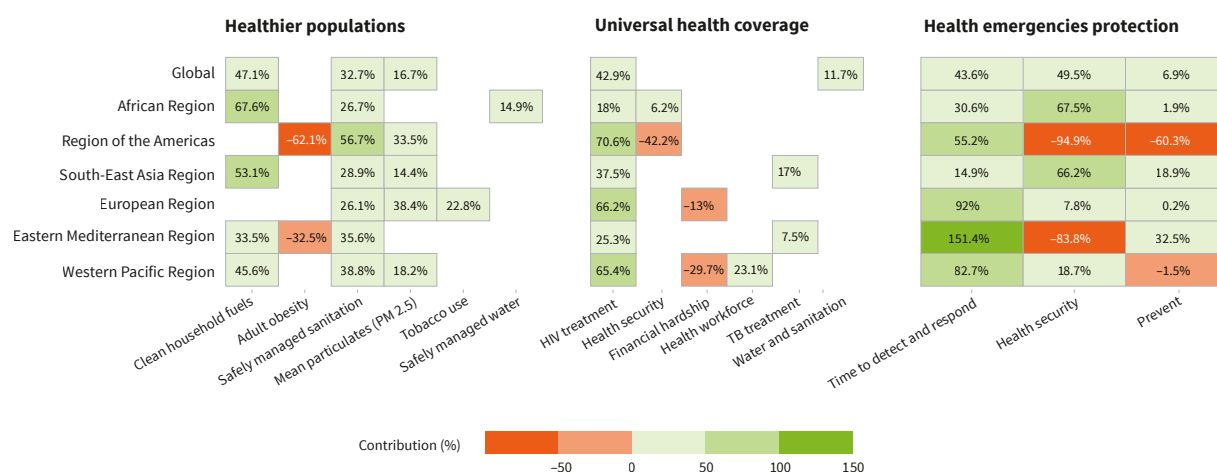
Current trends indicate that the world is on track to meet the goal of improving health and well-being for one more billion people by 2025. Compared with the baseline year of 2018, some 1.5 billion people are projected to be living healthier lives by 2025. Such progress is primarily due to reduced tobacco use, improved air quality, increased use of clean household fuels, and better access to water, sanitation and hygiene (Fig. 3.1). Yet, at the global level, significant challenges persist in the areas of rising rates of adult and childhood obesity.

Looking at the top contributing indicators by region (Fig. 3.2), the most striking features are worsening rates of adult obesity in the Eastern Mediterranean Region and Region of the Americas. These account for setbacks equal to 32.5% and 62.1% of the respective regional net contributions. Despite these difficulties, substantial progress has been made. Most regions are

seeing improvements in clean household fuel use as their primary marker of progress, with particularly large effects in the African Region (67.6%) and the South-East Asia Region (53.1%). In both these regions, improvements in clean fuel use represent over half of the additional people living healthier lives.

All regions showed substantial progress in safely managed sanitation, which ranked as the second most important contributor globally (32.7%). The top three indicators contributing to the Triple Billion targets in the European Region are improvements in air quality (38.4%), safely managed sanitation (26.1%) and tobacco use (22.8%). The African Region is unique in having safely managed water as a top contributor (14.9%), highlighting region-specific priorities and challenges in water infrastructure.

Figure 3.2 Top indicators with most contributions to the Triple Billion targets, 2018–2025^a



^a Percentage is calculated as the ratio of individual indicator contribution over magnitude of the overall contribution.



3.3 Progress in universal health coverage

Compared with the baseline year of 2018, about 500 million more people worldwide will enjoy access to essential health services without undue financial burden by 2025. This progress in achieving UHC has been supported by improvements in health workforce, enhanced coverage of hypertension treatment, and better access to family planning. However, the most substantial contributor to achieving UHC, and the progress made so far during the GPW13 period, is the expanded availability of antiretroviral therapy (ART) for HIV and AIDS. Challenges in access to health services continue to hinder faster progress, particularly in diabetes management and mitigating financial hardships incurred by utilizing health services. The declining coverage of DTP3 immunization is especially concerning.

Nearly all regions have seen progress in expanding access to HIV treatment (Fig. 3.2). However, there have been setbacks in progress in health security and affordability. In the Region of the Americas, the strong progress is substantially offset by deterioration in health security (–42.2%). Both the European and Western Pacific regions have seen significant increases in financial hardship. Rising financial hardship in the Western Pacific Region has significantly offset other UHC improvements, reducing progress by nearly 30%.

Most regions show contributions from multiple factors, indicating diverse approaches to managing health-care access. We see smaller, yet still substantial, gains in the health workforce in the Western Pacific Region (23.1%), as well as better tuberculosis treatment in the South-East Asia (17%) and Eastern Mediterranean regions (7.5%).

3.4 Progress in health emergencies protection

According to updated information for the outcome indicators, substantial advancements have been made towards better safeguarding an additional one billion people from health emergencies by 2025. Crucial amendments to the International Health Regulations, adopted by the World Health Assembly in 2024, represent a major step forward in global health security and pandemic preparedness. In addition, significant strides in early disease detection, mobilizing emergency response teams and ensuring equitable access to essential medical supplies, are key drivers of global resilience to health emergencies.

Health emergencies protection shows the greatest amount of regional variation in top indicator contributions (Fig. 3.2). The Eastern Mediterranean Region reported exceptional progress in detection and response times (151.4%) but was undermined by worsening health security (–83.8%). Meanwhile, the Region of the Americas faces challenges in both health security (–94.9%) and prevention (–60.3%). The European Region made strong gains in detection and response, with a positive profile across indices (92% time to detect and respond, 7.8% health security). These contrasting outcomes highlight regional differences in emergency preparedness and response strategies.

3.5 Forecast to 2030 and scenarios

The Triple Billion target projections through 2030 showed encouraging trends towards a healthier global population. Under the current trajectory, 2.0 billion more people were expected to live healthier by 2030 compared with the 2018 baseline. Figure 3.3 presents an alternative scenario in which all countries meet the annualized targets set by the 2030 Agenda using the same set of 46 outcome indicators (4). In this case, 5.6 billion more people would live healthier lives by 2030 – nearly triple the gain of 2.0 billion under the current trajectory. Similarly, achieving the 2030 SDG and other global targets would result in 2.1 billion more people gaining access to UHC without financial hardship, compared with 0.8 billion under current trends. The gap is smaller for health emergency protection; meeting the 2030 targets would still safeguard 1.0 billion more people against health emergencies, compared with 0.8 billion under the current trajectory.

Figure 3.4 illustrates the potential for additional progress if within-region inequalities were to be addressed. It

shows the gap between forecasted contributions and a scenario where all countries are able to match their regional top performers by 2030. The large gaps highlight significant missed opportunities in progress both now and in the future.

The regional disparities are most striking for health emergencies protection. By 2030, an additional 30.6% of people in the Region of the Americas could be better protected if all countries matched their regional leaders – a sharp increase from 10.6% in 2025. Similarly, the African Region could extend protection to an additional 28% of its population, up from 9.6% in 2025.

For UHC, closing these gaps would mean an additional 13.8% of people in the European Region and over 15.5% more people in the Region of the Americas would have affordable access to essential health care. These figures underscore how reducing within-region disparities could dramatically accelerate progress toward the Triple Billion targets.

Figure 3.3 Progress needed to achieve the Triple Billion and 2030 global targets

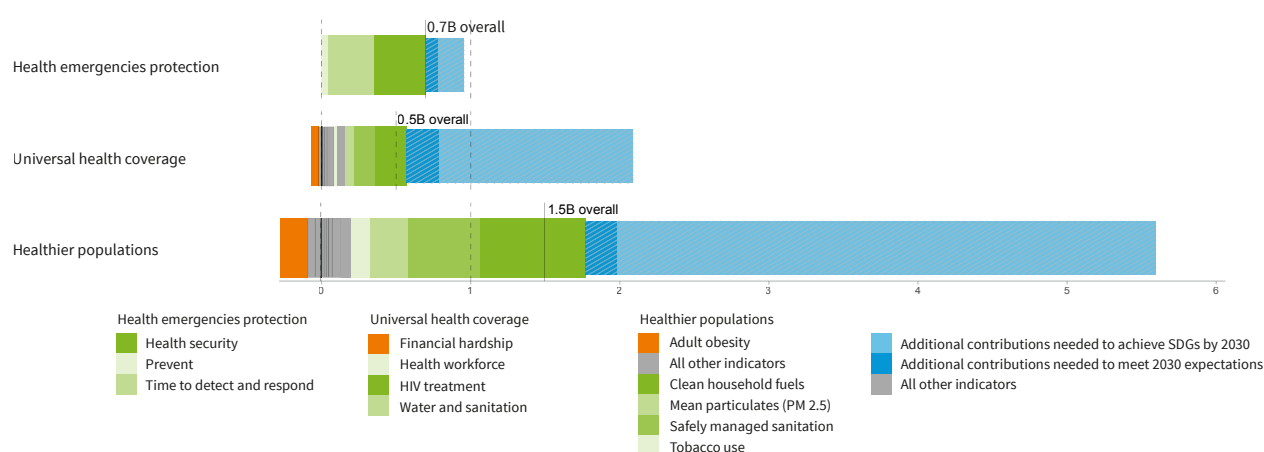
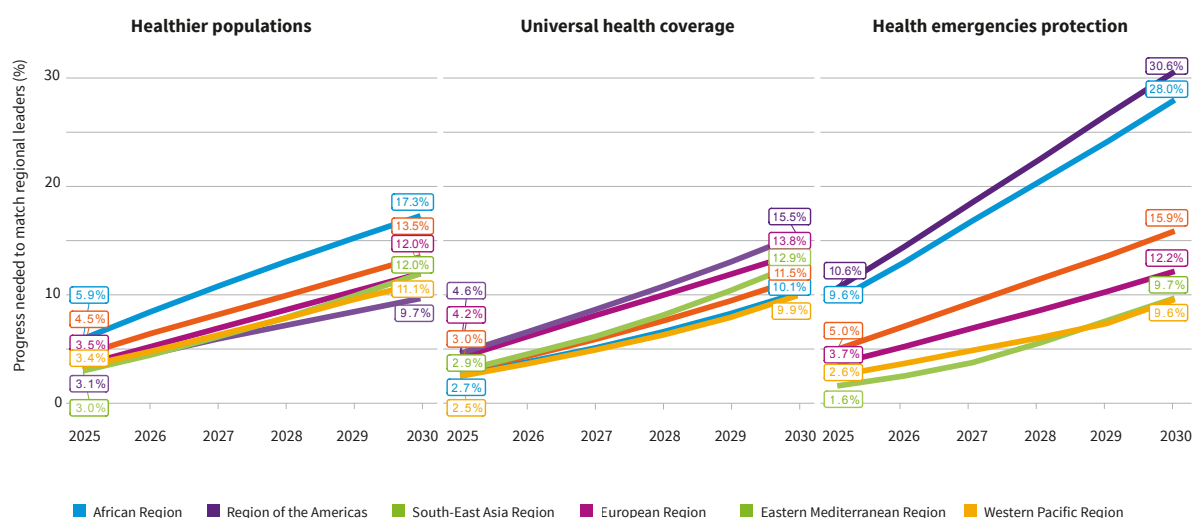




Figure 3.4 Progress required for all countries to match the top performers in their region, on top of the Triple Billion target projections, ^{a,b} 2025–2030



^a Matching the top performers means achieving the values of the indicators from the 10% best-performing countries in the same WHO region, by 2030.

^b Percentages are calculated as the ratio of additional persons required to match the top performers over the total 2030 population in the region.

Box 3.1 Transition from GPW13 to GPW14

The WHO GPW13 (2019–2023, extended to 2025), strengthened WHO’s effectiveness and accountability in improving global health outcomes. Triple Billion targets, a key component of GPW13, were designed to enhance accountability and drive measurable health improvements worldwide. By strategically focusing on three target areas and leveraging existing data sources, WHO aimed to deliver tangible health benefits to billions of people.

In September 2023, the United Nations Heads of State and Government Meeting called for doubling the rate of progress on the SDGs to meet the 2030 Agenda (5). In response, WHO is developing the fourteenth general programme of work (GPW14, 2025–2028) to address two fundamental challenges: slow progress and insufficient data.

Building on GPW13, GPW14 focuses on promoting, providing and protecting health while strengthening the global health ecosystem to advance the SDGs. Additionally, it seeks to enhance WHO’s organizational performance across all three levels – global, regional and country.

The GPW14 results framework translates health goals into measurable targets through a structured results chain. This chain links WHO’s work to broader health and development outcomes. As in GPW13, progress will be assessed using the Triple Billion indices through outcome indicators in three overarching areas: promote, provide and protect.

Triple Billion targets recalibration

Under GPW13, the Triple Billion targets for 2025 were defined in terms of relative changes – the additional number of people who achieved specific health status compared to 2018:

- 1 billion more people living healthier lives (promote)
- 1 billion more people benefitting from universal health coverage (provide)
- 1 billion more people better protected from health emergencies (protect)

For GPW14, the targets are recalibrated to reflect absolute population coverage by 2028, accounting for improved measurement methods and setting ambitious yet realistic goals:

- 6 billion people living healthier lives (promote)
- 5 billion people benefitting from universal health coverage without financial hardship (provide)
- 7 billion people protected from health emergencies (protect)

The revised Triple Billion targets offer several benefits:

- Better alignment with global targets, including the SDGs and World Health Assembly (WHA) resolutions
- Enable scenario analysis for assessing health interventions at regional and country levels
- Facilitate direct statistical forecasting using past trends and country-specific data
- Support benchmarking assessments to identify best practices at regional and global levels
- Balance ambition with realism, ensuring achievable yet aspirational goals
- Account for uncertainty in observed data and future projections
- Promote equity, ensuring that no one is left behind or unprotected.

3.6 Conclusion

The WHO Triple Billion framework has served as both a strategic guide and a practical tool for accountability and accelerating progress in global health. By providing a structured approach to tracking advancements in indicators across regions and countries, it offers decision-makers actionable insights on where to focus resources and policy interventions. The 2025 projections, developed using the most recent available data, reveal both encouraging progress and gaps that require targeted action. Regional analysis shows how varied approaches to reaching these targets reflect different priorities, challenges and opportunities across the six WHO regions.

Looking ahead, as WHO transitions from GPW13 to GPW14 in 2025, the core structure of the Triple Billion targets remains in place. However, the list of outcome indicators has been updated to reflect emerging health challenges and strategic priorities set collectively by the global community for GPW14 (see Box 3.1). Through GPW14, WHO reaffirms its unwavering commitment to addressing emerging global health priorities and tackling challenges, as mandated by its Member States, despite an evolving global landscape.



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4

Inequality in immunization

Ensuring equitable access to immunization is vital for achieving public health goals and ensuring that every person, regardless of their background and circumstances, fully benefits from vaccines to improve health and well-being. This chapter presents the state of inequality in immunization coverage. It shows how inequalities in immunization have changed in the last decade and discusses barriers to achieving equitable immunization coverage.

4.1 The importance of vaccine equity

Vaccines are life-saving interventions that have dramatically improved global health outcomes. Immunization currently prevents 3.5 million to 5 million deaths every year from diseases such as diphtheria, tetanus, pertussis, measles and influenza (1). Since the launch of the WHO Expanded Programme on Immunization in the mid-1970s, childhood vaccination coverage has improved significantly, contributing to a substantial reduction in U5MR (2). This underscores the critical role of vaccines in safeguarding public health and highlights the importance of continuing efforts to ensure equitable access to immunization worldwide.

Equity in immunization is a cornerstone of the Immunization Agenda 2030 (IA2030), which envisions a world where everyone, everywhere, at every age fully benefits from vaccines to improve health and well-being (3). This global strategy aims to leave no one behind by increasing equitable access to vaccines, ensuring that immunization programmes are integrated into primary health care and contributing to achieving universal health coverage.

Global inequalities persist in vaccine access, financing, availability and distribution – both between countries and within countries. Low-income countries often face challenges in securing sufficient vaccine supplies due to limited financial resources, disruptions in health-care services, logistical challenges and inequities in access to services (4).

Within countries, inequalities in vaccine access and coverage can be influenced by factors such as geographic location, socioeconomic status and education levels. Rural and remote areas may have lower access to vaccination services compared with urban centres, and those affected by social disadvantage may face barriers to immunization due to lack of information or logistical challenges. Addressing inequalities requires targeted efforts to strengthen health systems, improve vaccine delivery infrastructure and ensure that immunization programmes are inclusive and accessible to all segments of the population.

4.2 Inequalities in immunization coverage persist within countries

The delivery of three doses of the combined DTP3 vaccine among children under the age of 1 year is used as a proxy indicator for monitoring the capacity of a health system to reliably reach children through routine vaccination channels. Measuring inequalities in DTP3 immunization coverage within countries reveals where gaps exist in a routinely delivered vaccine and helps to inform appropriate approaches to reach under-vaccinated populations.

Figure 4.1 shows median inequality in DTP3 immunization coverage across LICs and middle-income countries with a recent household survey (from 2014–2023) by economic status, education and place of residence. For economic status and education, inequality is measured as the difference along the socioeconomic spectrum, while taking the composition of every socioeconomic subgroup into consideration.¹ For place of residence, inequality is measured as the difference between urban and rural subgroups. Medians are calculated across all countries

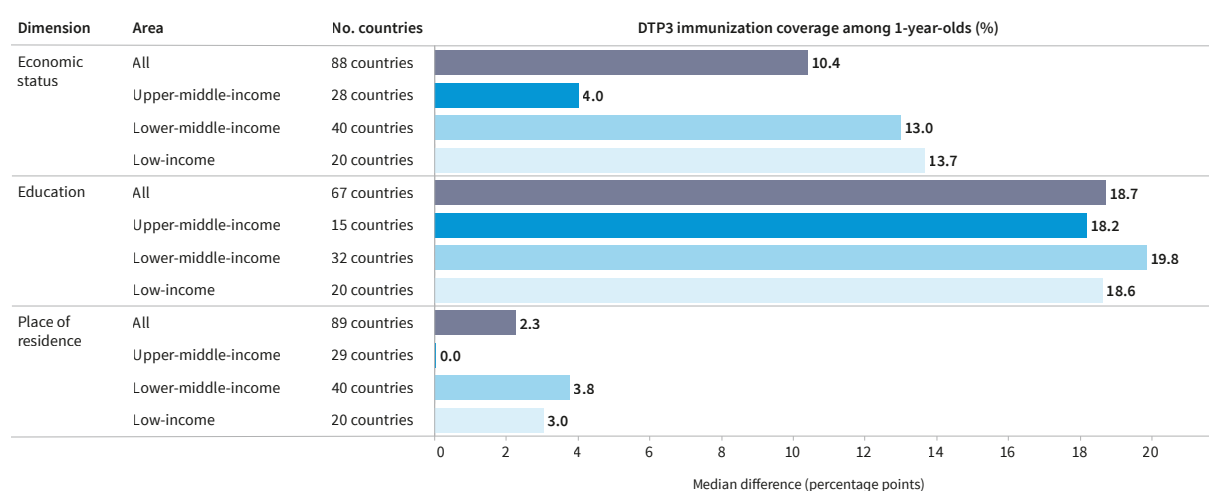
¹ Also referred to as the slope index of inequality.

with data available, as well as by World Bank country income grouping.

The results indicate inequalities favouring those living in richer households (median economic-related inequality of 10.4 percentage points across 88 countries), having more education (median education-related inequality of 18.7 percentage points across 67 countries) and living in urban areas (median difference of 2.3 percentage

points across 89 countries). Economic-related inequality in DTP3 immunization coverage is most pronounced in LICs and LMICs, where there are median inequalities of 13.7 and 13.0 percentage points in coverage along the economic status spectrum, respectively. Education-related inequality was high across UMICs, LMICs and LICs, while differences between urban and rural areas were low overall across country income groupings.

Figure 4.1 Median economic-related, education-related and place of residence inequality in DTP3 immunization coverage, by World Bank country income group, 2014–2023



DTP3: diphtheria–tetanus–pertussis vaccine third dose. The median difference for economic status and education reflects the difference along the whole socioeconomic spectrum, while taking the composition of every socioeconomic subgroup into consideration. The median difference for place of residence reflects the difference between urban and rural subgroups.

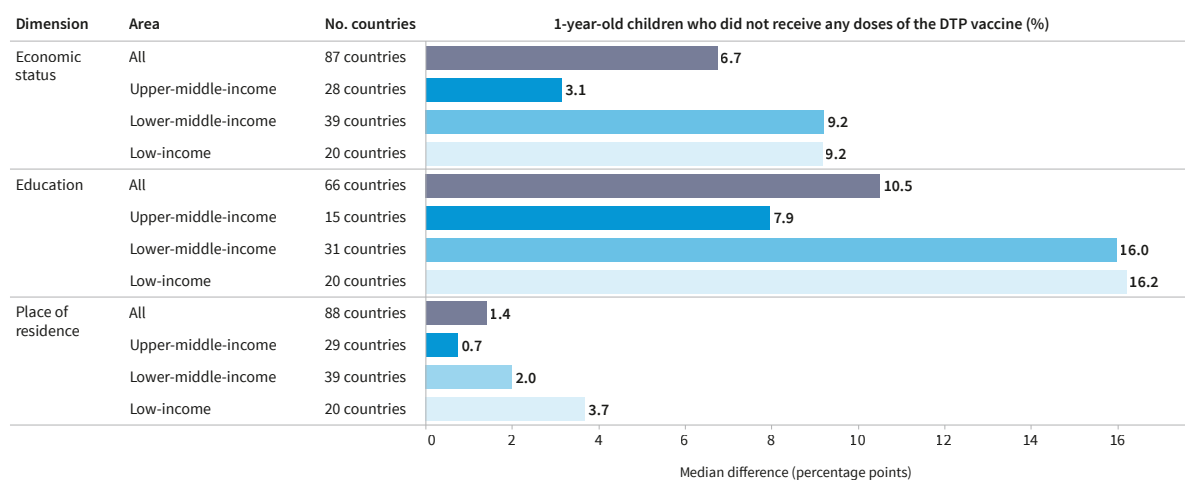
Source: derived from the WHO Health Inequality Data Repository, Childhood Immunization dataset, with data sourced from the most recent Demographic and Health Survey or Multiple Indicator Cluster Survey between 2014 and 2023 (5).

The number of children who have not received any routine vaccines (referred to as zero-dose children) reflects the accessibility of basic health services, including immunization services. The IA2030 includes an objective to extend immunization services to regularly reach zero-dose children, with a target of a 50% reduction in zero-dose children between 2019 and 2030 (3). An analysis of inequalities in the proportion of 1-year-old children who had not received any doses of the DTP vaccine, using household survey data across 88 countries, reveals substantial economic- and education-related inequalities (Fig. 4.2). Data from 87 countries showed a

median economic-related inequality of 6.7 percentage points in zero-dose DTP prevalence, but inequalities were particularly pronounced across LMICs and LICs, where median economic-related inequality was greater than 9 percentage points. Across 66 countries, median education-related inequality in zero-dose prevalence was 10.5 percentage points but inequalities were 16 percentage points in LMICs and LICs. Overall, there was low median inequality between rural and urban areas in 88 countries, but zero-dose prevalence was slightly higher in rural than urban areas in LMICs and LICs.



Figure 4.2 Median economic-related, education-related and place of residence inequality in zero-dose DTP prevalence, by World Bank country income group, 2014–2023



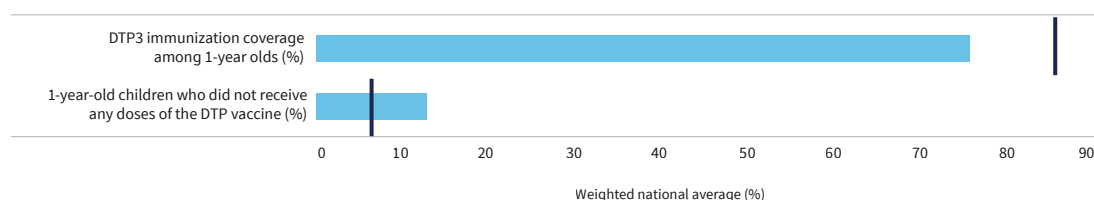
DTP3: diphtheria–tetanus–pertussis vaccine third dose. The median difference for economic status and education reflects the difference along the whole socioeconomic spectrum, while taking the composition of every socioeconomic subgroup into consideration. The median difference for place of residence reflects the difference between rural and urban subgroups.

Source: derived from the WHO Health Inequality Data Repository, Childhood Immunization dataset, with data sourced from the most recent Demographic and Health Survey or Multiple Indicator Cluster Survey between 2014 and 2023 (5).

Eliminating economic-related inequality in DTP3 coverage and zero-dose prevalence has the potential for many more children to benefit from immunization. The potential for improvement in each country is premised on all children achieving the same outcome as the richest 20% of households. If economic-related inequality in DTP3 coverage were eliminated across the

88 countries with household survey data from 2014–2023, the weighted national average across these countries would improve from 76.0% to 85.5% (Fig. 4.3). If countries reduced the prevalence of zero-dose children to that of the richest subgroup, the weighted national average prevalence would be more than halved, from 13.1% to 6.3%.

Figure 4.3 Potential improvement in national average DTP3 coverage and zero-dose children by eliminating economic-related inequality, 88 countries, 2014–2023



DTP3: three doses of diphtheria–tetanus–pertussis vaccine.

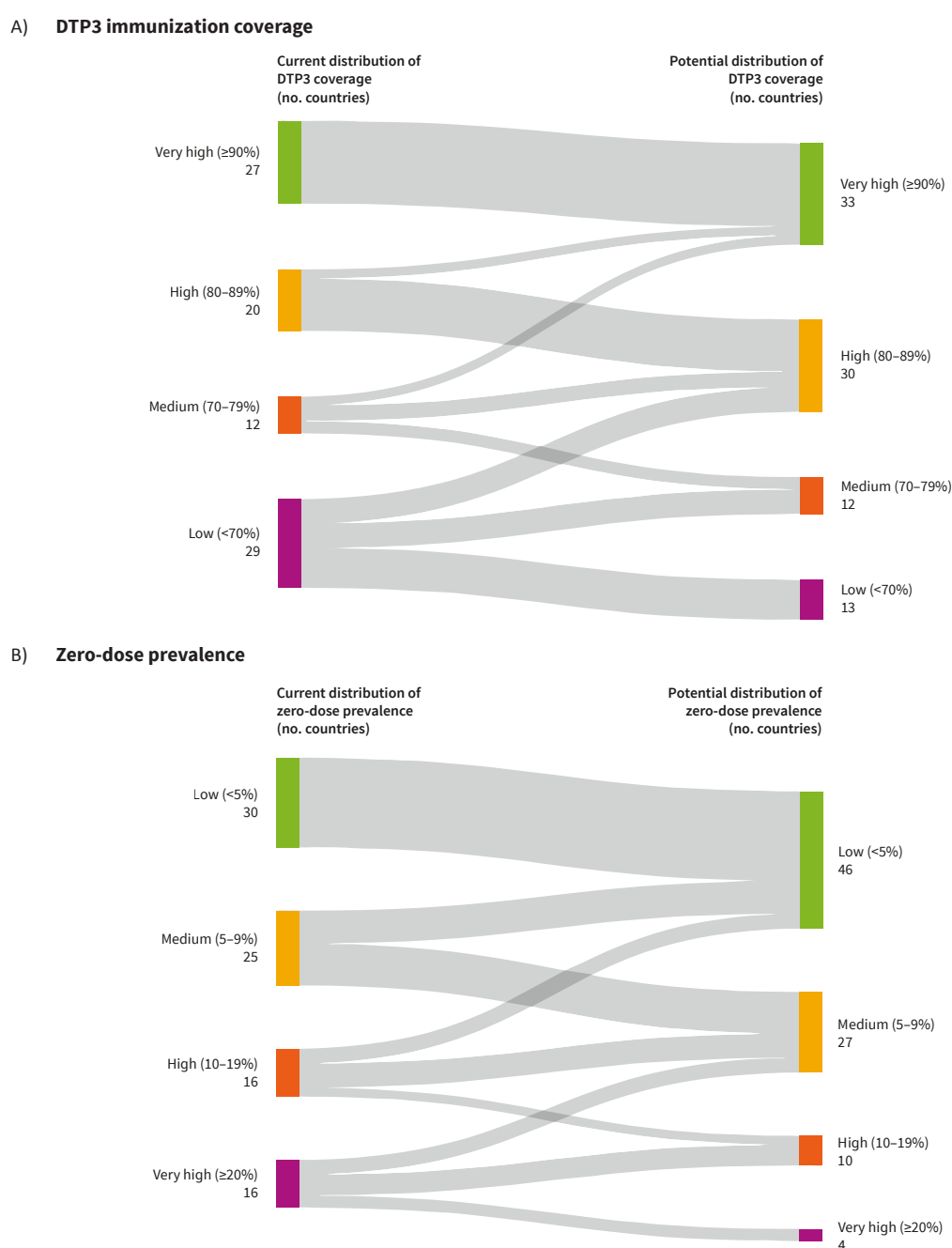
The potential for improvement (dark blue vertical line) represents the overall weighted average that would be possible if, in each country, all 1-year-olds had the same level of coverage as the most advantaged subgroup (richest quintile). The current weighted average is indicated by the light blue bar.

Source: derived from the WHO Health Inequality Data Repository, Childhood Immunization dataset, with data sourced from the most recent Demographic and Health Survey or Multiple Indicator Cluster Survey between 2014 and 2023 (5).

Of 88 countries with household survey data from 2014–2023, only 27 had DTP3 coverage of 90% or more, while coverage was 80–89% in 20 countries, 70–79% in 12 countries and below 70% in 29 countries (Fig. 4.4A). After considering the potential improvement in national average by eliminating economic-related inequality, 6 additional countries would have coverage greater than 90% and 10 additional countries would have coverage of 80–89%. If economic-related

inequality in zero-dose prevalence were eliminated, 16 additional countries would reduce zero-dose prevalence to below 5% (Fig. 4.4B). The link between economic status and immunization coverage is complex and, as Box 4.1 shows, inequalities can emerge at various points of contact with the health system – underscoring the need for exploration of the drivers of inequalities and context-specific strategies to address them.

Figure 4.4 Potential improvement in national average by eliminating economic-related inequality, 88 countries, 2014–2023



The potential improvement reflects all 1-year-olds having the same outcome as the richest 20% of the population.

Source: derived from the WHO Health Inequality Data Repository, Childhood Immunization dataset, with data sourced from the most recent Demographic and Health Survey or Multiple Indicator Cluster Survey between 2014 and 2023 (5).



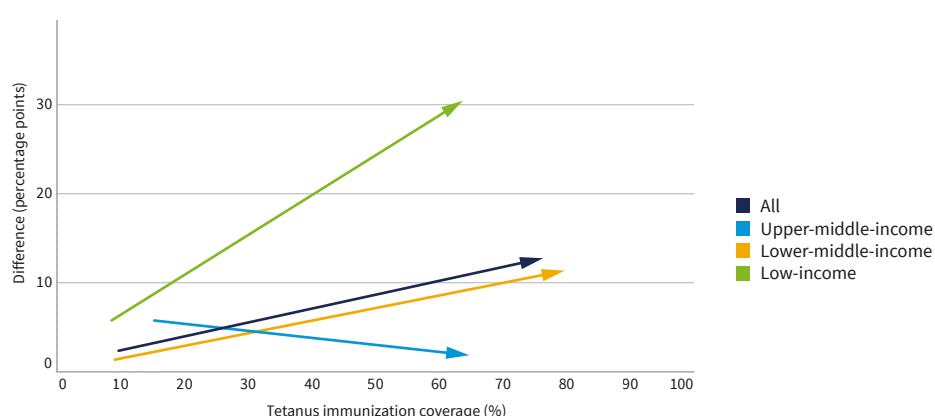
Box 4.1 Inequalities can be introduced through interventions: the example of tetanus immunization coverage at birth

Maternal and neonatal tetanus (MNT) is a form of tetanus that affects women during pregnancy or within six weeks of the end of pregnancy, and infants during their first 28 days of life (6). MNT constitutes a major public health concern, as neonatal case-fatality rates are upwards of 80% and approach 100% when untreated (7). Immunization of pregnant women against tetanus is a key strategy for reducing tetanus morbidity and mortality while also achieving the goal of maternal and neonatal tetanus elimination. An infant's tetanus protection at birth (PAB) is comprised of immunization received by the woman during and before her pregnancy (e.g. through childhood vaccination, booster doses or mass vaccination campaigns, or during prior pregnancies).

Global inequalities in tetanus immunization coverage at birth are evident across various dimensions, including wealth, maternal age, education and place of residence (8). For instance, poorer households, younger maternal age, lower education of the mother and residence in rural areas tend to have lower rates of PAB. These inequalities are particularly pronounced in LICs and LMICs, but they also persist in UMICs, especially concerning education level of the mother and rural residence.

Findings from a study of 71 LICs and middle-income countries suggest that the majority of tetanus immunization PAB for first births is the result of vaccinations received during pregnancy. The extent of economic-related inequality in PAB increases during pregnancy, as more women from richer households than women from poorer households are vaccinated during pregnancy (9). Economic-related inequality is measured as the difference along the socioeconomic spectrum, while taking the composition of every socioeconomic subgroup into consideration. Overall, LICs and middle-income countries had similar levels of economic-related inequality and tetanus immunization coverage before pregnancy. However, increases in tetanus immunization coverage at birth were coupled with increases in inequality in LICs, overall, in contrast with UMICs (Fig. 4.5). Inequalities in coverage at birth in this group were introduced during pregnancy, likely driven in part by inequalities in antenatal care access and utilization. Addressing these inequalities requires targeted interventions to improve access to and uptake of tetanus vaccines among women from poorer households, ensuring that all pregnant women and newborns receive adequate protection against this preventable disease.

Figure 4.5 Economic-related inequality in maternal tetanus immunization coverage and average coverage level before pregnancy and at birth, by World Bank country income group, 71 countries, 2013–2022



The start of the arrow indicates the level of economic-related inequality and tetanus immunization coverage before pregnancy and the end of the arrow indicates the level of inequality and coverage at birth. For each country, economic-related inequality is measured as the difference along the whole socioeconomic spectrum while taking the composition of every socioeconomic subgroup into consideration. For each World Bank income group, the weighted mean difference is calculated across countries. Based on 18 upper-middle-income, 33 lower-middle-income and 20 low-income countries.

Source: derived from Johns et al. (9) with data sourced from the most recent Demographic and Health Survey or Multiple Indicator Cluster Survey between 2013 and 2022.

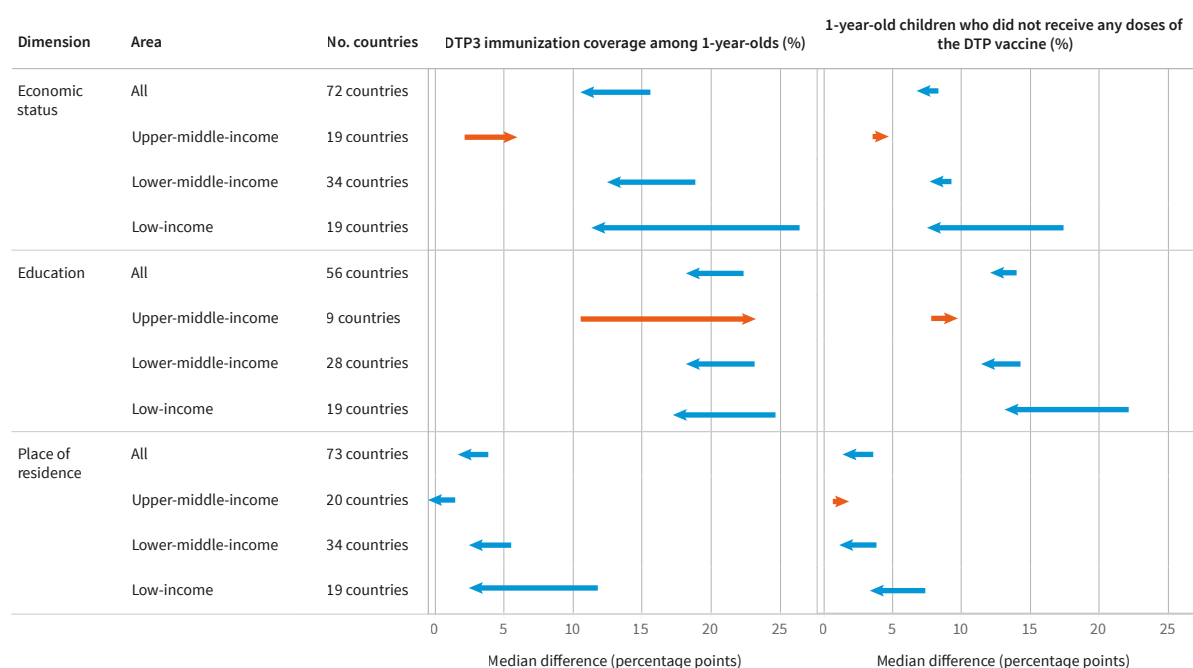
4.3 How inequalities have changed over time

Over the past two decades, significant efforts have been made to address inequalities in immunization coverage. IA2030 and its predecessor, the Global Vaccine Action Plan for 2011–2020, focus on reaching zero-dose children and ensuring equitable access to vaccines. Collaborations between WHO, UNICEF and Gavi, the Vaccine Alliance, have significantly impacted the number of zero-dose children in lower-income countries. For example, these efforts have helped to reduce the number of zero-dose children from 15.8 million in 2010 to 12.8 million in 2019 (10). However, the COVID-19 pandemic resulted in setbacks to vaccination programmes, resulting in an increase in the number of zero-dose children to 14.5 million in 2023. In response to disruptions caused by the COVID-19 pandemic, WHO and UNICEF launched catch-up and recovery programmes to restore immunization services. Targeted vaccination campaigns

in low-coverage regions aim to close immunization gaps and prevent outbreaks.

The overall change in inequality in DTP3 immunization coverage and zero-dose prevalence can be measured using the median difference along the socioeconomic spectrum (while taking the composition of every socioeconomic subgroup into consideration). Between 2004–2013 and 2014–2023, overall economic- and education-related inequalities in DTP3 immunization coverage decreased (Fig. 4.6). Across 72 countries, economic-related inequality in DTP3 coverage reduced from a median of 15.4 to 11.0 percentage points, and education-related inequality across 56 countries reduced from a median of 22.1 to 18.7 percentage points. Overall, economic- and education-related inequalities in zero-dose prevalence only decreased slightly over this period.

Figure 4.6 Change in median economic, education and place of residence inequality in DTP3 immunization coverage and zero-dose prevalence between 2004–2013 and 2014–2023, by World Bank country income group



DTP3: three doses of diphtheria–tetanus–pertussis vaccine.

The start of the arrow indicates the median level of inequality in 2004–2013 and the end of the arrow indicates the median level of inequality in 2014–2023. The median difference for economic status and education reflects the difference along the whole socioeconomic spectrum, while taking the composition of every socioeconomic subgroup into consideration. The median difference for place of residence reflects the difference between urban and rural subgroups for DTP3 coverage and between rural and urban subgroups for zero-dose prevalence. Blue indicates a decrease in inequality, while orange indicates an increase in inequality.

Source: derived from the WHO Health Inequality Data Repository, Childhood Immunization dataset, with data sourced from the most recent Demographic and Health Survey or Multiple Indicator Cluster Survey between 2014 and 2023 and between 2004 and 2013 (5).



Changes in immunization inequality related to economic status, education and place of residence varied across country income groupings. Broadly, in the last decade there has been a convergence in inequalities across LICs and LMICs, with inequalities diminishing in LICs to the point that their median level of inequality in 2014–2023 was similar to (or better than) that of LMICs. In particular,

across 19 LICs, economic-related inequality in DTP3 coverage and zero-dose prevalence reduced by more than half, from 26.2 to 11.8 percentage points and from 17.2 to 8.0 percentage points, respectively. In UMICs, inequality remained the same for zero-dose prevalence but economic- and education-related inequality in DTP3 coverage increased.

4.4 Barriers to immunization

The Equity Reference Group for Immunization² has identified several thematic priorities to address inequities in immunization coverage, including urban poor areas, remote/rural areas, children affected by conflict, and gender-related inequities and barriers (11). In such contexts, different strategies are needed to understand coverage levels and address inequities in immunization coverage. Surveys are rarely carried out to enable the accurate assessment of coverage rates in the above contexts, and when they are, they generally lack granularity. This results in a scarcity of disaggregated data and inadequate monitoring of inequalities. Consequently, it is challenging to identify disadvantaged groups within these contexts and implement programmes that prioritize reaching them.

In urban poor areas, overcrowding and poverty can significantly hinder vaccine delivery and uptake. These areas often lack adequate health-care facilities and resources, making it challenging to maintain consistent immunization services. Remote rural regions face similar issues, with limited health-care infrastructure and difficult terrain further complicating efforts to reach all children. For instance, a study in Malawi found that proximity to a health facility providing vaccination services in rural areas was associated with increased likelihood of vaccine receipt, and that remote rural children were more likely to be under-vaccinated (12). Additionally, children affected by conflict experience unique barriers to immunization. Conflict zones often see disrupted health services, displacement of populations and heightened insecurity, all of which contribute to lower immunization rates. These children may miss out on routine vaccinations due to instability and lack of access to health care.

Gender-related barriers can hinder immunization coverage by limiting women's autonomy and decision-making power, which affects their ability to access health services for themselves and their children. Additionally, cultural norms and practices may restrict women's mobility and access to information, further reducing vaccination coverage in communities with high gender inequality. Women with higher social independence, as measured by the Survey-based Women's Empowerment Global Index, are more likely to ensure their children receive vaccinations (13). In contrast, DTP3 coverage tends to be lower, and zero-dose prevalence higher, among children of women with lower social independence.

At the subnational and national levels, areas with higher gender inequality tend to have lower immunization coverage. In a study of 162 countries, those with high gender inequality (favouring men), as measured using the Gender Inequality Index,³ had DTP3 immunization coverage that was 13 percentage points lower (median value of 81% versus 94%) and zero-dose prevalence 7 percentage points higher (median value of 10% versus 3%) than countries with lower inequality (14). A study of 702 subnational regions across 57 countries also found that regions with greater gender inequality had lower DTP3 coverage and higher zero-dose prevalence (15). Subnational regions with higher gender inequality (favouring men), as measured by the Subnational Gender Development Index, had DTP3 immunization coverage that was 21.6 percentage points lower, while zero-dose prevalence was 13.4 percentage points higher (median values).

² The Equity Reference Group for Immunization is an action-oriented think tank that consists of senior experts in global health working with WHO, Gavi, the World Bank, Bill & Melinda Gates Foundation and UNICEF; academic experts in metrics, gender, and health systems development; and senior leaders from the ministries of health in Indonesia, Ethiopia and the United Republic of Tanzania.

³ The Gender Inequality Index is a composite metric of gender inequality using three dimensions: reproductive health, empowerment and the labour market. It shows the loss in potential human development due to inequality between female and male achievements in these dimensions.

4.5 Data availability to monitor inequalities in immunization

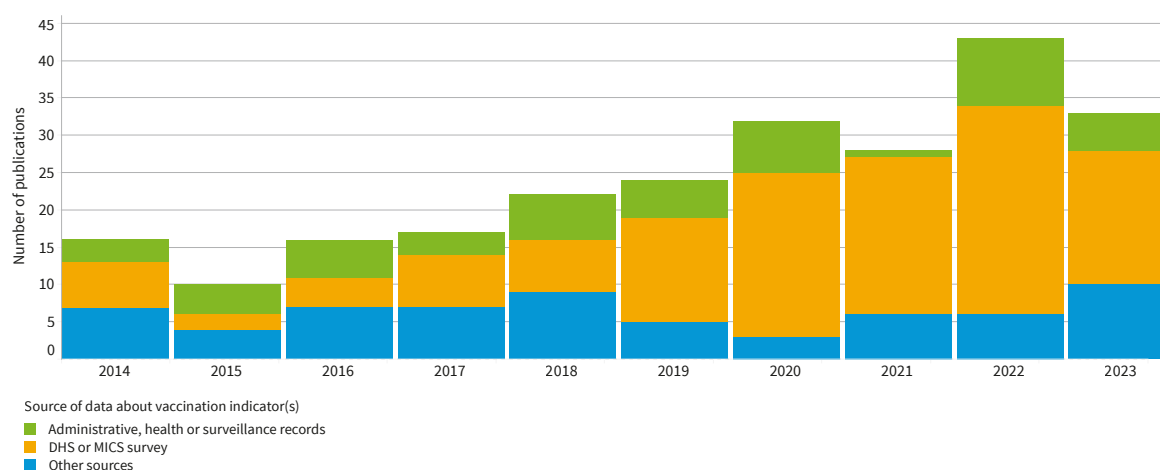
Monitoring inequalities in immunization relies on various data sources, each with its own advantages and disadvantages (16). Administrative data are routinely collected by health facilities and can provide timely, comprehensive information on immunization coverage. These data are cost-effective and cover large populations, making them useful for tracking trends over time. However, administrative data can suffer from inaccuracies due to reporting errors, incomplete records, sparse information about dimensions of inequality and uncertainty about target population numbers. Household survey data, such as those from DHS and Multiple Indicator Cluster Surveys (MICS), offer detailed, population-based insights into immunization coverage. These surveys are valuable for capturing data on hard-to-reach populations and for validating administrative data. They also provide rich contextual information, such as socioeconomic and demographic factors. However, household surveys are expensive, time-consuming and conducted infrequently, which can limit their utility for real-time monitoring.

Data availability can be a challenge, particularly in LICs where resources for comprehensive data collection and reporting are often limited. These countries may face difficulties in maintaining accurate and up-to-date records, which can hinder efforts to monitor

and address immunization inequalities effectively. Logistical challenges, such as limited access to remote areas and insufficient funding for large-scale surveys, further complicate data collection efforts. Inaccurate estimates of the target population for vaccination can exacerbate these difficulties, particularly when monitoring subnational inequalities and subpopulations of interest. Misestimations can lead to skewed coverage rates, masking inequalities and making it challenging to identify and address areas with the greatest need (17). Addressing these challenges requires investment in strengthening health information systems to ensure reliable data are available for informed decision-making and targeted interventions.

A scoping review of publications in academic journals analysing inequalities in immunization coverage among children revealed a rise in the use of DHS and MICS data between 2014 and 2023 (Fig. 4.7) (18). Across the 10-year period, 129 papers (54%) had utilized DHS and/or MICS data, demonstrating the utility of these data sources for monitoring inequalities in immunization. The number of publications using administrative, health or surveillance records remained relatively stable, as did those based on other sources such as non-routine, study-specific or small-scale surveys.

Figure 4.7 Publications of studies of inequalities in childhood vaccination coverage between 2014 and 2023 by data source for vaccination indicators



Source: derived from Lyons et al. (18).



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Annexes

Annex 1.

Progress assessment of selected health-related SDG indicators by WHO region

A summary of progress of selected health-related SDG indicators is presented in Table A1.1 for Goal 3 (Health and Well-being) indicators and Table A1.2 for those outside of Goal 3.

An indicator is included in the assessment if it meets the following criteria:

- There is an explicit numerical SDG or other global framework target to be achieved by a defined year, applicable at both global and regional levels.
- Estimates are available for the baseline year and at least one other later year to enable a trend assessment.

For each indicator, the numbers shown in the global and regional cells are the latest available estimated values. The reference year of these values is indicated in the corresponding column.

Progress is assessed based on the latest available value, and the projected value for the target year based on

current trends. Unless otherwise noted, current trends are defined by the average annual rates of change between the baseline year and the reference year of the latest available value. Progress is described in three categories as follows:

- “Target reached”: target is already achieved
- “Progress on track”: target will be achieved by the target year based on current trends
- “Acceleration needed”: target will not be met by the target year based on current trends

It is noted that this method of assessment may differ from those used by the United Nations Statistics Division or by the custodian agencies of individual indicators, leading to different assessment results.

Tables A1.3 and A1.4 provide the list of health-related SDG indicators that are not included in Tables A1.1 and A1.2.

Table A1.1 Latest values^a and progress assessment^b of selected SDG 3 indicators, ^c by WHO region

Indicator	Notes	Year			Target	Most recent estimate	Global	African Region	Region of the Americas	South-East Asia Region	European Region	Eastern Mediterranean Region	Pacific Region
		Baseline	Target	2025	2030	2024	2025	2026	2027	2028	2029	2030	2031
3.1.2 Proportion of births attended by skilled health personnel (%)		2015	2025	2030	2035	2024	90	87	74	96	90	85	98
3.3.1 New HIV infections (per 1000 uninfected population)	d	2010	2030	2035	2040	2023	Reduce new infections by 90%	0.17	0.55	0.16	0.06	0.09	0.07
3.3.2 Tuberculosis incidence (per 100 000 population)		2015	2030	2035	2040	2023	Reduce by 80%	134	206	33	234	116	97
3.3.3 Malaria incidence (per 1000 population at risk)		2015	2030	2035	2040	2023	Reduce by 90%	60.4	226.8	3.6	2.3	17.9	2.3
3.3.4 Hepatitis B surface antigen (HBsAg) prevalence among children under 5 years (%)	e	2015	2030	2035	2040	2020	0.1	0.9	2.5	0.1	0.4	0.8	0.3
3.3.5 Reported number of people requiring interventions against NTDs (millions)	f	2010	2030	2035	2040	2023	Reduce by 90%	1495	518.6	30.7	787.7	78.0	74.0
3.4.1 Probability of dying between exact ages 30 and 70 from any of cardiovascular disease, cancer, diabetes or chronic respiratory illness (%)		2015	2030	2035	2040	2019	Reduce by 1/3	18.0	21.0	13.9	22.9	22.7	15.7
3.4.2 Suicide mortality rate (per 100 000 population)		2015	2030	2035	2040	2021	Reduce by 1/3	9.2	7.3	9.9	10.1	12.4	9.5
3.5.2 Total alcohol per capita (aged 15 years and older) consumption (litres of pure alcohol)		2010	2030	2035	2040	2022	Reduce by 20%	5.0	3.5	8.0	3.6	9.1	5.2
3.6.1 Death rate due to road traffic injuries (per 100 000 population)	g	2021	2030	2035	2040	2021	Halve the number of deaths	14.9	19.5	14.2	15.4	16.3	15.4
3.a.1 Age-standardized prevalence of current tobacco use among persons 15 years and older (%)	h	2010	2025	2030	2035	2022	Reduce by 30%	20.9	9.5	16.6	26.5	17.9	22.5
3.b.1 Diphtheria-tetanus-pertussis third-dose (DTP3) immunization coverage among 1-year-olds (%)		2019	2030	2035	2040	2023	90	84	74	86	90	79	92
3.b.1 Measles-containing-vaccine second-dose (MCV2) immunization coverage by the locally recommended age (%)		2019	2030	2035	2040	2023	90	74	49	75	85	73	90
3.b.1 Pneumococcal conjugate vaccine third-dose (PCV3) immunization coverage among 1-year-olds (%)	i	2019	2030	2035	2040	2023	90	65	70	76	75	52	26
3.b.1 Human papillomavirus (HPV) immunization coverage estimates among 15-year-old girls (%)	i	2019	2030	2035	2040	2023	90	20	32	55	8	2	7

NTDs: neglected tropical diseases.

Notes:

a The numbers shown in the global and regional cells are the latest available estimated values.

b Progress is assessed on the basis of the latest available value, and the projected value for the target year based on the average rates of change since the baseline year, unless otherwise stated.

c An SDG 3 indicator is included if: (1) there is an explicit numerical SDG or other global framework target to be achieved by a defined year, applicable at global and regional levels, and (2) data are available for the baseline year and at least one other later year.

d Progress assessment is based on the change in the total number of new infections.

e Projection takes into account preliminary updates to the estimates.

f Projection is based on multi-disease country-level data and planned interventions in the next years.

g Projection for the 2030 number of deaths is based on the change in the number of deaths since 2010, which was the baseline of the first United Nations Decade of Action for Road Safety 2010–2020.

h No region or country will be assessed as “Target reached” before 2025 data are available.

i Projection takes into account recent introduction of the vaccine in several countries.

Source: Births attended by skilled health personnel (1), HIV (2,3), tuberculosis (4), malaria (5), hepatitis (6), NTD (7), noncommunicable disease premature mortality (8), suicide mortality (9), alcohol (9), road traffic mortality (8,10), tobacco (11), immunization (12).



**Table A1.2 Latest values^a and progress assessment^b of selected health-related SDG indicators (excluding Goal 3),^c by WHO region**

Indicator	Notes	Year			Global	African Region	Region of the Americas	South-East Asia Region	European Region	Eastern Mediterranean Region	Western Pacific Region
		Baseline	Target	Most recent estimate							
2.2.1 Prevalence of stunting among children under 5 years of age (%)	d	2012	2030	2024	23.2	31.7	9.9	29.7	4.6	26.0	10.2
2.2.2 Prevalence of wasting among children under 5 years of age (%)	d	2012	2030	2024	6.6	5.6	0.7	13.9	1.0	6.3	1.9
2.2.2 Prevalence of overweight among children under 5 years of age (%)	d	2012	2030	2024	5.5	4.2	9.1	3.3	7.5	4.5	9.7
2.2.3 Prevalence of anaemia in women aged 15 to 49 years (%)	d	2012	2030	2023	30.7	36.5	18.3	46.4	21.3	35.8	16.5
6.1.1 Proportion of population using safely managed drinking water services (%)		2015	2030	2022	73	33	81		92	67	
6.2.1 Proportion of population using safely managed sanitation services (%)		2015	2030	2022	57	26	66	49	78	55	68
6.2.1 Proportion of population using a handwashing facility with soap and water (%)	e	2015	2030	2022	75	26		76		72	95
7.1.2 Proportion of population with primary reliance on clean fuels and technology (%)		2015	2030	2023	74	24	93	73	97	74	87

Notes:

a The numbers shown in the global and regional cells are the latest available estimated values.

b Progress is assessed on the basis of the latest available value, and the projected value for the target year based on the average rates of change since the baseline year, unless otherwise stated.

c An indicator is included if: (1) there is an explicit numerical SDG or other global framework target to be achieved by a defined year, applicable at global and regional levels, and (2) data are available for the baseline year and at least one other later year.

d Methodology based on the WHO–UNICEF Methodology for monitoring progress towards the 2030 global nutrition targets. Global Nutrition Targets Tracking Tool <https://www.who.int/tools/global-targets-tracking-tool>.

e Projection for the Western Pacific Region is based on the average annual rate of change since 2017, the earliest year for which the regional estimate is available.

Source: Child malnutrition (13), anaemia (14), water, sanitation and hygiene (15), clean fuels (16).

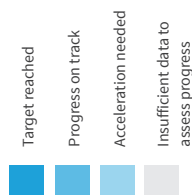


Table A1.3 List of SDG 3 indicators not included in the assessment

Indicator	Reason for exclusion
3.1.1 Maternal mortality ratio	Target set at the global level; progress not assessed at the regional level
3.2.1 Under-5 mortality rate	Target set at the country level; progress not assessed at the regional level
3.2.2 Neonatal mortality rate	Target set at the country level; progress not assessed at the regional level
3.5.1 Coverage of treatment interventions (pharmacological, psychosocial and rehabilitation and aftercare services) for substance use disorders	No explicit numerical target
3.7.1 Proportion of women of reproductive age (aged 15–49 years) who have their need for family planning satisfied with modern methods	No explicit numerical target
3.7.2 Adolescent birth rate	No explicit numerical target
3.8.1 Coverage of essential health services	No explicit numerical target
3.8.2 Proportion of population with large household expenditures on health as a share of total household expenditure or income	No explicit numerical target
3.9.1 Mortality rate attributed to household and ambient air pollution	No explicit numerical target
3.9.2 Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene	No explicit numerical target
3.9.3 Mortality rate attributed to unintentional poisoning	No explicit numerical target
3.b.3 Health product access index	No explicit numerical target
3.c.1. Health worker density and distribution	No explicit numerical target
3.d.1 International Health Regulations (IHR) capacity and health emergency preparedness	No explicit numerical target
3.d.2 Percentage of bloodstream infections due to selected antimicrobial-resistant organisms	No explicit numerical target

Table A1.4 List of health-related SDG indicators (excluding Goal 3) not included in the assessment

Indicator	Reason for exclusion
1.a.2 Domestic general government health expenditure (GGHE-D) as percentage of general government expenditure (GGE) (%)	No explicit numerical target
5.2.1 Proportion of ever-partnered women and girls aged 15 years and older subjected to physical, sexual or psychological violence by a current or former intimate partner in the previous 12 months (%)	Trend data not available
5.2.2 Proportion of women and girls aged 15 years and older subjected to sexual violence by persons other than an intimate partner in lifetime (%)	Trend data not available
6.3.1 Proportion of domestic and industrial wastewater flows safely treated (%)	Baseline data not available
11.6.2 Annual mean concentrations of fine particulate matter (PM _{2.5}) in urban areas (µg/m ³)	No explicit numerical target
16.1.1 Mortality rate due to homicide (per 100 000 population)	No explicit numerical target



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Annex 2.

Summary of methodology

Chapter 1

The analysis presented in Chapter 1 is based on the WHO Global health estimates 2021 (GHE2021), which included the latest estimates on life expectancy, healthy life expectancy, causes of deaths and disability globally, by region and country, and by age and sex. The GHE2021 were produced using data from multiple sources, including national vital registration data, latest estimates from WHO technical programmes, United Nations partners and inter-agency groups, and the Institute for Health Metrics and Evaluation Global Burden of Disease and other scientific studies. The GHE2021 was reviewed by WHO Member States through consultation with national focal points and WHO country and regional offices. The GHE2021 summary of methodology can be accessed online (1).

Secondary analysis was performed using well-established demographic methods to decompose the change and difference in healthy life expectancy over time or between different populations, by age and cause, using GHE2021 data (2,3).

Chapter 2

The analysis presented in Chapter 2 is based on data available from the global monitoring of health-related SDG indicators as of April 2025. Data were compiled from publications and databases produced and managed by WHO or United Nations partner entities. The methodology for each indicator is available from the cited references for the chapter. The data reference years for each indicator differ as data series are updated on different timelines and with different lags between the data reference year and the publication year. Unless otherwise stated, the assessment of progress towards reaching a target is determined by whether the target will be met if the current average annual rate of change applies until the target year.

Chapter 3

The Triple Billion estimates in Chapter 3 were calculated following the WHO GPW13 measurement framework, developed through extensive review and consultation with experts, WHO regional offices and Member States (4). Estimates were derived from 46 outcome indicators (5), using the latest available country-level data as of March 2025. Indicator-level projections were generated using statistical forecasting models that account for historical trends, data sparsity and recent developments, according to methods in Concept and methodologies in estimation and forecasting of Triple Billion targets and improving the WHO results framework (6).

Additionally, two hypothetical scenarios are presented. The first shows estimated progress if all countries were to achieve the annualized targets set by the 2030 Agenda (5,7). This scenario was constructed by calculating a population-weighted global average of expected 2030 values for each indicator. Each country's expected 2030 value was adjusted by the proportion needed to close the gap between the global average and the SDG target/indicator. The second scenario shows potential gains in progress if countries within each WHO region were to match the expected progress of top countries in their respective regions by 2030. Region-specific targets are defined as the top decile of the regional distribution of 2030 values for each indicator. Country-specific trajectories for the scenario were calculated by finding the annualized rate of change required to meet regional-targets, from a baseline year of 2023.

Chapter 4

Economic status is measured using a wealth index and is composed of five subgroups, each accounting for 20% of the population. Education refers to the highest level of education attained by the mother and is composed of three subgroups (no education, primary education and secondary or higher education). For economic status and education, inequality was measured as the difference along the socioeconomic spectrum, while taking the composition of every socioeconomic subgroup into consideration. This is also known as the slope index of inequality (SII). Subgroups were ranked from the most disadvantaged to the most advantaged, factoring in their population sizes, and the indicator value regressed against these relative ranks. SII represents the absolute difference between the two extremes of this regression line. For place of residence, inequality was measured as the difference between urban and rural subgroups (for DTP3 coverage) and the difference between rural and urban subgroups (for zero-dose prevalence). Since place of residence is categorized into two subgroups (urban and rural), the inequality measured by the difference might be less than when there are more subgroups (as is the case with economic status and education). This is because a larger number of groupings typically captures more heterogeneity. Globally and within World Bank country income groups, inequality was calculated as the median difference across countries. Change in inequality over time was measured using countries with two data points, from 2004 to 2013 and from 2014 to 2023 (spanning approximately 10 years).



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Retrieval-Augmented Generation for Large Language Models: A Survey

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Abstract—Large Language Models (LLMs) showcase impressive capabilities but encounter challenges like hallucination, outdated knowledge, and non-transparent, untraceable reasoning processes. Retrieval-Augmented Generation (RAG) has emerged as a promising solution by incorporating knowledge from external databases. This enhances the accuracy and credibility of the generation, particularly for knowledge-intensive tasks, and allows for continuous knowledge updates and integration of domain-specific information. RAG synergistically merges LLMs’ intrinsic knowledge with the vast, dynamic repositories of external databases. This comprehensive review paper offers a detailed examination of the progression of RAG paradigms, encompassing the Naive RAG, the Advanced RAG, and the Modular RAG. It meticulously scrutinizes the tripartite foundation of RAG frameworks, which includes the retrieval, the generation and the augmentation techniques. The paper highlights the state-of-the-art technologies embedded in each of these critical components, providing a profound understanding of the advancements in RAG systems. Furthermore, this paper introduces up-to-date evaluation framework and benchmark. At the end, this article delineates the challenges currently faced and points out prospective avenues for research and development.

Index Terms—Large language model, retrieval-augmented generation, natural language processing, information retrieval

I. INTRODUCTION

LARGE language models (LLMs) have achieved remarkable success, though they still face significant limitations, especially in domain-specific or knowledge-intensive tasks [1], notably producing “hallucinations” [2] when handling queries beyond their training data or requiring current information. To overcome challenges, Retrieval-Augmented Generation (RAG) enhances LLMs by retrieving relevant document chunks from external knowledge base through semantic similarity calculation. By referencing external knowledge, RAG effectively reduces the problem of generating factually incorrect content. Its integration into LLMs has resulted in widespread adoption, establishing RAG as a key technology in advancing chatbots and enhancing the suitability of LLMs for real-world applications.

RAG technology has rapidly developed in recent years, and the technology tree summarizing related research is shown

in Figure 1. The development trajectory of RAG in the era of large models exhibits several distinct stage characteristics. Initially, RAG’s inception coincided with the rise of the Transformer architecture, focusing on enhancing language models by incorporating additional knowledge through Pre-Training Models (PTM). This early stage was characterized by foundational work aimed at refining pre-training techniques [3]–[5]. The subsequent arrival of ChatGPT [6] marked a pivotal moment, with LLM demonstrating powerful in context learning (ICL) capabilities. RAG research shifted towards providing better information for LLMs to answer more complex and knowledge-intensive tasks during the inference stage, leading to rapid development in RAG studies. As research progressed, the enhancement of RAG was no longer limited to the inference stage but began to incorporate more with LLM fine-tuning techniques.

The burgeoning field of RAG has experienced swift growth, yet it has not been accompanied by a systematic synthesis that could clarify its broader trajectory. This survey endeavors to fill this gap by mapping out the RAG process and charting its evolution and anticipated future paths, with a focus on the integration of RAG within LLMs. This paper considers both technical paradigms and research methods, summarizing three main research paradigms from over 100 RAG studies, and analyzing key technologies in the core stages of “Retrieval,” “Generation,” and “Augmentation.” On the other hand, current research tends to focus more on methods, lacking analysis and summarization of how to evaluate RAG. This paper comprehensively reviews the downstream tasks, datasets, benchmarks, and evaluation methods applicable to RAG. Overall, this paper sets out to meticulously compile and categorize the foundational technical concepts, historical progression, and the spectrum of RAG methodologies and applications that have emerged post-LLMs. It is designed to equip readers and professionals with a detailed and structured understanding of both large models and RAG. It aims to illuminate the evolution of retrieval augmentation techniques, assess the strengths and weaknesses of various approaches in their respective contexts, and speculate on upcoming trends and innovations.

Our contributions are as follows:

- In this survey, we present a thorough and systematic review of the state-of-the-art RAG methods, delineating its evolution through paradigms including naive RAG,

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¹Resources are available at <https://github.com/Tongji-KGLLM/RAG-Survey>

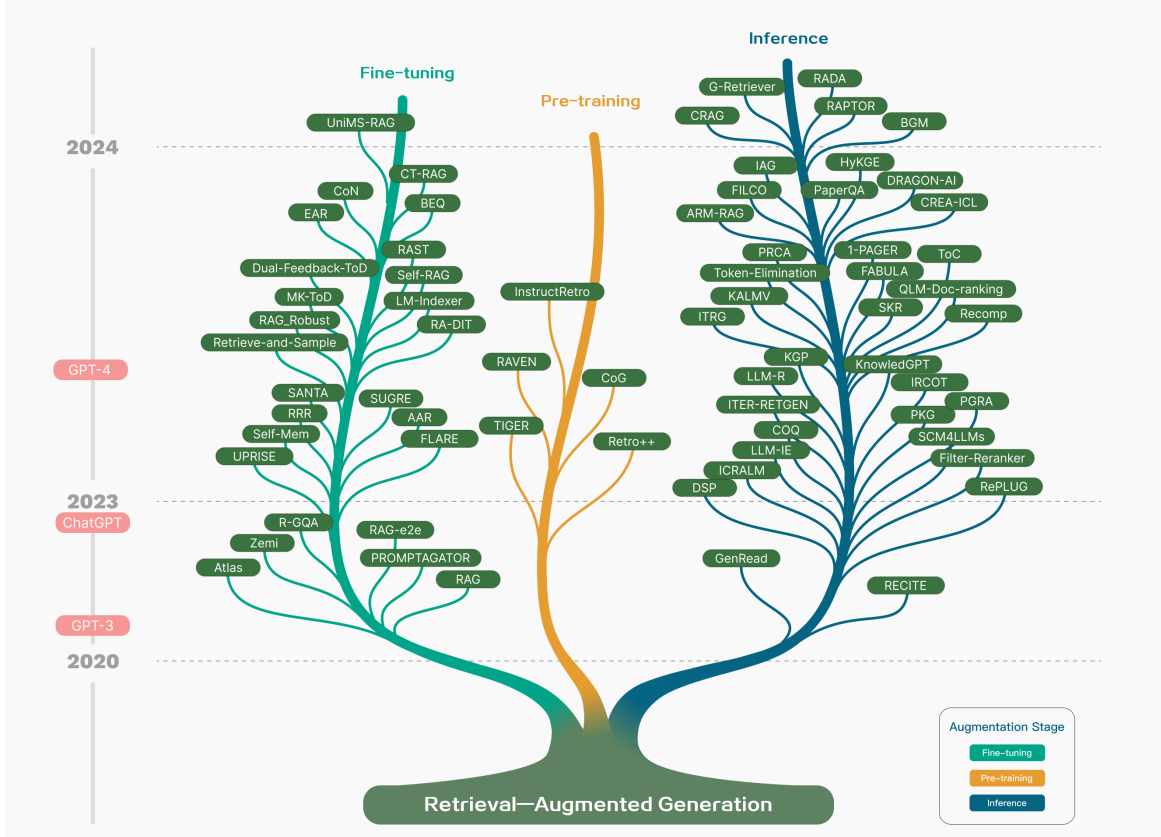


Fig. 1. Technology tree of RAG research. The stages of involving RAG mainly include pre-training, fine-tuning, and inference. With the emergence of LLMs, research on RAG initially focused on leveraging the powerful in-context learning abilities of LLMs, primarily concentrating on the inference stage. Subsequent research has delved deeper, gradually integrating more with the fine-tuning of LLMs. Researchers have also been exploring ways to enhance language models in the pre-training stage through retrieval-augmented techniques.

advanced RAG, and modular RAG. This review contextualizes the broader scope of RAG research within the landscape of LLMs.

- We identify and discuss the central technologies integral to the RAG process, specifically focusing on the aspects of “Retrieval”, “Generation” and “Augmentation”, and delve into their synergies, elucidating how these components intricately collaborate to form a cohesive and effective RAG framework.
- We have summarized the current assessment methods of RAG, covering 26 tasks, nearly 50 datasets, outlining the evaluation objectives and metrics, as well as the current evaluation benchmarks and tools. Additionally, we anticipate future directions for RAG, emphasizing potential enhancements to tackle current challenges.

The paper unfolds as follows: Section II introduces the main concept and current paradigms of RAG. The following three sections explore core components—“Retrieval”, “Generation” and “Augmentation”, respectively. Section III focuses on optimization methods in retrieval, including indexing, query and embedding optimization. Section IV concentrates on post-retrieval process and LLM fine-tuning in generation. Section V analyzes the three augmentation processes. Section VI focuses on RAG’s downstream tasks and evaluation system. Section VII mainly discusses the challenges that RAG currently

faces and its future development directions. At last, the paper concludes in Section VIII.

II. OVERVIEW OF RAG

A typical application of RAG is illustrated in Figure 2. Here, a user poses a question to ChatGPT about a recent, widely discussed news. Given ChatGPT’s reliance on pre-training data, it initially lacks the capacity to provide updates on recent developments. RAG bridges this information gap by sourcing and incorporating knowledge from external databases. In this case, it gathers relevant news articles related to the user’s query. These articles, combined with the original question, form a comprehensive prompt that empowers LLMs to generate a well-informed answer.

The RAG research paradigm is continuously evolving, and we categorize it into three stages: Naive RAG, Advanced RAG, and Modular RAG, as showed in Figure 3. Despite RAG method are cost-effective and surpass the performance of the native LLM, they also exhibit several limitations. The development of Advanced RAG and Modular RAG is a response to these specific shortcomings in Naive RAG.

A. Naive RAG

The Naive RAG research paradigm represents the earliest methodology, which gained prominence shortly after the

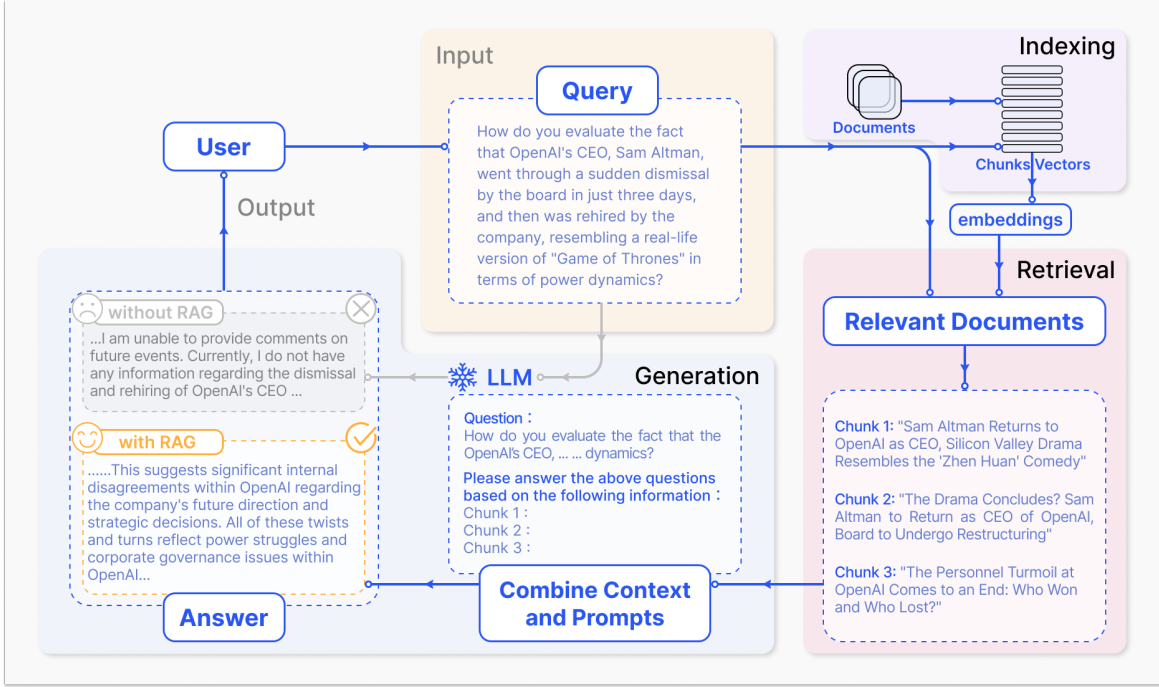


Fig. 2. A representative instance of the RAG process applied to question answering. It mainly consists of 3 steps. 1) Indexing. Documents are split into chunks, encoded into vectors, and stored in a vector database. 2) Retrieval. Retrieve the Top k chunks most relevant to the question based on semantic similarity. 3) Generation. Input the original question and the retrieved chunks together into LLM to generate the final answer.

widespread adoption of ChatGPT. The Naive RAG follows a traditional process that includes indexing, retrieval, and generation, which is also characterized as a “Retrieve-Read” framework [7].

Indexing. starts with the cleaning and extraction of raw data in diverse formats like PDF, HTML, Word, and Markdown, which is then converted into a uniform plain text format. To accommodate the context limitations of language models, text is segmented into smaller, digestible chunks. Chunks are then encoded into vector representations using an embedding model and stored in vector database. This step is crucial for enabling efficient similarity searches in the subsequent retrieval phase.

Retrieval. Upon receipt of a user query, the RAG system employs the same encoding model utilized during the indexing phase to transform the query into a vector representation. It then computes the similarity scores between the query vector and the vector of chunks within the indexed corpus. The system prioritizes and retrieves the top K chunks that demonstrate the greatest similarity to the query. These chunks are subsequently used as the expanded context in prompt.

Generation. The posed query and selected documents are synthesized into a coherent prompt to which a large language model is tasked with formulating a response. The model’s approach to answering may vary depending on task-specific criteria, allowing it to either draw upon its inherent parametric knowledge or restrict its responses to the information contained within the provided documents. In cases of ongoing dialogues, any existing conversational history can be integrated into the prompt, enabling the model to engage in multi-turn dialogue interactions effectively.

However, Naive RAG encounters notable drawbacks:

Retrieval Challenges. The retrieval phase often struggles with precision and recall, leading to the selection of misaligned or irrelevant chunks, and the missing of crucial information.

Generation Difficulties. In generating responses, the model may face the issue of hallucination, where it produces content not supported by the retrieved context. This phase can also suffer from irrelevance, toxicity, or bias in the outputs, detracting from the quality and reliability of the responses.

Augmentation Hurdles. Integrating retrieved information with the different task can be challenging, sometimes resulting in disjointed or incoherent outputs. The process may also encounter redundancy when similar information is retrieved from multiple sources, leading to repetitive responses. Determining the significance and relevance of various passages and ensuring stylistic and tonal consistency add further complexity. Facing complex issues, a single retrieval based on the original query may not suffice to acquire adequate context information.

Moreover, there’s a concern that generation models might overly rely on augmented information, leading to outputs that simply echo retrieved content without adding insightful or synthesized information.

B. Advanced RAG

Advanced RAG introduces specific improvements to overcome the limitations of Naive RAG. Focusing on enhancing retrieval quality, it employs pre-retrieval and post-retrieval strategies. To tackle the indexing issues, Advanced RAG refines its indexing techniques through the use of a sliding window approach, fine-grained segmentation, and the incorporation of metadata. Additionally, it incorporates several optimization methods to streamline the retrieval process [8].

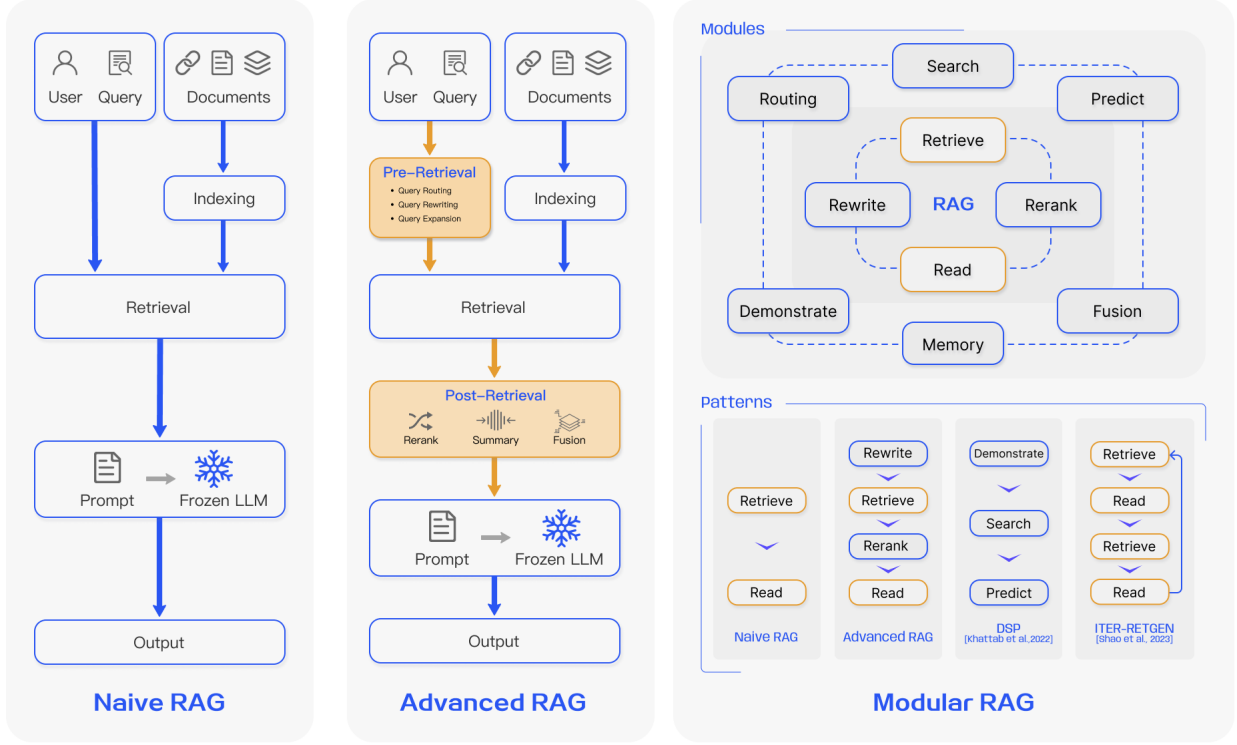


Fig. 3. Comparison between the three paradigms of RAG. (Left) Naive RAG mainly consists of three parts: indexing, retrieval and generation. (Middle) Advanced RAG proposes multiple optimization strategies around pre-retrieval and post-retrieval, with a process similar to the Naive RAG, still following a chain-like structure. (Right) Modular RAG inherits and develops from the previous paradigm, showcasing greater flexibility overall. This is evident in the introduction of multiple specific functional modules and the replacement of existing modules. The overall process is not limited to sequential retrieval and generation; it includes methods such as iterative and adaptive retrieval.

Pre-retrieval process. In this stage, the primary focus is on optimizing the indexing structure and the original query. The goal of optimizing indexing is to enhance the quality of the content being indexed. This involves strategies: enhancing data granularity, optimizing index structures, adding metadata, alignment optimization, and mixed retrieval. While the goal of query optimization is to make the user’s original question clearer and more suitable for the retrieval task. Common methods include query rewriting, query transformation, query expansion and other techniques [7], [9]–[11].

Post-Retrieval Process. Once relevant context is retrieved, it’s crucial to integrate it effectively with the query. The main methods in post-retrieval process include rerank chunks and context compressing. Re-ranking the retrieved information to relocate the most relevant content to the edges of the prompt is a key strategy. This concept has been implemented in frameworks such as LlamaIndex², LangChain³, and HayStack [12]. Feeding all relevant documents directly into LLMs can lead to information overload, diluting the focus on key details with irrelevant content. To mitigate this, post-retrieval efforts concentrate on selecting the essential information, emphasizing critical sections, and shortening the context to be processed.

C. Modular RAG

The modular RAG architecture advances beyond the former two RAG paradigms, offering enhanced adaptability and versatility. It incorporates diverse strategies for improving its components, such as adding a search module for similarity searches and refining the retriever through fine-tuning. Innovations like restructured RAG modules [13] and rearranged RAG pipelines [14] have been introduced to tackle specific challenges. The shift towards a modular RAG approach is becoming prevalent, supporting both sequential processing and integrated end-to-end training across its components. Despite its distinctiveness, Modular RAG builds upon the foundational principles of Advanced and Naive RAG, illustrating a progression and refinement within the RAG family.

1) New Modules: The Modular RAG framework introduces additional specialized components to enhance retrieval and processing capabilities. The Search module adapts to specific scenarios, enabling direct searches across various data sources like search engines, databases, and knowledge graphs, using LLM-generated code and query languages [15]. RAG-Fusion addresses traditional search limitations by employing a multi-query strategy that expands user queries into diverse perspectives, utilizing parallel vector searches and intelligent re-ranking to uncover both explicit and transformative knowledge [16]. The Memory module leverages the LLM’s memory to guide retrieval, creating an unbounded memory pool that

²<https://www.llamaindex.ai>

³<https://www.langchain.com/>

aligns the text more closely with data distribution through iterative self-enhancement [17], [18]. Routing in the RAG system navigates through diverse data sources, selecting the optimal pathway for a query, whether it involves summarization, specific database searches, or merging different information streams [19]. The Predict module aims to reduce redundancy and noise by generating context directly through the LLM, ensuring relevance and accuracy [13]. Lastly, the Task Adapter module tailors RAG to various downstream tasks, automating prompt retrieval for zero-shot inputs and creating task-specific retrievers through few-shot query generation [20], [21]. This comprehensive approach not only streamlines the retrieval process but also significantly improves the quality and relevance of the information retrieved, catering to a wide array of tasks and queries with enhanced precision and flexibility.

2) *New Patterns*: Modular RAG offers remarkable adaptability by allowing module substitution or reconfiguration to address specific challenges. This goes beyond the fixed structures of Naive and Advanced RAG, characterized by a simple “Retrieve” and “Read” mechanism. Moreover, Modular RAG expands this flexibility by integrating new modules or adjusting interaction flow among existing ones, enhancing its applicability across different tasks.

Innovations such as the Rewrite-Retrieve-Read [7] model leverage the LLM’s capabilities to refine retrieval queries through a rewriting module and a LM-feedback mechanism to update rewriting model., improving task performance. Similarly, approaches like Generate-Read [13] replace traditional retrieval with LLM-generated content, while Recite-Read [22] emphasizes retrieval from model weights, enhancing the model’s ability to handle knowledge-intensive tasks. Hybrid retrieval strategies integrate keyword, semantic, and vector searches to cater to diverse queries. Additionally, employing sub-queries and hypothetical document embeddings (HyDE) [11] seeks to improve retrieval relevance by focusing on embedding similarities between generated answers and real documents.

Adjustments in module arrangement and interaction, such as the Demonstrate-Search-Predict (DSP) [23] framework and the iterative Retrieve-Read-Retrieve-Read flow of ITER-RETGEN [14], showcase the dynamic use of module outputs to bolster another module’s functionality, illustrating a sophisticated understanding of enhancing module synergy. The flexible orchestration of Modular RAG Flow showcases the benefits of adaptive retrieval through techniques such as FLARE [24] and Self-RAG [25]. This approach transcends the fixed RAG retrieval process by evaluating the necessity of retrieval based on different scenarios. Another benefit of a flexible architecture is that the RAG system can more easily integrate with other technologies (such as fine-tuning or reinforcement learning) [26]. For example, this can involve fine-tuning the retriever for better retrieval results, fine-tuning the generator for more personalized outputs, or engaging in collaborative fine-tuning [27].

D. RAG vs Fine-tuning

The augmentation of LLMs has attracted considerable attention due to their growing prevalence. Among the optimization

methods for LLMs, RAG is often compared with Fine-tuning (FT) and prompt engineering. Each method has distinct characteristics as illustrated in Figure 4. We used a quadrant chart to illustrate the differences among three methods in two dimensions: external knowledge requirements and model adaption requirements. Prompt engineering leverages a model’s inherent capabilities with minimum necessity for external knowledge and model adaption. RAG can be likened to providing a model with a tailored textbook for information retrieval, ideal for precise information retrieval tasks. In contrast, FT is comparable to a student internalizing knowledge over time, suitable for scenarios requiring replication of specific structures, styles, or formats.

RAG excels in dynamic environments by offering real-time knowledge updates and effective utilization of external knowledge sources with high interpretability. However, it comes with higher latency and ethical considerations regarding data retrieval. On the other hand, FT is more static, requiring retraining for updates but enabling deep customization of the model’s behavior and style. It demands significant computational resources for dataset preparation and training, and while it can reduce hallucinations, it may face challenges with unfamiliar data.

In multiple evaluations of their performance on various knowledge-intensive tasks across different topics, [28] revealed that while unsupervised fine-tuning shows some improvement, RAG consistently outperforms it, for both existing knowledge encountered during training and entirely new knowledge. Additionally, it was found that LLMs struggle to learn new factual information through unsupervised fine-tuning. The choice between RAG and FT depends on the specific needs for data dynamics, customization, and computational capabilities in the application context. RAG and FT are not mutually exclusive and can complement each other, enhancing a model’s capabilities at different levels. In some instances, their combined use may lead to optimal performance. The optimization process involving RAG and FT may require multiple iterations to achieve satisfactory results.

III. RETRIEVAL

In the context of RAG, it is crucial to efficiently retrieve relevant documents from the data source. There are several key issues involved, such as the retrieval source, retrieval granularity, pre-processing of the retrieval, and selection of the corresponding embedding model.

A. Retrieval Source

RAG relies on external knowledge to enhance LLMs, while the type of retrieval source and the granularity of retrieval units both affect the final generation results.

1) *Data Structure*: Initially, text is the mainstream source of retrieval. Subsequently, the retrieval source expanded to include semi-structured data (PDF) and structured data (Knowledge Graph, KG) for enhancement. In addition to retrieving from original external sources, there is also a growing trend in recent researches towards utilizing content generated by LLMs themselves for retrieval and enhancement purposes.

TABLE I
SUMMARY OF RAG METHODS

Method	Retrieval Source	Retrieval Data Type	Retrieval Granularity	Augmentation Stage	Retrieval process
CoG [29]	Wikipedia	Text	Phrase	Pre-training	Iterative
DenseX [30]	FactoidWiki	Text	Proposition	Inference	Once
EAR [31]	Dataset-base	Text	Sentence	Tuning	Once
UPRISE [20]	Dataset-base	Text	Sentence	Tuning	Once
RAST [32]	Dataset-base	Text	Sentence	Tuning	Once
Self-Mem [17]	Dataset-base	Text	Sentence	Tuning	Iterative
FLARE [24]	Search Engine, Wikipedia	Text	Sentence	Tuning	Adaptive
PGRA [33]	Wikipedia	Text	Sentence	Inference	Once
FILCO [34]	Wikipedia	Text	Sentence	Inference	Once
RADA [35]	Dataset-base	Text	Sentence	Inference	Once
Filter-rerank [36]	Synthesized dataset	Text	Sentence	Inference	Once
R-GQA [37]	Dataset-base	Text	Sentence Pair	Tuning	Once
LLM-R [38]	Dataset-base	Text	Sentence Pair	Inference	Iterative
TIGER [39]	Dataset-base	Text	Item-base	Pre-training	Once
LM-Indexer [40]	Dataset-base	Text	Item-base	Tuning	Once
BEQUE [9]	Dataset-base	Text	Item-base	Tuning	Once
CT-RAG [41]	Synthesized dataset	Text	Item-base	Tuning	Once
Atlas [42]	Wikipedia, Common Crawl	Text	Chunk	Pre-training	Iterative
RAVEN [43]	Wikipedia	Text	Chunk	Pre-training	Once
RETRO++ [44]	Pre-training Corpus	Text	Chunk	Pre-training	Iterative
INSTRUCTRETRO [45]	Pre-training corpus	Text	Chunk	Pre-training	Iterative
RRR [7]	Search Engine	Text	Chunk	Tuning	Once
RA-e2e [46]	Dataset-base	Text	Chunk	Tuning	Once
PROMPTAGATOR [21]	BEIR	Text	Chunk	Tuning	Once
AAR [47]	MSMARCO, Wikipedia	Text	Chunk	Tuning	Once
RA-DIT [27]	Common Crawl, Wikipedia	Text	Chunk	Tuning	Once
RAG-Robust [48]	Wikipedia	Text	Chunk	Tuning	Once
RA-Long-Form [49]	Dataset-base	Text	Chunk	Tuning	Once
CoN [50]	Wikipedia	Text	Chunk	Tuning	Once
Self-RAG [25]	Wikipedia	Text	Chunk	Tuning	Adaptive
BGM [26]	Wikipedia	Text	Chunk	Inference	Once
CoQ [51]	Wikipedia	Text	Chunk	Inference	Iterative
Token-Elimination [52]	Wikipedia	Text	Chunk	Inference	Once
PaperQA [53]	Arxiv, Online Database, PubMed	Text	Chunk	Inference	Iterative
NoiseRAG [54]	FactoidWiki	Text	Chunk	Inference	Once
IAG [55]	Search Engine, Wikipedia	Text	Chunk	Inference	Once
NoMIRACL [56]	Wikipedia	Text	Chunk	Inference	Once
ToC [57]	Search Engine, Wikipedia	Text	Chunk	Inference	Recursive
SKR [58]	Dataset-base, Wikipedia	Text	Chunk	Inference	Adaptive
ITRG [59]	Wikipedia	Text	Chunk	Inference	Iterative
RAG-LongContext [60]	Dataset-base	Text	Chunk	Inference	Once
ITER-RETGEN [14]	Wikipedia	Text	Chunk	Inference	Iterative
IRCoT [61]	Wikipedia	Text	Chunk	Inference	Recursive
LLM-Knowledge-Boundary [62]	Wikipedia	Text	Chunk	Inference	Once
RAPTOR [63]	Dataset-base	Text	Chunk	Inference	Recursive
RECITE [22]	LLMs	Text	Chunk	Inference	Once
ICRALM [64]	Pile, Wikipedia	Text	Chunk	Inference	Iterative
Retrieve-and-Sample [65]	Dataset-base	Text	Doc	Tuning	Once
Zemi [66]	C4	Text	Doc	Tuning	Once
CRAG [67]	Arxiv	Text	Doc	Inference	Once
1-PAGER [68]	Wikipedia	Text	Doc	Inference	Iterative
PRCA [69]	Dataset-base	Text	Doc	Inference	Once
QLM-Doc-ranking [70]	Dataset-base	Text	Doc	Inference	Once
Recomp [71]	Wikipedia	Text	Doc	Inference	Once
DSP [23]	Wikipedia	Text	Doc	Inference	Iterative
RePLUG [72]	Pile	Text	Doc	Inference	Once
ARM-RAG [73]	Dataset-base	Text	Doc	Inference	Iterative
GenRead [13]	LLMs	Text	Doc	Inference	Iterative
UniMS-RAG [74]	Dataset-base	Text	Multi	Tuning	Once
CREA-ICL [19]	Dataset-base	Crosslingual, Text	Sentence	Inference	Once
PKG [75]	LLM	Tabular, Text	Chunk	Inference	Once
SANTA [76]	Dataset-base	Code, Text	Item	Pre-training	Once
SURGE [77]	Freebase	KG	Sub-Graph	Tuning	Once
MK-ToD [78]	Dataset-base	KG	Entity	Tuning	Once
Dual-Feedback-ToD [79]	Dataset-base	KG	Entity Sequence	Tuning	Once
KnowledGPT [15]	Dataset-base	KG	Triplet	Inference	Muti-time
FABULA [80]	Dataset-base, Graph	KG	Entity	Inference	Once
HyKGE [81]	CMeKG	KG	Entity	Inference	Once
KALMV [82]	Wikipedia	KG	Triplet	Inference	Iterative
RoG [83]	Freebase	KG	Triplet	Inference	Iterative
G-Retriever [84]	Dataset-base	TextGraph	Sub-Graph	Inference	Once

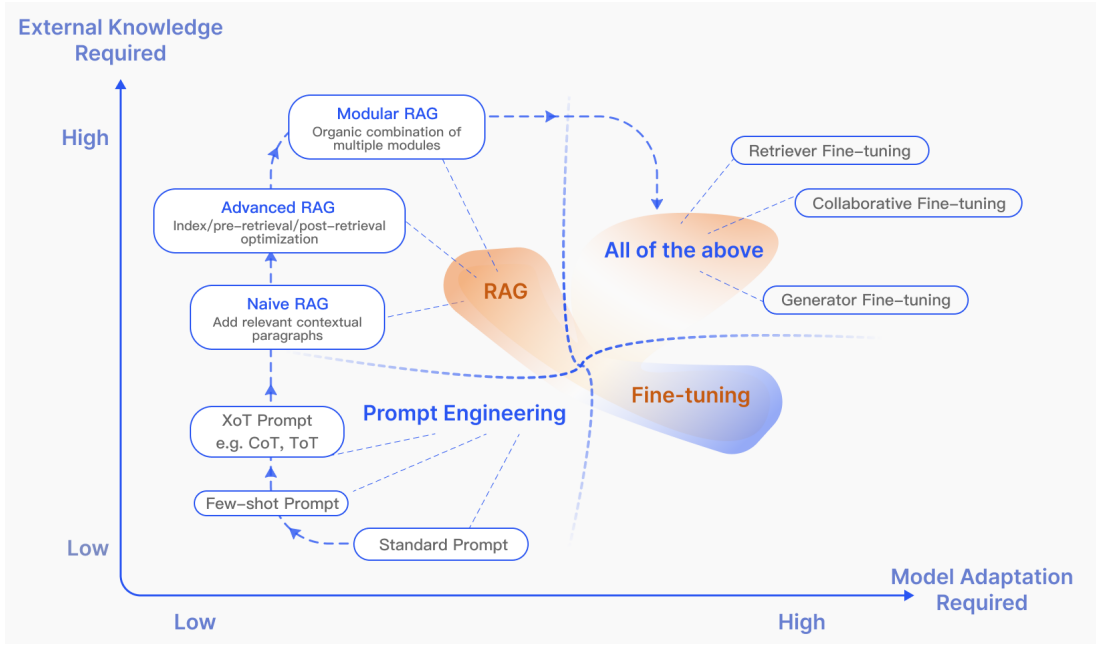


Fig. 4. RAG compared with other model optimization methods in the aspects of “External Knowledge Required” and “Model Adaption Required”. Prompt Engineering requires low modifications to the model and external knowledge, focusing on harnessing the capabilities of LLMs themselves. Fine-tuning, on the other hand, involves further training the model. In the early stages of RAG (Naive RAG), there is a low demand for model modifications. As research progresses, Modular RAG has become more integrated with fine-tuning techniques.

Unstructured Data, such as text, is the most widely used retrieval source, which are mainly gathered from corpus. For open-domain question-answering (ODQA) tasks, the primary retrieval sources are Wikipedia Dump with the current major versions including HotpotQA⁴ (1st October, 2017), DPR⁵ (20 December, 2018). In addition to encyclopedic data, common unstructured data includes cross-lingual text [19] and domain-specific data (such as medical [67] and legal domains [29]).

Semi-structured data, typically refers to data that contains a combination of text and table information, such as PDF. Handling semi-structured data poses challenges for conventional RAG systems due to two main reasons. Firstly, text splitting processes may inadvertently separate tables, leading to data corruption during retrieval. Secondly, incorporating tables into the data can complicate semantic similarity searches. When dealing with semi-structured data, one approach involves leveraging the code capabilities of LLMs to execute Text-2-SQL queries on tables within databases, such as TableGPT [85]. Alternatively, tables can be transformed into text format for further analysis using text-based methods [75]. However, both of these methods are not optimal solutions, indicating substantial research opportunities in this area.

Structured data, such as knowledge graphs (KGs) [86], which are typically verified and can provide more precise information. KnowledGPT [15] generates KB search queries and stores knowledge in a personalized base, enhancing the RAG model’s knowledge richness. In response to the limitations of LLMs in understanding and answering questions about textual graphs, G-Retriever [84] integrates Graph Neural Networks

(GNNs), LLMs and RAG, enhancing graph comprehension and question-answering capabilities through soft prompting of the LLM, and employs the Prize-Collecting Steiner Tree (PCST) optimization problem for targeted graph retrieval. On the contrary, it requires additional effort to build, validate, and maintain structured databases. On the contrary, it requires additional effort to build, validate, and maintain structured databases.

LLMs-Generated Content. Addressing the limitations of external auxiliary information in RAG, some research has focused on exploiting LLMs’ internal knowledge. SKR [58] classifies questions as known or unknown, applying retrieval enhancement selectively. GenRead [13] replaces the retriever with an LLM generator, finding that LLM-generated contexts often contain more accurate answers due to better alignment with the pre-training objectives of causal language modeling. Selfmem [17] iteratively creates an unbounded memory pool with a retrieval-enhanced generator, using a memory selector to choose outputs that serve as dual problems to the original question, thus self-enhancing the generative model. These methodologies underscore the breadth of innovative data source utilization in RAG, striving to improve model performance and task effectiveness.

2) *Retrieval Granularity*: Another important factor besides the data format of the retrieval source is the granularity of the retrieved data. Coarse-grained retrieval units theoretically can provide more relevant information for the problem, but they may also contain redundant content, which could distract the retriever and language models in downstream tasks [50], [87]. On the other hand, fine-grained retrieval unit granularity increases the burden of retrieval and does not guarantee semantic integrity and meeting the required knowledge. Choosing

⁴<https://hotpotqa.github.io/wiki-readme.html>

⁵<https://github.com/facebookresearch/DPR>

the appropriate retrieval granularity during inference can be a simple and effective strategy to improve the retrieval and downstream task performance of dense retrievers.

In text, retrieval granularity ranges from fine to coarse, including Token, Phrase, Sentence, Proposition, Chunks, Document. Among them, DenseX [30] proposed the concept of using propositions as retrieval units. Propositions are defined as atomic expressions in the text, each encapsulating a unique factual segment and presented in a concise, self-contained natural language format. This approach aims to enhance retrieval precision and relevance. On the Knowledge Graph (KG), retrieval granularity includes Entity, Triplet, and sub-Graph. The granularity of retrieval can also be adapted to downstream tasks, such as retrieving Item IDs [40] in recommendation tasks and Sentence pairs [38]. Detailed information is illustrated in Table II.

B. Indexing Optimization

In the Indexing phase, documents will be processed, segmented, and transformed into Embeddings to be stored in a vector database. The quality of index construction determines whether the correct context can be obtained in the retrieval phase.

1) *Chunking Strategy*: The most common method is to split the document into chunks on a fixed number of tokens (e.g., 100, 256, 512) [88]. Larger chunks can capture more context, but they also generate more noise, requiring longer processing time and higher costs. While smaller chunks may not fully convey the necessary context, they do have less noise. However, chunks leads to truncation within sentences, prompting the optimization of a recursive splits and sliding window methods, enabling layered retrieval by merging globally related information across multiple retrieval processes [89]. Nevertheless, these approaches still cannot strike a balance between semantic completeness and context length. Therefore, methods like Small2Big have been proposed, where sentences (small) are used as the retrieval unit, and the preceding and following sentences are provided as (big) context to LLMs [90].

2) *Metadata Attachments*: Chunks can be enriched with metadata information such as page number, file name, author, category timestamp. Subsequently, retrieval can be filtered based on this metadata, limiting the scope of the retrieval. Assigning different weights to document timestamps during retrieval can achieve time-aware RAG, ensuring the freshness of knowledge and avoiding outdated information.

In addition to extracting metadata from the original documents, metadata can also be artificially constructed. For example, adding summaries of paragraph, as well as introducing hypothetical questions. This method is also known as Reverse HyDE. Specifically, using LLM to generate questions that can be answered by the document, then calculating the similarity between the original question and the hypothetical question during retrieval to reduce the semantic gap between the question and the answer.

3) *Structural Index*: One effective method for enhancing information retrieval is to establish a hierarchical structure for the documents. By constructing In structure, RAG system can expedite the retrieval and processing of pertinent data.

Hierarchical index structure. File are arranged in parent-child relationships, with chunks linked to them. Data summaries are stored at each node, aiding in the swift traversal of data and assisting the RAG system in determining which chunks to extract. This approach can also mitigate the illusion caused by block extraction issues.

Knowledge Graph index. Utilize KG in constructing the hierarchical structure of documents contributes to maintaining consistency. It delineates the connections between different concepts and entities, markedly reducing the potential for illusions. Another advantage is the transformation of the information retrieval process into instructions that LLM can comprehend, thereby enhancing the accuracy of knowledge retrieval and enabling LLM to generate contextually coherent responses, thus improving the overall efficiency of the RAG system. To capture the logical relationship between document content and structure, KGP [91] proposed a method of building an index between multiple documents using KG. This KG consists of nodes (representing paragraphs or structures in the documents, such as pages and tables) and edges (indicating semantic/lexical similarity between paragraphs or relationships within the document structure), effectively addressing knowledge retrieval and reasoning problems in a multi-document environment.

C. Query Optimization

One of the primary challenges with Naive RAG is its direct reliance on the user's original query as the basis for retrieval. Formulating a precise and clear question is difficult, and imprudent queries result in subpar retrieval effectiveness. Sometimes, the question itself is complex, and the language is not well-organized. Another difficulty lies in language complexity ambiguity. Language models often struggle when dealing with specialized vocabulary or ambiguous abbreviations with multiple meanings. For instance, they may not discern whether "LLM" refers to *large language model* or a *Master of Laws* in a legal context.

1) *Query Expansion*: Expanding a single query into multiple queries enriches the content of the query, providing further context to address any lack of specific nuances, thereby ensuring the optimal relevance of the generated answers.

Multi-Query. By employing prompt engineering to expand queries via LLMs, these queries can then be executed in parallel. The expansion of queries is not random, but rather meticulously designed.

Sub-Query. The process of sub-question planning represents the generation of the necessary sub-questions to contextualize and fully answer the original question when combined. This process of adding relevant context is, in principle, similar to query expansion. Specifically, a complex question can be decomposed into a series of simpler sub-questions using the least-to-most prompting method [92].

Chain-of-Verification(CoVe). The expanded queries undergo validation by LLM to achieve the effect of reducing hallucinations. Validated expanded queries typically exhibit higher reliability [93].

2) *Query Transformation*: The core concept is to retrieve chunks based on a transformed query instead of the user’s original query.

Query Rewrite. The original queries are not always optimal for LLM retrieval, especially in real-world scenarios. Therefore, we can prompt LLM to rewrite the queries. In addition to using LLM for query rewriting, specialized smaller language models, such as RRR (Rewrite-retrieve-read) [7]. The implementation of the query rewrite method in the Taobao, known as BEQUE [9] has notably enhanced recall effectiveness for long-tail queries, resulting in a rise in GMV.

Another query transformation method is to use prompt engineering to let LLM generate a query based on the original query for subsequent retrieval. HyDE [11] construct hypothetical documents (assumed answers to the original query). It focuses on embedding similarity from answer to answer rather than seeking embedding similarity for the problem or query. Using the Step-back Prompting method [10], the original query is abstracted to generate a high-level concept question (step-back question). In the RAG system, both the step-back question and the original query are used for retrieval, and both the results are utilized as the basis for language model answer generation.

3) *Query Routing*: Based on varying queries, routing to distinct RAG pipeline, which is suitable for a versatile RAG system designed to accommodate diverse scenarios.

Metadata Router/ Filter. The first step involves extracting keywords (entity) from the query, followed by filtering based on the keywords and metadata within the chunks to narrow down the search scope.

Semantic Router is another method of routing involves leveraging the semantic information of the query. Specific approach see Semantic Router [6]. Certainly, a hybrid routing approach can also be employed, combining both semantic and metadata-based methods for enhanced query routing.

D. Embedding

In RAG, retrieval is achieved by calculating the similarity (e.g. cosine similarity) between the embeddings of the question and document chunks, where the semantic representation capability of embedding models plays a key role. This mainly includes a sparse encoder (BM25) and a dense retriever (BERT architecture Pre-training language models). Recent research has introduced prominent embedding models such as AngIE, Voyage, BGE, etc [94]–[96], which are benefit from multi-task instruct tuning. Hugging Face’s MTEB leaderboard [7] evaluates embedding models across 8 tasks, covering 58 datasets. Additionally, C-MTEB focuses on Chinese capability, covering 6 tasks and 35 datasets. There is no one-size-fits-all answer to “which embedding model to use.” However, some specific models are better suited for particular use cases.

1) *Mix/hybrid Retrieval* : Sparse and dense embedding approaches capture different relevance features and can benefit from each other by leveraging complementary relevance information. For instance, sparse retrieval models can be used

to provide initial search results for training dense retrieval models. Additionally, pre-training language models (PLMs) can be utilized to learn term weights to enhance sparse retrieval. Specifically, it also demonstrates that sparse retrieval models can enhance the zero-shot retrieval capability of dense retrieval models and assist dense retrievers in handling queries containing rare entities, thereby improving robustness.

2) *Fine-tuning Embedding Model*: In instances where the context significantly deviates from pre-training corpus, particularly within highly specialized disciplines such as healthcare, legal practice, and other sectors replete with proprietary jargon, fine-tuning the embedding model on your own domain dataset becomes essential to mitigate such discrepancies.

In addition to supplementing domain knowledge, another purpose of fine-tuning is to align the retriever and generator, for example, using the results of LLM as the supervision signal for fine-tuning, known as LSR (LM-supervised Retriever). PROMTAGATOR [21] utilizes the LLM as a few-shot query generator to create task-specific retrievers, addressing challenges in supervised fine-tuning, particularly in data-scarce domains. Another approach, LLM-Embedder [97], exploits LLMs to generate reward signals across multiple downstream tasks. The retriever is fine-tuned with two types of supervised signals: hard labels for the dataset and soft rewards from the LLMs. This dual-signal approach fosters a more effective fine-tuning process, tailoring the embedding model to diverse downstream applications. REPLUG [72] utilizes a retriever and an LLM to calculate the probability distributions of the retrieved documents and then performs supervised training by computing the KL divergence. This straightforward and effective training method enhances the performance of the retrieval model by using an LM as the supervisory signal, eliminating the need for specific cross-attention mechanisms. Moreover, inspired by RLHF (Reinforcement Learning from Human Feedback), utilizing LM-based feedback to reinforce the retriever through reinforcement learning.

E. Adapter

Fine-tuning models may present challenges, such as integrating functionality through an API or addressing constraints arising from limited local computational resources. Consequently, some approaches opt to incorporate an external adapter to aid in alignment.

To optimize the multi-task capabilities of LLM, UP-RISE [20] trained a lightweight prompt retriever that can automatically retrieve prompts from a pre-built prompt pool that are suitable for a given zero-shot task input. AAR (Augmentation-Adapted Retriever) [47] introduces a universal adapter designed to accommodate multiple downstream tasks. While PRCA [69] add a pluggable reward-driven contextual adapter to enhance performance on specific tasks. BGM [26] keeps the retriever and LLM fixed, and trains a bridge Seq2Seq model in between. The bridge model aims to transform the retrieved information into a format that LLMs can work with effectively, allowing it to not only rerank but also dynamically select passages for each query, and potentially employ more advanced strategies like repetition. Furthermore, PKG

⁶<https://github.com/aurelio-labs/semantic-router>

⁷<https://huggingface.co/spaces/mteb/leaderboard>

introduces an innovative method for integrating knowledge into white-box models via directive fine-tuning [75]. In this approach, the retriever module is directly substituted to generate relevant documents according to a query. This method assists in addressing the difficulties encountered during the fine-tuning process and enhances model performance.

IV. GENERATION

After retrieval, it is not a good practice to directly input all the retrieved information to the LLM for answering questions. Following will introduce adjustments from two perspectives: adjusting the retrieved content and adjusting the LLM.

A. Context Curation

Redundant information can interfere with the final generation of LLM, and overly long contexts can also lead LLM to the “Lost in the middle” problem [98]. Like humans, LLM tends to only focus on the beginning and end of long texts, while forgetting the middle portion. Therefore, in the RAG system, we typically need to further process the retrieved content.

1) *Reranking*: Reranking fundamentally reorders document chunks to highlight the most pertinent results first, effectively reducing the overall document pool, severing a dual purpose in information retrieval, acting as both an enhancer and a filter, delivering refined inputs for more precise language model processing [70]. Reranking can be performed using rule-based methods that depend on predefined metrics like Diversity, Relevance, and MRR, or model-based approaches like Encoder-Decoder models from the BERT series (e.g., SpanBERT), specialized reranking models such as Cohere rerank or bge-ranker-large, and general large language models like GPT [12], [99].

2) *Context Selection/Compression*: A common misconception in the RAG process is the belief that retrieving as many relevant documents as possible and concatenating them to form a lengthy retrieval prompt is beneficial. However, excessive context can introduce more noise, diminishing the LLM’s perception of key information.

(Long) LLMingua [100], [101] utilize small language models (SLMs) such as GPT-2 Small or LLaMA-7B, to detect and remove unimportant tokens, transforming it into a form that is challenging for humans to comprehend but well understood by LLMs. This approach presents a direct and practical method for prompt compression, eliminating the need for additional training of LLMs while balancing language integrity and compression ratio. PRCA tackled this issue by training an information extractor [69]. Similarly, RECOMP adopts a comparable approach by training an information condenser using contrastive learning [71]. Each training data point consists of one positive sample and five negative samples, and the encoder undergoes training using contrastive loss throughout this process [102].

In addition to compressing the context, reducing the number of documents also helps improve the accuracy of the model’s answers. Ma et al. [103] propose the “Filter-Reranker” paradigm, which combines the strengths of LLMs and SLMs.

In this paradigm, SLMs serve as filters, while LLMs function as reordering agents. The research shows that instructing LLMs to rearrange challenging samples identified by SLMs leads to significant improvements in various Information Extraction (IE) tasks. Another straightforward and effective approach involves having the LLM evaluate the retrieved content before generating the final answer. This allows the LLM to filter out documents with poor relevance through LLM critique. For instance, in Chatlaw [104], the LLM is prompted to self-suggestion on the referenced legal provisions to assess their relevance.

B. LLM Fine-tuning

Targeted fine-tuning based on the scenario and data characteristics on LLMs can yield better results. This is also one of the greatest advantages of using on-premise LLMs. When LLMs lack data in a specific domain, additional knowledge can be provided to the LLM through fine-tuning. Huggingface’s fine-tuning data can also be used as an initial step.

Another benefit of fine-tuning is the ability to adjust the model’s input and output. For example, it can enable LLM to adapt to specific data formats and generate responses in a particular style as instructed [37]. For retrieval tasks that engage with structured data, the SANTA framework [76] implements a tripartite training regimen to effectively encapsulate both structural and semantic nuances. The initial phase focuses on the retriever, where contrastive learning is harnessed to refine the query and document embeddings.

Aligning LLM outputs with human or retriever preferences through reinforcement learning is a potential approach. For instance, manually annotating the final generated answers and then providing feedback through reinforcement learning. In addition to aligning with human preferences, it is also possible to align with the preferences of fine-tuned models and retrievers [79]. When circumstances prevent access to powerful proprietary models or larger parameter open-source models, a simple and effective method is to distill the more powerful models (e.g. GPT-4). Fine-tuning of LLM can also be coordinated with fine-tuning of the retriever to align preferences. A typical approach, such as RA-DIT [27], aligns the scoring functions between Retriever and Generator using KL divergence.

V. AUGMENTATION PROCESS IN RAG

In the domain of RAG, the standard practice often involves a singular (once) retrieval step followed by generation, which can lead to inefficiencies and sometimes is typically insufficient for complex problems demanding multi-step reasoning, as it provides a limited scope of information [105]. Many studies have optimized the retrieval process in response to this issue, and we have summarised them in Figure 5.

A. Iterative Retrieval

Iterative retrieval is a process where the knowledge base is repeatedly searched based on the initial query and the text generated so far, providing a more comprehensive knowledge

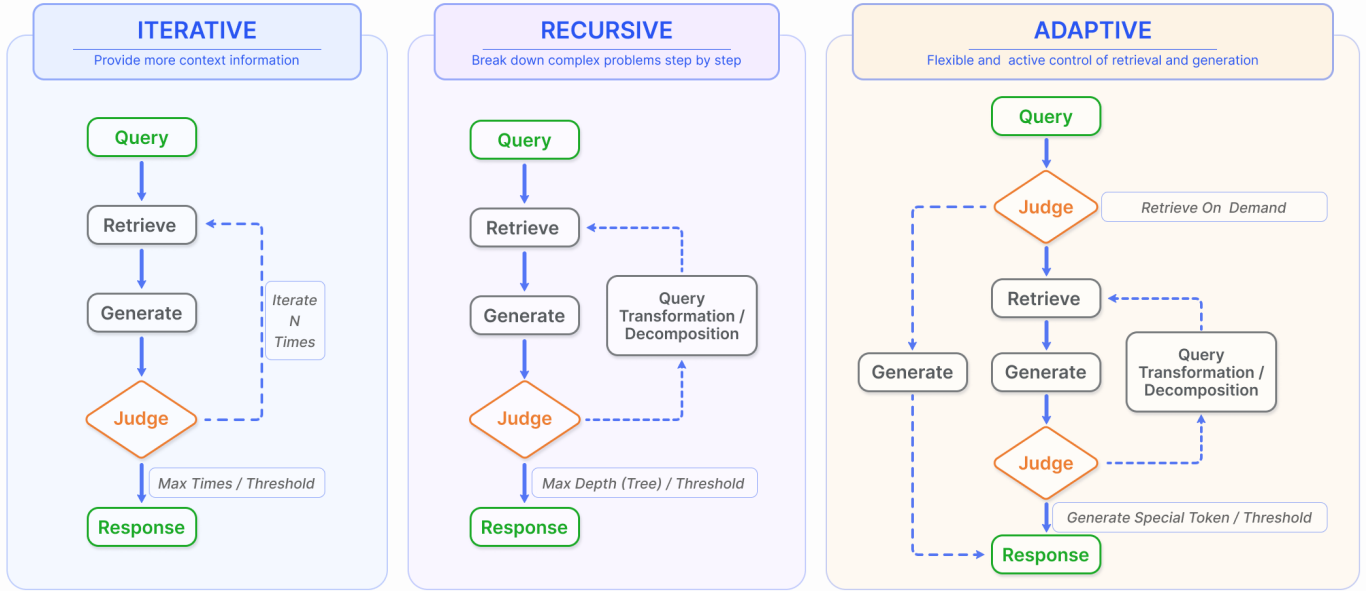


Fig. 5. In addition to the most common once retrieval, RAG also includes three types of retrieval augmentation processes. (left) Iterative retrieval involves alternating between retrieval and generation, allowing for richer and more targeted context from the knowledge base at each step. (Middle) Recursive retrieval involves gradually refining the user query and breaking down the problem into sub-problems, then continuously solving complex problems through retrieval and generation. (Right) Adaptive retrieval focuses on enabling the RAG system to autonomously determine whether external knowledge retrieval is necessary and when to stop retrieval and generation, often utilizing LLM-generated special tokens for control.

base for LLMs. This approach has been shown to enhance the robustness of subsequent answer generation by offering additional contextual references through multiple retrieval iterations. However, it may be affected by semantic discontinuity and the accumulation of irrelevant information. ITER-RETGEN [14] employs a synergistic approach that leverages “retrieval-enhanced generation” alongside “generation-enhanced retrieval” for tasks that necessitate the reproduction of specific information. The model harnesses the content required to address the input task as a contextual basis for retrieving pertinent knowledge, which in turn facilitates the generation of improved responses in subsequent iterations.

B. Recursive Retrieval

Recursive retrieval is often used in information retrieval and NLP to improve the depth and relevance of search results. The process involves iteratively refining search queries based on the results obtained from previous searches. Recursive Retrieval aims to enhance the search experience by gradually converging on the most pertinent information through a feedback loop. IRCot [61] uses chain-of-thought to guide the retrieval process and refines the CoT with the obtained retrieval results. ToC [57] creates a clarification tree that systematically optimizes the ambiguous parts in the Query. It can be particularly useful in complex search scenarios where the user’s needs are not entirely clear from the outset or where the information sought is highly specialized or nuanced. The recursive nature of the process allows for continuous learning and adaptation to the user’s requirements, often resulting in improved satisfaction with the search outcomes.

To address specific data scenarios, recursive retrieval and multi-hop retrieval techniques are utilized together. Recursive

retrieval involves a structured index to process and retrieve data in a hierarchical manner, which may include summarizing sections of a document or lengthy PDF before performing a retrieval based on this summary. Subsequently, a secondary retrieval within the document refines the search, embodying the recursive nature of the process. In contrast, multi-hop retrieval is designed to delve deeper into graph-structured data sources, extracting interconnected information [106].

C. Adaptive Retrieval

Adaptive retrieval methods, exemplified by Flare [24] and Self-RAG [25], refine the RAG framework by enabling LLMs to actively determine the optimal moments and content for retrieval, thus enhancing the efficiency and relevance of the information sourced.

These methods are part of a broader trend wherein LLMs employ active judgment in their operations, as seen in model agents like AutoGPT, Toolformer, and Graph-Toolformer [107]–[109]. Graph-Toolformer, for instance, divides its retrieval process into distinct steps where LLMs proactively use retrievers, apply Self-Ask techniques, and employ few-shot prompts to initiate search queries. This proactive stance allows LLMs to decide when to search for necessary information, akin to how an agent utilizes tools.

WebGPT [110] integrates a reinforcement learning framework to train the GPT-3 model in autonomously using a search engine during text generation. It navigates this process using special tokens that facilitate actions such as search engine queries, browsing results, and citing references, thereby expanding GPT-3’s capabilities through the use of external search engines. Flare automates timing retrieval by monitoring the confidence of the generation process, as indicated by the

probability of generated terms [24]. When the probability falls below a certain threshold would activates the retrieval system to collect relevant information, thus optimizing the retrieval cycle. Self-RAG [25] introduces “reflection tokens” that allow the model to introspect its outputs. These tokens come in two varieties: “retrieve” and “critic”. The model autonomously decides when to activate retrieval, or alternatively, a predefined threshold may trigger the process. During retrieval, the generator conducts a fragment-level beam search across multiple paragraphs to derive the most coherent sequence. Critic scores are used to update the subdivision scores, with the flexibility to adjust these weights during inference, tailoring the model’s behavior. Self-RAG’s design obviates the need for additional classifiers or reliance on Natural Language Inference (NLI) models, thus streamlining the decision-making process for when to engage retrieval mechanisms and improving the model’s autonomous judgment capabilities in generating accurate responses.

VI. TASK AND EVALUATION

The rapid advancement and growing adoption of RAG in the field of NLP have propelled the evaluation of RAG models to the forefront of research in the LLMs community. The primary objective of this evaluation is to comprehend and optimize the performance of RAG models across diverse application scenarios. This chapter will mainly introduce the main downstream tasks of RAG, datasets, and how to evaluate RAG systems.

A. Downstream Task

The core task of RAG remains Question Answering (QA), including traditional single-hop/multi-hop QA, multiple-choice, domain-specific QA as well as long-form scenarios suitable for RAG. In addition to QA, RAG is continuously being expanded into multiple downstream tasks, such as Information Extraction (IE), dialogue generation, code search, etc. The main downstream tasks of RAG and their corresponding datasets are summarized in Table III.

B. Evaluation Target

Historically, RAG models assessments have centered on their execution in specific downstream tasks. These evaluations employ established metrics suitable to the tasks at hand. For instance, question answering evaluations might rely on EM and F1 scores [7], [45], [59], [72], whereas fact-checking tasks often hinge on Accuracy as the primary metric [4], [14], [42]. BLEU and ROUGE metrics are also commonly used to evaluate answer quality [26], [32], [52], [78]. Tools like RALLE, designed for the automatic evaluation of RAG applications, similarly base their assessments on these task-specific metrics [160]. Despite this, there is a notable paucity of research dedicated to evaluating the distinct characteristics of RAG models. The main evaluation objectives include:

Retrieval Quality. Evaluating the retrieval quality is crucial for determining the effectiveness of the context sourced by the retriever component. Standard metrics from the domains

of search engines, recommendation systems, and information retrieval systems are employed to measure the performance of the RAG retrieval module. Metrics such as Hit Rate, MRR, and NDCG are commonly utilized for this purpose [161], [162].

Generation Quality. The assessment of generation quality centers on the generator’s capacity to synthesize coherent and relevant answers from the retrieved context. This evaluation can be categorized based on the content’s objectives: unlabeled and labeled content. For unlabeled content, the evaluation encompasses the faithfulness, relevance, and non-harmfulness of the generated answers. In contrast, for labeled content, the focus is on the accuracy of the information produced by the model [161]. Additionally, both retrieval and generation quality assessments can be conducted through manual or automatic evaluation methods [29], [161], [163].

C. Evaluation Aspects

Contemporary evaluation practices of RAG models emphasize three primary quality scores and four essential abilities, which collectively inform the evaluation of the two principal targets of the RAG model: retrieval and generation.

1) *Quality Scores:* Quality scores include context relevance, answer faithfulness, and answer relevance. These quality scores evaluate the efficiency of the RAG model from different perspectives in the process of information retrieval and generation [164]–[166].

Context Relevance evaluates the precision and specificity of the retrieved context, ensuring relevance and minimizing processing costs associated with extraneous content.

Answer Faithfulness ensures that the generated answers remain true to the retrieved context, maintaining consistency and avoiding contradictions.

Answer Relevance requires that the generated answers are directly pertinent to the posed questions, effectively addressing the core inquiry.

2) *Required Abilities:* RAG evaluation also encompasses four abilities indicative of its adaptability and efficiency: noise robustness, negative rejection, information integration, and counterfactual robustness [167], [168]. These abilities are critical for the model’s performance under various challenges and complex scenarios, impacting the quality scores.

Noise Robustness appraises the model’s capability to manage noise documents that are question-related but lack substantive information.

Negative Rejection assesses the model’s discernment in refraining from responding when the retrieved documents do not contain the necessary knowledge to answer a question.

Information Integration evaluates the model’s proficiency in synthesizing information from multiple documents to address complex questions.

Counterfactual Robustness tests the model’s ability to recognize and disregard known inaccuracies within documents, even when instructed about potential misinformation.

Context relevance and noise robustness are important for evaluating the quality of retrieval, while answer faithfulness, answer relevance, negative rejection, information integration, and counterfactual robustness are important for evaluating the quality of generation.

TABLE II
DOWNSTREAM TASKS AND DATASETS OF RAG

Task	Sub Task	Dataset	Method
QA	Single-hop	Natural Qustion(NQ) [111]	[26], [30], [34], [42], [45], [50], [52], [59], [64], [82] [3], [4], [22], [27], [40], [43], [54], [62], [71], [112] [20], [44], [72] [13], [30], [34], [45], [50], [64]
		TriviaQA(TQA) [113]	[4], [27], [59], [62], [112] [22], [25], [43], [44], [71], [72]
		SQuAD [114]	[20], [23], [30], [32], [45], [69], [112]
		Web Questions(WebQ) [115]	[3], [4], [13], [30], [50], [68]
		PopQA [116]	[7], [25], [67]
		MS MARCO [117]	[4], [40], [52]
	Multi-hop	HotpotQA [118]	[23], [26], [31], [34], [47], [51], [61], [82] [7], [14], [22], [27], [59], [62], [69], [71], [91]
		2WikiMultiHopQA [119]	[14], [24], [48], [59], [61], [91]
		MuSiQue [120]	[14], [51], [61], [91]
	Long-form QA	ELI5 [121]	[27], [34], [43], [49], [51]
		NarrativeQA(NQA) [122]	[45], [60], [63], [123]
		ASQA [124]	[24], [57]
		QMSum(QM) [125]	[60], [123]
	Domain QA	Qasper [126]	[60], [63]
		COVID-QA [127]	[35], [46]
		CMB [128], MMCU_Medical [129]	[81]
	Multi-Choice QA	QuALITY [130]	[60], [63]
		ARC [131]	[25], [67]
		CommonsenseQA [132]	[58], [66]
	Graph QA	GraphQA [84]	[84]
Dialog	Dialog Generation	Wizard of Wikipedia (WoW) [133]	[13], [27], [34], [42]
	Personal Dialog	KBP [134]	[74], [135]
		DuleMon [136]	[74]
	Task-oriented Dialog	CamRest [137]	[78], [79]
		Amazon(Toys,Sport,Beauty) [138]	[39], [40]
IE	Event Argument Extraction	WikiEvent [139]	[13], [27], [37], [42]
		RAMS [140]	[36], [37]
	Relation Extraction	T-REx [141], ZsRE [142]	[27], [51]
Reasoning	Commonsense Reasoning	HellaSwag [143]	[20], [66]
	CoT Reasoning	CoT Reasoning [144]	[27]
	Complex Reasoning	CSQA [145]	[55]
Others	Language Understanding	MMLU [146]	[7], [27], [28], [42], [43], [47], [72]
		WikiText-103 [147]	[5], [29], [64], [71]
		StrategyQA [148]	[14], [24], [48], [51], [55], [58]
	Fact Checking/Verification	FEVER [149]	[4], [13], [27], [34], [42], [50]
		PubHealth [150]	[25], [67]
	Text Generation	Biography [151]	[67]
	Text Summarization	WikiASP [152]	[24]
		XSum [153]	[17]
		VioLens [154]	[19]
	Text Classification	TREC [155]	[33]
		SST-2 [156]	[20], [33], [38]
	Sentiment		
	Code Search	CodeSearchNet [157]	[76]
	Robustness Evaluation	NoMIRACL [56]	[56]
	Math	GSM8K [158]	[73]
	Machine Translation	JRC-Acquis [159]	[17]

TABLE III
SUMMARY OF METRICS APPLICABLE FOR EVALUATION ASPECTS OF RAG

	Context Relevance	Faithfulness	Answer Relevance	Noise Robustness	Negative Rejection	Information Integration	Counterfactual Robustness
Accuracy	✓	✓	✓	✓	✓	✓	✓
EM					✓		
Recall	✓						
Precision	✓			✓			
R-Rate							✓
Cosine Similarity			✓				
Hit Rate	✓						
MRR	✓						
NDCG	✓						
BLEU	✓	✓	✓				
ROUGE/ROUGE-L	✓	✓	✓				

The specific metrics for each evaluation aspect are summarized in Table III. It is essential to recognize that these metrics, derived from related work, are traditional measures and do not yet represent a mature or standardized approach for quantifying RAG evaluation aspects. Custom metrics tailored to the nuances of RAG models, though not included here, have also been developed in some evaluation studies.

D. Evaluation Benchmarks and Tools

A series of benchmark tests and tools have been proposed to facilitate the evaluation of RAG. These instruments furnish quantitative metrics that not only gauge RAG model performance but also enhance comprehension of the model’s capabilities across various evaluation aspects. Prominent benchmarks such as RGB, RECALL and CRUD [167]–[169] focus on appraising the essential abilities of RAG models. Concurrently, state-of-the-art automated tools like RAGAS [164], ARES [165], and TruLens⁸ employ LLMs to adjudicate the quality scores. These tools and benchmarks collectively form a robust framework for the systematic evaluation of RAG models, as summarized in Table IV.

VII. DISCUSSION AND FUTURE PROSPECTS

Despite the considerable progress in RAG technology, several challenges persist that warrant in-depth research. This chapter will mainly introduce the current challenges and future research directions faced by RAG.

A. RAG vs Long Context

With the deepening of related research, the context of LLMs is continuously expanding [170]–[172]. Presently, LLMs can effortlessly manage contexts exceeding 200,000 tokens⁹. This capability signifies that long-document question answering, previously reliant on RAG, can now incorporate the entire document directly into the prompt. This has also sparked discussions on whether RAG is still necessary when LLMs

are not constrained by context. In fact, RAG still plays an irreplaceable role. On one hand, providing LLMs with a large amount of context at once will significantly impact its inference speed, while chunked retrieval and on-demand input can significantly improve operational efficiency. On the other hand, RAG-based generation can quickly locate the original references for LLMs to help users verify the generated answers. The entire retrieval and reasoning process is observable, while generation solely relying on long context remains a black box. Conversely, the expansion of context provides new opportunities for the development of RAG, enabling it to address more complex problems and integrative or summary questions that require reading a large amount of material to answer [49]. Developing new RAG methods in the context of super-long contexts is one of the future research trends.

B. RAG Robustness

The presence of noise or contradictory information during retrieval can detrimentally affect RAG’s output quality. This situation is figuratively referred to as “Misinformation can be worse than no information at all”. Improving RAG’s resistance to such adversarial or counterfactual inputs is gaining research momentum and has become a key performance metric [48], [50], [82]. Cuconasu et al. [54] analyze which type of documents should be retrieved, evaluate the relevance of the documents to the prompt, their position, and the number included in the context. The research findings reveal that including irrelevant documents can unexpectedly increase accuracy by over 30%, contradicting the initial assumption of reduced quality. These results underscore the importance of developing specialized strategies to integrate retrieval with language generation models, highlighting the need for further research and exploration into the robustness of RAG.

C. Hybrid Approaches

Combining RAG with fine-tuning is emerging as a leading strategy. Determining the optimal integration of RAG and fine-tuning whether sequential, alternating, or through end-to-end joint training—and how to harness both parameterized

⁸https://www.trulens.org/trulens_eval/core_concepts_rag_triad/

⁹<https://kimi.moonshot.cn>

TABLE IV
SUMMARY OF EVALUATION FRAMEWORKS

Evaluation Framework	Evaluation Targets	Evaluation Aspects	Quantitative Metrics
RGB [†]	Retrieval Quality Generation Quality	Noise Robustness	Accuracy
		Negative Rejection	EM
		Information Integration	Accuracy
		Counterfactual Robustness	Accuracy
RECALL [†]	Generation Quality	Counterfactual Robustness	R-Rate (Reappearance Rate)
RAGAS [‡]	Retrieval Quality Generation Quality	Context Relevance	*
		Faithfulness	*
		Answer Relevance	Cosine Similarity
ARES [‡]	Retrieval Quality Generation Quality	Context Relevance	Accuracy
		Faithfulness	Accuracy
		Answer Relevance	Accuracy
TruLens [‡]	Retrieval Quality Generation Quality	Context Relevance	*
		Faithfulness	*
		Answer Relevance	*
CRUD [†]	Retrieval Quality Generation Quality	Creative Generation	BLEU
		Knowledge-intensive QA	ROUGE-L
		Error Correction	BertScore
		Summarization	RAGQuestEval

[†] represents a benchmark, and [‡] represents a tool. * denotes customized quantitative metrics, which deviate from traditional metrics. Readers are encouraged to consult pertinent literature for the specific quantification formulas associated with these metrics, as required.

and non-parameterized advantages are areas ripe for exploration [27]. Another trend is to introduce SLMs with specific functionalities into RAG and fine-tuned by the results of RAG system. For example, CRAG [67] trains a lightweight retrieval evaluator to assess the overall quality of the retrieved documents for a query and triggers different knowledge retrieval actions based on confidence levels.

D. Scaling laws of RAG

End-to-end RAG models and pre-trained models based on RAG are still one of the focuses of current researchers [173]. The parameters of these models are one of the key factors. While scaling laws [174] are established for LLMs, their applicability to RAG remains uncertain. Initial studies like RETRO++ [44] have begun to address this, yet the parameter count in RAG models still lags behind that of LLMs. The possibility of an Inverse Scaling Law [10], where smaller models outperform larger ones, is particularly intriguing and merits further investigation.

E. Production-Ready RAG

RAG’s practicality and alignment with engineering requirements have facilitated its adoption. However, enhancing retrieval efficiency, improving document recall in large knowledge bases, and ensuring data security—such as preventing

inadvertent disclosure of document sources or metadata by LLMs—are critical engineering challenges that remain to be addressed [175].

The development of the RAG ecosystem is greatly impacted by the progression of its technical stack. Key tools like LangChain and LLamaIndex have quickly gained popularity with the emergence of ChatGPT, providing extensive RAG-related APIs and becoming essential in the realm of LLMs. The emerging technology stack, while not as rich in features as LangChain and LLamaIndex, stands out through its specialized products. For example, Flowise AI prioritizes a low-code approach, allowing users to deploy AI applications, including RAG, through a user-friendly drag-and-drop interface. Other technologies like HayStack, Meltano, and Cohere Coral are also gaining attention for their unique contributions to the field.

In addition to AI-focused vendors, traditional software and cloud service providers are expanding their offerings to include RAG-centric services. Weaviate’s Verba [11] is designed for personal assistant applications, while Amazon’s Kendra [12] offers intelligent enterprise search services, enabling users to browse various content repositories using built-in connectors. In the development of RAG technology, there is a clear trend towards different specialization directions, such as: 1) Customization - tailoring RAG to meet specific requirements. 2) Simplification - making RAG easier to use to reduce the

¹⁰<https://github.com/inverse-scaling/prize>

¹¹<https://github.com/weaviate/Verba>

¹²<https://aws.amazon.com/cn/kendra/>

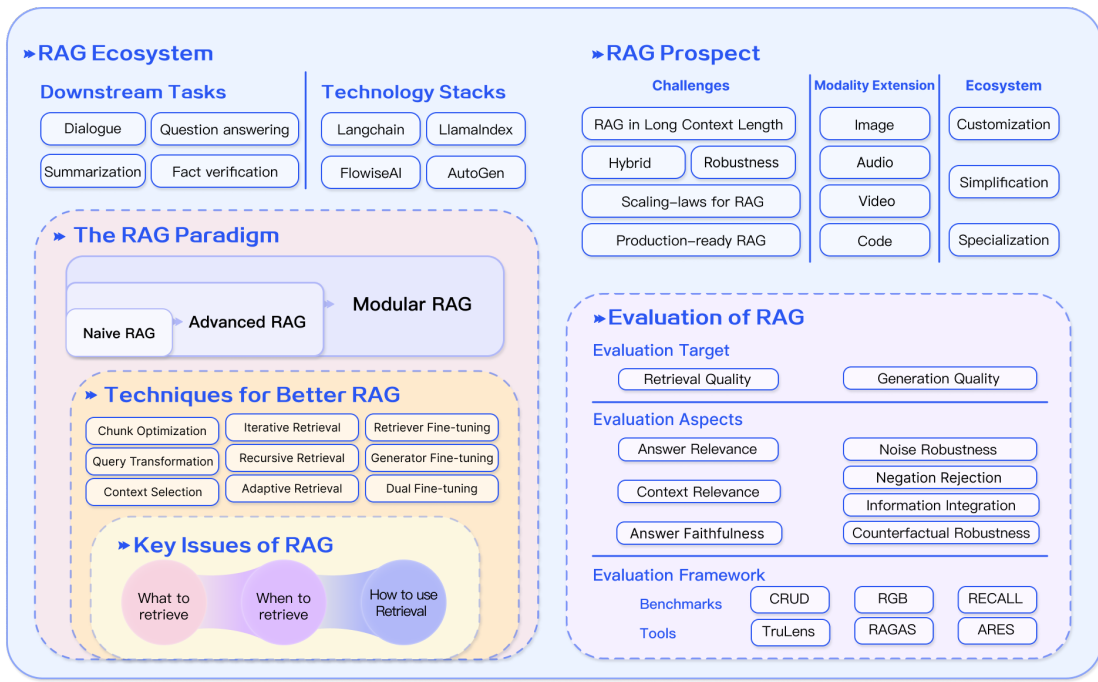


Fig. 6. Summary of RAG ecosystem

initial learning curve. 3) Specialization - optimizing RAG to better serve production environments.

The mutual growth of RAG models and their technology stacks is evident; technological advancements continuously establish new standards for existing infrastructure. In turn, enhancements to the technology stack drive the development of RAG capabilities. RAG toolkits are converging into a foundational technology stack, laying the groundwork for advanced enterprise applications. However, a fully integrated, comprehensive platform concept is still in the future, requiring further innovation and development.

F. Multi-modal RAG

RAG has transcended its initial text-based question-answering confines, embracing a diverse array of modal data. This expansion has spawned innovative multimodal models that integrate RAG concepts across various domains:

Image. RA-CM3 [176] stands as a pioneering multimodal model of both retrieving and generating text and images. BLIP-2 [177] leverages frozen image encoders alongside LLMs for efficient visual language pre-training, enabling zero-shot image-to-text conversions. The “Visualize Before You Write” method [178] employs image generation to steer the LM’s text generation, showing promise in open-ended text generation tasks.

Audio and Video. The GSS method retrieves and stitches together audio clips to convert machine-translated data into speech-translated data [179]. UEOP marks a significant advancement in end-to-end automatic speech recognition by incorporating external, offline strategies for voice-to-text conversion [180]. Additionally, KNN-based attention fusion leverages audio embeddings and semantically related text embeddings to refine ASR, thereby accelerating domain adaptation.

Vid2Seq augments language models with specialized temporal markers, facilitating the prediction of event boundaries and textual descriptions within a unified output sequence [181].

Code. RBPS [182] excels in small-scale learning tasks by retrieving code examples that align with developers’ objectives through encoding and frequency analysis. This approach has demonstrated efficacy in tasks such as test assertion generation and program repair. For structured knowledge, the CoK method [106] first extracts facts pertinent to the input query from a knowledge graph, then integrates these facts as hints within the input, enhancing performance in knowledge graph question-answering tasks.

VIII. CONCLUSION

The summary of this paper, as depicted in Figure 6, emphasizes RAG’s significant advancement in enhancing the capabilities of LLMs by integrating parameterized knowledge from language models with extensive non-parameterized data from external knowledge bases. The survey showcases the evolution of RAG technologies and their application on many different tasks. The analysis outlines three developmental paradigms within the RAG framework: Naive, Advanced, and Modular RAG, each representing a progressive enhancement over its predecessors. RAG’s technical integration with other AI methodologies, such as fine-tuning and reinforcement learning, has further expanded its capabilities. Despite the progress in RAG technology, there are research opportunities to improve its robustness and its ability to handle extended contexts. RAG’s application scope is expanding into multimodal domains, adapting its principles to interpret and process diverse data forms like images, videos, and code. This expansion highlights RAG’s significant practical implications for AI deployment, attracting interest from academic and industrial sectors.

The growing ecosystem of RAG is evidenced by the rise in RAG-centric AI applications and the continuous development of supportive tools. As RAG’s application landscape broadens, there is a need to refine evaluation methodologies to keep pace with its evolution. Ensuring accurate and representative performance assessments is crucial for fully capturing RAG’s contributions to the AI research and development community.

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